

**STUDIES ON DRY ROOT ROT (*Rhizoctonia
bataticola* Taub (Butler)) OF CHICKPEA (*Cicer
arietinum*).**

Ph.D. THESIS

by

PUSHPANJALI WAGH

**DEPARTMENT OF PLANT PATHOLOGY
COLLEGE OF AGRICULTURE
INDIRA GANDHI KRISHI VISHWAVIDYALAYA
RAIPUR (Chhattisgarh)**

2015

**STUDIES ON DRY ROOT ROT (*Rhizoctonia
bataticola* Taub (Butler)) OF CHICKPEA (*Cicer
arietinum*).**

Thesis

Submitted to the

Indira Gandhi Krishi Vishwavidyalaya, Raipur

by

PUSHPANJALI WAGH

**IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR
THE DEGREE OF**

Doctor of Philosophy

in

Agriculture

(Plant Pathology)

Roll No. 15426

ID No.: 130111005

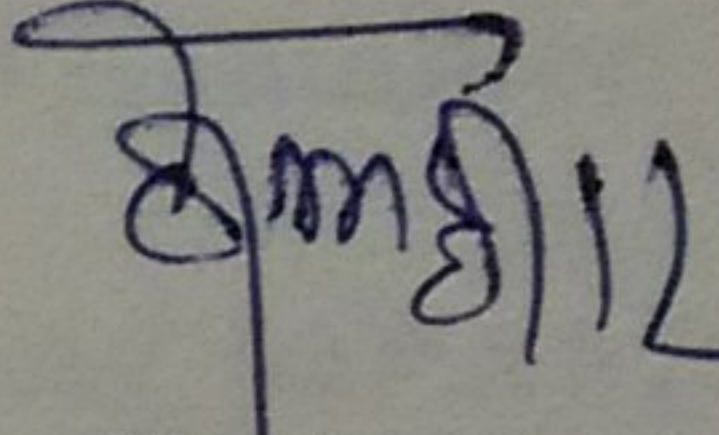
2015

CERTIFICATE – I

This is to certify that the thesis entitled “studies on dry root rot (*Rhizoctonia bataticola* Taub (Butler)) of chickpea (*Cicer arietinum*)” submitted in partial fulfilment of the requirements for the degree of “**Doctor of Philosophy**” of the Indira Gandhi Krishi Vishwavidyalaya, Raipur, is a record of the bonafide research work carried out by **Pushpanjali Wagh** under my guidance and supervision. The subject of the thesis has been approved by student’s Advisory Committee and the Director of Instructions.

No part of the thesis has been submitted for any other degree or diploma (certificate awarded etc.) or has been published / published part has been fully acknowledged. All the assistance and help received during the course of the investigations have been duly acknowledged by her.

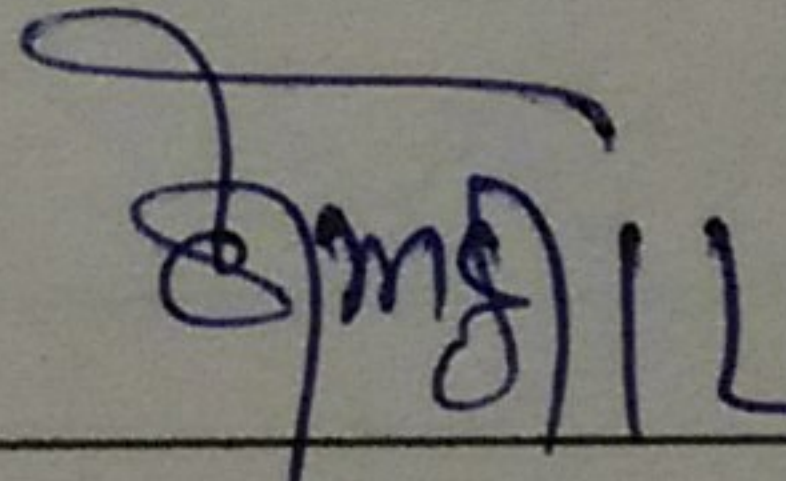
Date: 22/5/15


Chairman

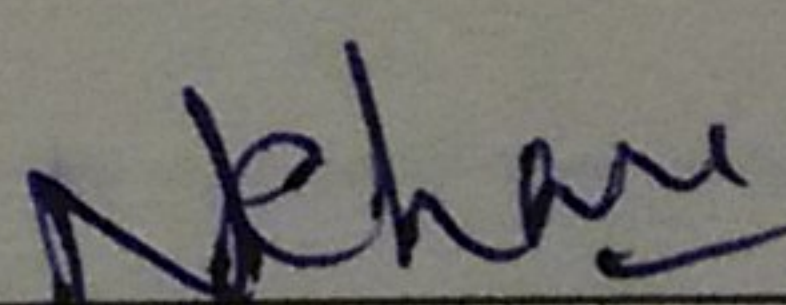
(Advisory Committee)

THESIS APPROVED BY THE STUDENT’S ADVISORY COMMITTEE

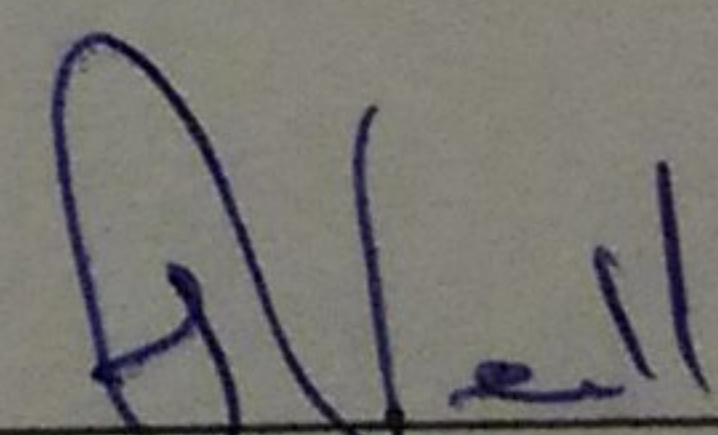
Chairman: **Dr. R. K. Dantre**



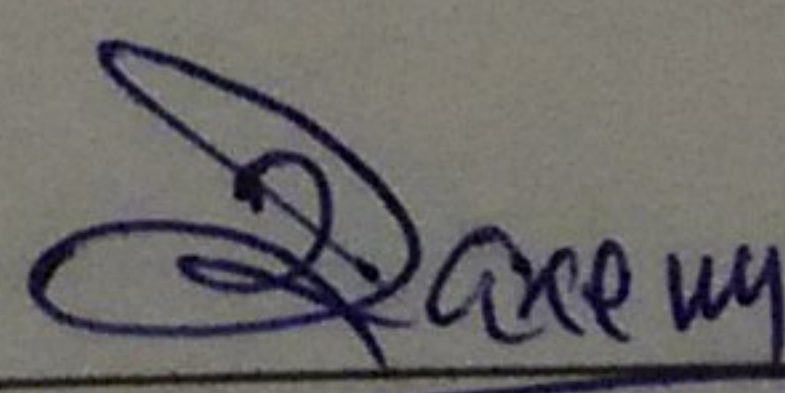
Member: **Dr. N. Khare**



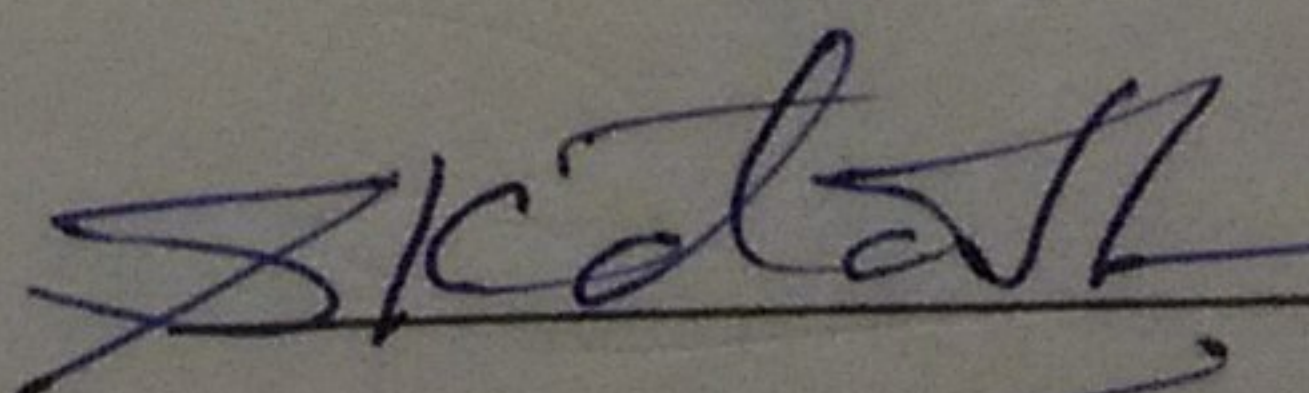
Member: **Dr. S. B. Verulkar**



Member: **Dr. R. R. Saxena**


22/5/15

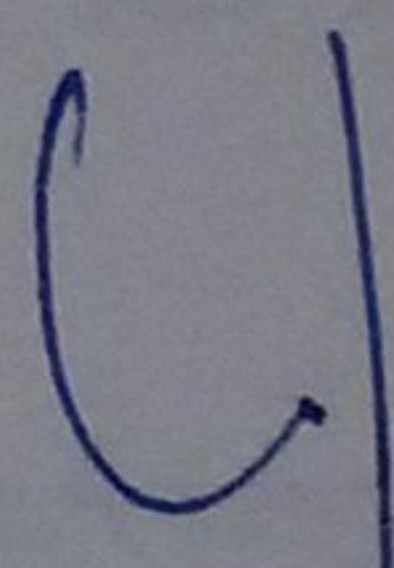
Member: **Dr. A. S. Kotasthane**



CERTIFICATE-II

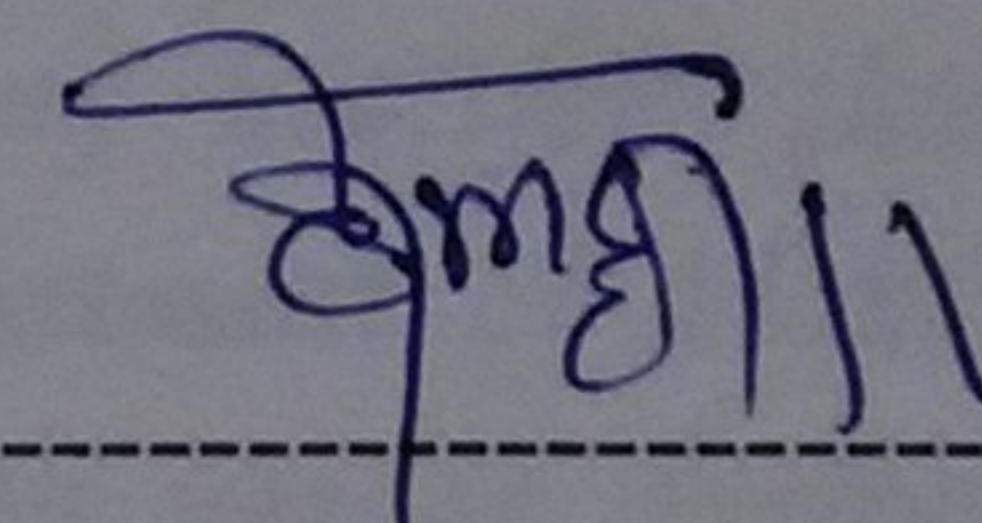
This is to certify that the thesis entitled "**Studies on dry root rot** (*Rhizoctonia bataticola* Taub (Butler)) of chickpea (*Cicer arietinum*)" by **Pushpanjali Wagh** to the Indira Gandhi Krishi Vishwavidyalaya, Raipur in partial fulfilment of the requirements for the degree of "Doctor of Philosophy" in the **Department of Plant Pathology** has been approved by the external examiner and student's Advisory Committee after oral examination.

Date: 14/9/15

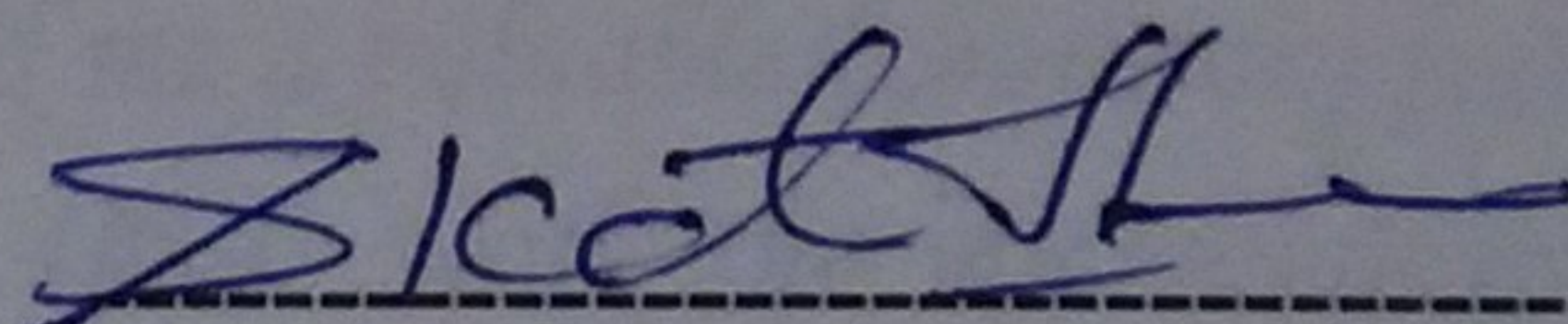

14 Sep 2015
External Examiner

(Name: C. CHATTOPADHYAY)

Major Advisor



Head of the Department



Dean/Dean Faculty

Approved/Not approved

Director of Instructions

ACKNOWLEDGMENT

On the accomplishment of the present study, I would like to take this opportunity to extend my deepest sense of gratitude and words of appreciation towards those, who helped me during the pursuit of study. I shall ever, remain thankfully indebted to all those learned souls, my present and former teachers, known and unknown hands who directly or indirectly motivated me to achieve my goal and enlightened me with the touch of their knowledge and constant encouragement. I feel this is an extremely significant and joyous opportunity bestowed upon me by the goddess of learning, to think about and thank all those persons.

*Words are inadequate in the available lexicon to express my gratitude and sincere thanks to my major advisor **Dr. R. K. Dantre** Professor, Department of Plant Pathology for his scholastic guidance, prudent planning, keen interest, constant critical supervision, cooperation and invaluable counsel throughout the pursuit of this study and preparation of this manuscript. His very cordial behavior would always remain beacon light for me in future also.*

*I am equally grateful to the members of advisory committee **Dr. N. Khare** Principal scientist Department of Plant Pathology, **Dr. S.B. Verulkar**, Professor and H.O.D of PMBB and **Dr. R. R. Saxena** Associate Professor, Department of Statistics, for providing me constant guidance, valuable suggestions and supportive attitude.*

*I take this opportunity to express my heartfelt thanks to **Dr. A. S. Kostasthane**, Professor and Head of the Department of Plant Pathology, for his guidance, encouragement and all necessary help needed during the course of the investigation. Especially for who provided me a valuable laboratory facility, rendering help to carry out the experiment for my thesis work the microbiology laboratory of PMBB.*

*I am deeply obliged to **Dr. Om Gupta** Dean, Professor and Head and **Dr. R. K. Verma** Principal scientist, Department of Plant Pathology, JNKVV, Jabalpur for his valuable help and suggestions.*

*My literal power is too less to express my thankfulness to respected teachers **Dr. V.S. Thrimurthy**, **Dr. G.K. Awadhiya**, **Dr. C.P. Khare**, **Dr. C.S. Shukla**, **Dr. K.P. Verma**, **Dr. N. Lakpale**, **Dr. P. K. Tiwari** and **Mr. H. K. Singh** Department of Plant Pathology, College of Agriculture, IGKV, Raipur. I extend my thanks to **Shri S.R. Tikariha**, **Shri Siyaram Sahu**, **Shri B. L. Sinha**, **Budharu uncle**, **Dhani Bhaiya**, **Bansi Bhaiya** and non-teaching staff of our department for their kind help, cooperation and encouragement.*

I wish to present my explicit thanks to Dr. S.K. Patil, Hon'ble Vice Chancellor, Dr. J.S. Urkurkar, Director Research Services, Dr. S.S. Shaw, Director of Instructions, Dr. M. P. Thakur, Director Extension Services and Dr. S. R. Patel Dean, College of Agriculture, IGKV, Raipur for providing necessary facilities, technical and administrative supports for conductance of my research work.

I wish to express my sincere thanks to all my dear friends Sandeep Bhaiya, Swati, Anu mam, Yogesh, Deepti and Priyadarshani for co-operation, help, encouragement and support. I feel pleasure in expressing my thanks.

Especial words of thanks to my seniors and all juniors for their timely help, support and wholehearted co-operation during the entire course of study.

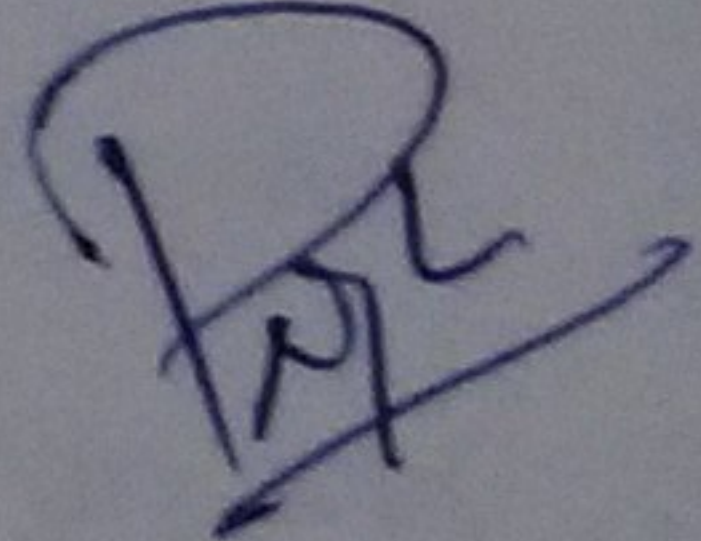
At this inexplicable moment of joy, I deem it a proud privilege to recall all the cooperation, contribution and the encouragement, critical counsel and inspiration of my husband Akash Dhok.

It is like a drop in the ocean of words that can never reach its mark to acknowledge infinite love, blessings, sacrifices and constant encouragement of all my family members, younger brothers Sankalp and Siddhant, and father and mother in-law who have been the sole source of inspiration for me to proceed ahead this course of time. I can't forget to give my special sincere thanks to my grandparents. I take this opportunity to dedicate this work to my beloved parents Mrs. Kavita Wagh and Mr. Sudhir Wagh.

Last but not the least, I thank all the individuals who have in one any way been associated with the completion of this work but have not been mentioned so far.

Place: Raipur

Date: 22nd / July / 15'



(Pushpanjali Wagh)

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO.
	ACKNOWLEDGEMENT	i
	TABLE OF CONTENTS	iii
	LIST OF TABLES	vii
	LIST OF FIGURE	xi
	LIST OF PLATES	xiii
	LIST OF ABBREVIATIONS	xiv
	ABSTRACT	xv
I	Introduction	1
II	Review of literature	7
	2.1 Chickpea (<i>Cicer arietinum</i> L.)	7
	2.1.1 Morphology	7
	2.1.2 Centre of origin and distribution	8
	2.1.3 Season	9
	2.1.4 Nutrition	9
	2.1.5 Yield and losses	10
	2.2 <i>Rhizoctonia bataticola</i>	11
	2.2.1 History and occurrence of <i>R. bataticola</i>	11
	2.2.2 Distribution and Economic Importance of <i>R. bataticola</i>	14
	2.2.3 Taxonomic position of <i>R. bataticola</i>	16
	2.2.4 Symptomatology	16
	2.2.5 Disease cycle	17
	2.2.6 Multiplication of the pathogen	18
	2.3 Morphological and cultural characters of <i>Rhizoctonia bataticola</i>	18
	2.4 Aggressiveness of <i>Rhizoctonia bataticola</i>	24
	2.5 Anastomosis	25
	2.5.1 Hyphal anastomosis	25
	2.5.2 Microscopic analysis of hyphal anastomosis	28
	2.6 Screening of chickpea entries	33
	2.7 Study of soil factors affecting development of dry root rot	36
	2.8 Identification of bio agent	39
III	Materials and methods	47
	3.1 Materials	47
	3.1.1 Experimental site	47
	3.1.2 Source of materials	47
	3.1.3 Source of seeds	47
	3.1.4 Cleaning and washing of glassware	48
	3.2 Methods	50
	3.2.1 Survey and collection of dry root rot infected plants	50

3.2.2 Isolation of pathogen from diseased samples	50
3.2.3 Purification and identification of isolates of <i>R. bataticola</i>	51
3.3 Variability among isolates of <i>R. bataticola</i>	51
3.3.1 Cultural and morphological characteristics of the isolates	51
3.3.1.1 Mycelial growth	51
3.3.1.2 Morphological variability	51
3.4 Aggressiveness test of <i>R. bataticola</i> isolates	51
3.4.1 Blotter paper technique (Nene <i>et al.</i> , 1981)	51
3.4.2 Mass multiplication of <i>R. bataticola</i> inoculum for soil application	52
3.4.3 <i>In vitro</i> Aggressiveness test	52
3.4.4 <i>In vivo</i> pot experiment	53
3.5 Anastomosis	53
3.6 Screening of chickpea entries	54
3.6.1 Field screening	53
3.6.2 <i>In vitro</i> screening	55
3.7 Soil factors affecting dry root rot	55
3.7.1 Effect of soil pH	55
3.7.2 Effect of soil type	56
3.7.3 Soil texture	56
3.8 Identification of bio control agent <i>Trichoderma</i> spp. as potential candidate against <i>R. bataticola</i>	57
3.8.1 Collection of soil samples	57
3.8.2 Isolation of <i>Trichoderma</i> spp. from collected soil sample	57
3.8.3 Morphological Characterization of the isolates of <i>Trichoderma</i> spp.	58
3.8.4 Microscopic Characterization of the isolates of <i>Trichoderma</i> spp.	58
3.8.5 Screening of <i>Trichoderma</i> spp. isolates	58
3.8.5.1 Dual culture interaction	58
3.8.5.2 <i>In vivo</i> efficacy of <i>Trichoderma</i> spp. under glass house condition	59
3.9 Identification of fluorescent <i>Pseudomonas</i> bio control agent as potential candidate against <i>R. bataticola</i>	60
3.9.1 Isolation of Fluorescent <i>Pseudomonas</i>	60
3.9.1.1 Collection of soil samples	60
3.9.1.2 Isolation of fluorescent <i>Pseudomonas</i> from collected soil sample by serial dilution (Aneja, 1993)	60
3.9.1.3 Isolation of fluorescent <i>Pseudomonas</i> and designation of isolates	60
3.9.2 Morphological and Biochemical characterization of the fluorescent <i>Pseudomonas</i> isolates	61
3.9.2.1 Morphological characterization	61

	3.9.2.1.1 Gram reaction	61
	3.9.2.2 Biochemical characterization	61
	3.9.2.2.1 Gelatin hydrolysis test	62
	3.9.2.2.2 Nitrate reduction test	62
	3.9.2.2.3 Starch Hydrolysis Test	63
	3.9.2.2.4 Casein hydrolysis	63
	3.9.2.2.5 Lipolytic activity with Tween 80	63
	3.9.2.2.6 Growth at 4°C and 42°C	64
	3.9.2.2.7 Antibiotic sensitivity test	64
	3.9.2.2.8 Triple Sugar Iron agar test	64
	3.9.2.2.9 Hydrogen sulphide gas production test	65
	3.9.3 Screening for antagonistic activity of fluorescent <i>Pseudomonas</i> against <i>Rhizoctonia bataticola</i>	65
	3.9.3.1 Dual culture interaction	65
	3.9.3.2 <i>In vivo</i> efficacy of fluorescent <i>Pseudomonas</i> under glass house condition	66
	3.10 Statistical analysis	67
IV	Results and Discussion	68
	4.1 Survey of chickpea crops to assess the incidence of the disease.	69
	4.2 Symptomatology	73
	4.3 Collection isolation purification and identification of the pathogen	74
	4.4 Cultural and morphological variability	75
	4.4.1 Mycelial growth	75
	4.4.2 Hyphal width	75
	4.4.3 Hyphal branching	76
	4.4.4 Type of mycelial growth	76
	4.4.5 Colony colour	77
	4.4.6 Sclerotial size	79
	4.4.7 Sclerotial shape	79
	4.4.8 Number of sclerotia per microscopic field	80
	4.4.9 Sclerotial development	80
	4.5 Pathogenicity test	81
	4.5.1 Aggressiveness of <i>R. bataticola</i> isolates	81
	4.5.1.1 <i>In vitro</i> blotter paper technique	81
	4.5.2.2 <i>In vivo</i> sick pot method	82
	4.6 Anastomosis Behaviour	86
	4.7 Screening of chickpea entries	87
	4.7.1 Field screening	87
	4.7.2 <i>In vitro</i> screening	111
	4.8 Soil factors affecting development of dry root rot caused by <i>R. bataticola</i> in chickpea	117
	4.8.1 Effect of Soil pH on dry root rot of chickpea	118
	4.8.2 Effect of Soil type on dry root rot of chickpea	121
	4.8.3 Effect of Soil texture on dry root rot of chickpea	124

4.9	Potentiality of bio control agent <i>Trichoderma</i> spp. against <i>R. bataticola</i>	128
4.9.1	Isolation of <i>Trichoderma</i> isolates	128
4.9.2	Cultural and Morphological characterization of <i>Trichoderma</i> isolates	129
4.9.2.1	Colony colour	129
4.9.2.2	Growth pattern	129
4.9.2.3	Appearance	129
4.9.2.4	Pustules	129
4.9.2.5	Pigments	129
4.9.2.6	Odour	129
4.9.3	Evaluation of antagonistic efficacy of <i>Trichoderma</i> spp. by dual culture method	131
4.9.4	Screening for mycoparasitic activity <i>Trichoderma</i> spp. isolates against <i>R. bataticola</i>	135
4.9.5	Mechanism of antagonism of <i>Trichoderma</i> spp. on <i>R. bataticola</i> causing dry root rot of chickpea.	136
4.9.6	Efficacy of <i>Trichoderma</i> against <i>R. bataticola</i> under <i>in vivo</i> condition	139
4.10	Potentiality of bio control agent fluorescent <i>Pseudomonas</i> against <i>R. bataticola</i>	145
4.10.1	Isolation of fluorescent <i>Pseudomonas</i> isolates	145
4.10.2	Screening of fluorescent <i>Pseudomonas</i>	145
4.10.3	Biochemical tests for characterization of fluorescent <i>Pseudomonas</i>	146
4.10.3.1	Gelatin hydrolysis test	146
4.10.3.2	Nitrate test	147
4.10.3.3	Starch Hydrolysis Test	147
4.10.3.4	Casein hydrolysis	148
4.10.3.5	Lypolytic activity	148
4.10.3.6	Growth at 4°C and 42°C	148
4.10.3.7	Antibiotic sensitivity test	149
4.10.3.8	Triple sugar iron agar test	149
4.10.3.9	Hydrogen sulphide test	149
4.10.4	Evaluation of antagonistic efficacy of fluorescent <i>Pseudomonas</i> by dual culture method	153
4.10.5	Efficacy of fluorescent <i>Pseudomonas</i> against <i>R. bataticola</i> under <i>in vivo</i> condition	157
V	SUMMARY AND CONCLUSION	165
	REFFERNCES	174
	APPENDIX	204
	VITA	206

LIST OF TABLES

TABLE	TITLE	PAGE
3.1	Composition of different media used in present investigation	48
3.2	Description of the rating scale for scoring	52
3.3	Reaction type and its description to group	54
3.4	IIPR rating scale	55
3.5	Reaction for Triple sugar iron agar test	65
4.1	Incidence of chickpea dry root rot in different location of Chhattisgarh	70
4.2	Isolates of <i>R. bataticola</i> , collected from different chickpea growing locations of Chhattisgarh	74
4.3	Cultural and Morphological characteristics of various isolates of <i>Rhizoctonia bataticola</i>	77
4.4	Variability in morphology of sclerotia produce by different isolates of <i>R. bataticola</i>	80
4.5	Aggressiveness of different isolates of <i>Rhizoctonia bataticola</i> under <i>in vitro</i> condition by blotter paper method	82
4.6	Aggressiveness of <i>Rhizoctonia bataticola</i> isolates causing dry root rot of chickpea under glass house condition	83
4.7	Effect of <i>Rhizoctonia bataticola</i> isolates on germination, shoot and root length of chickpea under sick pot soil	84
4.8	Screening of chickpea entries (Desi I) against dry root rot during 2012-2013	88
4.9	Reaction of chickpea entries (Desi I) against dry root rot during 2012-2013	89
4.10	Screening of chickpea entries (Desi II) against dry root rot during 2012-2013	89
4.11	Reaction of chickpea entries (Desi II) against dry root rot during 2012-2013	89
4.12	Screening of chickpea entries (Desi III) against dry root rot during 2012-2013	90
4.13	Reaction of chickpea entries (Desi III) against dry root rot during 2012-2013	91
4.14	Screening of chickpea entries (late sown I) against dry root rot during 2012-2013	91
4.15	Reaction of chickpea entries (late sown I) against dry root rot during 2012-2013	92
4.16	Screening of chickpea entries (late sown II) against dry root rot during 2012-2013	92

4.17	Reaction of chickpea entries (late sown II) against dry root rot during 2012-2013	93
4.18	Screening of chickpea entries (Rain fed I) against dry root rot during 2012-2013	93
4.19	Reaction of chickpea entries (Rain fed I) against dry root rot during 2012-2013	94
4.20	Screening of chickpea entries (Rain fed II) against dry root rot during 2012-2013	94
4.21	Reaction of chickpea entries (Rain fed II) against dry root rot during 2012-2013	95
4.22	Screening of chickpea entries (national nursery for elite lines) against dry root rot during 2012-2013	95
4.23	Reaction of chickpea entries (national nursery for elite lines) against dry root rot during 2012-2013	96
4.24	Screening of chickpea entries (Desi I) against dry root rot during 2013-2014	96
4.25	Reaction of chickpea entries (Desi I) against dry root rot during 2013-2014	97
4.26	Screening of chickpea entries (Desi II) against dry root rot during 2013-2014	97
4.27	Reaction of chickpea entries (Desi II) against dry root rot during 2013-2014	98
4.28	Screening of chickpea entries (late sown I) against dry root rot during 2013-2014	98
4.29	Reaction of chickpea entries (late sown I) against dry root rot	99
4.30	Screening of chickpea entries (late sown II) against dry root rot during 2013-2014	99
4.31	Reaction of chickpea entries (late sown II) against dry root rot during 2013-2014	100
4.32	Screening of chickpea entries (Rain fed I) against dry root rot during 2013-2014	100
4.33	Reaction of chickpea entries (Rain fed I) against dry root rot during 2013-2014	101
4.34	Screening of chickpea entries (Rain fed II) against dry root rot	101
4.35	Reaction of chickpea entries (Rain fed II) against dry root rot during 2013-2014	102
4.36	Screening of chickpea entries (Kabuli I) against dry root rot during 2013-2014	102
4.37	Reaction of chickpea entries (Kabuli I) against dry root rot during 2013-2014	103
4.38	Screening of chickpea entries (Kabuli II) against dry root rot during 2013-2014	103
4.39	Reaction of chickpea entries (Kabuli II) against dry root rot during 2013-2014	104
4.40	Screening of chickpea entries (Kabuli III) against dry root rot during 2013-2014	104

4.41	Reaction of chickpea entries (Kabuli III) against dry root rot during 2013-2014	105
4.42	Screening of chickpea entries (ELSK I) against dry root rot during 2013-2014	106
4.43	Reaction of chickpea entries (ELSK I) against dry root rot during 2013-2014	106
4.44	Screening of chickpea entries (ELSK II) against dry root rot during 2013-2014	106
4.45	Reaction of chickpea entries (ELSK II) against dry root rot during 2013-2014	107
4.45	Screening of chickpea entries (ELSK III) against dry root rot during 2013-2014	107
4.47	Reaction of chickpea entries (ELSK III) against dry root rot during 2013-2014	108
4.48	Screening of chickpea entries (national nursery for elite lines) against dry root rot during 2013-2014	109
4.49	Reaction of chickpea entries (national nursery for elite lines) against dry root rot during 2013-2014	110
4.50	<i>In vitro</i> screening of chickpea by blotter paper method against <i>R. bataticola</i> (RB5)	112
4.51	Reaction of the entries under <i>in vitro</i> condition by blotter paper method	116
4.52	Effect of different pH on disease development of dry root rot of chickpea	117
4.53	Effect of pH on germination, root and shoot length of chickpea in inoculated soil with <i>R. bataticola</i>	118
4.54	Effect of different soil type on disease development of dry root rot of chickpea	121
4.55	Effect of soil type on germination, shoot and root length of chickpea in inoculated soil with <i>R. bataticola</i>	122
4.56	Effect of different soil texture on disease development of dry root rot of chickpea	124
4.57	Effect of soil texture on germination, shoot and root length of chickpea in inoculated soil with <i>R. bataticola</i>	125
5.58	Isolates of <i>Trichoderma</i> spp. collected from different locations of Chhattisgarh	128
4.59	Morphological characteristics of <i>Trichoderma</i> isolates based on key characters	130
4.60	Antagonistic efficacy of <i>Trichoderma</i> spp. against isolates of <i>R. bataticola</i> in dual culture technique	133
4.61	Screening for mycoparasitic activity of <i>Trichoderma</i> spp. isolates against <i>R. bataticola</i>	136
4.62	Efficacy of <i>Trichoderma</i> spp. against dry root rot of chickpea	140
4.63	Effect of <i>Trichoderma</i> spp. on germination, shoot and root length of chickpea	143

4.64	Isolates of fluorescent <i>Pseudomonas</i> collected from different locations of Chhattisgarh	146
4.65	Biochemical tests for characterization of fluorescent <i>Pseudomonas</i>	150
4.66	Antibiotic sensitivity test to confirm the identity of fluorescent pseudomonas	151
4.67	Biochemical test to confirm the identity of fluorescent <i>Pseudomonas</i> through Triple Sugar Iron Agar test and Hydrogen Sulphide Test	152
4.68	Antagonistic efficacy of fluorescent <i>Pseudomonas</i> against isolates of <i>R. bataticola</i> in by dual culture technique	155
4.69	Efficacy of fluorescent <i>Pseudomonas</i> on the incidence of dry root rot of chickpea	158
4.70	Effect of fluorescent <i>Pseudomonas</i> on germination, shoot and root length of chickpea under sick pot soil	161

LIST OF FIGURES

FIGURE	TITLE	PAGE
4.1	Disease incidence in different chickpea growing areas of Chhattisgarh	71
4.2	Disease incidence in different soil types of Chhattisgarh	715
4.3	Disease incidence in different types of seed in Chhattisgarh	72
4.4	Disease incidence influenced by availability of moisture (irrigated and rainfed) in Chhattisgarh	72
4.5	Mycelial growth rate of different <i>R. bataticola</i> isolates	78
4.6	Aggressiveness of different isolates of <i>Rhizoctonia bataticola</i> under <i>in vitro</i> condition by blotter paper method	82
4.7	Per cent mortality of chickpea plants by <i>R. bataticola</i> isolates causing dry root rot under glass house condition	85
4.8	Disease rating of different isolates of <i>R. bataticola</i> under glass house condition	85
4.9	Effect of <i>R. bataticola</i> isolates on germination, root and shoot length of chickpea under pot condition	85
4.10	Per cent mortality of chickpea due to dry root rot under glass house condition at different soil pH	120
4.11	Disease rating of chickpea dry root rot at different pH	120
4.12	Effect of pH on germination, root and shoot length of chickpea in inoculated soil with <i>R. bataticola</i>	120
4.13	Per cent mortality of chickpea due to dry root rot of under glass house condition at different soil type	123

4.14	Disease rating of chickpea dry root rot of at different soil type	123
4.15	Effect of soil type on germination, root and shoot length of chickpea in inoculated soil with <i>R. bataticola</i>	123
4.16	Per cent mortality of chickpea due to dry root rot under glass house condition at different soil texture	127
4.17	Disease rating of chickpea dry root rot at different soil texture	127
4.18	Effect of different soil texture on germination, root and shoot length of chickpea in inoculated soil with <i>R. bataticola</i>	127
4.19	Efficacy of <i>Trichoderma</i> spp. against dry root rot of chickpea	141
4.20	Disease rating scored from different <i>Trichoderma</i> isolates against <i>R. bataticola</i> (RB4) causing dry root rot of chickpea	141
4.21	Effect of <i>Trichoderma</i> spp. on germination, root and shoot length of chickpea	144
4.22	Efficacy of fluorescent <i>Pseudomonas</i> on the incidence of dry root rot of chickpea	159
4.23	Disease rating scored from different fluorescent <i>Pseudomonas</i> isolates against <i>R. bataticola</i> (RB4) causing dry root rot of chickpea	159
4.24	Effect of fluorescent <i>Pseudomonas</i> on germination, root and shoot length of chickpea	162

LIST OF PLATES

PLATE	TITLE	PAGE NO.
1.	Multiplication of <i>Rhizoctonia bataticola</i> inoculum on potato dextrose broth	51-52
2.	Disease rating scale	52-53
3.	Symptoms of dry root rot under natural field condition	73-74
4.	Cultural and morphological variability of <i>R. bataticola</i> isolates	78-79
5.	Pattern of hyphal branching in <i>R. bataticola</i>	78-79
6.	Microphotograph of sclerotia of different <i>R. bataticola</i> isolates	80-81
7.	Microphotograph of sclerotial development	80-81
8.	Extent of root damage in <i>in vitro</i> aggressiveness of <i>R. bataticola</i> isolates through blotter paper technique	82-83
9.	<i>In vivo</i> aggressiveness of <i>R. bataticola</i> isolates under glass house pot condition	85-86
10.	Extent of root damage in <i>in vivo</i> aggressiveness test of <i>R. bataticola</i> isolates	85-86
11.	Anatomosis among <i>R. bataticola</i> isolates exhibiting self – compatibility (reaction type 3)	86-87
12.	<i>In vitro</i> screening of chickpea against <i>R. bataticola</i> by blotter paper method	116-117
13.	Effect of different soil pH on disease development of dry root rot of chickpea	120-121
14.	Effect of different soil type on disease development of dry root rot of chickpea	123-124
15.	Effect of different soil texture on disease development of dry root rot of chickpea	127-128
16.	Cultural and phialide morphology of different <i>Trichoderma</i> isolates	130-131
17.	Best antagonistic efficacy of <i>Trichoderma</i> spp. by dual culture against <i>R. bataticola</i>	134-135
18.	Intercoiling and lysis of mycelium of <i>R. bataticola</i> with <i>Trichoderma</i>	138-139
19.	Best results of chickpea seeds bioprimered <i>Trichoderma</i> with in sick pots	141-142
20.	Antagonistic efficacy of fluorescent <i>Pseudomonas</i> by dual culture against <i>R. bataticola</i>	156-157
21.	Best results of chickpea seeds bioprimered with fluorescent <i>Pseudomonas</i> in sick pots	159-160

LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
BOD	Biological oxygen demand
B.C.	Before christ
cm	Centimeter
<i>et al</i>	Co worker
CD	Critical difference
cv	Cultivar
DAI	Days after inoculation
DAS	Days after sowing
°C	Degree Celsius
Fig	Figure
G	Gram
Ha	Hectare
Kg	Kilogram
L	Liter
Mt	Metric tonne
ml	Milliliter
m ha	Million hectare
viz.	Namely
No.	Number
ppm	Part per million
%	Per cent
/	Per, also means and or
S.N.	Serial number
SEm±	Standard error of means
i.e.	That is

THESIS ABSTRACT

Title of the Thesis	Studies on dry root rot (<i>Rhizoctonia bataticola</i> Taub (Butler)) of chickpea (<i>Cicer arietinum</i>).
Full Name of the Student	Pushpanjali Wagh
Major Subject	Plant Pathology
Name and Address of the Major Advisor	Dr. R. K. Dantre, Professor, Department of Plant Pathology, College of Agriculture, IGKV, Raipur (C.G.)
Degree to be Awarded	Doctor of Philosophy

Signature of Student

Signature of Major Advisor

Date: _____

**Signature of Head of
Department**

ABSTRACT

Studies on dry root rot (*Rhizoctonia bataticola* Taub (Butler)) of chickpea (*Cicer arietinum*).

Pushpanjali Wagh

Nine isolates of *Rhizoctonia bataticola* causing dry root rot in chickpea were collected, isolated and purified from different chickpea growing area of Chhattisgarh. The isolates differed markedly in terms of types of mycelia growth, colony colour, sclerotial number, shape, size and number of sclerotia per microscopic field. Isolates when paired (in all possible combinations), exhibited self-compatibility that is fusion of wall and membrane and showed homologous anastomosis (reaction type 3). Aggressiveness of the isolates [*in vitro* (blotter paper method) and *in vivo* (sick soil method)] revealed differences in pathogenic ability. Under *in vitro* condition the Raigarh isolate (RB 5) was found more virulent, however, under *in vivo* condition the Kawardha isolate (RB 4) was highly virulent. Isolate originating from Raigarh (RB 5) expressed more virulence when pathogenicity assays were followed *in vitro* (blotter paper method), whereas isolate originating from Kawardha (RB4) expressed virulence when pathogenicity assays were followed *in vivo* (sick soil method). Inoculum (mass multiplied in sand maize media) of *Rhizoctonia bataticola* isolate (RB 4) was used to screen 118 and 220 chickpea lines for tolerance to dry root rot during *rabi* 2012 and 2013, respectively. During first season 2012- 2013 in Desi III only one entry was moderately resistant (BG 3033), three entries NDG 11-12, RKG 2020 22 and H 09-90 were tolerant. Whereas, Phule G 0752 was moderately resistant and JG 33 and CSJ 741 was tolerant while, remaining entries were susceptible to dry root rot. Two entries RSG 9056 and CSJ 8023 were found tolerant, while remaining all the entries were susceptible to dry root rot. While during second season (2013-14) CSJ 809 was found moderately resistant whereas, JG 17, CSJ 808, CSJ 556, JG 24, CSJ 815, GLK 28127, JG 27, JG 36 were tolerant. From the set of accessions of 220 chickpea entries which were undertaken during second *rabi* season 2014- 2015, 132 entries were screened *in vitro* through blotter paper method against *R. bataticola* isolate RB 5 (Raigarh) which was most virulent under *in vitro* aggressiveness test. Variety L550 of chickpea was used as susceptible check on blotter paper technique (Nene *et al.* 1981). Out of 132 entries, 21 entries categorized as resistant, followed by 19 entries categorized as moderately While, 21 entries grouped as tolerant, 53 entries were susceptible followed by 18 entries were highly susceptible .

Significantly least mortality per cent (12.5%) was recorded at pH 8.0 followed by pH 4 with mortality per cent of 41.67%.Extent of root damage varied from 3-9 grades and least disease grade was also recorded at pH 8.0 with minimum root damage and infection that is, 3 rating, With regard to soil type, significantly least mortality per cent (16.66%) was recorded in Matasi followed by dorsa with mortality per cent of 20.83% which were at par with each other. Much less root damage was too observed in Matasi with minimum infection that is, 1.6 rating, followed by Dorsa and Bhata with disease rating 5.6. In case of soil texture

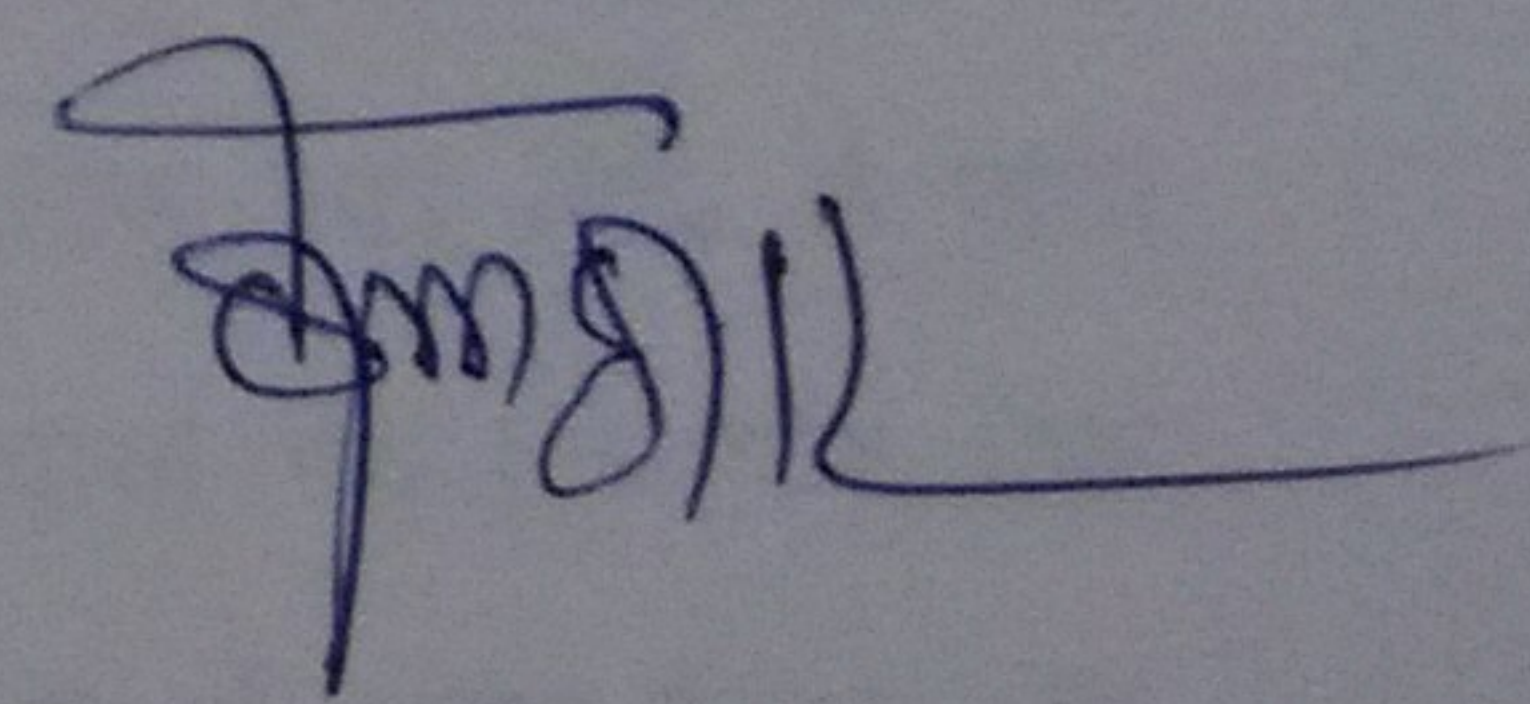
damage was too observed in matasi with minimum infection that is, 1.6 rating, followed by dorsa and bhata with disease rating 5.6. In case of soil texture significantly least mortality per cent (20.8%) was recorded in sandy loam texture followed by sandy texture with mortality per cent of 25.0% which were at par with each other. Extent of root damage varied greatly from 1-7 grades and lowest disease grade was recorded in sandy loam with minimum root damage and infection that is, 1.6 rating, followed by sandy and loam with disease rating 3.6 and 5 respectively.

Isolates of *Trichoderma* spp. (15 isolates) and fluorescent *Pseudomonas* (22 isolates) were isolated from dry root rot infested areas. For isolation of *Trichoderma* and *Pseudomonas* soil was derived from chickpea rhizosphere. Both the all *R. bataticola* isolates were screened against collected *Trichoderma* and *Pseudomonas*.

Fifteen isolates of *Trichoderma* were evaluated for their efficacy against nine isolates of *R. bataticola*. Isolate T40 was found significantly superior among all the isolates of *Trichoderma* in reducing the mycelial growth of *R. bataticola* by 67.65% over control, followed by T36 (64.81%), T14 (64.07%) and T29 (63.21%) which were at par with each other. The average mortality of chickpea in sick pots after 30 days of challenge inoculation was minimum in T40 (Bemetara) and T 23 (Bilaspur) that is 20.83 per cent. Diseases grade observed was 4.3 out of 9 in both the isolates whereas, maximum in T23 (Bilaspur) isolate that is 8.3.

Fluorescent *Pseudomonas* were screened for confirmation with following biochemical test gelatin hydrolysis test, nitrate test, starch hydrolysis test, lypolytic test, casein hydrolysis test, growth test at 4°C and 42°C, hydrogen sulphide gas production, triple sugar iron test and antibiotic sensitivity test. Fluorescent *Pseudomonas* inhibited the mycelial growth by the antagonistic activity across all isolates of *R. bataticola*. The maximum growth inhibition was recorded by P33 (Lohara) isolate (63.21%) followed by P40 (Bemetara) isolate (62.35%) and P36 (Dhamdha) isolate (61.91%). The average mortality of chickpea in sick pots after 30 days of challenge inoculation with RB4 (Kawardha) was recorded minimum in P3 (Rajnandgoan), P8 (Raigarh) and P40 (Bemetara), which were significantly superior over control, with 12.5 per cent mortality and disease grading on the basis of extent of root damage were 3.66, 5.66 and 5.66 respectively.

Date : 22/2/15



Dr. R. K. Dantre
(Major Advisor)
(Raipur (C.G.))

सारांश

चने(साइसर एरीटीनमे) का सूखा जड़ सड़न (राइजोक्टोनिया बटाटीकोला टौबे) (बटलर))

का अध्ययन

पुष्पाजली वाघ

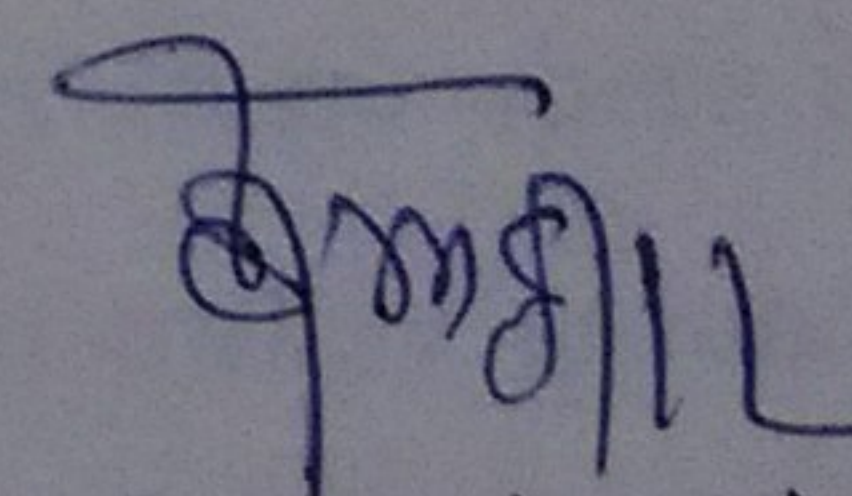
वर्तमान षोध शीर्षक “चने (साइसर एरीटीनमे) में सूखा जड़ सड़न(राइजोक्टोनिया बटाटीकोला) का अध्ययन” के अंतर्गत छत्तीसगढ़ में चना उत्पादन क्षेत्र से चने में सूखा जड़ सड़न रोग के रोगजनक (राइजोक्टोनिया बटाटीकोला) की नौ पृथक्कृतों का एकत्रीकरण, पृथक्करण एवं शुद्धीकरण किया गया, उपरोक्त पृथक्कृतों के कवकजाल वृद्धि, प्रारंभिक वृद्धि के रंग, आकार, आकृति एवं बीजाणु संख्या प्रति सूक्ष्मदर्शी क्षेत्र में भिन्नतायें पाई गयी। पृथक्कृतों का मिलन (समस्त संभव संयोजन में) स्वयं संगमता प्रदर्शित करते हुये पाया गया जो कि भित्ती एवं झिल्ली का संलग्न है एवं एक समान शाखा मिलन प्रदर्शित करता है। प्रयोगशाला के अन्दर (ब्लोटर पेपर विधि) एवं प्रयोगशाला के बाहर (रोगग्रसित मृदा विधि) में पृथक्कृतों का उग्रता अध्ययन रोगजनक की रोगजनन क्षमता में भिन्नतायें पाई गयी। समस्त पृथक्कृतों में से रायगढ़ पृथक्कृत (RB5) प्रयोगशाला के अन्दर अधिक उग्र पाया गया, जबकि प्रयोगशाला के बाहर कवर्धा पृथक्कृत (RB4) में अधिक उग्रता परिलक्षित हुआ। प्रयोगशाला में रोगजनकता विप्लेषण करने पर रायगढ़ (RB5) से प्राप्त पृथक्कृत अधिक उग्रता प्रदर्शित करते हुये अवलोकित किया गया, जबकि प्रयोगशाला के बाहर रोगजनकता विप्लेषण करने पर कवर्धा (RB4) से प्राप्त पृथक्कृत अधिक उग्रता व्यक्त करते पाया गया। रबी 2012 एवं 2013 के दौरान चने की 118 एवं 220 प्रजाति (इंट्री) की रोगसहनशीलता जाँच हेतु राइजोक्टोनिया बाटाटीकोला (RB4) का निवेश द्रव्य (रेत मक्का माध्यम में समूह गुणन) प्रयोग किया गया। प्रथम वर्ष 2012–2013 के दौरान केवल देशी तीन प्रजाति (इंट्री) (BG-3033) में सामान्य प्रतिरोधकता पाया गया एवं तीन प्रजाति (इंट्री) क्रमशः NDH 11-12, RKG 2020-22 तथा H 09-90 सहनशील पाया गया, जबकि फुले (G-0752) में सामान्य प्रतिरोधक क्षमता पाया गया एवं परंतु शेष स प्रजाति (इंट्री) सूखा जड़ सड़न रोग हेतु संवेदनशील पाया गया। दो प्रजाति (इंट्री) (RSG 9056) एवं (CSJ 8023) सहनशील पाया गया जबकि शेष समस्त प्रजाति (इंट्री) शुष्क सूखा जड़ सड़नरोग के प्रति संवेदनशील पाया गया। वर्ष 2013–2014 के दौरान (CSJ 809) में सामान्य प्रतिरोधकता देखा गया जबकि (JG 17, CSJ 808, CSJ 556, JG 24, CSJ 815, GLK 28127, GJ 27, JG 36) सहनशील पाया गया। रबी 2014–2015 के दौरान प्रयोगशाला के अंदर राइजोक्टोनिया बाटाटीकोला के विपरीत 220 प्रजाति (इंट्री) में से 132 प्रजाति (इंट्री) की ब्लोटर पेपर विधि से जाँच की गई जिसमे पृथक्कृत (RB5) रायगढ़ उग्रता परीक्षण करने पर अधिक उग्र पाया गया। ब्लोटर पेपर विधि में चने की (L 550) किस्म को संवेदनशील किस्म को संवेदनशील चिन्ह के रूप में प्रयोग किया गया, (नने एवं सहकार्यकर्ता 1981)। समस्त 132 प्रजाति (इंट्री) मे से 56 प्रजाति (इंट्री) की 1–5 रोग श्रेणी अंक के साथ प्रतिरोधक संवर्ग में रखा गया इसके पश्चात् 19 प्रजाति (इंट्री) को 6 रोग श्रेणी अंक के साथ संवेदनशील संवर्ग में रखा गया जबकि 58 प्रजाति (इंट्री) को 7–9 रोग श्रेणी अंक के साथ अतिसंवेदनशील की श्रेणी में रखा गया।

पी. एच. मान 8.0 पर न्यूनतम मर्त्यता (12.5 प्रतिषत) एवं पी. एच. मान 4 पर 41.67 प्रतिषत मर्त्यता पाया गया। मूल रोग का विस्तार 3-9 श्रेणी के बीच एवं पी. एच. मान 8.0 पर न्यूनतम रोग श्रेणी 4 के साथ न्यूनतम मर्त्यता 16.66 प्रतिषत के साथ मटासी मृदा में देखा गया इसके पश्चात् डोरसा में 20.83 प्रतिषत मर्त्यता पाया गया एवं सभी एक दुसरे के समतुल्य थे। मटासी मृदा में न्यूनतम रोग 1.6 श्रेणी वर्ग के साथ क्षति पाया गया, जिसके पश्चात् भाटा एवं डोरसा मृदा में 5.6 रोग श्रेणी वर्ग पाया गया। मृदा कणाकार के आधार पर अध्ययन करने पर रेतीली दोमट कणाकार में न्यूनतम मर्त्यता (20.8 प्रतिषत) इसके पश्चात् 25.0 प्रतिषत मर्त्यता रेतीली कणाकार में पाया गया जो कि एक दुसरे के समतुल्य थे। मूल क्षति का विस्तार 1-7 श्रेणी के मध्य पाया गया एवं बलुई दोमट मृदा में 1.6 श्रेणी वर्ग न्यूनतम रोग श्रेणी के साथ न्यूनतम मूल रोग एवं संक्रमण अवलोकित किया गया जिसके पश्चात् रेतीली एवं बलुई दोमट में क्रमशः 3.6 एवं 5 रोग श्रेणी वर्ग पाया गया।

सूखा जड़ सड़न संक्रमित क्षेत्र से *ट्राइकोडर्मा* प्रजाति (15 पृथक्कृत) एवं प्रतिदिप्त *स्यूडोमोनास* (22 पृथक्कृत) का पृथक्करण किया गया। पृथक्करण हेतु चने की जड़ सतह से मृदा नमूना एकत्र किया गया। *स्यूडोमोनास* एवं *ट्राइकोडर्मा* के विपरित समस्त *राइजोक्टोनिया बाटाटीकोला* को जॉचा गया। *ट्राइकोडर्मा* (15 पृथक्कृत) का *राइजोक्टोनिया बाटाटीकोला* (9 पृथक्कृतों) के विपरित प्रभाव जॉचा गया, जिसमें *ट्राइकोडर्मा* की टी 40 पृथक्कृत *राइजोक्टोनिया बाटाटीकोला* की कवकजाल वृद्धि की कम करने में 67.65 प्रतिषत घटाव के साथ श्रेष्ठ पाया गया इसके पश्चात् टी 36 (64.81 प्रतिषत) टी 14 (64.07 प्रतिषत) एवं टी 29 (63.21 प्रतिषत) श्रेष्ठ पाया गया एवं सभी एक दुसरे के समतुल्य पाये गये। टी 40 (बेमेतरा) एवं टी 23 बिलासपुर में अतः क्रमशः के 30 दिन पश्चात् चने की मर्त्यता 20.83 प्रतिषत अवलोकित किया गया दोनो पृथक्कृतों को 9 मे से 4.3 श्रेणी प्रदान किया गया जबकि टी 23 बिलासपुर को अधिकतम 8.3 श्रेणी प्रदान किया गया।

पुष्टीकरण हेतु प्रतिदिप्त *स्यूडोमोनास* को विभिन्न जैवरासायनिक परीक्षणों जैसे कि जिलेटिन हाइड्रोलासीस परीक्षण, 4°C एवं 42°C पर विकास परीक्षण हाइड्रोजन सल्फाइड गैस परीक्षण, ट्रीपल आयरन सुगर परीक्षण एवं प्रतिजैविक संवेदनशीलता परीक्षण द्वारा जॉचा गया। प्रतिदिप्त *स्यूडोमोनास* प्रतिद्वन्दी क्रियाकलाप द्वारा *राइजोक्टोनिया बाटाटीकोला* के समस्त पृथक्कृतों की कवकजालों वृद्धि को रोकते हुये देखा गया, सर्वाधिक कवकजाल वृद्धि निषेध इसके पश्चात् पी 33 (लोहारा) पृथक्कृत में पाया गया, इसके पश्चात् पी 40 (बेमेतरा) पृथक्कृतों में (62.35 प्रतिषत) एवं पी 36 (धमधा) पृथक्कृतों में (61.91 प्रतिषत) पाया गया। पी 3 राजनांदगाँव के अंतः क्रमण के 30 दिन पश्चात् चने की औसत मर्त्यता RB 4 (कवर्धा) पी 3 (राजनांदगाँव), पी 8 (रायगढ़) एवं पी 40 बेमेतरा में मूल क्षति विस्तार के आधार पर 12.5 प्रतिषत मर्त्यता के साथ न्यूनतम पाया गया, जो कि नियंत्रित पौधे से श्रेष्ठ रहा।

दिनांक 22/1/15
रायपुर (छ.ग.)


डॉ. आर. के. दान्त्रे
(प्रमुख मार्गदर्शक)

CHAPTER-I

INTRODUCTION

Among the grain legumes, chickpea/ bengal gram (*Cicer arietinum* L.) has occupied a prominent position. It is the third most important pulse crop of the world, next to beans (*Phaseolus vulgaris*) and peas (*Pisum sativum*). Out of total production of pulses, chickpea accounts for 37 per cent and area wise 28.28 per cent. India contributes the major share to the global chickpea area (65%) and production (68%). Indian Council of Medical Research (ICMR) has recommended an intake of 50 g of pulse/ capita/ day.

Pulses including chickpea are increasingly being used in health conscious diet to promote well being and reduce the risk of illness. Chickpea is low in fat and most of this is polyunsaturated. One hundred grams of mature boiled chickpeas contains 164 calories, 2.6g of fat (of which only 0.27g is saturated), 7.6g of dietary fibre and 8.9g of protein. Chickpeas also provide dietary calcium (49–53mg/100g). According to the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) chickpea seeds contain, 23% protein, 64% total carbohydrates (47% starch, 6% soluble sugar), 5% fat, 6% crude fiber and 3% ash. High mineral content has been reported for phosphorus (340mg/100g), calcium (190mg/100g), magnesium (140mg/100g), iron (7mg/100g) and zinc (3mg/100g). Therefore, chickpea is an excellent heart healthy food that may be beneficial of coronary and cardiovascular disease and by reducing blood lipids also help in some serious complication of diabetes.

Chickpea (*Cicer arietinum*) is a legume of the family *Fabaceae*, subfamily *Faboideae*. Chickpea, a self-pollinating diploid ($2n=2x=16$) with a genome size of 740 Mbp. It has been thought to originate in south-eastern Turkey from where it has spread to other countries of the world. It was first grown in turkey around 7,500 B.C. Chickpea is a hardy, deep-rooted, dryland crop sown on marginal lands, which can grow to full maturity in conditions that would be unsuitable for most crops.

It ensures nutritional security besides being a rich source of protein it is also important in substantial agriculture as it improve /enhance physical, chemical and biological properties of soil by fixing atmospheric nitrogen symbiotically.

Chickpea is a self-pollinated crop and cross-pollination is a rare event; only 0-1 % is reported (Singh, 1987). It is usually grown as a rainfed cool weather crop or as a dry climate crop in semi-arid regions. Optimum conditions include 18-26° C day and 21-29° C night temperatures and annual rainfall of 600-1000 mm (Duke, 1981). It is generally grown on heavy black or red soils (pH 5.5-8.6).

In India chickpea production has increased during the past 30 years from 6.58 million tonnes (1980-1981) to 9.19 million tonnes (2010-11) because of increase in grain yields from 657 to 895 kg/ha during this period. In India area under chickpea cultivation was 8.29 m ha in the year 2011-12, with production of 7.70 million tonnes and average yield of 928 kg/ha. Production of chickpea in year 2013-2014 and 2014-2015 was 9.53 million tonnes and 8.28 million tonnes, respectively (Directorate of Economic and Statistics, Department of Agriculture and Co-operation).

As per the latest report of Food and Agriculture Organization (FAO) for 2011, India is the largest producer and consumer of chickpeas in the world. Chickpea is grown in winter season starting after monsoon. South Asia, India, Nepal, Pakistan, Bangladesh, Myanmar are the main countries, where area of this crop is highest next to major cereals. Legumes are known to be main precursor of high yield for other crop. The importance to include chickpea in crop rotation lies due to its capacity to bind free atmosphere nitrogen in soil, enhance the soil aeration through their well developed root system.

Chickpea is cultivated throughout the state of Chhattisgarh. The total geographical area of Chhattisgarh is 13.6 m ha, out of which gross sown area is 5.54 million ha i.e. 35.5 per cent of total geographical area and the area under chickpea cultivation was 0.241 million ha in the year 2011-12, with production of 0.240 million tonnes and productivity of 995kg/ ha (Directorate of Economic and Statistics, Department of Agriculture and Co-operation, 2012).

Chickpea is mostly grown in kanhar soil in Chhattisgarh plains, particularly Durg, Rajnandgaon, Kawardha, Bilaspur and Raipur district. However, chickpea

productivity is low due to susceptibility of crop to various biotic and abiotic stresses. In rice based farming system, the widespread abiotic stress is the moisture stress. Terminal drought stress, which occurs during the pod-filling phase, is a common yield reducer in chickpea. Cold is the second most important abiotic stress and susceptibility to cold is greater at the late vegetative stage than at the seedling stage.

Regarding biotic stress diseases, insects, nematodes and parasitic weeds account for major losses. For example, extent of yield losses due to wilt and collar rot disease are more in the event of moist and high temperature in the country. Though 67 fungi, 3 bacteria, 22 viruses and 80 nematodes are reported to adversely affect the growth and productivity of the crop (Nene, 1979 and Ali and Dennis, 1992), only a few among them causes economic losses (Haware, 1998).

One of the major constraints limiting agriculture production is the difficult in managing disease caused by soil borne pathogens. This problem further compounds when the disease is incited by more than one pathogen. Major soil borne pathogens infecting chickpea are wilt, collar and root rot pathogen caused by *Fusarium oxysporum* f. sp. *ciceri*, *Sclerotium rolfsii* and *Rhizoctonia* spp. These pathogen cause significant loss in yield and primarily responsible for wide gap in yield level in farmer field.

Among the several soil borne fungal diseases of chickpea, dry root rot (DRR) caused by *Rhizoctonia bataticola* Taub (Butler) is the most severe diseases in the central and southern zone, where the crop is mostly grown under rainfed.

The disease generally appears around flowering and podding stage. Most conspicuous symptom is sudden drying of the whole plant scattered in the field. Symptoms include yellowing, drooping of the petioles and leaflets only on the tips. Leaves and stems of the affected plants are straw colored; in some cases, lower leaves turn brown. The tap root turns black, shows signs of rotting, and is devoid of lateral and finer roots. A grayish mycelium can sometimes be seen on the tap root. The dead roots are quite brittle and show shredding of bark and lateral root. Minute sclerotia can be seen on the exposed wood of the root and inner side of the bark or whenever split open at the collar region vertically.

Some strains of *R. bataticola* produce the pycnidial state of *Macrophomina phaseolina*. Because of its extraordinary host range geographical distribution and environmental adaptability, *R. bataticola* has become one of the really important plant pathogen. This pathogen does cause substantial mortality and loss in a crop which gets caught in higher ambient temperatures (30 °C and above) in the post – flowering stage. In the Indian situation, this occurs more in central and southern India. It is insignificant in northern India, where cooler temperatures extend through March and by the time temperatures rise, the crop is ready for harvest. *R. bataticola* is also known to survive for long periods in soil in the form of mycelia and sclerotia associated with plant debris (Mayer *et al.*, 1997). Sclerotial populations of *M. phaseolina* decline rapidly under high soil moisture and in soils with high C:N ratio amendments. Reduction in number of viable sclerotia may be brought about in the field by keeping soil moisture at 60% moisture-holding capacity for 3 to 4 weeks at 30°C or above (Dhingra and Sinclair, 1978).

DRR is caused by necrotrophic fungus *Rhizoctonia bataticola* (Taub) Butler [Pycnidial stage – *Macrophomina phaseolina* (Tassi) Goid]. This species morphology is composed of hyphae and sclerotia (hyphal propagules). The pathogen has a predominant sclerotial stage and less frequently, a pycnidial stage in its life cycle. The sclerotia are viable for more than 12 months in the soil. It survives between crop seasons on infected plant debris. These sclerotia serve as a primary means of survival. The sclerotia float freely on soil surface when field is flooded for irrigation and become primary inoculum for emerging seedlings.

Dry root rot caused by sclerotial form of *R. bataticola* causes up to 60 % yield loss. It is a diverse omnipresent soil borne species with a wider host range involving more than 500 species in 75 families. It mainly produces micro sclerotia (primary source of inoculum) or pycnidia. It is both seed and soil borne and infects plants from seedling stage to maturity. In chickpea up to 5-20% yield losses have been recorded due to dry root rot caused by *R. bataticola* (Sundravadana *et al.*, 2012).

Owing to the highly polykaryotic characteristics of the mycelium, *Rhizoctonia bataticola* appears to be highly variable in nature. Variations are observed in growth rate among the isolates from the same plant and between

isolates from various plants on different media. The pathogen is non-specific but varies in their aggressiveness. Survey reports in India showed that there are a lot of variability on cultural character, growth rate, and morphology of sclerotia, pycnidia, pycniospore and disease severity indicating that the pathogen might be variable in nature (Dhingra and Sinclair, 1973).

Anastomosis is one of the phenomenon to know variability among the fungi. Similar fungi leads to fusion between branches of the same or different hyphae. Hence the bifurcating fungal hyphae can form true reticulating networks. By sharing materials in the form of dissolved ions, hormones, and nucleotides, the fungus maintains bidirectional communication with itself. Two different haploid mating types - if compatible - merge. Somatically, they form a morphologically similar mycelial wave front that continues to grow and explore.

Root disease severity is strongly influenced by soil type, soil pH, soil temperature, moisture, and the biological activity of suppressive microorganism. Organic matter is known to affect soil structure, aeration, drainage, moisture holding capacity, nutrient availability, and microbial ecology (Davey, 1996).

Cultivation of a resistant variety is an economical approach for the management of dry root rot of chickpea. However, in chickpea, the Desired level of resistance to dry root rot is not known till now and often resistance identified at one location breaks down at another location, indicating the possibility of variability in the pathogen. Thus, an understanding of the pathogenicity and genetic diversity of the pathogen is an important prerequisite in developing and deploying varieties with durable resistance to dry root rot disease. Thus, development of a cost effective and user-friendly diagnostic tool to detect *R. bataticola* isolates in chickpea is the need of the hour. Understanding of the pathogen variability will help in developing multiple disease resistant varieties and their strategic deployment for the durability of resistance in molecular breeding programs.

Biological control using antagonistic microorganisms is also an alternative method to the fungicides and provides an ecologically based approach to integrated pest management in sustainable agriculture in crop production systems. The use of

bio control agents and organic amendments for controlling fungal diseases in plants is considered as an interesting alternative to synthetic fungicides due to their eco friendly effect on the environment as well as they are economically feasible. They can reduce cost of disease management and also mitigate the harm done by excessive and repeated use of pesticide.

Keeping in view, the severity and losses caused by dry root rot disease, present investigation is undertaken entitled “Studies on dry root rot (*Rhizoctonia bataticola* Taub (Butler)) of chickpea (*Cicer arietinum*).”

The specific objectives of the present study:

- 1) Isolation and characterization (pathogenicity, AG grouping) of *Rhizoctonia bataticola* isolates from different geographical locations of Chhattisgarh.
- 2) Identification of potential candidate isolate / bio-control agent against *Rhizoctonia bataticola*.
- 3) Screening of chickpea varieties / wild relatives for resistance to dry root rot.
- 4) Study of soil factors affecting disease development of dry root rot in chickpea.

CHAPTER II

REVIEW OF LITERATURE

2.1 Chickpea (*Cicer arietinum* L.)

Chickpea is the third most important legume crop with a worldwide production of about 13.1 mt (Million tonnes). In India chickpea is the most important legume crop as revealed by the India's contribution (accounts for ~60%) towards the world's total production as well as area under cultivation (6700 thousand hectares). Though, chickpea is grown and locally consumed, India is also the 17 world's largest importer of chickpeas accounting for about 20% of global imports. These figures reflect on the growing demand for chickpea as well as the immense strain on crop production and yield.

2.1.1 Morphology

Chickpea (*Cicer arietinum* L.) has a deep tap root system, which enhances its capacity to withstand drought conditions; it is well adapted to areas having relatively cooler climatic conditions and a low level of rainfall. The aerial portion is profusely branched, erect or spreading, reaching a height of 0.2-1 m, appearing glandular pubescent, olive, dark green or bluish green in color. Leaves are imparipinnate, glandular-pubescent with 3-8 pairs of leaflets with rachis ending in a terminal leaflet. Leaflets are ovate to elliptic, 0.6-2.0 cm long, 0.3-1.4 cm wide; margin serrate, apex acuminate to aristae, base cuneate; stipules 2-5 toothed or absent. The inflorescence consists of solitary flowers, sometimes 2 per inflorescence and borne on 0.6-3 cm long peduncles, 7-10 mm long calyx; while the bracts are triangular or tripartite; the corolla is 0.8-1.2 cm long and varies from white, pink, purplish (fading to blue), or blue. The staminal column is diadelphous (9-1) with a sessile, inflated and pubescent ovary (Duke, 1981; Cubero, 1987; van der Maesen, 1987). The seeds (1-2 or maximum 3) are contained in a pod which is rhomboid ellipsoid, inflated and glandular-pubescent. The seed color varies from cream, yellow, brown, black, or green. Seeds may be rounded to angular with a

smooth or wrinkled, or tuberculate seed coat, which is laterally compressed with a median groove around two thirds of the seed forming a beak at the anterior end; during the cryptocotylar germination cotyledon tips remain in the seed coat in intimate contact with the endosperm (Duke, 1981; Cubero, 1987 van der Maesen, 1987).

2.1.2 Centre of origin and distribution

Chickpea is an ancient crop that marks its origin well before 10,000 B.C. when it was used by the 'hunter-gatherer' for sustaining their communities. The regions of Turkey and the ancient city of Jericho domesticated this crop around 7,500 B.C., which is considered as its centre of origin by Ladizinsky (1975) while Van der Maesen (1987) recognized the south-eastern part of Turkey adjoining Syria as the possible center of origin of chickpea based on the presence of the closely related annual species. *C. reticulatum*, *C. echinospermum* and *C. reticulatum* is regarded as the progenitor of cultivated chickpea. Botanical and archaeological evidence shows that chickpea was first domesticated in the Middle East and were widely cultivated in India, Mediterranean area, the Middle East, and Ethiopia since antiquity. In the New World, it is important in Mexico, Argentina, Chile, Peru, U.S. and also in Australia. Wild species are most abundant in Turkey, Iran, Afghanistan, and Central Asia (Duke, 1981). Chickpea was brought to the Western Europe and was known in many areas in the Bronze Age, most popularly, Italy and Greece. People in these regions consumed chickpea in various forms like roasted as snacks, raw, carbonized or in broth. Many past writings have also been found telling the uses and importance of chickpeas both medical and as a food crop. With time, many other varieties of chickpea were developed as it spreads in many areas of Asia and Australia. During the First World War, Germany cultivated chickpea as a coffee substitute. Chickpea has majority of its cultivation in dry areas of the Indian subcontinent (Saxena, 1990) and India is the principal chickpea producing country with a share of 90% in Asian region. Presently, the most important chickpea producing countries are India (9.6 mha, 8.83 mt), Pakistan (9.92mha, 0.75mt), Turkey (0.42mha, 0.50mt), Iran (0.5mha, 0.29mt), Mexico (0.11mha, 0.20mt), Myanmar (0.33mha, 0.49mt), Ethiopia (0.12mha, 0.24mt),

Australia (0.5mha, 0.81mt), and Canada (0.07mha, 0.16mt), respectively (FAOSTAT).

2.1.3 Season

The yields from chickpea is maximum when grown on sandy, loam soils having an appropriate drainage system as this crop is very sensitive to excess water. The production of chickpea or 'chana' is also affected in excessive cold conditions. Chickpea is sown in the months of September to November in India and is considered as a *rabi* (post rainy/monsoon) crop. The Desi type chickpea reaches physiological maturity by 95-105 days and Kabuli type by 100- 110 days. The plant is harvested when its leaves start drying and shedding and can be done manually or with the help of a harvester. In India, it is harvested in February, March and April. This crop is often cultivated as a sole crop but sometimes it is also grown rotationally with other crops such as sorghum, pear millet, wheat and coriander. Pale yellow, dark brown or reddish chickpea are some of the varieties that are grown today.

2.1.4 Nutrition

Chickpea (*Cicer arietinum* L.) belongs to the family leguminosae and is commonly known as chana, bengal gram or garbanzo. It is a self-pollinating crop with a diploid genetic content of $2n=16$, with a C value of $1C=931\text{Mbp}$. Chickpea is mainly employed for human consumption and also a small proportion forms, the part of animal and poultry feed. Chickpea has one of the highest nutritional compositions of any dry edible legume and is not reported to contain any specific major anti-nutritional factors (Williams and Singh, 1987). On an average, chickpea seed contains 23% protein, 64% total carbohydrates, 47% starch, 5% fat, 6% crude fiber, 6% soluble sugar and 3% ash. The mineral component is high in phosphorus (340 mg/100 g), calcium (190 mg/100 g), magnesium (140 mg/100g), iron (7 mg/100 g) and zinc (3 mg/100 g). Chickpea protein has the highest digestibility when compared to other dry edible legumes. The lipid fraction is high in unsaturated fatty acids, primarily linoleic and oleic acids. They are also a good source of calcium, magnesium, potassium, phosphorus, iron, zinc and manganese (Ibrikci *et al.*, 2003). Chickpeas do not contain high amounts of isoflavones as soybeans do (USDA-ARS, 2005), but provide more beneficial carotenoids such as

β -carotene than genetically engineered “Golden Rice”. Thus, chickpea is considered a functional food or nutraceutical (Agharkar, 1991; McIntosh and Topping, 2000; Charles *et al.*, 2002). While it is a cheap source of protein and energy in the developing world, it is also an important food to the affluent populations to alleviate major food-related health problems. However, more research is necessary to elucidate and extend the food and nutraceutical benefit of this important food legume through breeding. Having a capacity to stand in drought conditions, this crop doesn't have the requirement of being fed with nitrogen fertilizers. Chickpea through its BNF (Biological Nitrogen Fixing) capability meets 80% of its nitrogen requirement and can fix up to 140 kg N/ha from air. It leaves substantial amount of residual nitrogen behind for subsequent crops and adds much needed organic matter to maintain and improve soil health, longterm fertility and sustainability of the agro-ecosystems. Commercially, the species is grouped into Desi (small, dark colored seed coat) and Kabuli (large, white or cream seed coat) types. To a certain extent this classification overlaps with the macrosperma and microsperma races proposed by Moreno and Cubero (1978). Classification also reflects utilization: whereas Kabulis are usually utilized as whole grains, Desis as whole seeds, de-hulled splits (dhal) or flour. Seeds are ground to flour and used in confectionery. Young shoots or green pods, shelled for the peas and eaten as a snack or vegetable. Chickpea is also known for its use in herbal medicine and cosmetics. An acrid liquid from the glandular hairs of the plant is collected by spreading a cloth over the crop at night, which absorbs the exudation with the dew. The exudates contain about 94% malic acid and 6% oxalic acid and is used medicinally. Kabuli $\times$ Desi crosses are used in many breeding programs to combine genes for cold tolerance, bold seededness, resistance to Ascochyta blight and long vegetative growth more frequently found in Kabuli types, while genes for heat and drought tolerance, resistance to Fusarium wilt and early flowering contributed by the Desi types (Singh, 1987).

2.1.5 Yield and losses

The potential seed yield of about 5 t/ha has been reported in chickpea. However, the realized seed yield over around 850 kg/ha (world average $\approx$ 0.8 t/ha, FAO STAT, 2005) which has stagnated over the years. Series of biotic and abiotic

stresses reduce the yield and yield stability leaving room for only marginal improvements. This affects development of widely adapted cultivars and susceptibility to several biotic and abiotic stresses. Generally the crop produces excessive vegetative growth under high input conditions and is unable to translate the biomass into high seed yields. The major abiotic constraints to productivity include drought, heat, cold and salinity and the key biotic constraints are Ascochyta blight (*Ascochyta rabe*), Fusarium wilt (*Fusarium oxysporum* f. sp. *ciceri*), Dry root rot (*R. bataticola*), Botrytis grey mould (*Botrytis cinerea*), Collar rot (*Sclerotium rolfsii*), Root-knot nematode (*Meloydogyne incognita* and *M. javanica*), Stunt-virus, Pod borer (*Helicoverpa armigera*), and Cutworm (*Agrotis ipsilon*). Amongst the causal agents of biotic stresses, about 67 fungi, 3 bacteria, 22 viruses and 80 nematodes have been reported on chickpea (Nene *et al.*, 1996) but only few of these cause economically important diseases (Haware, 1998). There has been an increase in different chickpea pathogens like fungi, bacteria and viruses over a period of past 17 years. The maximum number of pathogens has been reported from India alone with the number rising to 89 pathogens in 1995 from 35 in 1978 (Nene *et al.*, 1996). The insect *Helicoverpa armigera* which feeds on foliage, flowers and developing seeds, is the most important pest of chickpea, while stunt is the most important and prevalent viral disease in most chickpea growing regions of the world.

2.2 *Rhizoctonia bataticola*

2.2.1 History and occurrence of *R. bataticola*

R. bataticola (Taub.) Butler as a plant pathogen was recognized by Halsted (1890). The first report of occurrence of dry root rot in chickpea alongwith wilt was made by Padwick (1948). In India, Shaw (1912) described a sterile fungus with black sclerotia causing seedling disease in jute, cowpea, groundnut and cotton. He identified the fungus as *R. solani*. Taubhanus (1913) gave the name of genus *Sclerotium* because of absence of spores and the species name as *bataticola* because it was pathogenic to *Ipomea bataticola* (L.) Lam. Briton- Jones (1925) transferred the fungus to the genus *Rhizoctonia* based on the identification of cultures by Butler (1918). Ashby (1927) accepted *Macrophomina* and rejected the binomial *Macrophomina phaseoli* and proposed a new binomial *M. phaseolina* as

the pycnidial stage of *R. bataticola*. He was not aware that Tassi had earlier described *M. phaseolina*. Haigh (1930) suggested that *R. bataticola* be used for sclerotial isolate and pycnidial strains should be called *M. phaseolina*.

The form genus *Rhizoctonia* was established in 1815, to include two parasitic species, both characterized in part by the production of a mat of mycelium infesting the affected roots or other submerged member. The serious roots diseases due to these organisms have received consideration by many mycologists since that time. The first mention of a plant disease which may be referred with certainty to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberoïdes*. It was more than 50 years later, Bondaroy (1785), another French investigator mentioned that *Asparagus* when grown on land in which diseased saffron bulbs had been previously grown suffered from the same disease. Bulliard in 1791 named the fungus on *Crocus sativus*, *Tuber parasiticum*, as he considered the *Sclerotium homologus* with the tuffe. Persoon (1801) however did not accept Bulliard's disposition of the fungus and named *Sclerotium crocorum*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic discussion he first employed Persoon's name for the fungus. Later after careful study of the characteristics and parasitism of the species, he established in 1815, the genus *Rhizoctonia* to include two species *R. crocorum* DC. on *Crocus* and other hosts and *R. medicaginis* DC. on alfalfa. He also considered a doubtful species *R. meli*, reported on apple. Nees (1817) referred to a fungus attacking *Crocus* as *Thanatophytum crocum*. According to Peltier (1916) Nees's fungus was probably *Rhizoctonia*.

Fries (1823) recognized *Sclerotium crocorum* of Persoon as a synonym of *R. crocorum* D.C and placed it in Sclerotiaceae. A new species of *Rhizoctonia* was described in France by Duby (1830) as *R. ascalonicum*. Leveille (1843) noted a similar *Rhizoctonia* on *Rubia ascalonicum*, *Solanum tuberosum*, *Phaseolus* and Tulips. Incidentally this appears to be the first record of *Rhizoctonia* on *Phaseolus*.

Rhizoctonia on *Crocus* was also reported from Germany by Kuhn (1858). He also found the same fungus, which he identified as *R. medicaginis* on a sugar

beet. At the same time he described a new species of *Rhizoctonia* on potato, which he clearly distinguished from the above species and to which he gave the name *R. solani* Kuhn.

The most important taxonomic work on the genus *Rhizoctonia* in the 19th century is that of Tulasne brother's (1862) they published the first extensive morphological study on the genus *Rhizoctonia*. They reduced *R. crocorum* DC and *R. medicaginis* DC to a single species to which they applied a new name *R. violaceae* Tul. This reduction to a single form was made after a most careful morphological study of the fungus at all stages. Shaw (1912) described a sterile fungus with minute black sclerotia as the cause of a seedling disease of jute, cowpea, groundnut and cotton at Pusa, India. He reported it erroneously to *R. solani* Kuhn, which is now regarded as the sterile stage of *Corticium solani* (Prill and Del.) Bourd. and Galz.

Taubenhaus (1913) described a charcoal rot disease of potato from USA and named the fungus *Sclerotium bataticola* Taub. Later, Martin (1917) also described a disease of chillies due to *S. bataticola*. Briton Jones (1921-1922) found in Egypt, a *Rhizoctonia* parasitizing cotton, beans and *Vigna sinensis* Briton Jones isolates was compared at the Imperial Bureau of Mycology with Taubenhaus's *S. bataticola* and found to be identical with it.

Small (1924) described *Rhizoctonia* which develop black sclerotia of variable size and shape in the diseased roots of many woody plants from Uganda and he named it *Rhizoctonia lamellifera*. Butler (1925), however, considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub). Briton Jones (1925) gave the name *Rhizoctonia bataticola* (Taub) Butler to a sclerotium causing a root rot of cotton, french beans and cowpea from Egypt and this binomial was based on a comparison by Butler of pure cultures of the forms from India, Egypt and America which showed them to be identical. Small (1926) recognized the identity of his fungus with the forms from India, Egypt and America and abandoned the name *Rhizoctonia lamellifera*.

Among the taxonomic and pathological studies on *Rhizoctonia* in India, the most important is that of Shaw (1912) he had studied two species of *Rhizoctonia*, one of which was characterized by its formation of small black sclerotia attacking

jute, cotton, groundnut, cowpea, potato and mulberry. This fungus was identified as *R. solani* Kuhn Shaw continued his work on *Rhizoctonia* and in 1915 he and Ajrekar again recorded *R. solani* and opined that it was the cause of disease of 16 hosts including *Phaseolus aureus*, Butler (1918) attributed the *Rhizoctonia* disease of the above hosts to *Rhizoctonia* spp. Indicating thereby that the fungus had been incorrectly ascribed to *R. solani* fact which had been pointed to by other worker.

Goidanich (1947) examined the original material of Tassi and compared with *Macrophomina phaseoli*, *M. corchori*, *M. cajani*, *M. sesami*, *M. philippinesis*, *Dothorela cajani* and *D. phaseoli* and found all of them identical. He correlated the mistake made by Ashby and according to the International code of Botanical Nomenclature the binomial *M. phaseolina* is the valid name for pycnidial stage of *R. bataticola*.

Macrophomina phaseolina (Tassi) Goid the incitant of root rot of mungbean, also causes leaf blight (Philip *et al.*, 1969) is one of the most important disease. The pathogen causes root rot in a number of crops including groundnut, sesamum, soybean and kidney bean (Dhingra and Sinclair, 1977). The pathogen was also reported to cause leaf blight, seedling blight and charcoal rot of guar, moth bean, urdbean, mungbean, groundnut, soyabean and sunflower (Singh, 1953., Kumar *et al.*, 1969 and; Senecha and Srivastava, 1982; Shukla and Bhargava, 1976; Philip *et al.*, 1969; Shanmugam and Govindaswamy, 1971; Gangopadhyay *et al.*, 1973, Udit, 1983; and Grover and Sakhuja, 1981).

2.2.2 Distribution and economic importance of *R. bataticola*

Distributed worldwide, *Rhizoctonia* fungi occur in agricultural and non-agricultural fields, living as plant pathogens, symbionts or saprophytes. Because some groups in *Rhizoctonia* spp. can severely damage major crops (Kuninaga, 2002, 2003; Naito and Sugimoto, 1978; Ogoshi, 1987; Sharon *et al.*, 2006, 2008), control of diseases resulting from infection by *Rhizoctonia* spp. is a major concern for farmers and plant pathologists. *R. bataticola* (Taub) Butler is a soil borne pathogen with a wide host range (Dhingra and Sinclair, 1978) and causes charcoal rot (seedling phase), root rot and stem blight in more than 500 species of plants (Sinclair, 1982).

The pathogen has been found to occur throughout the world particularly in tropical and sub-tropical regions, namely India (Butler, 1918), Uganda (Hansford, 1943), United States (Haffmaster *et al.*, 1943), Australia (Anonymous, 1965), Phillipines (Yang, 1977) and Pakistan (Shahzed and Ghaffer, 1986). Nene *et al.* (1996) reported comprehensive list contains the occurrence of the pathogen in world. *R. bataticola* was found to be distributed in world across many country namely Australia, Bangladesh, China, Egypt, Ethiopia, India, Iran, Kenya, Lebanon, Mexico, Myanmar, Pakistan, Spain, Sri Lanka, Sudan, Syria, Tanzania, Thrkey, Uganda, USA, Zambia.

Kataria and Grover (1977) reported a yield loss of 10.8 per cent and 24.1 per cent due to *R. bataticola* in mungbean from the states of Haryana and Rajasthan respectively. The disease development is influenced by dry soil conditions, especially at flowering causing the plants to sudden drying (Singh and Mehrotra, 1982). The incidence of dry root rot of chickpea has been reported to be as high as 24.70 per cent in Madhya Pradesh by Sharma *et al.* (1999). Ahmed and Mohammad (1986) reported a yield loss of 50 to 71 per cent due to *R. bataticola* in chickpea growing areas of Bihar. Leaf blight of mungbean and other crops, caused by *M. phaseolina*, is a serious disease responsible for heavy loss in yield.

Kaushik *et al.* (1987) reported 2.2-15.7 per cent infection by *R. bataticola* (*Macrophomina phaseolina*) causing 10.8 per cent reduction in grain yield and 12.3 per cent reduction in protein content of the seed.

Pineda and Avila (1993) reported a yield loss of 37 to 79 per cent due to *Macrophomina phaseolina* in sunflower under rainfed conditions. Plant losses upto 77 per cent have been reported due to *Rhizoctonia bataticola* in soybean (Muthuswamy and Mariappan, 1991). Singh and Sirohi (2003) evaluated the incidence of chickpea diseases in the Himachal Pradesh and reported that the incidence of dry root rot was highest in Una (4.86%) and Sirmour (3.04%) districts.

Hwang *et al.* (2003) studied the impact of seedling blight and root rot caused by *R. solani* Ag-4 on nodulation and seed yield of chickpea and reported that emergence and dry matter production declined and root rot severity increased with increasing inoculum concentrations under green house conditions. Gurha and

Trivedi (2008) conducted a survey in all the chickpea growing districts of Karnataka and reported that *R. bataticola* was found as the predominant pathogen which infected 60.0 per cent plants in the fields of Gulbarga.

2.2.3 Taxonomic position of *R. bataticola*

Kingdom: Fungi, Subkingdom: Diakarya, Phylum: Basidiomycota, Subphylum: Aquaricomycotina, Class: Basidiomycetes, Subclass: Aquaricomycetidae, Order: Polyporales, Family: Corticiaceae, Genus: *Rhizoctonia*, Specific descriptor: *bataticola*, Scientific name: *Rhizoctonia bataticola*.

2.2.4 Symptomatology

Philip *et al.* (1969) and Grover and Sakhuja (1981) reported leaf blight phase of mungbean with the information that disease usually makes its appearance when the crop is 4-6 weeks old and the older leaves are first affected. Initially small, circular to irregular brown to reddish brown lesions appear on or near the margins which enlarge and coalesce. Under hot and humid conditions the entire plant may be blighted. The dried lesions are yellowish brown and become papery in texture. Severely affected leaves fall off prematurely. Nene *et al.* (1978) observed continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristic symptoms of dry root rot of chickpea. Sanecha and Srivastava (1982) reported that in affected young seedlings of cowpea, discolouration and rotting occurred from young root tip and proceed backwards. The cotyledonary leaves are completely blighted and necrotic. Stem decay occurs in advance stage

Baldev *et al.* (1988) reported symptoms of dry root rot of chickpea which consisted of straw colored leaves and stems. Tap root of infected plants was observed to be dry and devoid of lateral and finer roots. Roots were turned dark and showed signs of rotting. The dead root was observed brittle towards the tip and showed shredding of bark. Haware (1990) reported that dry root rot of chickpea appears around flowering and podding time in the form of scattered dried plants but seedlings are also infected. The symptoms induced were drooping of petioles and leaflets which were confined to top of the plants. Shredding of bark in form of flakes was observed. Singh *et al.* (1990) observed that *R. bataticola* inoculated

roots of chickpea, upon microscopic examination, showed disintegration of cortical tissue and plugging of xylem vessels with mycelial and sclerotial bodies of fungus.

Rangaswamy (1996) reported field symptoms which include yellowing of plants with drooping of leaves. Due to decay of roots, plants can be easily uprooted. Root portion appears brownish from outside. The stem and root below the region shows rotting with frequently pinkish white mycelial growth. Dried plants scattered throughout the field are indicative of root rot incidence.

Singh (1999) reported that dry root rot of chickpea occurred from flowering to podding stage. Infected plants were suddenly dried in the field. The tap roots turned dark brittle and devoid of lateral roots. Khalid and Ilyas (2000) considered the presence of root lesions and sclerotium plugging in xylem vessels of roots and collar region as symptoms of dry root rot of chickpea for screening of germplasm against this disease. Singh and Agarwal (2002) observed the withering and drying of chickpea plants in the field due to the infestation of *R. bataticola*.

2.2.5 Disease cycle

R. bataticola can survive for many years by producing small (1 to 3-mm diameter), irregular-shaped, brown to black structures (called sclerotia) in soil and on plant tissue. *R. bataticola* have evolved the ability to produce sclerotia with a thick outer layer that allows them to float and survive in water. *R. bataticola* also survives as mycelium by colonizing soil organic matter as a saprophyte, particularly as a result of plant pathogenic activity. Sclerotia and/or mycelium present in soil and/or on plant tissue germinate to produce vegetative threads (hyphae) of the fungus that can attack a wide range of food and fiber crops. The fungus is attracted to the plant by chemical stimulants released by actively growing plant cells and/or decomposing plant residues. As the attraction process proceeds, the fungal hypha will come in contact with the plant and become attached to its external surface. After attachment, the fungus continues to grow on the external surface of the plant and will cause disease by producing a specialized infection structure (either an appressorium or infection cushion) that penetrates the plant cell and releases nutrients for continued fungal growth and development. The infection process is promoted by the production of many different extracellular enzymes that degrade various components of plant cell walls (e.g. cellulose, cutin and pectin).

As the fungus kills the plant cells, the hyphae continue to grow and colonize dead tissue, often forming sclerotia. New inoculum is produced on or in host tissue, and a new cycle is repeated when new substrates become available.

2.2.6 Multiplication of the pathogen

For various studies, mass or bulk production of mycelium and sclerotia is necessary to create artificial epiphytotics as the large number of sclerotia is required to study their survival in nature. Mayee and Garud (1978) used sorghum grain for mass culturing of *M. phaseolina*. Dandnaik *et al.* (1986) used sorghum, bajra, mung, wheat, maize and rice grain for mass culturing of sorghum charcoal rot fungus *M. phaseolina*. A mixture of sand and maize was used for mass inoculum production of *R. bataticola* (Byadgi and Hegde, 1988, Muthukrishnan *et al.*, 1995). Grover and Sakhuja (1981) used Czapek's nutrient solution for mass production of mycelial growth for the establishment of the disease. The pathogen *R. bataticola* was mass multiplied on sand maize medium (Sajeena *et al.*, 2004; Kataria *et al.*, 2007). Sterilized sorghum grains were used for mass culturing of *R. bataticola* (Rajeswari *et al.*, 1999; Durai, 2004).

2.3 Morphological and cultural characters of *R. bataticola*.

Small (1924) observed the sclerotia of *M. phaseoli*, which exhibited some differentiation, smooth, black and hard, and variable in size and form. In the tissue of herbaceous plant they range from 50-150 μ m in diameter, but in the roots of woody plants 80-100 μ m, in culture the variation is from 50-200 μ m. Haigh (1930) divided 27 strains of *M. phaseolina* in the three groups according to their mean sclerotial diameter in culture. First group 120 μ m or small, second group 120-200 μ m or medium and third group 200 μ m or large in the diameter. All the strains were from tree and woody plants. Uppal *et al.* (1936) reported variation in growth rate and colony character for two sorghum strains of *M. phaseolina* from two different localities, one from Broach farm and another from Mohol farm in Maharashtra. Cultures of Broach strain showed concentric rings on PDA, while that of Mohol strain produced aerial mycelium.

Sulaiman and Patil (1966) studied nineteen isolates of *M. phaseolina* of cotton and identified two races on the basis on cultural character and pathogenicity on three hosts. Race-1 formed olive green mycelium while race-2 had dark gray

mycelium with abundant sclerotia. Ahmed and Ahmed (1969) reported that the cultural characteristics and growth rates of eight different jute isolates of *M. phaseolina* appeared to be related to their pathogenicity.

Philip *et al.* (1969) noticed the differences in morphology of *Macrophomina phaseolina* isolates from various parts of host plants. They include the pycnidial development *in vitro* on mung bean roots infected with *Macrophomina*, as black, globose or depressed and 150-200 µm in diameter. Pycnidiospores were oval or elliptical, hyaline, non-septate, thin walled with 10-24 x 6-10 µm in size. The pycnidiospores develop into *R. bataticola* in culture at room temperature. Occurrence of numerous small black sclerotia of this pathogen under *in vitro* conditions was also reported by Jain *et al.* (1973). According to him, sclerotial morphology of *Macrophomina phaseolina* of urdbean was found to be different when isolated from different plant parts. The soil and seed isolate developed small sclerotia which were more pathogenic than leaf isolate which developed larger sclerotia.

Dhingra and Sinclair (1973) studied the variability in *R. bataticola*. Colour of cultures on PDA, ability to sporulate in infected host plants and pycnidial size also have been reported to vary greatly in *M. phaseolina*. Also noticed variation in growth rate among the isolates from same plant and also between isolates from different plants. Owing to the highly polykaryotic characteristics of the mycelium, *R. bataticola* (*M. phaseolina*) appears to be highly variable in nature.

Jain *et al.* (1973) studied isolates of *R. bataticola* obtained from urd plant parts (root, stem, leaf, pod and seeds) and soil. The isolates from various plant parts and soil showed differences in virulence. The soil isolate was most pathogenic. The isolates differed in their growth pattern and sclerotial size. The leaf isolate developed the largest sclerotia and the seed whereas, soil isolates developed the smallest sclerotia. When grown on different media the isolates varied in growth pattern and growth rate. The soil isolate showed the least amount of growth in almost all media.

Dhingra and Sinclair (1977) studied fifteen isolates of *M. phaseolina* from root, stem, petiole, pod and seed of soybean plants. They observed that sclerotia

from root isolates ranged maximum from 122- 188 x 77-127 μm , while stem isolates range was minimum 84-100 x 60-71 μm .

Byadgi and Hegde (1985) obtained the isolates of *R. bataticola* from different plants, which differed in cultural characteristics, growth rate, morphology of sclerotia and virulence. *Gliricidia* (*Gliricidia muculata*) isolate produced biggest sclerotia 101.51 μm while cowpea isolate the smallest sclerotia 66.88 μm .

Jharia and Khare (1985) isolated *R. bataticola* from different host. The species of *R. bataticola* from different host differ in their morphological and cultural characters and even differences was observed in the isolates, isolated from various parts of same host. Variation was recorded in their growth rate. The soil isolate was fast (90 mm) followed by root (89 mm) and seed (67 mm) isolates on potato dextrose agar medium. Maximum width of hyphae (4.5 μm) was found in the root isolate followed by soil isolate (3.5 μm) and seed isolate (3.0 μm). The sclerotia were largest in the root isolate (84x93 μm) in comparison to soil isolate (90 x 75 μm) and seed isolate (85x70 μm).

Raut and Ingle (1989) studied isolates of *R. bataticola* from fifteen host crops (mung, chickpea, guar, black gram, cowpeas, pigeon peas, soybean, sorghum, safflowers, sunflowers, cotton, okra, sesamum, chilli and citrus. Pathogen varied in their growth rate in terms of sclerotial size, production of pycnidia and size of pycnidiospores. Differences were evident between isolates from crops belonging to the same family. On observation it was found that cotton and cowpea isolates formed comparatively more sclerotia. The sclerotia of cotton isolate were the largest (mean diameter 115.06 μm) and soybean isolate the smallest (56.33 μm). Cluster bean, pigeonpea and black gram isolates measured between 61 and 70 μm . Safflower, okra, green, chillies, sunflower and citrus isolates between 71 and 80 μm and cow pea sorghum, chickpea and sesamum isolates between 81 and 90 μm .

Than *et al.* (1991) conducted colony compatibility tests on thirty seven isolates of chickpea along with two isolates from different hosts, collected from several locations in Myanmar and reported the possible occurrence of more than one type of *R. bataticola* isolate in a field and the existence of closely related isolates at different places.

Monga and Sheo (1994) studied seven isolates of *R. bataticola* and observed that the isolates were distinct in cultural characteristics, sclerotial number and size. The size of sclerotial varied from 58.83 to 126.63 μm .

Ratnoo *et al.* (1997) observed the large variation in *R. bataticola*, they differ in cultural, morphological and pathogenic characters. Studied two isolates, from which it was found that M1 isolate showed dark gray mycelium, fluffy growth, hyphae thin, sclerotia formation abundant, blackish globose to round and measure 49- 76 μm in diameter. M2 isolate was gray cottony and fast growth of mycelium, hyphae thin dark brown, round and measured 45-53 μm in diameter.

Mandal *et al.* (1998) studied the sclerotial morphology and pathological reaction of thirteen isolates of *M. phaseolina* obtained from jute growing regions of India. Found that sclerotial size ranged from 66.14 to 128.25 μm and the colour varied from light to dark brown. Sclerotial shape in most cases was irregular except for few which were round to elongate. Prameela and Singh (1998) collected 56 isolates of *M. phaseolina* from root, stems, pod and seed of black gram and green gram from eleven different locations of India. Divided the fifty six isolates into six morphologically different groups.

Lokesha *et al.*, (2004) evaluated cultural variability of isolates of *M. phaseolina*. The differences in the growth of four isolates, upon incubation exhibited in terms of the maximum mean mycelial weight i.e. 231.0 mg was recorded, on 18th day of incubation and the isolate from ICRISAT recorded lowest mycelial weight (190.9 mg) and rest remained significantly superior to other isolates. Potato Dextrose Agar (PDA) and Czapek's agar media were found best for the growth of all the isolates (88.4 mm each). The sclerotia production was maximum on PDA (123.3/microscopic field of 10x). The ICRISAT isolate recorded maximum sclerotia production (71.2), followed by Bidar isolate (68.68). Purkayastha *et al.* (2004) stated that the isolates with fast mycelial growth and abundant sclerotial production were more pathogenic on clusterbean (*Cyamopsis tetragonoloba*) seedlings than isolates growing more slowly in PDA.

Shekhar *et al.* (2006) divided seven isolates into four groups, on the basis of colony colour, namely grayish white, blackish gray, dark black and cottony white colonies while working with charcoal rot of maize. Aghakhani and Dubey

(2009) studied morphological and pathogenic variation among twenty three isolates of *R. bataticola*, collected from ten different major chickpea growing state of India. They found isolates were highly variable in their morphology. The majority of isolates (nineteen) produced suppressed mycelium and rest produced aerial mycelium. Cultural character were too found to varied in colour, as white to dull white creamy, gray and black. The sclerotia formed in different isolates were dark black to black and highly variable in size (40-600 μm).

Kanchan and Biswas (2009) evaluated morphological variability of *R. bataticola*, causal agent of leaf spot and blight disease of pigeon pea. The results revealed that the nature of mycelium varied from fluffy dark brown to partially fluffy mycelium colony with smooth margin.

Manjunatha and Naik (2011) studied the cultural and morphological diversity, in isolates of *R. bataticola*, causing dry root rot of chickpea. The results revealed that the isolates were highly variable. Based on cultural and morphological characters, the isolates were categorized into three groups. The first group comprised of all isolates from Bidar (RB7) which exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur and Gulbarga and also an isolate from Bidar (RB7) showed brownish black with flat sclerotia which were uniformly distributed. Third group comprised of only RG3 isolates from Gulbarga which showed flat colony and brown coloured sclerotia formed at the center of the culture.

Mohan and Balabaskar (2012) conducted survey on the incidence of groundnut root rot disease in cuddalore district of Tamil Nadu and assessed the cultural characters and pathogenicity of *Macrophomina phaseolina*. The isolates of *M. phaseolina* exhibited cultural and pathogenic variability among them. The isolate (MP18) with faster mycelial growth, maximum sclerotial production and maximum sclerotial size, recorded the maximum incidence of root rot disease under pot culture condition.

Sundravadana *et al.* (2012) studied the variability in morphology of *R. bataticola* isolates, causing dry root rot of pulses. The results signify that the

eleven isolates varied in synthesized pink pigment and their intensity viz., low (+), medium (++) and high (+++).

Gupta *et al.* (2012) evaluated diversity in isolates of *R. bataticola* causing dry root rot in chickpea collected from four major chickpea growing states of central India. The isolates were highly variable, in their morphological and cultural characteristics and were grouped into 24 groups, out of which six isolates produced highest number of sclerotia (18-23 μm) of bigger size (103.3 to 117.2 x 90.1- 106.5 μm) Whereas 23 isolates produced minimum number of sclerotia (8-13) that too with smaller size (72.7-87.5 x 57.1-73.5 μm). Based on sclerotial morphology, two groups of isolates could be formed, one with oblong shape having irregular edges and the other being round with regular edges. Culturally the isolates varied in terms of mycelial growth (submerged, partially submerged and fluffy), colour (dark black to greyish cottony) and size of colony (growth rate).

Sharma *et al.* (2012) made attempt to find out the diversity in *R. bataticola* populations in India, a total of ninety four isolates collected from *R. bataticola* infected chickpea plants from different agro climatic regions of India, and were analyzed for different morphological properties. *R. bataticola* isolates were found to be very diverse in respect to their cultural and morphological parameters like colony colour (light black to black and light grey to grey), growth pattern (radial to irregular), growth rate (fast and slow), mycelial characters (colour, low and high), sclerotial initiation time, sclerotial intensity (high, moderate and low) and morphology of the sclerotia (oblong, ellipsoid, irregular and round).

Varma and Pathe (2013) evaluated morphological and cultural variability of *R. bataticola* responsible for charcoal rot of soybean twenty two isolates of *R. bataticola* obtained from soybean from different places of Madhya Pradesh state showed considerable variability in cultural and morphological characters. Cultural studies of all twenty two isolates of *R. bataticola* revealed that isolates do not differ in the type of mycelial growth and growth rate on potato dextrose agar medium to be favorable for luxuriant growth for all the isolates. Morphological studies of different isolates of *R. bataticola* revealed variability in size of sclerotia (72.1-99.9x57.1-106.5 μm) and number of sclerotia (9.2-22.2 μm). On the basis of

measurement, the sclerotia were categorized into three groups i.e small, medium and large.

2.4 Aggressiveness of *R. bataticola*

The two isolate of *M. phaseolina* were catergorized into two groups virulent and avirulent according to production of pectolytic and cellulolytic enzymes. These enzyme were important in further development of the disease (Chan and Sackston, 1972). Pectolytic and cellulolytic enzymes seem to play important role during pathogenesis (Bateman, 1963; Goodman *et al.*, 1967; Wood, 1965) and their different pathogenic ability to produce these enzyme in host issue (Hancock and Miller, 1965; Chan and Sackston, 1971; Dhingra *et al.*, 1974).

Hooda and Grover (1982) isolates of *Macrophomina phaseolina* obtained from different plant spp. and plant parts of the same host (*Vigna radiata*) differed in their morphological and cultural characteristics. There was no correlation between these characteristics and their pathogenicity on *Vigna radiata*. Ratnoo *et al.* (1997) found pathogenic variation in the isolates of *M. phaseolina* from different cowpea growing areas of Udaipur.

Sobti and Sharma (1992) isolated *R. bataticola*, purified and named according to the location at which it was collected. Cultural characteristics, pycnidial formation and virulence of the isolates were investigated. No correlation between radial growth and virulence or virulence and pycnidial formation was found. Varied range (13 to 63 per cent) of disease incidence on groundnut rot root with different isolates of *R. bataticola* recorded.

Gupta *et al.*, (2006) studied the virulence of fourty chickpea isolates tested on susceptible variety BG 212 through blotter paper technique and grouped them into two pathotypes which were related to central region of the country. The disease reaction of the isolates varied from 6.6 – 9.0 on a scale of 1-9. Based on the disease rating and maximum mortality after inoculation, all the isolates caused varying degree of incidence on variety and found two pathotypes *i.e.* virulent (02 isolates from Chhattisgarh) where the disease rating ranged from below 7.0 score and highly virulent (38 isolates from Madhya Pradesh, Chhattisgarh, Maharashtra and Gujarat), disease rating score was 7.0 to 9.0.

Kanchan and Biswas (2009) studied pathogenic variability of *R. bataticola*, causal agent of leaf spot and blight disease of pigeon pea. The potency of infection of different isolates of *R. bataticola* was varied on four different cultivars of pigeon pea viz. Prabhat, Bahar, T-7 and T-21. The *R. bataticola* isolates T-4 and T-5 showed maximum disease severity with 58.3% each on Bahar. Whereas, on T-7 and Prabhat the disease severity was recorded 54.4 % and 53.3%, respectively. Among the pigeon pea varieties, T- 21 was most susceptible with disease severity range 30.6% to 49.3% whereas, Bahar and T-7 were quite resistant. The maximum disease severity was recorded in the plants that had attained the age of 47 days followed by 57 and 67 days old plant. The early age group of plant showed low disease severity as compare to middle and above middle age group.

Mohan and Balabaskar (2012) conducted the pot culture experiment by artificial inoculation of the pathogen which revealed varied levels of pathogenicity with different isolates. Among the twenty isolates of *M. phaseolina* collected from different conventionally groundnut growing areas of Cuddalore district of Tamil Nadu, the isolate from Vengatakuppam (MP18) was found to be the most virulent. The highest incidence of 67.34 per cent at harvest was recorded and the isolate MP5 collected from Pattampakkam was the least virulent with the lowest disease incidence of 35.60 per cent.

Kumar *et al.* (2013) tested eight isolates of *M. phaseolina* causing root rot, for their pathogenicity using susceptible variety viz. RMO-225 of moth bean, RMG-62 of mung bean, Varsha uphar of okra and M-83 of clusterbean, had varying disease incidence levels in pot culture study using sterilized and unsterilized soil. The pathogen was inoculated to test virulence, isolate Bikaner on moth bean, isolate Hisar on mungbean, isolate Hanumangarh on okra and isolate Nohar on cluster bean and found that these isolate were highly virulent in both types of soil. The incidence of disease appeared first in okra (50.0 to 96.6% and 30.0 to 70.0%) followed by cluster bean (46.6 to 93.3% and 30.0 to 66.6%), moth bean (33.3 to 90.0% and 26.7 to 73.3%) and mung bean (30.0 to 90.0 %and 23.3 to 66.6%) sterilized and unsterilized soil, respectively.

2.5 Anastomosis

2.5.1 Hyphal anastomosis

Hyphal anastomosis is comparable to cell fusion events in other organisms, such as fertilization between egg and sperm, and muscle differentiation. In fungi that outcross, fusion between opposite mating-type cells is a prerequisite for entry into sexual reproduction (Coppin *et al.*, 1997). Although vegetative hyphal fusion has been observed in filamentous fungi since the earliest days of mycology. Buller (1933) was the first to outline the process of anastomosis from start to finish, surprisingly little has been added to Buller's framework in the past 60 or so years (Gregory 1984; Rayner 1996). Mechanistically, it is useful to view hyphal anastomosis as three distinct physiological states of participating hyphae pre-contact, post-contact, and post-fusion. Hyphae at the leading edges of the colony exhibit negative autotropism or avoidance which keeps a suitable distance between growing tips Prosser (1994). The mechanism of avoidance is unclear, although suggestions of both a negative response to accumulating metabolites and positive chemotropism to oxygen levels have been made by Gooday (1975). Behind the growing colony margin, hyphae involved in anastomosis exhibit positive autotropism or attraction of hyphal tips (Bhuiyan and Arai 1993; Buller 1933; Garnjobst and Wilson 1956; McCabe *et al.*, 1999). The pre-contact initiation of new tips (very short specialized fusion branches called pegs) and redirection of hyphal growth are most likely due to diffusible chemical signals Gooday (1975). Only a few exceptions to the rule of fusion between hyphal tips have been reported most notably hyphae to spore fusions in the nematode trapping fungi *Arthrobotrys conoides* and *Arthrobotrys cladodes* (Hay, 1995). Hyphal tips show growth arrest after physical contact prior to hyphal fusion. There are both temporal and spatial aspects to the fusion event. Initially, the cell wall is broken down, presumably by the delivery of hydrolytic enzymes to the contact point, and then a new cell wall bridge is formed between the two hyphae, presumably by the delivery of cell wall material to this area. Galun *et al.* (1981) reported that chitin could be visualized in the anastomosis bridges, in peg-like projections, as well as in hyphal tips. Post-contact events of hyphal fusion, involving plasma membrane fusion and cytoplasmic mixing, are also virtually uncharacterized, although ultrastructural studies on anastomosis in *Cryphonectria parasitica* showed numerous

microtubules and mitochondria across newly formed cytoplasmic bridges (Newhouse and MacDonald, 1991).

An individual hypha (a multinucleate, multicellular filament with incomplete crosswalls, or septa) grows by hyphal tip extension and branching. Hyphal tip growth is believed to be mediated by delivery of the components for cell wall extension to hyphal tips by a vesicle supply center, or “Spitzenkörper” (Girbardt, 1957 and Riquelme *et al.*, 1998). Numerous mutants have been identified that affect hyphal tip growth and branching (Perkins *et al.*, 2000) including mutants in signaling pathways (Lee *et al.*, 1998) and the cytoskeleton (Xiang and Morris, 1999). As fungi grow fusions between hyphae are continuously formed (a process called anastomosis), yielding a network of inter-connected hyphae or mycelium that makes up the fungal individual. Networked hyphae are presumably important in intrahyphal communication and homeostasis in an individual colony during growth and reproduction.

Growth in filamentous fungi thus consists of three balanced processes: hyphal tip extension, branching, and fusion. Of these the molecular genetics and physiological control of hyphal anastomosis is the least understood. Hyphal fusions are a way to increase protoplasmic flow restricted by septa, which may be important in influencing hyphal pattern formation in the mycelium (Buller, 1933; Davidson *et al.*, 1996 and Rayner, 1996). Different fungal individuals are capable of undergoing hyphal fusion with each other to form a heterokaryon, whereby genetically different nuclei may coexist in a common cytoplasm. Heterokaryon formation has potential benefits of functional diploidy and mitotic genetic exchange (“parasexual cycle”) (Pontecorvo, 1956.) or may increase biomass for cooperation in physiological efforts such as resource exploitation or asexual/sexual reproduction (Buller, 1933; Gregory, 1984). Although there are obvious benefits to filamentous fungi to form heterokaryons, a genetic mechanism exists that restricts heterokaryon formation between two genetically different individuals. If individuals differ in specificity at one or more (heterokaryon incompatibility also called vegetative incompatibility) loci, heterokaryotic fusion cells are usually compartmentalized and undergo death by a lytic process (Garnjobst and Wilson, 1956), a phenomenon called vegetative incompatibility or sometimes referred to as

heterogenic, heterokaryon, or somatic incompatibility (Esser and Blauch, 1994; Glass and Kulda, 1992; Leslie, 1993). Heterokaryosis by hyphal fusion is believed to be virtually excluded in nature by the action of het genes (Caten and Jinks, 1966; Mylyk, 1976; Pandit and Maheshwari, 1996; Puhalla and Speith, 1985). Vegetative incompatibility reduces the risks of transmission of infectious cytoplasmic elements, such as virus-like dsRNAs (Caten and Jinks, 1966; Debets and Griffith 1998), and of exploitation by aggressive genotypes (Debets and Griffiths, 1994) this self-defence mechanism may be important in filamentous fungi because they lack cellular compartmentation.

Rhizoctonia species represents an amalgam of taxonomically diverse group that differ in many significant features, including their sexual stages, asexual stages and other characters (Talbot, 1970 and Tu and Kimbrough, 1978). The AG system is considered the most useful grouping method for *R. solani*, however, the behavior of this fungus cannot be completely understood solely in terms of AGs. Since its inception, the anastomosis group concept has represented the single most important advance toward understanding genetic diversity in *Rhizoctonia*, initially within *Thanaterophorus* (Parmeter *et al.*, 1969) and more recently within *Ceratobasidium* (Ogoshi, 1978, 1985) and (Oniki *et al.*, 1985). Affinities for hyphal fusion (anastomosis) (Parmeter, 1970; Parmeter and Whitney, 1970 and Ogoshi, 1985) and DNA base sequence homologies (Kuninaga, 1980; Kuninaga and Yokosawa, 1980) has been used to assess genotypic relationship among isolates of *R. solani*, *R. zae*, *R. oryzae*, *R. repens* and binucleate *Rhizoctonia* spp. with *Ceratobasidium* teleomorphs.

2.5.2 Microscopic analysis of hyphal anastomosis

Hyphal anastomosis is comparable to cell fusion events in other organisms. In fungi that out-cross, fusion between opposite mating-type cells is a prerequisite for entry into sexual reproduction Coppin *et al.* (1997). Although vegetative hyphal fusion has been observed in filamentous fungi since the earliest days of mycology, Buller (1933). There are both temporal and spatial aspects to the fusion event. Mechanistically, it is useful to view hyphal anastomosis as three distinct physiological states of participating hyphae: pre-contact, post-contact, which were observed during the course of present investigation. The pre-contact initiation of

new tips (very short specialized fusion branches called pegs) and redirection of hyphal growth are most likely due to diffusible chemical signals Gooday (1975). Hyphal tips show growth arrest after physical contact, prior to hyphal fusion was observed. The cell wall at the point of contact was dissolved (presumably by the delivery of hydrolytic enzymes to the contact point), followed by the formation of a new cell wall bridge.

2.5.3 Four reaction types

Reaction Type 0: No anastomosis or repulsion. Neutral response. Occasional hyphal contact/ parallel growth but only for a short distance.

Reaction type 1: Hyphal contact and apparent contact of wall but no membrane to contact. Both mycelium must grow in close contact with each other. Occasional hyphal contact/ parallel growth but only for a short distance. Hyphae at the leading edges of the colony exhibit negative autotropism, or avoidance, which keeps a suitable distance between growing tips (Prosser, 1994). The mechanism of avoidance is unclear, although suggestions of both a negative response to accumulating metabolites and positive chemotropism to oxygen levels have been made (Gooday, 1975).

Reaction Type 2: Wall to Wall connection, pore formation and incompatible fusion of cytoplasm leading to death of the fusing and adjoining cells. Although there are obvious benefits to filamentous fungi to form heterokaryons, a genetic mechanism exists that restricts heterokaryons formation between two genetically different individuals. If individuals differ in specificity at one or more het (heterokaryon incompatibility, also called for vegetative incompatibility) loci, heterokaryotic fusion cells are usually compartmentalized and undergo death by a lytic process (Garnjobst and Wilson, 1956) a phenomenon called vegetative incompatibility or sometimes referred to as heterogenic, heterokaryon, or somatic incompatibility (Esser and Blaich, 1994; Glass and Kulda, 1992; Leslie, 1993). Heterokaryosis by hyphal fusion is believed to be virtually excluded in nature by the action of het genes (Caten and Jinks, 1966; Mylyk, 1976; Pandit and Maheshwari, 1996; Puhalla and Speith, 1985). Vegetative incompatibility reduces the risk of transmission of infectious cytoplasmic elements, such as virus-like dsRNAs (Caten and Jinks, 1966; Debets *et al.*, 1994) and of exploitation by

aggressive genotypes (Debets and Griffiths, 1998) this self-defence mechanism may be important in fungi because the lack of cellular compartmentation.

Reaction type 3: Fusion of wall and membrane and was Designated as compatible reaction. As fungi grow, fusions between hyphae, or mycelium, that makes up the fungal individual. Networked hyphae are presumably important in intrahyphal communication and homeostasis in an individual colony during growth and reproduction. Hyphal fusions are a way to increase protoplasmic flow restricted by septa, which may be important in influencing hyphal pattern formation in the mycelium (Buller, 1933; Davidson *et al.*, 1996; Rayner, 1996). Different fungal individuals are capable of undergoing hyphal fusion with each other to form a heterokaryon, whereby genetically different nuclei may coexist in a common cytoplasm. Heterokaryon formation has potential benefits of functional diploidy and mitotic genetic exchange (“parasexual cycle”) (Pontecorvo, 1956) or may increase biomass for cooperation in physiological efforts such as resource exploitation or asexual/sexual reproduction (Buller, 1933; Gregory, 1984).

Self- anastomosis reported to occurs commonly in culture of *R. solani* and it is often assumed that cross anastomosis between cultures also occurs commonly. Schultz (1936) and Richter and Schneider (1953) paired a large number of cultures and, on the assumption that successful anastomosis indicated a close genetic relation, they attempted to group their cultures according to their success in anastomosing.

Matsumoto *et al.* (1932) early recognized that anastomosis in *R. solani* is a complex process, and that there are varying degrees of anastomosis. They found that some isolates fused readily to produce mixing of the cytoplasm. Other isolates showed cell wall fused but no mixing of the cytoplasms of the fused cells. Yet other isolates failed to fuse at all.

Kernakamp *et al.* (1952) have also shown that hyphae of the singal spore isolates from the same parent may fail to anastomosis. In light of these observations plus our nearly complete ignorance of the mechanisms underlying the complex process of anastomosis in *R. solani*, particularly with regard to the genetics of success or failure, it is clear that the failure to anastomosis cannot be used as a criterion for species distinction.

Whitney and Parmeter (1963) investigated heterokaryon formation following anastomosis between single-spore cultures derived from one parent. They found that two distinctive single-spore cultures, when paired regularly gave to a culture type, different from that of the single-spore culture but similar to the original parent. Ten other distinctive single-spore isolates of same parent was then paired with each of the first two single-spore cultures. All ten give rise to a new culture type, resembling the parent, with one or other, but not with both of the first two single-spore cultures.

Parmeter *et al.* (1969) suggested several methods to determine the AG of a field isolates of *R. solani*. The field isolates can be paired with a known AG tester isolates on 2% water agar; on the membrane placed over nutrient agar that can be removed, placed on the glass slide and examined with a compound microscope; or on the a glass slide coated with nutrient agar. If on fusion occur at the junction of the hyphae of the two isolates, the field isolate is considered to be in a different AG that the tester isolates, they are considered to be in the same AG. After fusion of hyphae of the two isolates, a very interesting and important phenomenon occurs.

Ogoshi (1975) indicated that there may be pathological, ecological and morphological differentiation in *R. solani* and it is suggested that this differentiation can be seen in anastomosis groups. In Japan *R. solani* consists of 5 such groups. Ogoshi (1987) described the characterization of anastomosis and intraspecific groups of *R. solani* based on pathological, morphological, vitamin requirement, serological, isoenzyme and DNA-base differences. Gonzalez *et al.* (2001) determined whether each anastomosis group corresponds to a distinct species or to divergent populations from a single species. Several characters, such as morphology, biochemistry, cytology, and ultrastructure, are used with recent phylogenetic studies using (nuclear ribosomal DNA). *R. solani* is a destructive fungal pathogen with a wide host range. The *R. solani* complex species includes several divergent groups delimited by affinities for hyphal anastomosis (Mohammadi *et al.*, 2003).

Sen *et al.* (1999) studied common hyphal and colony morphological traits, displayed by the Finnish binucleate *Rhizoctonia* isolates and a range of Canadian orchid root endophytes enabled them to be placed in the anamorphic genus

Ceratorhiza. Five main anastomosis groups, were identified and included groups that contained different combinations of scots pine, *G. repens* and Canadian *Ceratorhiza* spp. isolates. Two scots pine root endophytes anastomosed with a phytopathogenic Japanese tester isolate, confirming their membership of anastomosis group I, which is known to include the teleomorphic species *Ceratobasidium cornigerum*.

Rauf *et al.* (2007) an investigation was conducted to study the occurrence of *R. solani* anastomosis groups (AGs) on potato. 127 *R. solani* isolates were recovered, all of them were found multinucleate. Anastomosis group determination revealed 81.89% isolates belonging to AG 3 followed by AG 5, 8.66% and AG 4, 5.5%.

Mikhail *et al.* (2010) accounted the current *R. solani* anastomosis groups in Egypt. Workers isolated 51 isolates from different host (cotton, peanut, chickpea, flax, tomato, watermelon, potato, cantaloupe, pepper and lupine.) Microscopic examination revealed hyphal anastomosis (method opted was described by Windels and Nabben, 1989) of all the tested isolates which were divided into four anastomosis groups. 17 isolates (33.33%) belonged to AG-2-2, 17 isolates to AG-4-HG-I, 17 isolates (13.73%) to AG-4-HG-II while 10 isolates (19.61%) belonged to AG-5.

Rashad *et al.* (2012) reported diversity among some Egyptian isolates of *R. solani* based on anastomosis grouping. Nine isolates were obtained from different host (common bean, broad bean, bell pepper, tomato and cucumber). The obtained isolates were assigned to AG according to hyphal anastomosis. Of these, six isolates belonged to AG 2-2 IIIB, while the other three isolates belonged to sAG 4 HG-I.

Tuncer and Eken (2013) studied anastomosis grouping behaviour of *R. solani* isolated from Pepper. Isolated ninety eight isolates of *Rhizoctonia* spp. from roots of pepper (*Capsicum annuum* L.). The most prevalent multinucleate anastomosis groups (AG) were AG-4 (85.2%), followed by AG-2 type 1 (7.4%), AG-6 (5.0%), and AG-3 (2.5%). The population of binucleate *Rhizoctonia* spp. comprised AG-A (82.4%), AG-K (11.8%) and AG-G (5.9%). *R. solani* AG-3 and

AG-6, as well as binucleate *Rhizoctonia* spp. AG-G and AG-K on pepper (*Capsicum annuum*) were determined.

2.6 Screening of chickpea entries

Krishnamohan *et al.* (1981) tested twenty chickpea entries screened under field conditions and artificially in the green house, using a 1-6 score, BG 205 and BG 206 found moderately susceptible in the green house, and highly resistant under field conditions against *R. bataticola*. Singh *et al.* (1982) reported that three chickpea cultivars did not related to their resistance to *R. bataticola*. Carbohydrate content was higher in the susceptible cultivar in comparison to resistance cultivar.

Singh and Mehrotra (1982) reported few lines resistant to *R. bataticola*. Cultivars BG-203, G-543 and Hare chhole exhibited resistance reaction when grown in infested soil. Reddy *et al.* (1990) tested resistance to wilt and different root rots of chickpea and found that the mortality of variety J.G.-62 was 100% whereas, Avrodhi and ICC-48 showed 20% mortality.

Baker and Ahmed (1991) evaluated the resistance against 90 genotypes of chickpea in field infested with wilt, dry root rot pathogens and found that ICC 12263 was most resistant. Jayanti and Bhatt (1993) evaluated pre-germinated seed of 21 chickpea varieties, which were sown in contaminated soil. *R. bataticola* caused seed rot within 24 hours in NEC 874 and EG 234, Bold 2375, BG 209, JG 62, JG 315, ICC 3357 and JG 1133 developed necrotic lesion 3 to 5 cells deep on the hypocotyls region within seven days. RSG 44, AGC 677, NEC 41, GL 269, JG 74, ICC 8983 and ICC 5003 developed only superficial necrosis along the hypocotyle region. BGM 416, BG 416, ICC1376 and ICC 113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases.

Monga and Sheo (1995) assessed different cotton varieties against *R. bataticola* causing root rot. *Gossypium arboreum* varieties were more susceptible (43.9–63.1 % disease incidence) and minimum incidence was noted in G 1 (43.9%) followed by LD 327 (52%) and RG 8 (61.2%). In case of *Gossypium hirsutum* varieties, LH 900 showed minimum disease incidence (19.1%) followed by varieties LH 886 (28.1%) and F 505 (30.7%) with maximum in DS 1. Forty G.

hirsutum lines along with H 777 as check were screened. B 1371 (11.2%) showed relatively lower root rot incidence as compared to 45% in check variety H 777.

Prajapati *et al.* (2003) Forty-five chickpea cultivars were evaluated for resistance to dry root rot *R. bataticola* (*Macrophomina phaseolina*). Among these, the following cultivars were rated as resistant which are H 355, JG 315, Pusa 212, G 543, ICC 32, BG 244, Phule G5, ICC 2644, 10384, 11224, 10630 and 11332, ICCV10, ICC numbers 12441, 12263, ICCL 81010, 81002, Avrodhi, ILC 702 and ILC 200. Five cultivars were moderately resistant (GCP-9504, Phule G 96022, 96020 and GL 91059) and 9 were tolerant (JG 22 and 23, and Phule G -96022, 95123, 93208, 92028, BCP 91 and BCP 28).

Ingle *et al.* (2004) eight safflower cultivars (AKS 152, AKS 68, AKS 216, AKSF 13, Sharda, AKS 65, A 1 and Bhima) were evaluated for resistance to *R. bataticola* (*Macrophomina phaseolina*). Ten day old seedlings were inoculated with a suspension from a 7-day-old culture of *R. bataticola*. The inoculated seedlings were incubated at 27 degrees C under a day-night period of 12-12 h. Disease incidence was evaluated from the 5th to the 15th day after inoculation. AKS 152 and AKS 68 were found resistant, whereas AKS 216 and AKSF 13 were moderately resistant, Bhima and A 1 were highly susceptible to *R. bataticola*.

Pande *et al.*, (2004) screened twenty-nine chickpea germplasm accessions, ten cultivars, eight advanced breeding lines for resistance to dry root rot, caused by *R. bataticola* (*Macrophomina phaseolina*). Artificially inoculated 8 day old seedlings were incubated at 35 degrees C with 12 h photoperiod. The dry root rot was scored on a 1-9 rating scale on the 8th day after inoculation. Based on the disease score (DS), the accessions were grouped as immune (DS=1), resistant (DS>1 and <=3), moderately resistant (DS>3 and <=5), susceptible (DS>5 and <=7) and highly susceptible (DS>7). Out of the 47 lines, none were immune to dry root rot. One germplasm accession (ICC 14395), a cultivar (ICCV 2) and an advanced breeding line (ICCX830203-BH-BH-11H) were resistant to dry root rot. Of the remaining lines, 22 were moderately resistant, 19 susceptible and 3 highly susceptible. The disease severity in the two susceptible lines (BG 212 and ICC 12267) was used as the control and rated 9.

Mishra *et al.* (2005) have tested 470 chickpea germplasm lines against *R. bataticola* and found KG 86, KWR 4, KWR 108 and KWR 277 as resistant genotypes. Jayalakshmi *et al.* (2008) identified the sources of resistance to dry root rot in chickpea. Therefore, conducted a pot culture experiment for screening of chickpea with twelve breeders promising entries and one susceptible check JG 62. Out of twelve genotypes tested, four (GBM-6, GBM-2, ICCV-10, GCP-101) were found resistant, two (GMB 10 and AI) were moderately susceptible, three were susceptible and three were highly susceptible to dry root rot disease.

Chaturvedi and Dua (2009) have screened twenty five resistant cultivars. Cultivars KPG 59, Radhey and K 50 were found to be most promising against dry root rot of chickpea. Aghakhani and Dubey (2009) twenty three isolates of *R. bataticola* causing dry root rot of chickpea (*Cicer arietinum*) were collected from 10 different major chickpea growing states of India. Isolate were highly variable in their morphological and cultural characters as well as in their pathogenicity /virulence. The virulence analysis of the isolates on a set of chickpea cultivars namely ICC12441, ICC1224, ICC12450, Pusa 362, BGD112, Pusa1103, Pusa212, Pusa1088 were conducted *via.*, blotter paper method as well as sick soil method. Pathogens were grouped into 6 pathotypes. The pathotype groups were related to agro ecological region of the country. The most virulent isolate (RBI from Bangalore, Karnataka) was fast growing and produced largest sclerotia. A set of cultivars was proposed for the first time for differentiating the pathotypes of *R. bataticola* causing dry root rot of chickpea.

Gupta *et al.* (2012) assessed one hundred seventy chickpea genotypes for locating new and better sources of resistance against dry root rot through blotter paper technique and in multiple disease sick fields at JNKVV, Jabalpur during 2007 to 2010. The susceptible cultivar BG 212 was used which exhibited 100 per cent mortality *i.e.* rating 9 in 1-9 scale. Out of 170 accessions, sixty eight genotypes exhibited resistant reaction (<10% mortality), out of which twenty eight were the promising lines namely (JG 1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas) from JNKVV, Jabalpur. These lines

further evaluated for their performance in sick field for three consecutive years and revealed six lines viz., JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas as resistant exhibiting <10 per cent mortality, however fourteen lines showed moderately resistance reaction. Resistant genotypes had white healthy root system with greater number of lateral roots. All other lines having >20% mortality showed necrotic root lesions leading to extensive root rotting.

2.7 Soil factors affecting development of dry root rot

Shanmugam and Govindswamy (1973) studied the physiological parameter of *acrophomina phaseolina* causing groundnut root rot and found that, it grew best at pH of 5.0.

Papavizas (1977) reported 75% of the sclerotia of *Macrophomina phaseolina* survived for 1 year in natural soil kept at 26°C and at 50–55% of the soil moisture holding capacity. Although survivability was reduced in a very acid soil (pH 4.5) collected under a pine stand, 33% of the sclerotia survived for one year. Soil pH had very little or no effect on sclerotial survivability whereas, temperature and moisture content were the two most important factors affecting survivability of sclerotia.

Nene (1979) observed factor affecting dry root rot in chickpea and founded that vertisols seem to favour the disease more than alfisols.

Nene *et al.* (1979) found that alfisol extract medium supports less sclerotia production than vertisol extract medium. The dry root rot is observed more in vertisol at ICRISAT center in both pigeonpea and chickpea.

Srivastava and Dhawan (1980) conducted an experiment using two isolates of *M. phaseolina* (highly virulent Mp 8 and less virulent Mp 18) causing root rot of *Brassica juncea*. In case, of soil pH maximum plant mortality was observed in soil of pH 5.5 and disease intensity decreased with the increase in soil pH above 5.5. Plant mortality in soil of pH 4.5 and 8.5 was found to be poor. Although, mortality was very poor and slow in case of isolate Mp 18. Wyllie and Calvert (1969) reported that the higher disease severity caused by *M. phaseolina* in sandy soils across different hosts.

Satischandra *et al.* (1980) studied the various conditions affecting the survival of *R. bataticola* in black soil. Variation in pH and addition of salts did not affect the saprophytic activity of fungus whereas, it is adversely affected by soil moisture.

Taya *et al.* (1990) observed incidence of *R. bataticola* (*Macrophomina phaseolina*) on *Cicer arietinum* was higher on sandy than on clay soil. Low soil moisture was conducive to disease development. Disease was enhanced by increased levels of N fertilizer and decreased by increased P. When both fertilizers were added together the effect of N was masked by that of P.

Sugha *et al.*, 1994 investigated that the soil characteristic soil pH in relation to fusarium wilt in chickpea, found that the maximum wilt occurred at pH 5.2, with slight decline toward neutrality and significant reduction of disease with increasing soil alkalinity.

Chauhan *et al.* (1965) reported high wilt incidence in vertisol as it contain high clay content (64%). Krishna and Krishnappa (1996) reported *Fusarium oxysporum* f.sp. *ciceri* spore was high in vertisol (1300 CFU/g soil) when compared to alfisol (1153 CFU/g soil). The fungal population was high in vertisol which was significantly influenced by soil type with a higher severity in vertisol when compared to alfisol. Total mortality was observed on 20-24 days after sowing in vertisol while the wilt incidence was only 73.4% in alfisol.

Mathur and Sinha (1968) studied the disease development in gaur and gram attacked with *Sclerotium rolfsii*, observed that the infection in guar was maximum at pH 6.6 (54.2%) and in gram at 5.7 (89.6%). Alkaline condition reduced the disease in both the crops.

Chattopadhyay and Mustafee (1977) reported that *Sclerotium. Rolfsii* from jute, showed considerable increase in population at all pH levels tested, however, growth was good at pH 5 and 6.

Singh and Gandhi (1991) tested the effect of soil pH on seedling mortality of guar caused by *Sclerotium rolfsii* reported that *S. rolfsii* caused maximum mortality to guar (*Cyamopsis tetragonoloba*) seedlings at pH 6.1 and an decreases at pH 8.4 significantly reduced disease incidence. The efficacy of 4 seed treatments carboxin (as Vitavax), tolclofos-methyl (as Rizolex), captafol (as Difolatan) and

thiram (as Thiride) was affected by soil pH. At pH 6.1 disease incidence was significantly reduced while at 6.8 and 7.5 no significant differences were observed.

Shim and Star (1999) examined the effect of soil pH on sclerotial germination and pathogenicity of two isolates of *Sclerotium rolfsii* on peanut and reported that sclerotial germination for both isolates was greater ($P=0.05$) in acidic soil than at alkaline pHs. Similarly, percentage of peanut stems infected by *S. rolfsii* in greenhouse tests was greater at soil pH 5.6 than at alkaline soil pH ($P=0.05$), but disease did develop at soil pH 8.7 and 9.8. In contrast to a previous *in-vitro* study, these data confirm that sclerotia of *S. rolfsii* will germinate and initiate disease at soil pH > 7.0 .

Jimenez (2006) conducted an experiment and found that the soil texture had a significantly high impact on the *M. phaseolina* soil populations dynamics. There was variation in the populations across soil types but this variation was significantly different in the sandy soil texture in which populations were reduced. It was found an increasing number of colonies of *M. phaseolina* as the amounts of silt loam in each soil were increased in order to form the artificial soil texture. In the soil *M. phaseolina* populations may be indirectly influenced by clay content. There is some evidence to suggest that increasing levels of organic matter contained in the silt loam soil used to form artificial soil textures can support antagonistic microorganisms that compete with microsclerotia in the soil having a negative impact on soil populations. Decreasing soil populations in sandy soils also suggest that propagules require a certain level of soil nutrients to remain viable and active in soil. Sundar *et al.* (2009) tested effect of different soil type on fungicidal control of *Rhizoctonia solani* causing seedling mortality in mungbean. Three soil of different textural group viz. sandy, loam and silt clay were used. The result revealed that disease incidence was more in silt clay soil and loam soil than sandy soil.

Parmar (2013) studied the *in vitro* effect of different pH ranged from 6.0 to 8.0 on the growth of different isolates of *R. bataticola* the maximum growth of the isolates of *R. bataticola* was recorded at 6.5 pH (84.92mm) followed by pH 7.0 (81.33mm), pH 7.5 (76.61mm), pH 6.0 (66.25mm), and the minimum colony diameter was recorded at pH 8.0 (61.69mm).

2.8 Identification of bio agent

Microorganisms that can grow in the rhizosphere are ideal to use as biocontrol against fungal pathogen since the rhizosphere provides the front line defense for roots against the pathogen. Pathogens encounter the antagonism from rhizosphere micro-organism before and during primary infection and also during secondary spread on the root. The review on the evaluation of biocontrol agents viz., *Trichoderma harzianum*, *Trichoderma viride*, and *Pseudomonas fluorescens* quoted below.

T. harzianum and *T. viride* were found effective in reducing the mycelial growth of *R. bataticola* (Deshmukh and Raut, 1992; Anju and Verma, 2007; Mandhare and Suryawansi, 2008). *T. harzianum* found most effective in inhibiting the mycelial growth of *R. bataticola* (Haque *et al.*, 1990; Pant and Mukhopadhyay, 2001; Ammajamma *et al.*, 2009). Dinakaran *et al.* (1995) observed the effect of three biocontrol agents against root rot of sesame caused by *M. phaseolina*. Seed treatment with *T. viride*, *T. harzianum* and *Bacillus subtilis* reduced levels of root rot incidence. Sankar and Jeyarajan (1995) reported that *T. harzianum* and *T. viride* significantly reduced *M. phaseolina* root rot incidence in sesame to 10.1 and 12.8 per cent, respectively, compared with 60 per cent incidence in the control plots. Carbendazim treatment did not increase the rhizosphere and soil populations of the antagonists. The soil population of *Trichoderma* spp. was maximum in plots which were applied with *Trichoderma* spp. Seed treatment with *T. harzianum* significantly increased root length, shoot length, yield and oil content.

Raguchander *et al.* (1998) reported that *T. viride* was effective in inhibiting the mycelial growth of *R. bataticola* by greatly reducing the sclerotial number.

Jayashree *et al.* (2000) noticed that *P. fluorescens* strain pf1 effectively inhibited the mycelial growth of *M. phaseolina*, the pathogen causing dry root rot in black gram cv. Co 5 and sesame cv. Tmv 6. Application of pf1 as seed treatment (10 g/kg seed) followed by soil application (2.5 kg/ha) against root rot effectively supported higher plant

growth, better native rhizobium nodulation and grain yield. Sclerotial number and root rot incidence were also greatly reduced.

Kumar *et al.* (2002) reported that plant growth-promoting rhizobacterial strains belonging to fluorescent *Pseudomonads* were isolated from the rhizosphere of rice and sugarcane. Among forty strains that were confirmed as *Pseudomonas fluorescens*, eighteen exhibited strong antifungal activity against *R. bataticola*, mainly through the production of antifungal metabolites.

Sindhan *et al.* (2002) found that *T. harzianum*, *T. viride* and *P. fluorescens* inhibited the mycelial growth and sclerotial production in *R. bataticola*. Kaswate *et al.* (2003) evaluated the antagonistic effect of *Trichoderma viride* and *Pseudomonas fluorescens* against isolates of *R. bataticola* from pigeon pea, chickpea, green gram, cluster bean, field pea, cotton, okra and safflower. Plates inoculated with *R. bataticola* and each of the biological control agents were work out for the inhibition of mycelial growth. *T. viride* was the most effective in the inhibition of the various isolates of *R. bataticola* (100% inhibition) and *P. fluorescens* (73.98-78.94%).

Rajpurohit (2004) reported that seed treatment with *T. viride* at the rate 4 g/kg was highly effective against *M. phaseolina* and reduced the disease from 28.7 to 12.5 per cent and increased seed yield from 399 to 490 kg/ha.

Karthikeyan *et al.* (2006) evaluated three isolates of *T. viride*, and one isolate in each of *T. harzianum* and *P. fluorescens* *in vitro*, to assess their antagonistic potential against *M. phaseolina* causing root rot of groundnut. *T. viride* and *T. harzianum* were the most effective in reducing the mycelial growth and sclerotial formation of *M. phaseolina* culture filtrates of *T. viride* inhibited the growth of the pathogen as well as sclerotial germination to a greater extent.

Nair *et al.* (2006) conducted a pot experiment to determine the effects of biological control agents (*T. harzianum*, *T. viride* and *B. subtilis*) on *R. bataticola* (*M. phaseolina*) infecting sesame cv. var-1 and found that seeds treated with spore suspension of *T. harzianum* were superior over all

the other treatments. The germination percentage was maximum (92%) in case of *T. harzianum* followed by *T. viride* (89.33%).

Durai (2004) made efforts to manage charcoal rot of sesame caused by *M. phaseolina*. The most effective treatment was the combination of mancozeb (100 ppm) + soil application of bioagents viz., native isolate of *T. viride* (10 g/kg of soil) and *P. fluorescens* (20 ml/kg of soil). The treatment significantly reduced disease incidence (10.67%) and enhanced plant growth components.

Singh *et al.* (2005) evaluated fungal antagonists against *M. phaseolina* causing root rot of mungbean. Out of four bioagents (*Aspergillus sp.*, *Gliocladium virens*, *T. harzianum* and *T. viride*) tested against the root rot pathogen. *T. harzianum* proved most effective, which inhibited maximum mycelial growth (71.85%). Highest disease control 66.67% was recorded by *T. harzianum* followed by *G. virens* 50.00% incidence under green house conditions.

Lokesha and Benagi (2007) evaluated the efficacy of biocontrol agents *in vitro* and *in vivo* on *M. phaseolina* which causes dry root rot of pigeonpea. The results revealed that the both bioagents; *Trichoderma virens* and *Pseudomonas fluorescens* significantly inhibited the mycelial growth of *M. phaseolina* by 78.22 per cent and 76.66 per cent respectively. Talc based formulations of these bioagents were applied, seed treatment (4g/kg seed) along with soil application (2 g talc powder mixed with farm yard manure/pot) supported the maximum plant stand and less root rot incidence compared to other treatment and found significantly superior to chemical seed treatment (2g/kg seed). Seed treatment + soil application with *T. virens* recorded less root rot incidence (2.89%) among the four bioagents used.

Aly (2007) an *in vivo* experiment was conducted to determine the efficacy of *Trichoderma* spp. against *M. phaseolina*. Greenhouse evolution of the interaction between *M. phaseolina* isolates and *Trichoderma* spp. isolates revealed a very significantly effective result. *Trichoderma* play a role as source of variation in survival, plant height, and dry weight of plant, while *M. phaseolina* as source of variation in pre-emergence damping-off and post-emergence damping-off of plant.

Moi and Bhattacharyya (2008) attempted to control rootrot of sesame caused by *M. phaseolina* by suppressive organisms following *in vitro* and *in vivo* studies. Five isolates of *Bacillus subtilis* namely bs-12, bs-14, bs-17, bk-1, bm-1, *P. fluorescens* and *T. harzianum* were taken as antagonists. *In vitro* studies revealed that, in case of dual culture technique, highest mycelial inhibition was recorded in bs-17 (up to 65.56%). Under *in vivo* studies *M. phaseolina* was grown in sand maize meal medium and inoculated in the pots @ 200 g/pot. The symptom of rootrot disease appeared within 10 days of sowing and continued up to 30 days. *In vivo* studies revealed that all the antagonists reduced the rootrot of sesame significantly, when applied as soil drenching or in combination of seed soaking and soil drenching. Combination of seed soaking and soil drenching gave better results than the soil drenching alone. Among the antagonists maximum disease control was observed by the *T. harzianum* which gave up to 66.17 per cent of sesamerootrot disease control followed by isolate bs-12 (up to 65.27%) and bs.17 (up to 65.25%).

Ramezani (2008) evaluated the efficacy of four fungal bioagents viz., *T. hamatum*, *T. harzianum*, *T. polysporum* and *T. viride* under *in vitro* condition against the eggplant root - rot caused by *M. phaseolina*. Among the bioagents, *T. harzianum* produced the maximum inhibition zone of 18.20 per cent compared to the minimum of 7.30 per cent by *T. hamatum*. Soil application of talc - based formulation of *T. harzianum*, *T. polysporum* and *T. viride* effectively controlled the root- rot of egg-plant under field condition.

Usha *et al.* (2009) noticed that among the antagonists assay that, *T. viride*, *T. harzianum* and *B. subtilis* were found inhibitory to *M. phaseolina*. Further, the root rot incidence was low (14.1 per cent) with *T. viride* (seed treatment + soil application) followed by *T. viride* and *T. harzianum* (seed treatment).

Manjunatha and Naik (2010) evaluated several isolates of two bio-control agents, *T. viride* and *P. fluorescens*. Bio-control agents were assessed for their ability to reduce the growth of *M. phaseolina* under laboratory conditions and

subsequently used for field studies. The most effective isolate of each bio-control agent and the common chemical for seed treatments were too used as experimental trail. It was found that trial combining with soil application through bioagent enriched farm-yard manure, along with seed treated with the bio-control agent showed maximum germination, least root rot incidence and highest yield, as compared to the other biological or chemical seed treatments included.

Govindappa *et al.* (2010) isolated three biocontrol agents such as *T. harzianum*, *P. fluorescens* and *Bacillus subtilis* from safflower rhizosphere soil and tested individually for their effectiveness in controlling root-rot of safflower. Talc based formulations were prepared and treated the seeds at different concentrations, for assessing their ability to induce plant growth and in turn control root rot disease. Among bioagents, *P. fluorescens* and *T. harzianum* (10 g/kg) proved to be effective in controlling disease under laboratory, greenhouse and field conditions. The efficacy of these biocontrol agents are equivalent to the standard fungicide Bavistin (carbendazim). Seed treatment with these biocontrol agents enhanced the seed germination and growth parameters against root-rot disease.

Jyothirmai *et al.* (2011) evaluated the plant growth promoting microorganisms viz., *Trichoderma* spp. and fluorescent pseudomonads which was isolated from the rhizosphere of groundnut, exhibited the suppression of *R. bataticola*. *Trichoderma* spp. and fluorescens *Pseudomonas* isolate 1 effectively restricted the radial growth of *R. bataticola* by 69.92 % and 67.31 % respectively. In green house studies, talc based formulations of potential *Trichoderma* spp. and fluorescens *Pseudomonas* isolate 1 effectively suppressed the seedling disease and improved the plant growth, shoot and root dry weight and pod yield.

Patel *et al.* (2011) tested *in vitro* antagonistic activity of *P. fluorescens* via dual culture which exhibited significance reduction in the radial growth of *R. bataticola* over control (90 mm colony diameter) and in the field trial study seed treatment with four strains significantly reduced the disease incidence of dry root rot in all treatments in comparison to uninoculated control, however *P. fluorescence* strains BHUPf4 was found more effective in mycelial growth reduction as well as in disease reduction followed by strains BHUP5, BHUP6 and BHUPsb.

Govindappa *et al.* (2011) screened thirty eight isolates of *Pseudomonas fluorescens* for biological control of *M. phaseolina* root-rot of safflower. These results indicated that the seedling assay is more important for selection of promising biocontrol agents besides they also increased the plant growth by controlling the disease as evidenced by induction of defense enzymes.

Devadason and Subramanian (2012) evaluated the effects of biocontrol agents on *M. phaseolina* infecting *Gloriosa superba* a medicinal plant. Under *in vitro* conditions a commercial formulation of *T. viride* and *P. fluorescens* inhibited the mycelial growth of *M. phaseolina* isolates. Seven isolates of *M. phaseolina* was completely inhibited by the commercial formulation of *T. viride*. Inhibition by other isolates ranged from 22% to 83%. The other native isolates (1–10) were not effective against the seven isolates of *M. phaseolina*. Commercial formulation of the bacterial antagonist *P. fluorescens* was tested against *M. phaseolina* isolates under *in vitro* condition. Among the seven isolates of *M. phaseolina* 88.63 % inhibit on was recorded over the control whereas, *P. Fluorescence* isolate on other hand inhibited 47.95% to 73.75%.

Doley and Jite (2012) tested biological efficacy of *T. viride* against soil-borne plant pathogen *M. phaseolina* under *in-vitro* condition. The dual culture technique was followed in which *T. viride* showed significant antifungal activity by inhibiting the mycelial radial growth of *M. phaseolina* by 71.42%. The results showed variable mycelial growth rate for all fungal isolates which was determined after 6 days of incubation in which *T. viride* showed minimum of 4 days to cover the Petri plates completely and *M. phaseolina* 6.33 days.

Gejara *et al.* (2012) *In vitro* potentialities of seven species of *Trichoderma* were evaluated against phytopathogen *Macrophomina phaseolina* by dual culture techniques. The maximum growth inhibition of test pathogen was observed by antagonist *T. koningi* MTCC 796 (T4) (74.3%) followed by *T. harzianum* NABII Th 1 (T1) (61.4%) at 7 days after inoculation (DAI). Further, mycoparasitism of antagonists were observed upto 14 DAI. Pattern of growth inhibition of test fungus was continued with maximum 14.7% increases in T4 (85.2%) followed by 6.8% elevation in T1 (65.6%) antagonists during 7 to 14 DAI. Microscopic study showed that these two antagonists were capable of overgrowing and degrading *M.*

phaseolina mycelia, coiling around the hyphae with apressoria and hook-like structures. At 14 DAI, *T. koningi* MTCC 796 completely destroyed the host and sporulated. The growth inhibitions of pathogen during antagonism was positively correlated with coiling pattern of antagonists at 14 DAI.

Belkar and Gade (2013) isolated fifteen isolates of *Pseudomonas fluorescens* from acidic and alkaline rhizospheric soil from Vidarbha and Konkan region. All the cultural and biochemical studies confirmed them to be *P. fluorescens*. *In vitro* antagonistic study showed per cent inhibition by *P. fluorescens* (Pf3) against *R. bataticola* was 67.04%.

Hegde *et al.* (2013) an attempt was made for biological management of *Jatropha curcas*, root rot caused by *R. bataticola*. Efficacy of bio agent were evaluated. Among different bioagents evaluated against *R. bataticola*, *T. virens* (57.78%) was the most effective and at par with *T. harzianum* (57.23%), *T. viride* (55.83%) and *T. koningii* (54.45%) followed by *P. fluorescens* (30.11%).

Kumar *et al.* (2013) evaluated three bio agents, *T. harzianum*, *T. Viride* and *T. atroviride*, and one bacterium, *Bacillus subtilis* isolated from moth bean fields at Bikaner, were tested for their antagonistic activity against eight isolates of *M. phaseolina* causing dry root rot of moth bean. Inhibition of the mycelium growth of *M. phaseolina* isolates by *T. harzianum*, *T. viride*, *T. atroviride* and *B. subtilis* varied from 61.1 to 70.1%, 58.6 to 66.6%, 52.0 to 63.1% and 45.8 to 54.8%, respectively. Maximum growth inhibition was caused by *T. harzianum*. In a greenhouse study, these antagonists reduced root rot incidence in a susceptible moth bean cv. Rmo- 225 up to 69.0, 65.0, 62.5 and 58.3%, respectively, as compared to control where 100 % mortality was observed. *T. viride* was most effective among the antagonists trial in minimizing the disease followed by *T. harzianum*, *T. atroviride* and *B. subtilis*.

Manjunatha and Naik (2010) studied the mechanism of antagonism of two isolates of *Trichoderma* spp. viz. *T. viride* and *T. harzianum* were selected for intensive studies on the basis of their *in vitro* antagonistic activity towards *R. bataticola*. On microscopic examination, these fungal antagonists were found parasitic on *R. bataticola* as evidenced by coiling. Penetration, disintegration and overlapping on germinating sclerotia as well as hyphae of the pathogen. The

sclerotial production of *R. bataticola*, was drastically reduced in the presence of *Trichoderma viride* (tv-41), *T. harzianum* (th-r) and *T. viride* (tv-r) recording 1.00, 0.67 and 2.33 sclerotia per microscopic field respectively against check.

Ashwini *et al.* (2014) evaluated the efficacy of different antagonists viz. *Trichoderma viride* and *Pseudomonas fluorescens* against *M. phaseolina* of black gram by dual culture technique. Among these bioagents, *Pseudomonas fluorescens* proved best antagonist against *M. phaseolina* (62.41%) by inhibition mycelial growth whereas, lowest mycelial growth was observed against *T. viride* (44.94%).

Veena and Reddy (2014) studied the mode of parasitism between *Trichoderma* and *R. bataticola* under a microscope. Observed the formation of several loops and coiling around the hyphae of pathogen, forming a thick compact rope like structure followed by rupturing, twisting and leakage of hyphal protoplasm, air bubbling inside the cytoplasm, breaking of cytoplasmic continuity, aggregation of cytoplasm within cell leading to severe vacuolation.

Nagamani *et al.* (2014) conducted a study for efficient management of dry root rot caused by *R. bataticola* in chickpea. The results revealed that seed treatment with carbendazim and *T. viride* along with soil application of FYM (Farm Yard Manure) fortified with *T. viride* as the most effective in decreasing the mortality per cent.

Marisa *et al.* (2014) evaluated *Pseudomonas chlororaphis* subsp. *aurantiaca* for control of *M. phaseolina*. The bio agent caused a significant inhibition of *M. phaseolina in vitro* and reduced damping-off in the *in vivo* assays.

CHAPTER – III

MATERIALS AND METHODS

The methods and materials used during the course of present study are given below.

3.1 Materials

3.1.1 Experimental site

The present investigation was carried out at the Department of Plant Pathology and Microbial Biotechnology Laboratory, Department of Biotechnology, College of Agriculture, IGKV, Raipur.

3.1.2 Source of materials

The glassware, chemicals and other materials were obtained from Department of Plant Pathology, College of Agriculture, IGKV, Raipur.

The following instruments were used:-

1. Autoclave for sterilization
2. BOD incubator for incubation
3. Compound microscope
4. Hot air oven for glassware sterilization
5. Forceps, needle, cork borer, inoculation needle
6. Growth chamber
7. Laminar air flow for inoculation
8. Water bath
9. Electronic digital balance
10. Microwave for melting of media.

3.1.3 Source of seeds

Chickpea seeds were procured from AICRP (All India Coordinated Research Project) on Chickpea. IGKV, Raipur.

3.1.4 Cleaning and washing of glassware

All the glassware prior to use were cleaned and washed with detergent followed by rinsing with tap and distilled water. The glasswares were thereafter dried in oven at 60°C for half an hour. All the glasswares were further sterilized in hot air oven at 180°C for 2 hours. Plastic Petri plates were surface sterilized with alcohol, whereas inoculation needle, cork borer, scissors, knife and blade were sterilized by heating over the flame after dipping in 95% alcohol. Sterilization of media was done by autoclaving at 121°C temperature and 15lbs pressure for 20 minutes.

Table 3.1: Composition of different media used in present investigation

S.No.	Media	Ingredient	Content (g)
1	PDA (Potato Dextrose Agar)	Dextrose	20
		Agar- Agar	20
		Peeled potato	200
		Distilled Water	1000 ml
2	Potato dextrose broth	Dextrose	20
		Peeled potato	200
		Distilled Water	1000 ml
3	Casein hydrolysis	Peptone	5
		Skimmed milk powder	70
		Distilled Water	1000 ml
4	H ₂ S production test	Sodium thio-sulphate	0.025
		Ferrous ammonium sulphate	0.2
		Beef extract	3
		Agar	3
		Peptone	30
		Distilled Water	1000 ml
5	Gelatin broth	Yeast extract	3
		Peptone	5
		Gelatin	120
		Distilled Water	1000 ml

6	Lypolytic media	Calcium chloride	0.1
		Sodium chloride	5
		Peptone	10
		Agar	15
		Distilled Water	1000 ml
7	Nitrate reduction (broth)	Potassium nitrate	1
		Beef extract	3
		Peptone	5
		Distilled Water	1000 ml
8	King's 'B' medium	Cyclohexamide	75mg
		Novobiocin	45mg
		Penicillin	75mg
		Dipotassiumhydrogen phosphate	1.5
		Magnesium sulphate	1.5
		Protease peptone	20
		Agar	20
		Glycerol	8 ml
		Distilled Water	1000 ml
9	<i>Trichoderma</i> selective media	Rose Bengal	0.15
		Potassium nitrate	0.3
		Magnesium sulphate	0.4
		Di potassium phosphate	1.8
		Ammonium nitrate	2
		Glucose	3
		Agar	15
		Distilled water	1000 ml

3.2 Methods

3.2.1. Survey and collection of dry root rot infected plants

An intensive survey was conducted during *rabi* season, 2013-14, on the incidence of dry root rot in chickpea growing districts of Chhattisgarh. Chickpea plants showing typical symptoms of dry root rot that is, dead roots were quite brittle and show shredding of bark and lateral root, at flowering and pod formation stage were collected from different locations of Chhattisgarh state (Ambikapur, Dhamtari, Bemetara, Bhatapara, Balod, Durg, Dhamdha, Gandai, Kanker, Kawardha, Narayanpur, Patan, Raigarh, Raipur, Rajnandgaon, Saragaon and Sitapur) during January to February 2013. Samples of infected roots were collected from farmer's fields. Total 10-12 spots were selected randomly for taking root samples representing the whole field. Randomly 50 plants were selected in different locations in a field and numbers of plants infected were counted and the mean disease incidence was expressed in percentage. Wherever required, the complete infected plants were also collected for isolation of the pathogen and other studies. The per cent disease incidence was calculated by using the following formula.

$$\text{Per cent disease incidence} = \frac{\text{Number of plants affected}}{\text{Total number of plants observed}} \times 100$$

Each sample was kept in paper bag and tied with a rubber band and labelled immediately. Information pertaining to the locality, crop history, etc. was also obtained about the samples. Samples and roots were brought in the laboratory and analysed after collection. The roots of affected plant showing the symptoms of dry root rot were incised and isolation were made for the presence or absence of the causal agent.

3.2.2 Isolation of pathogen from diseased samples

The roots of diseased plant showing symptoms were washed thoroughly with tap water, small pieces of infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride (HgCl_2) solution for one minute followed by three washings with sterilized distilled water to remove traces of HgCl_2 . The pieces were then transferred aseptically to Petri plates containing PDA. Inoculated Petri plates were incubated at $25 \pm 2^\circ\text{C}$ for three

to five days and examined at frequent intervals to see the growth of the fungus developing from different pieces. As and when fungal colony appears they were transferred to PDA slant.

3.2.3 Purification and identification of isolates of *R. bataticola*

The cultures of *R. bataticola* were purified by frequent sub culturing by the hyphal tip method and maintained by mass transfer on potato dextrose agar medium at $25\pm 2^{\circ}\text{C}$. These isolates of *R. bataticola* were identified by observing the colony character, pattern of sclerotial formation and sclerotial size measured with the help of micrometer.

3.3 Variability among isolates of *R. bataticola*

3.3.1 Cultural and morphological characteristics of the isolates

3.3.1.1 Mycelial growth

Fifteen ml of the sterile PDA medium was poured into sterile Petri dishes and allowed to solidify. A five mm culture disc of *R. bataticola* obtained from actively growing region was aseptically placed at the centre of the dish and incubated. The colour of the colony and types of mycelial growth of different isolates were observed after seven days of incubation.

3.3.1.2 Morphological variability

The slides of various isolates were prepared in lactophenol from 10 days old culture for morphological studies. For morphological characteristics viz., hyphal width, size of sclerotia and shape of sclerotia were recorded after ten days of incubation (Dhingra and Sinclair, 1978).

3.4 Aggressiveness test of *R. bataticola* isolates

3.4.1 Blotter paper technique (Nene *et al.*, 1981)

Twenty seeds of chickpea (variety L 550) surface sterilized with 2.5 per cent sodium hypo-chlorite were sown in autoclaved riverbed sand in pots. Potato dextrose broth was prepared and 100 ml of broth was placed in 250 ml flasks. Thereafter they were autoclaved at 121°C temperature and 15 lbs pressure for 20 minutes. The broth was inoculated with fungus and incubated for 5 days at 25°C . Thereafter the mycelial mats were removed from flasks at the end of the incubation period. Two mycelia mats were macerated in a warring blender with 100 ml



A



B



C

Plate 1: A- Multiplication of *Rhizoctonia bataticola* inoculum on potato dextrose broth
 B- Root dipping of one weak old chickpea seedling in macerated mycelial mats of *R. bataticola* for 30 seconds
 C- Incubation of inoculated seedling, wrapped in moist blotter paper

sterilized distilled water. The five days old seedling were uprooted and the root system was washed in running water followed by rinsing in sterilized distilled water and then dipped in inoculum solution for 30 seconds. The inoculated seedlings were placed side by side on a germination paper towel. These germination paper towels were kept in a tray. These trays were placed in an incubator at 30°C for seven days having 12 hours artificial light. The germination paper towel was moistened with sterile distilled water every alternate day. At the end of incubation period (7 days) the seedlings were examined for the extent of root damage and for the mortality. Untreated seedlings were served as check (Plate 1).

3.4.2 Mass multiplication of *R. bataticola* inoculum for soil application

The isolates of the pathogen were multiplied in sand maize medium (Riker and Riker, 1936). Sand and ground maize seeds were mixed in the ratio of 19:1, moistened to 50 per cent level, filled in 500 ml conical flask and autoclaved at 121°C temperature and 15 lbs pressure for 20 minutes. Actively growing mycelial discs (9 mm) of the *R. bataticola* were inoculated into each flask under aseptic condition and the flasks were incubated at temperature 25±2°C for 15 days. The inoculum thus obtained was used for the pot experiments.

3.4.3 *In vitro* Aggressiveness test

Difference in aggressiveness among the nine isolates (collected from different chickpea growing areas of Chhattisgarh) of *R. bataticola* were tested using blotter paper technique as described by Nene *et al.* (1981) in 3.4.1. Seedlings were examined at three days intervals for extent of root damage. On pluse disease rating scale was used (Nene *et al.*, 1981) (Plate 2A).

Table 3.2: Description of the rating scale for scoring (Nene *et al.*, 1981)

Rating	Symptoms of dry root rot
1	No infection on roots
3	Very few small lesions on roots
5	Lesions on roots clear but small, new roots free from infection
7	Lesions on roots many, new roots generally free from lesions



A



B

Plate 2A: Disease rating scale 1 to 9 for *in vitro* aggressiveness test

Plate 2B: Disease rating scale for field screening

9	Roots infected and completely discoloured
---	---

3.4.4 *In vivo* pot experiment

In vivo aggressiveness test was conducted in poly house. The inoculum of each isolate of *R. bataticola*, mass multiplied on sand maize media was added to the sterilized soil. The sick pots were prepared by using the inoculum of the pathogen @ 30 gm/kg soil (Jayalakshmi *et al.*, 2008). The pots were left for 48 hrs for stabilization before sowing. Three replications for each treatment were maintained. Germination per cent, root length, shoot length and per cent mortality was observed.

Percent mortality was calculated by using the following formula:

$$\text{Per cent mortality} = \frac{\text{Number of plants affected}}{\text{Total number of plants observed}} \times 100$$

3.5 Anastomosis

In order to study anastomosis, disposable plastic Petri dishes were used. Each isolate of *R. bataticola* was paired with other *R. bataticola* isolates on 2% water agar coated slides with two replication in Petri dish (Windels and Nabben, 1989). Mycelia from the growing margins of young colony were plated 2 - 3 cm apart on water agar coated slide in a Petri dish and incubated at 25°C in the dark until advancing hyphae made contact and slightly overlapped (Mikhail *et al.*, 2010). The area of hyphal contact was then examined under the microscope. The branching habit of the isolate and the direction of hyphal growth were traced back to their origin in order to ensure that anastomosis has occurred between mycelium of two isolates. The grouping of the isolate was done on the basis of reaction types (Singh *et al.*, 1999) following microscopic observation.

Table 3.3: Reaction type and its description to group

Reaction Type	Description
-1	Repulsion before contact or just after contact, where at least one of them tends to move away from each other.
0	No anastomosis or repulsion. Neutral response, occasional hyphal contact/parallel growth but only for short distance.
1	Hyphal contact and apparent contact of wall but no membrane to membrane contact. Both mycelia must grow in close contact to each other.
2	Wall to wall connection, pore formation and incompatible fusion of cytoplasm leading to death of the fusing and adjoining cells.
3	Fusion of wall and membrane. Compatible.

3.6 Screening of chickpea entries

3.6.1 Field screening

Seeds were procured from AICRP (All India Coordinated Research Project) on Chickpea, Raipur. The field experiment was laid out during *rabi* season 2012-13 and 2013-2014 at the research farm, IGKV Raipur. Total 338 entries were screened against *R. bataticola*, out of which 118 entries during 2012-2013 were screened while, 220 entries were screened in 2013-2014. Each test entries was sown in a plot of two rows of 5 meter length 30 cm apart with one row of susceptible check variety L 550 after every two test entry and replicated twice in randomized block Design. Observations on emergence were recorded at ten and twenty DAS. Culture of *R. bataticola* was artificially inoculated in collar zone 15 days after sowing. Row of each replication was inoculated. Light irrigation was given just to activate the growth of fungus. Observations on per cent mortality were started from ten days after inoculation and recorded at five day intervals upto maturity, finally computed as follows.

Table 3.4: IIPR rating scale (Plate 2B)

S. No.	Reaction	Per cent mortality
1	R- Resistant	0-10
2	MR- Moderately Resistant	11-20
3	MS- Moderately Susceptible	21-30
4	S- Susceptible	31-50
5	HS- Highly Susceptible	51-100

Location severity index (LSI) was calculated by using the formula:

$$LSI = \frac{\text{Sum of all numerical rating of entries}}{\text{Total numbers of entries}}$$

3.6.2 *In vitro* screening

From the set of accessions of 220 chickpea entries which were undertaken during second *rabi* season 2014- 2015, 132 entries were screened *in vitro* through blotter paper method as described in 3.4.1 against *R. bataticola* isolate RB 5 (Raigarh) which was most virulent under *in vitro* aggressiveness test. Variety L550 of chickpea was used as susceptible check on blotter paper technique (Nene *et al.*, 1981). The disease rating scale 1 to 9 was used (Table 3.2). Untreated seedlings were served as check.

3.7 Soil factors affecting dry root rot disease of chickpea

3.7.1 Effect of soil pH

In order to determine the effect of different soil pH on development of disease, soil was collected from experimental field of Plant Pathology. Twenty gram of soil was taken and suspended in 40 ml sterile distilled water in 250 ml conical flask and stirred for 30 min. The soil pH was estimated 6.5. The soil pH was adjusted to 4.0, 5.0, 6.0, 7.0 and 8.0 by using NaOH and HCL (Kulkarni, 2007). The inoculum of *R. bataticola* was allowed to multiply on sand maize media. The inoculum of *R. bataticola* was added @ 30g/kg. Soil was mixed in each pH amended soil and left for 5days, meant for establishment of inoculum. Eight surface sterilized seeds of chickpea were sown in each pot and data on per cent mortality were recorded 30 days after sowing.

3.7.2 Effect of soil type

This experiment was conducted to know the effect of major soil type found in Chhattisgarh state on dry rot root development of chickpea. In order to determine the effect of four soil types viz., (Alfisol (Dorsa), Inceptisol (Matasi), Vertisol (Kanhra) and Entisol (Bhata)) were obtained from adjoining fields of College of Agriculture, Raipur. The inoculum of *R.bataticola* was allowed to multiply on sand maize media. The inoculum of *R. bataticola* was added @ 30g/kg in each soil type mixed thoroughly and left for 5days, meant for establishment of inoculum. Eight surface sterilized seeds of chickpea were sown in each pot and data on per cent mortality were recorded 30 days after sowing. Treatment details are given below -

S.No.	Treatment	Soil	Common name	Scientific name
1	T ₁	Dorsa	Clayloam	Alfisol
2	T ₂	Matasi	Sandyloam	Inceptisol
3	T ₃	Kanhra	Clayey	Vertisol
4	T ₄	Bhata	Gravely	Entisol

3.7.3 Soil texture

Soil sample was taken in a beaker and 15 ml sodium hexametaphosphate (10 %) was added and was stirred for 10 minutes in a 1000 ml beaker. The contents were transferred to 1000 ml measuring cylinder with volume make up. At 5 minutes and 5 hours, temperature and hydrometer reading were taken. Percentage of sand, silt and clay were determined with the help of Bouyoucos hydrometer (Piper, 1950) and the textural classes were determined by the help of textural diagram (International system). In the present study five textures were under taken viz., sandy, sandy loam, loam, clay and silt clay. The inoculum of *R.bataticola* was allowed to multiply on sand maize media. The inoculum of *R. bataticola* was added @ 30g/kg in each soil texture mixed thoroughly and left for 5days, meant for establishment of inoculum. Eight surface sterilized seeds of chickpea were sown in each pot and data on per cent mortality were recorded 30 days after sowing.

3.8 Potentiality of bio control agent *Trichoderma* spp. against *R. bataticola*

Studies were undertaken to identify the *Trichoderma* spp. a fungal bio control agents as potential candidates against *R. bataticola* causal agent of dry root rot.

3.8.1 Collection of soil samples

Soil samples were collected from different locations of Chhattisgarh (Abhanpur, Ambikapur, Bemetara, Bhatapara, Dhamtari, Dhamdha, Durg, Kanker, Khadsiya, Lohara, Raigarh, Raipur, Rajnandgaon, Sitapur, Thankhamariya, Uattai and Udaypur). Soil samples from the rhizoplane were collected from randomly selected and counted number of plants, which were gently uprooted using trowel to avoid damage and loss of root system. Loosely adhering soil particles were removed by gentle tapping and thereafter, roots were transferred to sterile polythene bags, brought to the laboratory and stored at 28°C. The rhizoplane soil was removed by mechanically scraping of root surface.

3.8.2 Isolation of *Trichoderma* spp. from collected soil sample

Trichoderma spp. were isolated from the collected soil samples by the following procedure developed for serial dilution method using Potato Dextrose Agar (PDA) medium. Rhizoplane soil samples collected were subjected to serial dilutions. One gram of soil was diluted to the strength of 10^{-1} by dissolving it in 10 ml distilled water (DW) (dilution of the original sample to 10 times). The soil suspended in the tube was shaken gently so as to thoroughly mix the soil particles and get them uniformly dispersed. One ml of the suspension from the first dilution (10^{-1}) was transferred aseptically to another tube containing 10 ml DW marked as 10^{-2} and thus diluting the original suspension to 100 times ($1/10 \times 1/10 = 1/100 = 10^{-2}$). This procedure was repeated till the original sample was diluted to 100000 times suspend. Soil once diluted to strength of 10^{-5} was inoculated on to the PDA plates by spread plate technique. The inoculated plates were incubated at 28°C and were observed for the appearance of *Trichoderma* spp. which were then aseptically transferred to PDA slants on which the growth appears within 1-2 days. The cultures were identified and confirmed as *Trichoderma* spp. through microscopic observation and selective growth on TSM (Trichoderma Selective Media) medium.

3.8.3 Morphological Characterization of the isolates of *Trichoderma* spp.

Cultural characters of all the single spore isolates under investigation were studied on 20 ml sterilized PDA in petri plates. Mycelial disc of approximately 5 mm diameter was transferred aseptically from culture at the centre of each Petri dish and incubated at $28 \pm 2^\circ\text{C}$ for 4 days. The Petri dishes with the culture were observed for colony colour, growth pattern (presence or absence of aerial mycelium or subdued growth etc), appearance, pustule, pigmentation secreted metabolite in the substrate medium against white background under sunlight and odour. The cultural characteristics were photographed using digital camera.

3.8.4 Microscopic Characterization of the isolates of *Trichoderma* spp.

Conidiophores and conidial morphology, branching pattern, critical for identification to the species level are best observed before the conidia are completely mature. Mounts from the actively growing (fungal growth from the growing colony margin) isolates of *Trichoderma* were prepared in lactophenol cotton blue on glass microscopic slide (preferably young tufts where the conidia just begins to develop pigment in actively growing cultures). A strip of cello tape was held in thumb and forefinger and the gum coated surface was placed gently against the sporulating growth of the isolate. The cello tape thus lifted the intact sporulating growth (including conidiophores, phialides, and sporulation) without forming any clump of mycelium and injuring the substrate. Microscopic morphological features of all the selected isolates were observed using a Leica ATC 2000 bi-nocular microscope and were micro photographed digitally using COSLAB DCE-2attachment at 100X magnification.

3.8.5 Screening of *Trichoderma* spp. isolates

3.9.5.1 Dual culture interaction

Mycoparasitic activity of fifteen isolates of *Trichoderma* spp. was studied *in vitro* following dual culture method (Bell *et al.*, 1982) against *R. bataticola*. For dual culture 5 mm diameter mycelial block derived from 7daysold culture of each isolates of *Trichoderma* spp. and *R. bataticola* were placed 3 cm apart on the PDA surface under aseptic conditions, while care was taken to avoid contamination.

Petri dishes containing dual cultures were incubated at $28\pm 2^{\circ}\text{C}$ under continuous light and were observed at regular intervals until mycelia growth of the oppositely paired cultures merged. Mycoparasitic activity of the respective isolates of *Trichoderma* spp. was assessed as follows:

- 1= *Trichoderma* completely overgrew the pathogen and covered the entire medium surface.
- 2= *Trichoderma* overgrew at least $2/3^{\text{rd}}$ of the medium surface.
- 3= *Trichoderma* and pathogen each colonized approximately $1/2$ of the medium surface (more than $1/3^{\text{rd}}$ and less than $2/3^{\text{rd}}$), and neither organism appeared to dominate the other.
- 4= The pathogen colonized at least $2/3^{\text{rd}}$ of the medium surface and appeared to withstand encroachment by *Trichoderma*.
- 5= The pathogen completely overgrew the *Trichoderma* and covered the entire medium surface.

In vitro antagonistic potential of different isolates of *Trichoderma* was assessed by calculating percentage inhibition growth of pathogen (*R. bataticola*) over control in presence of *Trichoderma* using the formula:

$$\text{Per cent inhibition} = \left(\frac{C-T}{C} \right) \times 100$$

Where,

C = Radial growth of pathogen in control.

T= Radial growth of pathogen in presence of *Trichoderma*.

3.9.5.2 *In vivo* efficacy of *Trichoderma* spp. under glass house condition

Pot experiments (CRD) was carried out under green house conditions. The plastic pots (12 cm diameter) filled with sterilized sandy loam soil, were further inoculated with two week old culture of *R. bataticola*(RB4) (prepared on sand maize medium) @ 30g/kg soil and allowed to stabilize for five days.

Seeds of chickpea were surface sterilized with 2 per cent sodium hypochlorite for three minutes, rinsed thoroughly in sterilized distilled water and dried aseptically. The seeds were coated with microbial antagonists viz., *Trichoderma* spp. by using 2 per cent sticker for 1 hour. Eight seeds per pot were sown in 12 cm diameter plastic pots, each containing 400 g of soil. Three replications of each treatment were maintained. Pots without antagonist, seed

coating material (untreated) were served as control. Root rot incidence was recorded up to 30 days after sowing. Pots were kept randomly in a green house. Ten treated seeds microbial antagonists were transferred in test tube containing 9 ml sterile distilled water. The test tubes were shaken and dilution series was made. One ml suspension was poured on PDA and fungal conidia / seed was calculated by using the formula: No. of cells or conidia x dilution factor.

Germination was recorded at ten days of sowing. The per cent mortality was observed after 30 days of sowing. Then plants were uprooted for measurement of root length and shoot length.

3.9 Potentiality of bio control agent fluorescent *Pseudomonas* against *R. bataticola*

Studies were undertaken to identify the fluorescent *Pseudomonas* a bio control agents as potential candidates against *R. bataticola* causal agent of dry root rot.

3.9.1 Isolation of fluorescent *Pseudomonas*

3.9.1.1 Collection of soil samples

Soil samples were collected from different locations of Chhattisgarh (Abhanpur, Ambikapur, Balod, Bilaspur, Bemetara, Bhatapara, Dhamtari, Dhamdha, Durg, Kanker, Kawardha, Khadsiya, Lohara, Patan, Raigarh, Raipur, Rajnandgaon, Saragoan, Sitapur, Thankhamariya, Uattai and Udaypur). Soil samples from the rhizoplane were collected from randomly selected and counted number of plants, which were gently uprooted using trowel to avoid damage and loss of root system. Loosely adhering soil particles were removed by gentle tapping and thereafter, roots were transferred to sterile polythene bags, brought to the laboratory and stored at 28°C. The rhizoplane was removed by mechanically scraping of root surface.

3.9.1.2 Isolation of fluorescent *Pseudomonas* from collected soil sample by serial dilution (Aneja, 1993)

Serial dilutions were prepared for each collected soil sample for isolation of fluorescent *Pseudomonas*. One gram of processed soil was diluted to the strength of 10^{-1} (dilution of the original sample 10 times i.e. $1/1 + 9 = 1/10 = 10^{-1}$). The soil suspended in the tube was shaken gently but thoroughly to mix soil particles and

get them uniformly dispersed. 1ml of the suspension from the first dilutions (10^{-1}) was aseptically transferred to another tube having 10 ml of sterile water marked as (10^{-2}) and thus diluting the original suspension to 100 times ($1/10 \times 1/10 = 1/100 = 10^{-2}$). This procedure was repeated till the original sample was diluted to 1000 (10^{-3}) times.

3.9.1.3 Isolation of fluorescent *Pseudomonas* and Designation of isolates

For isolation and enumeration of fluorescent *Pseudomonas* 10 μ l of soil suspension from the final dilution (10^{-3}) was transferred in sterilized plates. Twenty ml of medium was poured in each plate and gently rotated to ensure uniform distribution on Kings medium 'B' (KMB) (King's *etal.*, 1954). After solidification the plates were incubated at 28°C for 24 hrs and thereafter examined under on transilluminator against UV light. Colonies with yellow - green and blue-white color pigmentation were marked and recorded.

Individual fluorescent colony was picked up with the help of sterilized loop and inoculated on solidified KMB by zig-zag streaking. The plates were incubated at 28°C for 24 hrs. (colony growing at the last tip of the zig-zag line were transferred to KMB slants).

3.9.2 Morphological and Biochemical characterization of the fluorescent *Pseudomonas* isolates

3.9.2.1 Morphological characterization

3.9.2.1.1 Gram reaction

This stain is essential for differentiating bacteria in two broad groups; gram positive and gram negative. Gram staining characteristics tire related to structural and chemical properties of the cell wall. On a clean slide, bacterial film was formed by thin spreading a loopfull of the cells on grease free clean micro slide. The film was dried in air for a few minutes. Then the slide was lightly heated at the underside to fix the bacteria on it and stained properly and observed under oil immersion lens of a compound microscope. The gram reactions and the shape of cell was observed and recorded.

3.9.2.2 Biochemical characterization

3.9.2.2.1 Gelatin hydrolysis test

Gelatin is a protein which dissolves in warm water and exists as a liquid above 25°C and solidifies (gel) when cooled below 25° C. Gelatin hydrolysis was demonstrated by growing test organism in nutrient gelatin / gelatin agar medium. The gelatin hydrolysis was detected by observing liquefaction at low temperature (4°C) after incubation of the test organism in it at $28 \pm 2^\circ\text{C}$ for 48 hrs. Gelatin inoculated tubes that remained liquefied produced gelatinase and showed positive test for gelatin hydrolysis and tubes that remained solid were negative. Gelatin iron agar may also be used for detecting gelatin liquefaction and hydrogen sulphide production.

3.9.2.2.2 Nitrate reduction test

Bacterial species may be differentiated on the basis of their ability to reduce nitrate to nitrogenous gases. The reduction of nitrate may be coupled to anaerobic respiration in some species. The nitrate reduction test was performed using nitrate broth containing following ingredients: peptone 5g, beef extract 3g, potassium nitrate 1g, pH was adjusted to 7 prior to autoclaving. One litre nitrate broth was prepared by adding the component mentioned above and 20ml was distributed in large test tubes. It was then sterilized by autoclaving at 15 lbs pressure at 121°C for 30 minutes. The flasks were inoculated with a fresh culture of the selected isolates and incubated at $28 \pm 2^\circ\text{C}$ for 48 hrs in an incubator. Five drops each of reagents A (Sulfanillic acid: Himedia make) and B (N,N- dimethyl 1-1 naphthylamine: Himedia make) were added to the test tube containing culture to be tested and mixed properly by shaking. A distinct red or pink colour, which should develop within a few minutes indicated nitrate reduction. If the suspension turned pink-red before the addition of Zn powder, the reduction is positive and the test is completed. To the suspensions found colourless after the addition of reagents A and B small amount (“sharp knife point”) of zinc powder was added to the flasks and after shaking vigorously they were kept at room temperature for 10-15 min. If the medium remain colourless after the addition of Zn powder, the test result is positive. If the medium turns pink after the addition of Zn powder, the result is negative.

3.9.2.2.3 Starch Hydrolysis Test

Starch hydrolysis test is used to detect the enzyme amylase, which breaks down starch. Amylase hydrolyses starch (polysaccharide) into maltose, a disaccharide and some monosaccharide such as glucose. Single colony of isolates were streaked on starch agar medium and incubated at $28 \pm 2^\circ$ for 48 hrs. After required incubation, few drops of Lugol's Iodine solution were added. A yellow zone around the colony indicated starch hydrolysis, while blue/black area around them indicated or presence of starch.

3.9.2.2.4 Casein hydrolysis

Some microorganisms have the ability to degrade the protein casein by producing proteolytic exoenzyme called caseinase which break the peptide bond CO-NH by introducing water into the molecule, liberating smaller chains of amino acid called peptides which are later broken down into free amino acid by extracellular or intracellular peptidases. Skim milk agar medium was used for casein hydrolysis. After autoclaving the medium was poured in Petri plates the culture was inoculated and incubated at $28 \pm 2^\circ\text{C}$ or 48 hrs. Formation of clear zone around the colony indicated casein hydrolysis.

3.9.2.2.5 Lipolytic activity with Tween 80

The agar medium was prepared with 10.0gm bacteriological peptone, 5.0gm NaCl, 0.1gm CaCl_2 , 15.0gm agar, pH was set to 6.8 and volume was made up to 1000ml with distilled water. After the medium was autoclaved it was cooled to about 50°C and 5ml of autoclaved Tween 80 was added. The Petri dishes were filled with 25ml of the medium. The inoculated plates were incubated at $28 \pm 2^\circ\text{C}$ and were examined daily for 10 days.

The presence of a halo around an inoculated site on the Tween medium was viewed with transmitted light; this indicated a positive test and indicated that the *Pseudomonas* isolates produced an esterase. The hydrolysis of the Tween opacity medium is associated with the lipolytic enzyme produced by the respective *Pseudomonas* spp. that liberates fatty acids and bind with the calcium incorporated into the medium. The calcium complex is visible as insoluble crystal around the inoculated site.

3.9.2.2.6 Growth at 4°C and 42°C

Pseudomonas isolates were inoculated in Kings 'B' medium, incubated at 4°C and 42°C for 48 hrs and observation was taken.

3.9.2.2.7 Antibiotic sensitivity test

A rapid antibiotic sensitivity test was used to distinguish different species of fluorescent *Pseudomonas*. Antibiotic sensitivity studies were performed by the streak plate method.

3.9.2.2.8 Triple Sugar Iron agar test

Triple Sugar Iron agar media is used for the differentiation of microbes by ability to determine carbohydrate fermentation and hydrogen sulphide production. Organisms that ferment glucose produce a variety of acids, turning the colour of the medium from red to yellow. More amounts of acids are liberated in butt (fermentation) than in the slant (respiration). Growing bacteria also form alkaline product from the oxidative decarboxylation of peptone and these alkaline products neutralize the large amount of acid present in the butt. Thus the appearance of the alkaline (red) slant and an acid (yellow) butt after incubation indicates that the organism is a glucose fermenter but is unable to ferment lactose and or sucrose (or both), in addition of glucose, produce large amount of acid enables no reversion of pH in that region and thus bacteria exhibit an acid slant and acid butt. Gas production (CO₂) is detected by the presence of cracks of bubble in the medium, when the accumulated gas escapes. Thiosulphate is reduced to hydrogen sulphide by several species of bacteria and H₂S combines with ferric ions of ferric salts to produce the insoluble black precipitate of ferrous sulphide. Reduction of thiosulphate proceeds only in an acid environment and blackening usually occurs in the butt of the tube.

Triple sugar iron agar medium was prepared by dissolving 16.13gm TSI powder in 1000ml of distilled water and slants were prepared to set in sloped form with a butt about 1 inch long slants were streaked, incubated at $28 \pm 2^\circ\text{C}$ for 48 hrs and results were observed. The reaction can be summarized as follows-

Table 3.5: Reaction for Triple sugar iron agar test

S. No.	Reaction	Colour	Description
1.	Acidic Butt	Yellow	Only dextrose fermented
	Alkaline slant	Red	
2.	Acid Butt	Yellow	Glucose and sucrose fermented or glucose and lactose fermented or all the three sugar, glucose, lactose and sucrose fermented
	Acid slant	Yellow	
3.	Bubbles or cracks	-	Gas production
4.	Black precipitate	-	H ₂ S gas production

3.9.2.2.9 Hydrogen sulphide gas production test

This test was performed in sulphide indole motility (SIM) agar medium. All the ingredients of the medium were dissolved in 1000ml of distilled water, dispensed in tubes and autoclaved. The test organism was stab inoculated and incubated at $28\pm 2^{\circ}\text{C}$ for 48 hrs. The tubes were observed for the presence or absence of black coloration along the line of stab inoculation. The black colour indicated positive result. The black colour is due to the production of H₂S from sodium thiosulphate present in the medium that then combines with ferrous ammonium sulphate resulting in the formation of black insoluble ferrous sulphide.

3.9.3 Screening for antagonistic activity of fluorescent *Pseudomonas* against *R. bataticola*.

3.9.3.1 Dual culture interaction

Fluorescent *Pseudomonas* was screened via dual culture method to test the efficacy as bio-control agent against *R. bataticola*. Equal volume of sterilized potato dextrose agar and King's B medium were mixed in sterilized Petri plates. Inoculum from actively growing fluorescent *Pseudomonas* was applied as streak by a loop on the edge of Petri plates and inoculum of *R. bataticola* was placed in centre of the plate. Control was also maintained to see the difference. Observation was taken after three days of inoculation.

In vitro antagonistic potential of different isolates of fluorescent *Pseudomonas* was assessed by calculating percentage inhibition growth of pathogen (*R. bataticola*) over control in presence of fluorescent *Pseudomonas* using the formula:

$$\text{Per cent inhibition} = \left(\frac{C-T}{C} \right) \times 100$$

Where,

C = Radial growth of pathogen in control

T = Radial growth of pathogen in presence of *Trichoderma*

3.9.3.2 *In vivo* efficacy of fluorescent *Pseudomonas* under glass house condition

Pot experiments (CRD) was carried out under green house conditions. The plastic pots (12 cm diameter) filled with sterilized sandy loam soil, were further inoculated with two-week old culture of *R. bataticola* (RB4) (prepared on sand maize medium) @ 30g/kg soil and allowed to stabilize for five days.

Seeds of chickpea were surface sterilized with 2 per cent sodium hypochlorite for three minutes, rinsed thoroughly in sterilized distilled water and dried aseptically. The seeds were coated with microbial antagonist viz., fluorescent *Pseudomonas* by using 2 per cent sticker for 1 hour. Eight seeds per pot were sown in 12 cm diameter plastic pots, each containing 400 g of soil. Three replication of each treatment were maintained. Pots without antagonist, seed coating material (untreated) were served as control. Root rot incidence was recorded up to 30 days after sowing. Pots were kept randomly in a green house. Ten treated seeds with microbial antagonists were transferred in test tube containing 9 ml sterile distilled water. The test tubes were shaken and dilution series was made. One ml suspension was poured on PDA and bacterial cells /seed was calculated by using the formula: No. of cells x dilution factor.

Germination was recorded after ten days after sowing. The per cent mortality was observed after 30 days of sowing. Then plants were uprooted for measurement of root length and shoot length.

3.10 Statistical analysis

Statistical analysis was done for the interpretation of the results obtained. The experiments were laid out in CRD and analysis was done accordingly. The critical differences were worked out at 5 % probability level. The value in percentage was transformed by angular transformation.

CHAPTER - IV

RESULTS AND DISCUSSION

The low yield of chickpea can be attributed of biotic and abiotic constraints. Among biotic factors, diseases are the most dominant. Depending upon the crop variety, the losses due to diseases to pulse crops have been estimated to be as high as 44 per cent. Among the several soil borne fungal diseases of chickpea, dry root rot caused by *R. bataticola* Taub (Butler) is the most severe disease. *Rhizoctonia* occur throughout the world and cause losses. Apart from chickpea, it infect annual and perennial plants as well. The pathogen is distributed in diverse climatic conditions from arid to tropical regions and has a broad host range. It includes more than 500 hosts of the fungus including legume and cereal plants. *Rhizoctonia* is unspecialized fungus with no definite pathogenic race and yet not differentiate on the basis of anastomosis grouping. So, looking to the paucity of the information on the pathogen and disease in Chhattisgarh state an immediate attention from the research side was needed. Therefore, it was felt necessary to take up this problem.

Therefore, keeping in view, the severity and losses caused by dry root rot disease, the present investigation on “**Studies on dry root rot (*R. bataticola* Taub (Butler)) of chickpea (*Cicer arietinum*)**” was carried out in the laboratory as well as in the field during 2012-14 at Department of Plant Pathology, College of Agriculture, Raipur (C.G.).

During the present investigation, field observations were recorded to gather information on the occurrence of disease and screening of chickpea entries against dry root rot, in research farm, IGKV, Raipur (C.G.). Laboratory and poly house studies on isolation, pathogenicity, symptomatology, anastomosis, soil factors affecting disease development and evaluation of biocontrol agents against the pathogen under *in-vitro* and *in vivo* condition were carried out. The results thus obtained are presented in different sections under this chapter.

4.1 Survey of chickpea crops to assess the incidence of dry root rot

The survey for the occurrence and severity of chickpea dry root rot were undertaken at flowering and pod formation stage, from sixteen location of Chhattisgarh state (Ambikapur, Dhamtari, Bemetara, Bhatapara, Balod, Durg, Dhamdha, Gandai, Kanker, Kawardha, Narayanpur, Patan, Raigarh, Raipur, Rajnandgaon, Saragaon and Sitapur) during January to February 2013. Observations were recorded from farmer's fields under natural condition.

Results presented in Table 4.1 indicated that the disease occurred in all chickpea growing area of Chhattisgarh. Disease incidence ranged from 2.80 to 19.10 per cent. In respect to areas, maximum disease incidence was recorded in western region of Chhattisgarh (12.93 per cent) consist of three locations followed by northern region 9.10 per cent undertaking two locations and central region 8.27 per cent with comparatively more area with eight locations. Minimum disease incidence (6.85 per cent) was found in southern region of Chhattisgarh with two locations (Fig. 4.1).

Variation was also observed in soil types. Red, yellow and black soil types were found in two, three and eleven locations of Chhattisgarh respectively. Results presented in Table 4.1 and Fig. 4.2 indicated that the maximum average disease incidence was recorded in black soil (9.67 per cent) followed by red soil (9.10 per cent). Minimum disease incidence was recorded in yellow soil (6.86 per cent).

Observations of disease incidence were also recorded on improved and local chickpea varieties. Improved varieties were found in fourteen locations while, local varieties found in two locations. Results presented in Table 4.1 and Fig. 4.3 indicated that the disease incidence was more in improved varieties (9.30 per cent) in comparison to local varieties (7.20 per cent).

Results also indicated that the diseases incidence was more in irrigated condition (8.83 per cent) found in eleven locations. While, in rainfed conditions in two location disease incidence recorded was 6.75 per cent (Fig. 4.4).

Table 4.1: Incidence of chickpea dry root rot in different location of Chhattisgarh

S.No.	Area	District/Tehsil	Soil type	Seed Type	Rainfed/ Irrigated	Incidence of dry root rot (%)
1	Northern Area	Ambikapur	Red	Improved (Anna giri)	Irrigated	7.50
		Sitapur	Red	Improved (Yashoda 09)	Rainfed	10.70
2	Eastern Area	Raigarh	Black	Improved (Jaki)	Irrigated	7.60
3	Western Area	Gandai	Black	Improved (Vijay)	Irrigated	16.90
		Kawardha	Black	Improved (Vishal)	Irrigated	19.10
		Rajnandgaon	Black	Improved (JG-6)	Rainfed	2.80
4	Southern Area	Kanker	Yellow	Improved (JG- 74)	Irrigated	9.20
		Narayanpur	Yellow	Improved (JG- 226)	Irrigated	4.50
5.	Central Area	Balod	Black	Local	Irrigated	8.90
		Bhatapara	Black	Improved (JG- 74)	Irrigated	8.60
		Dhamdha	Black	Local	Irrigated	5.50
		Dhamtari	Yellow	Improved (JG- 74)	Irrigated	6.20
		Durg	Black	Improved (JG- 74)	Irrigated	3.70
		Patan	Black	Improved (JG-16)	Irrigated	14.80
		Raipur	Black	Improved (experimental seed)	Irrigated	12.20
		Saragaon	Black	Improved	Irrigated	6.30

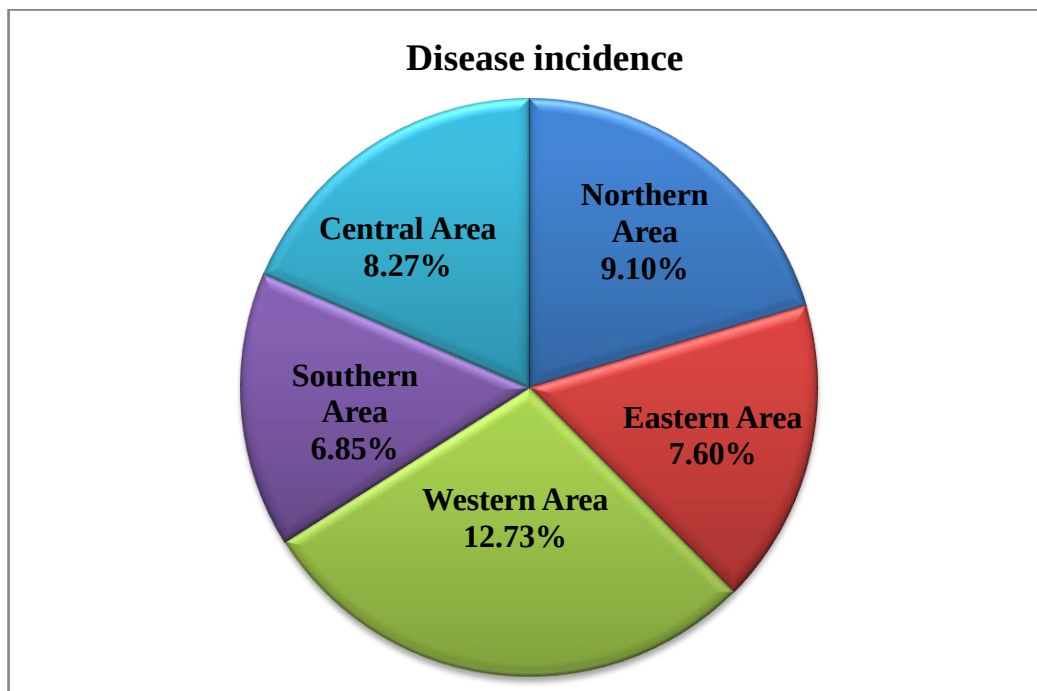


Fig. 4.1: Disease incidence in different chickpea growing areas of Chhattisgarh.

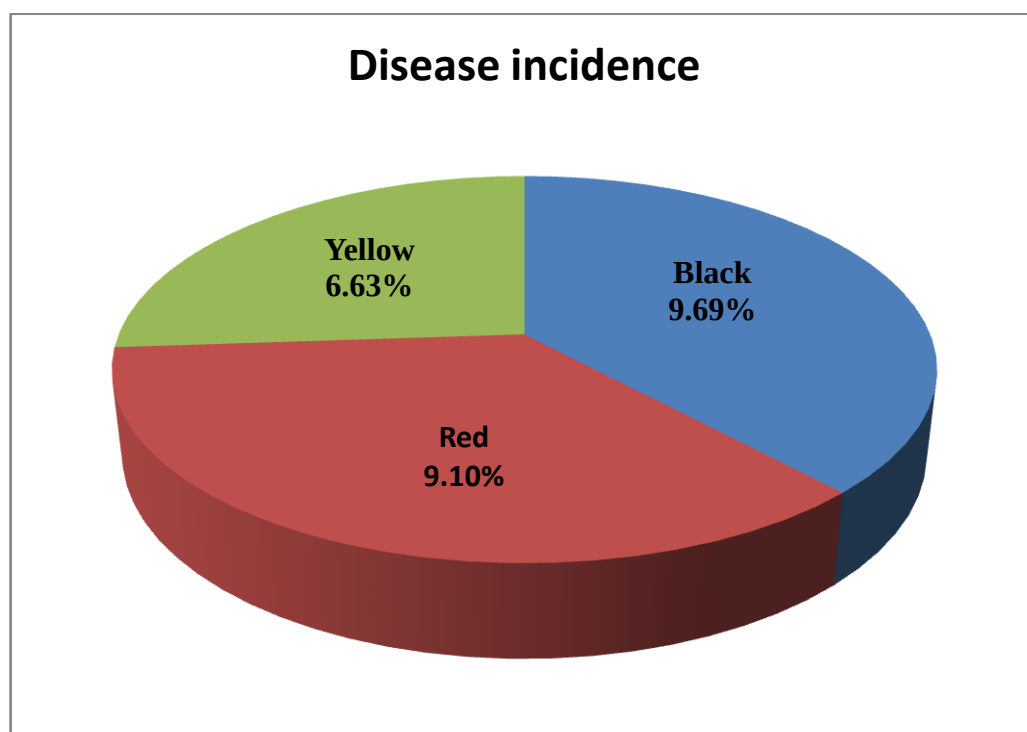


Fig. 4.2: Disease incidence in different soil types of Chhattisgarh.

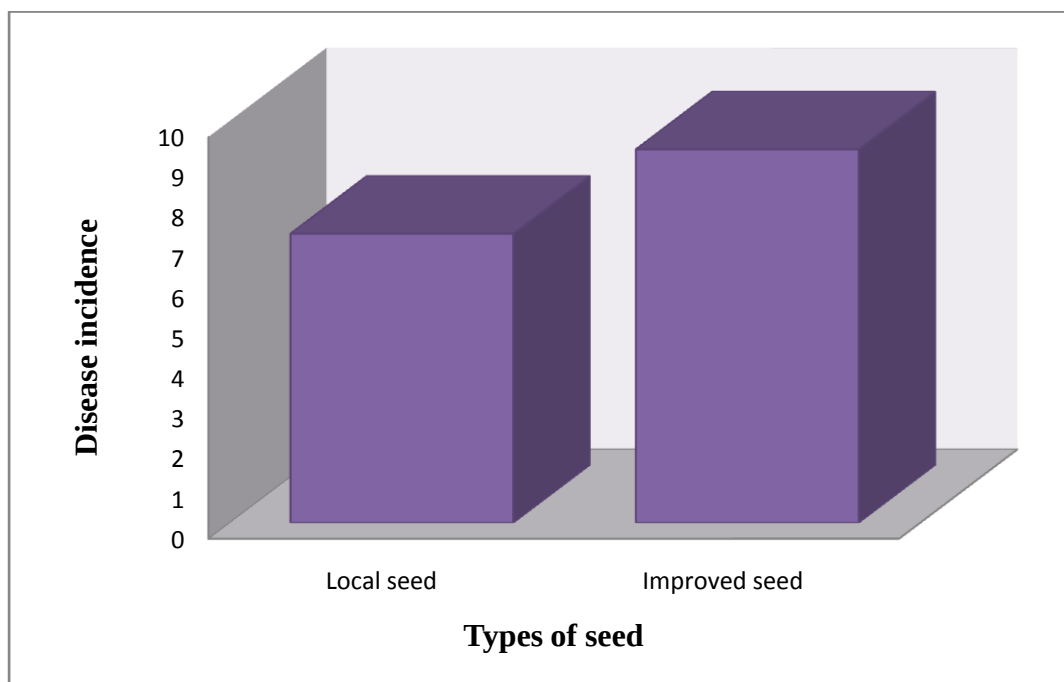


Fig. 4.3: Disease incidence in different types of seed in Chhattisgarh

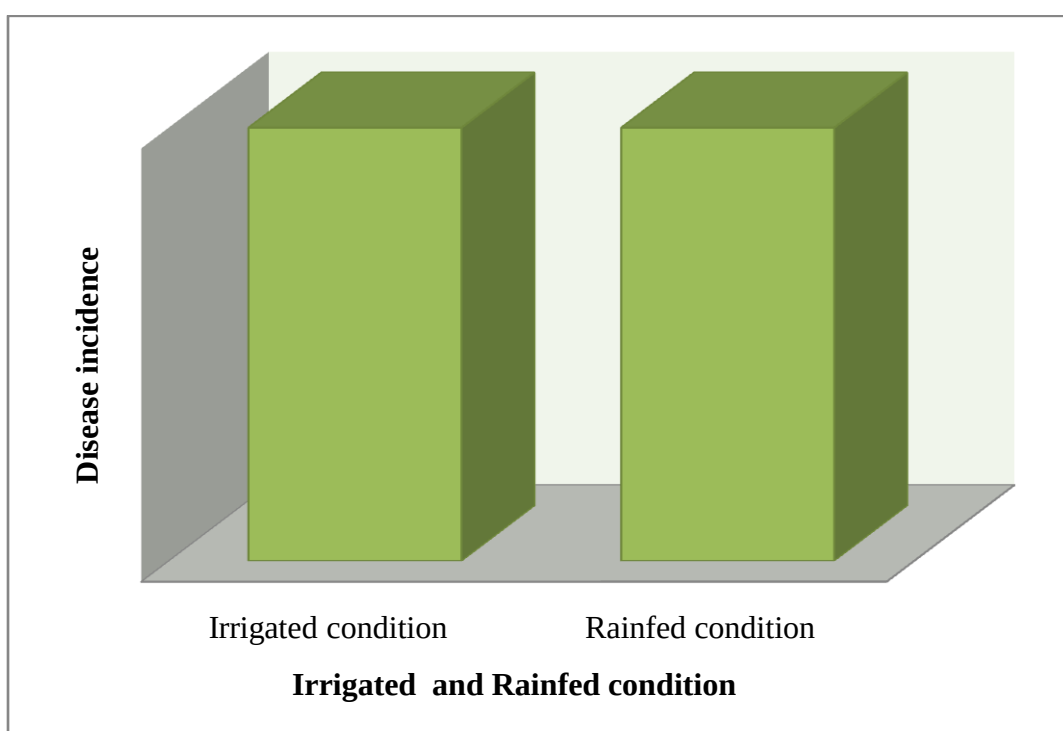


Fig. 4.4: Disease incidence influenced by availability of moisture (irrigated and rainfed) in Chhattisgarh

Similar report was found by Parmar (2013) where, maximum mean disease incidence (percent) was recorded from Shajapur district (10.8%), followed by Sehore district (9.3%) and minimum from Indore district (8.05). Manjunatha *et al.*, (2011) reported from Karnataka that the dry root rot incidence was more severe (9.8 %) in Gulbarga district as compared to Raichur (7.6 %) and Bidar (6.18 %) in chickpea. Khan *et al.*, (2012) reported the maximum disease was observed in village Shangus (40.0 %) of Kashmir division. Minimum disease was observed in Naina (4.11%).

4.2 Symptomatology

Following symptoms of dry root rot disease were observed in chickpea under natural field condition (Plate 3).

- Wilting without chlorosis of the whole plant scattered all over the field was observed.
- Drooping of petioles and leaflet was confined to those at very top of the plant was found. Sometimes when the rest of the plant was dry, the topmost leaves are chlorotic.
- The leaves of affected plants were straw colored, but in some cases the lower leaves and stem were also observed brown in colour.
- The prominent symptoms were found at the flowering and pre-podding stage.
- Wilted plants were seen more in flowering and pre-podding stage, in comparison to seedling stage.
- After uprooting blackening of the tap root with signs of extensive rotting and loss of lateral and finer roots were observed as characteristic symptoms.
- The dead roots were quite brittle and accompanied with shredding of bark and lateral roots.



A



B



C



D

Plate 3: Symptoms of dry root rot under natural field condition

A,B- Wilted plant

C,D- Blackening of the tap root with signs of extensive rotting and loss of lateral and finer roots

4.3 Collection, isolation, purification and identification of the pathogen

Symptomatic dry root rot infected plants were collected from sixteen chickpea growing areas of Chhattisgarh state (Ambikapur, Dhamtari, Bemetara, Bhatapara, Balod, Durg, Dhamdha, Gandai, Kanker, Kawardha, Narayanpur, Patan, Raigarh, Raipur, Rajnandgaon, Saragaon and Sitapur) for isolation and purification of the causal fungus in the laboratory. Out of sixteen locations *R. bataticola* isolated from nine locations (Raigarh, Kawardha, Rajnandgaon, Kanker, Balod, Dhamtari, Durg, Raipur and Saragaon) and Designated as mentioned in Table 4.2. The cultures were purified by single hyphal tip method and were maintained on PDA at $27\pm 2^{\circ}\text{C}$. These isolates were further under taken for study.

Table 4.2: Isolates of *R. bataticola*, collected from different chickpea growing locations of Chhattisgarh.

S.No.	Chickpea growing area	District/Tehsil	Isolate
1	Northern area	Ambikapur	-
		Sitapur	-
2	Eastern area	Raigarh	RB5
3	Western area	Gandai	-
		Kawardha	RB4
		Rajnandgaon	RB7
4	Southern area	Kanker	RB3
		Narayanpur	-
5	Central area	Balod	RB1
		Bhatapara	-
		Dhamdha	-
		Dhamtari	RB2
		Durg	RB8
		Raipur	RB6
		Patan	-
		Saragaon	RB9

- (not obtained)

4.4 Cultural and morphological variability

Variations in cultural and morphological characteristics were observed in all the isolates of *R. bataticola* and result is represented in Table 4.3 Plate 4.

4.4.1 Mycelial growth

Significant variations were observed in mycelial growth of *R. bataticola* isolates. The data presented in Table 4.3 Fig.4.5 indicated that the mycelial growth varied from 28mm to 68mm after 48 hours of incubation. Mycelial growth was recorded highest in RB4 (Kawardha) 68mm followed by RB8 (Durg) 60 mm and RB1 (Balod) 58mm which were at par with each other. While, the minimum growth was observed in RB3 (Kanker) isolate 28mm. Isolates B1 (Balod), RB4 (Kawardha), RB5 (Raigarh), RB6 (Raipur), RB7 (Rajnandgaon) and RB8 (Durg) exhibited full (fast) growth (90 mm) on fourth day. However, least (slow) growth was observed in RB3 (Kanker) with 56.6 mm on fourth day. Present findings coincide with Gupta *et al.* (2012) they categorized the *R. bataticola* with respect to colony diameter (growth rate) at 4th day of incubation as, slow growth (70-76 mm) and fast growth (84-90 mm). Monga and Sheo (1994) who observed that Rs-2 group of isolates was fastest in mycelial growth and Rs-4 was slow growing among all the four groups of isolates on potato dextrose agar medium.

4.4.2 Hyphal width

Based on the hyphal width of isolates, the pathogen was categorized in three groups. First group comprised of RB1 (Balod), RB3 (Kanker) and RB6 (Raipur), RB8 (Durg) and RB9 (Saragaon) isolates with hyphal width ranging between 5-6 μm . Second group include 6-7 μm , RB2 (Dhamtari) and RB7 (Rajnandgaon) as hyphal width and third group consist of RB4 (Kawarda) and RB5 (Raigarh) with hyphal width of more than 7 μm . However, maximum and minimum hyphal width observed in RB5 (Raigarh) (7.2 μm) and RB6 (Raipur) (5.2 μm) respectively. The diameter of vegetative hyphae and dimension of cells in vegetative hyphae are so similar to other fungi and so variable that they probably have little diagnostic value (Bracker and Butler, 1963). Peltier (1916) stated: "Cells varied in size at different ages and on different media." In agreement to present finding Gupta *et al.* (2012) observed a wide range of hyphal diameter they classified in three groups very thin (5.2-6.5 μm), thin (6.6-7.9 μm) size and thick

(8.0-9.3µm). These observations coincide with the observations of Sobti and Sharma (1992) who reported that the width of hyphae varied from 4.16-8.48µm. Jharia and Khare (1985) observed the hyphal width and found variation among the isolates root isolate 4.5 µm, soil isolate (3.5µm) and seed isolate (3.0 µm). Hyphal width 2.5-3 x 2-12 µm was given by (Taub) Butler (1925).

4.4.3. Hyphal branching

Branches of mature hyphae that served to identify isolated pathogen *R. bataticola* arose at right and acute angle that is near to 45 degree to main hyphae. A septum in the branch near the point of origin was also observed. In addition to ordinary vegetative hyphae moniloid cells also known as doliform cells, barrel-shaped cells, short cell, sclerotial cell were also observed. Moniloid cell arose as bud or blown out ends of pre existing cells (Parmeter, 1970) (Plate 5).

4.4.4 Type of mycelial growth

Based on mycelial growth pattern on PDA after seven days for incubation, the nine isolates were placed in three groups. The first group includes RB1 (Balod), RB6 (Raipur) and RB8 (Durg) isolates having fluffy mycelial growth. Whereas, RB3 (Kanker), RB4 (Kawarda) and RB7 (Rajnandgaon) isolates comprised of second group with partially submerged. Submerged mycelial growth pattern observed in RB2 (Dhamtari), RB5 (Raigarh) and RB9 (Saragaon) isolate which formed third group (Plate 4). In agreement to present finding Kanchan and Biswas (2009) evaluated morphological variability of *R. bataticola*, revealed that the nature of mycelium and colony varied from fluffy dark brown to partially fluffy mycelium colony with smooth margin. In accordance to present investigation Gupta *et al.* (2012) also reported mycelial growth as submerged, partially submerged and fluffy. Varma and Pathe (2013) categorized the isolate on the basis of mycelial growth into two groups as fluffy and submerged. These results confirm the observations made by Byadgi and Hegde (1985) who reported fluffy, dark brown mycelium, partial fluffy mycelium and appressed light brown mycelium growth.

4.4.5 Colony colour

Colony colour is divided in two fractions first black which include RB2 (Dhamtari), RB5 (Raigarh), RB7 (Rajnandgaon) and RB9 (Saragaon), second group include greyish colony colour RB1 (Balod), RB3 (Kanker), RB4 (Kawarda), RB6 (Raipur) and RB8 (Durg). Sulaiman and Patil (1966) reported olive green mycelium and dark gray mycelium of culture imparting the same colour to colony. Whereas, Mandal *et al.* (1998) found colony colour varied from light to dark brown. Shekhar *et al.* (2006) divided seven isolates into four groups, on the basis of colony colour, namely greyish white, blackish grey, dark black and cottony white colonies. Difference was observed by Aghakhani and Dubey (2009) in colony colour, white to dull white creamy, gray and black. Sharma *et al.* (2012) found variation in colony color like light black to black and light grey to grey. Present results confirm the observations made by Gupta *et al.* (2012) who reported colony colour varied from dark black to greyish cottony. Varma and Pathe (2013) observed colony colour as dark black.

Table 4.3: Cultural and Morphological characteristics of various isolates of *R. bataticola*

S.No.	Isolate	Location	Mycelial growth (mm)		Hyphal width (µm)*	Types of mycelial growth	Colony colour
			2 nd	4 th			
1	RB1	Balod	58	90	5.3	Fluffy	Greyish cottony
2	RB2	Dhamtari	50	84	6.5	Submerged	Dark black
3	RB3	Kanker	28	56.6	5.4	Partial submerged	Partial greyish
4	RB4	Kawardha	68	90	7.1	Partial submerged	Greyish
5	RB5	Raigarh	58	90	7.2	Submerged	Black
6	RB6	Raipur	50	90	5.2	Fluffy	Greyish
7	RB7	Rajnandgoan	45	90	6.3	Partial submerged	Black
8	RB8	Durg	60	90	5.8	Fluffy	Greyish
9	RB9	Saragoan	46	88	5.6	Submerged	Black
SEm±			3.65	0.93	0.27		
CD (5%)			16.7	4.16	0.83		

*Average of 10 hyphae

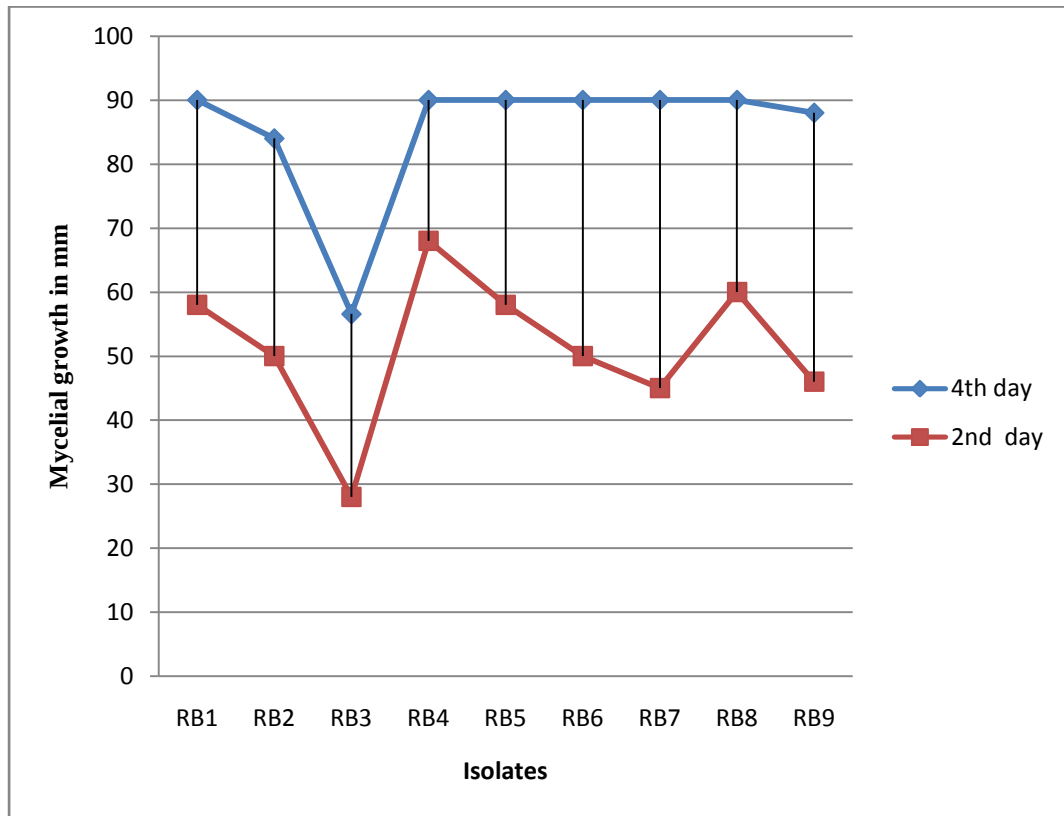


Fig 4.5: Mycelial growth rate of different *R. bataticola* isolates



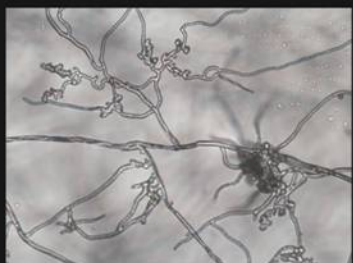
**Plate 4: Cultural and morphological variability of
R. bataticola isolates**



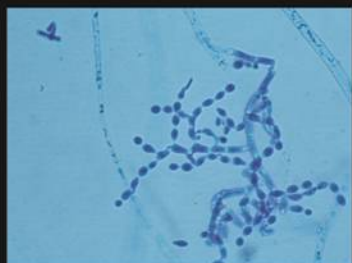
A



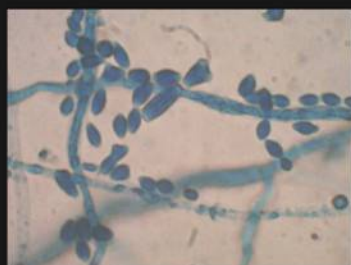
B



C



D



E

**Plate 5: Pattern of hyphal branching in
*R. bataticola***

4.4.6 Sclerotial size

On the basis of microscopic observations the size of sclerotia varied from 68-103 x 59-97 μm . Where RB6 (Raipur) found to have largest sclerotial size 103 x 97 μm followed by RB8 (84x78 μm), RB1 (82x74 μm), RB7 (79x70 μm) and RB3 (76x71 μm). The smallest sclerotial size was observed in RB2 (Dhamtari) 68 x 59 followed by RB4 (69x63 μm), RB9 (71x65 μm) (Table 4.4). Gupta *et al.* (2012) also found great variation in sclerotial size, bigger size (103.3 to 117.2 x 90.1- 106.5 μm) and smaller size (72.7-87.5 x 57.1-73.5 μm). On the basis of microscopic observations Varma and Pathe (2013) found the size of sclerotia varied from 72.2-117.2 x 57.1-106.5 μm . Different workers had given the wide range of dimensions of sclerotial size in herbaceous plants (50-150 μm), roots of woody plants (80-100 μm) and in culture (50-200 μm) (Small, 1924); 120 μm or small, 120-200 μm or medium and 200 μm or large (Haigh, 1930); 60-165 x 57 -114 μm and 36-99 x 36-81 μm (Hildebran *et al.*, 1945); 150-200 μm by Philip *et al.* (1969); biggest sclerotia 101.51 μm while smallest sclerotia 66.88 μm reported by Byadgi and Hegde (1985); Sclerotia from root isolates ranged from 122- 188 x 77-127 μm , and stem isolates 84-100 x 60-71 μm was observed by Dhingra and Sinclair (1977). Jharia and Khare (1985) found sclerotia in the root isolate (84x93 μm), soil isolate (90 x 75 μm) and seed isolate (85x70 μm). Monga and Sheo (1995) and Mandal *et al.* (1998) recorded the sclerotial size 58.83 to 126.63 μm and 66.14 to 128.25 μm respectively.

4.4.7 Sclerotial shape

On the basis of microscopic observations of sclerotial shape the isolates were classified in two groups that is round and oval. Round shape sclerotia recorded in isolates of RB3 (Kanker), RB4 (Kawarda), RB5 (Raigarh), RB6 (Raipur), RB8 (Durg) and RB9 (Saragaon) whereas, oval shape sclerotia were found in RB1 (Balod), RB2 (Dhamtari) and RB7 (Rajnandgaon) isolates (Table 4.4, Plate 6). Present finding coincide with Gupta *et al.* (2012) found sclerotial morphology, as oblong shape having irregular edges and the other being round with regular edges. Sharma *et al.* (2012) observed difference in sclerotial shape as oblong, ellipsoid, irregular and round. Mandal *et al.* (1998) observed shape in most cases was irregular except for few which were round to elongate

4.4.8 Number of sclerotia per microscopic field

Significant variation in number of sclerotia per microscopic field of *R. bataticola* was also observed in all isolates. Highest number of sclerotia per microscopic field was noted in RB 4 (235.00) followed by RB1 (208.38) which were at par with each other. Least sclerotia per microscopic field was formed in RB 7 (44.83) followed by RB 8 (46.83) and RB3 (58.66) (Table 4.4, Plate 6). Present results were compared with finding of Parmar (2013) he observed 470-500 sclerotia of *R. bataticola* per microscopic field. While, Varma and Pathe (2013) grouped the *R. bataticola* isolates according to number of sclerotia per microscopic field into three categories i.e. sparse, medium and abundant. The number of sclerotia ranged from 9.7 to 22.2.

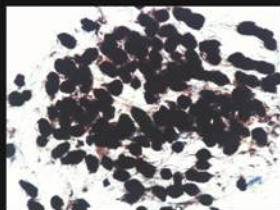
4.4.9 Sclerotial development

Sclerotial initial were formed by branching of hyphae followed by laying down of many more septa. The abundant septation resulted in the formation of short barrel shaped cells filled with dense content. Initial increased in size by repeated branching and formation of new cell by more septa. There was no rind formation and zonation in the mature sclerotium. Mature sclerotia were much less closely interwoven and there were no well identified zones. The basic colour of the mature sclerotia was typically brown to dark black (Parmeter, 1970) (Plate 7).

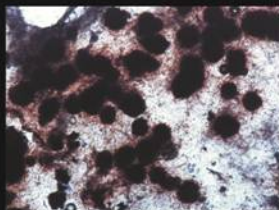
Table 4.4 Variability in morphology of sclerotia produced by different isolates of *R. bataticola*

Isolates no.	Isolates	Size of sclerotia (µm)**	Shape	No. of sclerotia per microscopic field
1	RB1	82x74	Oval	208.33
2	RB2	68x59	Oval	133.50
3	RB3	76x71	Round	58.666
4	RB4	69x63	Round	235.00
5	RB5	75x68	Round	173.16
6	RB6	103x97	Round	106.83
7	RB7	79x70	Oval	44.83
8	RB8	84x78	Round	46.83
9	RB9	71x65	Round	156.50
SEm±			12.37	
CD (5%)			35.23	

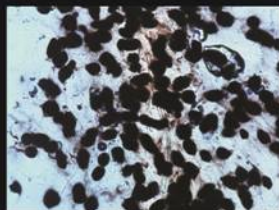
** Average of 10 sclerotia



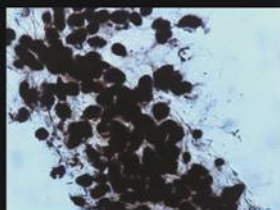
RB1



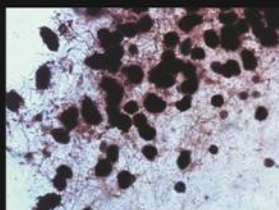
RB2



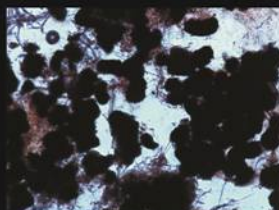
RB3



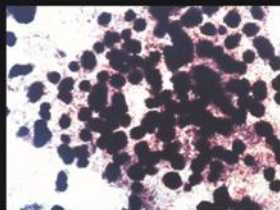
RB4



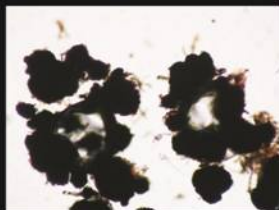
RB5



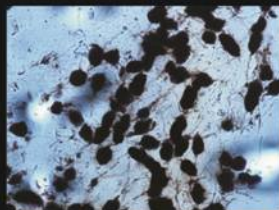
RB6



RB7



RB8



RB9

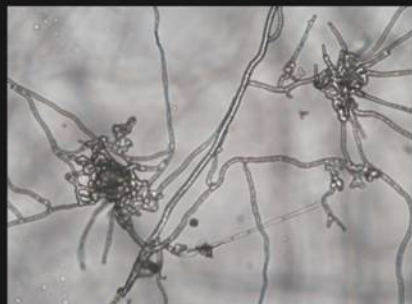
**Plate 6: Microphotograph of sclerotia of different
R. bataticola isolates**



A



B



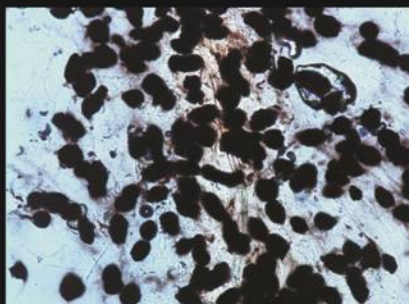
C



D



E



F

Plate 7: Microphotograph of sclerotial development

4.5 Pathogenicity test

All the isolates were found to pathogenic under artificially inoculated condition. Artificially inoculated plants expressed typical symptoms of dry root rot. The infected plant lost turbidity and showed chlorotic symptoms. The leave and stem of infect plants turned straw colour. Tap root become dark and rotting extended to entire root system. Morphologically on re-isolation of fungus confirmed the presence of *R. bataticola*.

4.5.1 Aggressiveness of *R. bataticola* isolates.

4.5.1.1 *In vitro* blotter paper technique

The data presented in Table 4.5, Fig. 4.6, Plate 8, revealed that the isolates were variable in their aggressiveness. The disease response on cultivar L 550 of different isolate varied between 5-9 grades. Based on their aggressiveness the isolates were classified in two group RB2 (Dhamtari), RB4 (Kawarda), RB6 (Raipur), RB7 (Rajnandgaon) and RB9 (Saragaon) isolates produced clear but small lesion on roots with new roots free from infection. Second group comprised of RB1 (Balod), RB3 (Kanker), RB5 (Raigarh) and RB8 (Durg) with many lesion on roots, completely discolour and destroyed. While, control plants without inoculation scored grade 1 with no infection on roots.

However, the most aggressiveness was found to be RB5 (Raigarh) isolate which scored 8.33 disease rate grade and comparatively least aggressive was found to be RB2 (Dhamtari) isolate scored 5.66 disease rate grade (Table 4.5).

In agreement to present investigation Gupta *et al.* (2006) varying degree of virulence among forty chickpea isolates, through blotter paper technique. The disease reaction of the isolates varied from 6.6 – 9.0 on a scale of 1-9.

Table 4.5: Aggressiveness of different isolates of *R. bataticola* under *in vitro* condition by blotter paper method

S.No.	Isolate	Disease rating
1	RB1	7.6
2	RB 2	5.6
3	RB 3	7.6
4	RB 4	7.0
5	RB 5	8.3
6	RB 6	7.0
7	RB 7	7.0
8	RB 8	7.6
9	RB 9	6.3
10	Control (uninoculated)	1.0

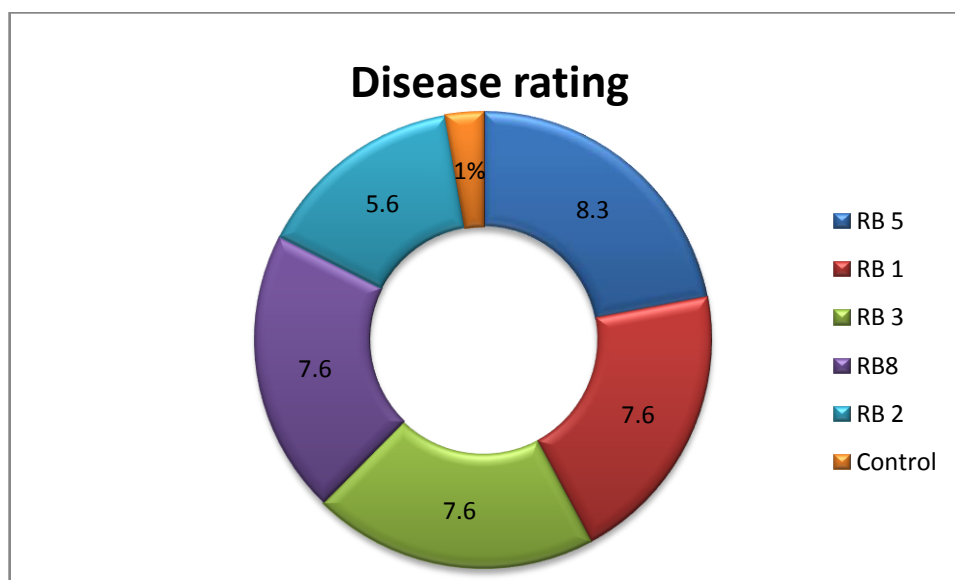


Fig 4.6: Aggressiveness of different isolates of *Rhizoctonia bataticola* under *in vitro* condition by blotter paper method

4.5.1.2 *In vivo* sick pot method

Aggressiveness of all the nine isolates were also tested by sick pot method. The data presented in Table 4.6, Fig. 4.7, Fig. 4.8, Plate 9 and 10, revealed that the isolates are comparatively variable in their virulence. The disease response on cultivar L 550 of different isolates varied between 4-9 grade with average mortality per cent of 55 to 100 %. Highest mortality 100% was observed in RB4 (Kawarda) with disease score 9, which was significantly superior than other isolates, followed



Plate 8: Extent of root damage in *in vitro* aggressiveness test of *R. bataticola* isolates through blotter paper technique

by RB8 (Durg) with great significant difference of 87.5% with disease score 6.3 and RB9 (Saragaon) 83.3% with disease rate 4.3. However, lowest mortality 58.33% was observed in RB3 (Kanker) with 5.66 disease rate. All the treatments, that is, inoculated with the pathogen exhibited mortality except the control pot (uninoculated), with disease rate 1 showed no infection on roots.

Table 4.6: Aggressiveness of *R. bataticola* isolates causing dry root rot of chickpea under glass house condition

S.No.	Isolate	Disease rating	Per cent mortality
1	RB1	8.3	70.8 (57.39)
2	RB 2	5.0	62.5 (52.39)
3	RB 3	5.6	58.3 (49.80)
4	RB 4	9.0	100 (90.00)
5	RB 5	7.0	62.5 (52.56)
6	RB 6	5.0	70.8 (57.90)
7	RB 7	5.6	66.6 (54.97)
8	RB 8	6.3	87.5 (69.26)
9	RB 9	4.3	83.3 (66.17)
10	Control (uninoculated)	1.0	10.8 (18.42)
SEm±4.02			
CD(5%)		11.95	

Average of three replication

Figures in parentheses are angular transformation

Effect of *R. bataticola* isolates on germination, root and shoot length of chickpea under sick pot soi was studied and the data presented in Table 4.7 and Fig. 4.9. Which revealed that *R. bataticola* isolates significantly affect the germination, root and shoot length of chickpea. Isolate RB1 was found significantly superior in reducing the germination per cent (70 per cent) over all the other isolates, followed by RB6 (Raipur) and RB7 (Rajnandgaon) that is 76.65 per cent. However, maximum germination per cent was recorded in RB3 (Kanker), RB5 (Raigarh) and RB9 (Saragoan) that is 86.66 per cent. Germination 93.33 per cent was recorded in control pot (uninoculated with pathogen).

Root length was found to vary from 7 cm to 14 cm. RB4 (Kawarda) was having minimum root length 7.6 cm, which was 53.37 % reduced over control followed by RB1 (Balod) 8.0 cm (50.92 % reduction over control), which exhibited significant difference with RB2 (Dhamtari) 10.0 cm (38.65 % reduction

over control) while, the maximum root length was observed in RB9 14 cm (14.11 % reduction over control). However, maximum root growth was recorded in control pot, that is, 16.3 cm.

Shoot length was found to vary from 24cm to 31 cm. RB1 (Balod) was having minimum shoot length 24.0 cm which was 29.41 % reduced over control followed by RB4 (Kawarda) 26.0 cm (23.52 % reduction over control) and RB2 (Dhamtari) 7.0 cm (20.58 % reduction over control) which were at par with each other. While, the maximum shoot length was observed in RB9 31 cm (7.64 % reduction over control). Maximum shoot length was recorded in control pot (uninoculated with pathogen) 34.0 cm.

Control pot (uninoculated with pathogen) showed highest germination per cent, root length and shoot length over other treatments.

Table 4.7: Effect of *R. bataticola* isolates on germination, root and shoot length of chickpea under sick pot soil

S.No.	Isolate	Germination (%) [*]	Root length (cm) [*]	Per cent reduction over control	Shoot length (cm) [*]	Per cent reduction over control
1	RB1	70.00 (8.42)	8.0	50.92	24.0	29.41
2	RB 2	83.33 (9.20)	10.0	38.65	27.0	20.58
3	RB 3	86.66 (9.36)	10.3	36.80	29.0	14.70
4	RB 4	83.33(9.20)	7.6	53.37	26.0	23.52
5	RB 5	86.66 (9.36)	11.6	28.83	28.3	16.76
6	RB 6	76.65 (8.77)	13.0	20.24	29.6	12.94
7	RB 7	76.65 (8.77)	12.0	26.38	26.0	23.52
8	RB 8	80.00 (9.00)	13.3	18.40	28.0	17.64
9	RB 9	86.66 (9.36)	14.0	14.11	31.3	7.64
10	Control (uninoculated)	93.33 (9.69)	16.3	0.00	34.0	0.00
SEm±		0.05	0.63		0.81	
CD		0.15	1.87		2.39	
(5%)						

^{*}Average of three replication

Figures in parentheses are square root transformation

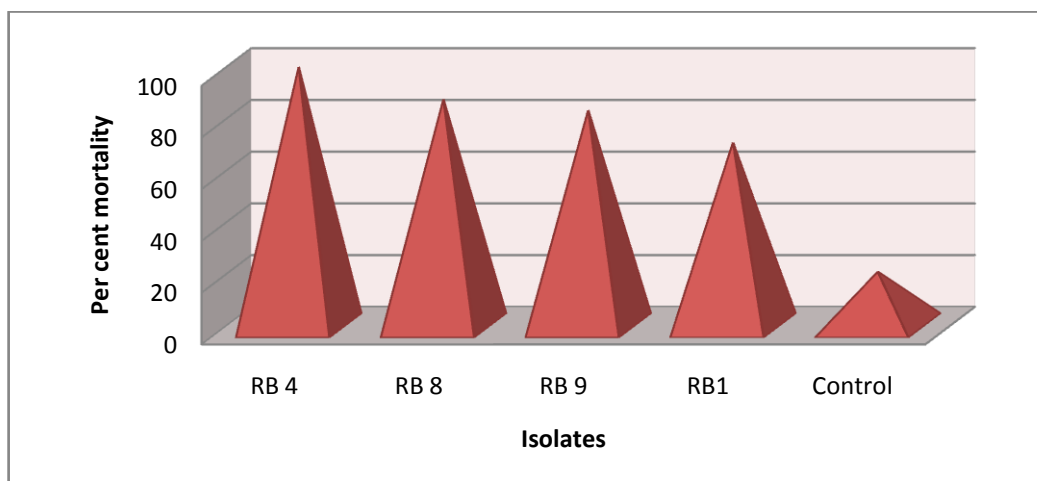


Fig 4.7: Per cent mortality of chickpea plants by *R. bataticola* isolates causing dry root rot under glass house condition

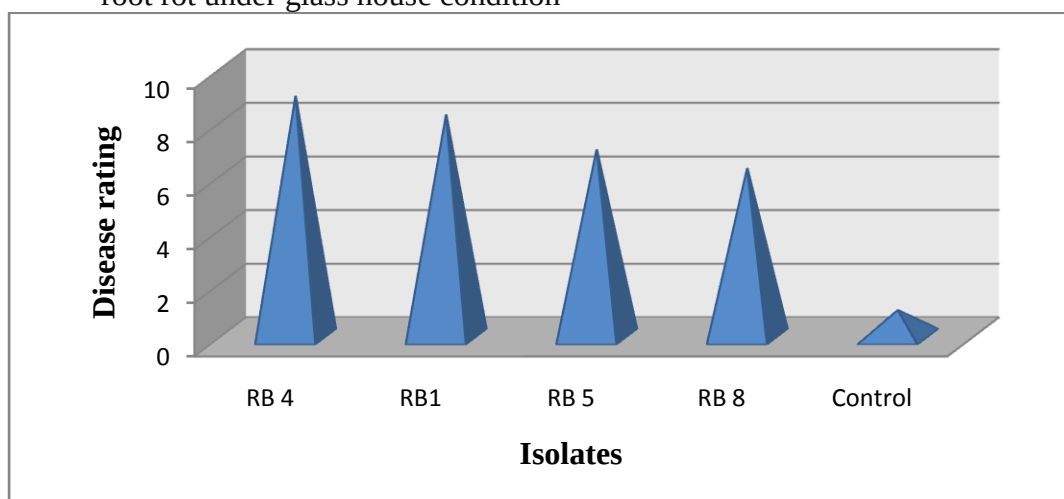


Fig 4.8: Disease rating of different isolates of *R. bataticola* under glass house condition

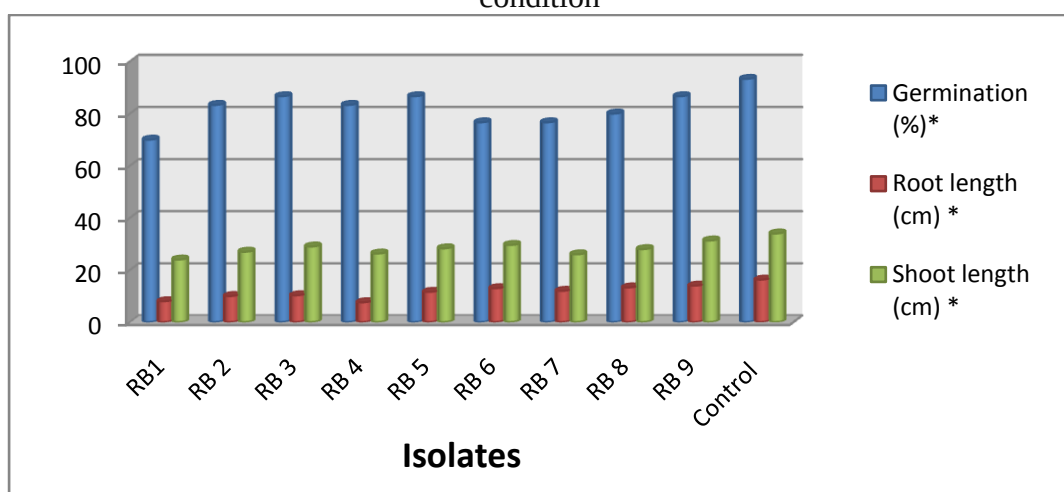


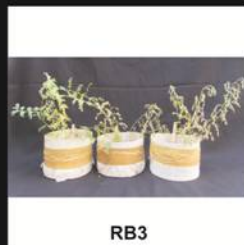
Fig 4.9: Effect of *R. bataticola* isolates on germination, root and shoot length of chickpea under pot condition



RB1



RB2



RB3



RB4



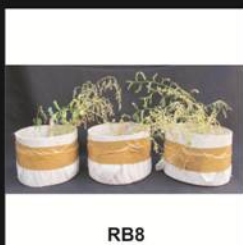
RB5



RB6



RB7



RB8

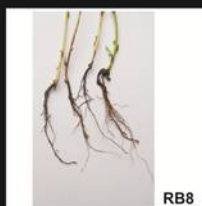
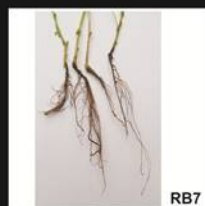


RB9



CONTROL

Plate 9: *In vivo* aggressiveness test of *R. bataticola* isolates under glass house pot condition



CONTROL

Plate 10: Extent of root damage in *in vivo* aggressiveness test of *R. bataticola* isolates

Mohan and Balabaskar (2012) revealed varied levels of pathogenicity with different isolates of *M. phaseolina*. The isolate from Vengatakuppam (MP18) was found to be the most virulent highest incidence of 67.34 per cent and the isolate MP5 collected from Pattampakkam was the least virulent with the lowest disease incidence of 35.60 per cent under *in vivo* condition. Subramanian *et al.* (2011) found great variation in aggressiveness of 65 isolates *M. phaseolina* against three cultivar of mungbean. Kanchan and Biswas (2009) observed the potency of infection of different isolates of *R. bataticola* which varied on four different cultivars of pigeon pea viz. Prabhat, Bahar, T-7 and T-21. The *R. bataticola* isolates T-4 and T-5 showed maximum disease severity. Similarly, variations in pathogenicity was observed by Beas *et al.* (2006) among *M. phaseolina* isolates taken from different hosts as well as from different parts of the same host. Ratnoo *et al.* (1997) found pathogenic variation in the isolates of *M. phaseolina* from different cowpea growing areas of Udaipur. In addition to present finding Sobti and Sharma (1994) observed varying virulence level which range (13 to 63 per cent) of disease incidence on groundnut causing rot root.

4.6 Anastomosis Behaviour

Anastomosis behavior of different isolates of *R. bataticola* with each other collected from different location of Chhattisgarh was investigated in different combination. All isolates exhibited self-compatibility that is fusion of wall and membrane and showed homologous anastomosis (reaction type 3) (Plate 11).

In accordance to above finding self- anastomosis reported to occurs commonly in culture of *R. solani* and it is often assumed that cross anastomosis between cultures also occurs commonly. Schultz (1936) paired a large number of cultures and, on the assumption that successful anastomosis indicated a close genetic relation, they attempted to group their cultures according to their success in anastomosing. Whitney and Parmeter (1963) investigated heterokaryon formation following anastomosis between single- spore cultures derived from one parent. They found that two distinctive single spores culture, when paired regularly gave to a culture type different from that of the single spore culture but similar to the original parent.

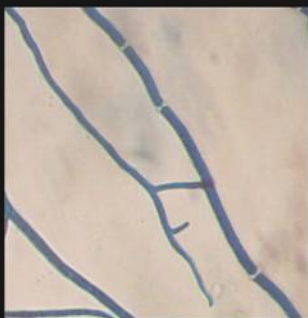
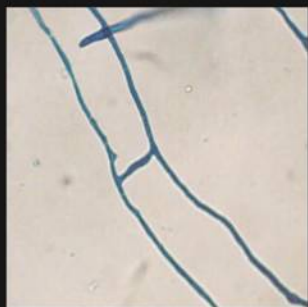


Plate 11: Anastomosis among *R. bataticola* isolates exhibiting self –compatibility (reaction type 3)

Fusion of wall and membrane and was Designated as compatible reaction (reaction type 3). As fungi grow, fusions between hyphae, or mycelium, that makes up the fungal individual. Networked hyphae are presumably important in intrahyphal communication and homeostasis in an individual colony during growth and reproduction. Hyphal fusions are a way to increase protoplasmic flow restricted by septa, which may be important in influencing hyphal pattern formation in the mycelium (Buller, 1933; Davidson *et al.*, 1996; Rayner, 1996). Different fungal individuals are capable of undergoing hyphal fusion with each other to form a heterokaryon, whereby genetically different nuclei may coexist in a common cytoplasm. Heterokaryon formation has potential benefits of functional diploidy and mitotic genetic exchange (“parasexual cycle”) Pontecorvo (1956) or may increase biomass for cooperation in physiological efforts such as resource exploitation or asexual/sexual reproduction (Buller, 1933 and Gregory, 1984).

4.7 Screening of chickpea entries

4.7.1 Field screening

To find out the source of resistance, chickpea entries were evaluated for their reaction against *R. bataticola* under natural condition. These were evaluated as per standard evaluation system. The reactions of the entries are depicted from Table 4.8 to 4.49.

During first *rabi* season 2012- 2013 in Desi III only one entry was found moderately resistant (BG 3033), while three entries NDG 11-12, RKG 2020 22 and H 09-90 were tolerant (Table 4.12 and 4.13). Whereas, in rainfed (R.F. II) sets of entries, Phule G 0752 was exhibited moderately resistant and JG 33 and CSJ 741 were tolerant (Table 4.20 and 4.21). Three entries namely, RSG 9056, JG 16 and CSJ 8023 were found tolerant in elites lines (Table 4.22 and 4.23). However, remaining all the entries in Desi I (Table 4.8 and 4.9), Desi II (Table 4.10 and 4.11), late sown I (Table 4.14 and 4.15), late sown II (Table 4.16 and 4.17) and rainfed I (Table 4.18 and 4.19) were showed susceptible and highly susceptible to dry root rot.

While during second *rabi* season (2013-14) CSJ 809 was found moderately resistant whereas, JG 17, CSJ 808, CSJ 556, JG 24, CSJ 815, GLK 28127, JG 27 and JG 36 were tolerant in national nursery for elite (Table 4.48 and

4.49). However, in Desi I (Table 4.24 and 4.25), Desi II (Table 4.26 and 4.27), late sown I (Table 4.28 and 4.29), late sown II (Table 4.30 and 4.31) and rainfed I (Table 4.32 and 4.33), rainfed II (Table 4.34 and 4.35), Kabuli I (Table 4.36 and 4.37), Kabuli II (Table 4.38 and 4.39), Kabuli III (Table 4.40 and 4.41), ELSK I (Table 4.42 and 4.43), ELSK II (Table 4.44 and 4.45) and ELSK III (Table 4.46 and 4.47) were found susceptible and highly susceptible to dry root rot.

100 per cent mortality was observed in the susceptible check (L-550) indicating the high and uniform sickness in the soil. Similar findings have been reported (Jayalakshami *et al.*, 2008). Chickpea genotypes have been screened and identified as resistant to dry root rot by various workers on the basis of *in vitro* and field conditions (Nene *et al.*, 1981; Anand Rao and Haware, 1987; Reddy *et al.*, 1990; Gupta, 1997; Khalid and Ilyas, 2000 and Gupta *et al.*, 2006).

Gupta *et al.*, (2012) evaluated 170 germplasm under *in vitro* condition and further evaluation of 26 promising lines under disease sick field, it was found that six lines viz., JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas were resistant, exhibiting less than 10 per cent mortality, however 14 lines showed moderately resistant reaction (JG 2003-14-16, JG 2001-12, JG 315, MPJG 2001-04, H02-14, CSJ 556, CSJ 558, GNG 1488, PG01108, HK- 02-201, HK 03-45, HK 03113, WCG 2000-07 and IPC 2005-28) against *R. bataticola* during the testing years (2007-2010) under high disease pressure.

Table 4.8: Screening of chickpea entries (Desi I) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1.	GNG15181	36.0	46.90
2.	GJG 0809	32.0	53.84
3.	PPG 5	27.0	42.71
4.	GNG 469	24.5	59.53
5.	GL 27104	27.5	59.48
6.	GPF 2	29.5	48.33
7.	GNG 1958	13.5	33.33
8.	L550 (susceptible check)	36.5	72.56

Table 4.9: Reaction of chickpea entries (Desi I) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score / reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	4	GNG15181, PPG 5, GPF 2, GNG 1958,
5.	Above 50	5 (HS)	3	GJG 0809, GNG 469, GL 27104,
Total entries			7	
LSI			4.42	

Table 4.10: Screening of chick pea entries (Desi II) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1.	Vishal	17.0	46.53
2.	Phule G 0204-4	34.5	42.06
3.	GCP 101	19.5	62.11
4.	GJG 0906	33.0	38.35
5.	RSG 888	24.0	43.53
6.	GJG 0907	33.5	59.55
7.	JAKI 9218	23.0	58.90
8.	L550 (susceptible check)	36.5	60.15

Table 4.11: Reaction of chickpea entries (Desi II) against dry root rot during 2012- 2013

S.No.	Per cent mortality	Score / reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	4	Vishal, Phule G 0204-4, GJG 0906, RSG 888,
5.	Above 50	5 (HS)	3	GCP 101, GJG 0907, JAKI 9218
Total entries			7	
LSI			4.42	

Table 4.12: Screening of chickpea entries (Desi III) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1	NDG 11-11	21.5	30.52
2	BGD 1073	39.5	43.01
3	NBEG 165	19.0	34.21
4	NDG 11-12	34.5	21.54
5	PG 0120	17.5	30.88
6	RKG 201-95	15.0	47.51
7	CSJ 563	14.5	51.90
8	BGD 1074	19.0	30.28
9	RKG 020-22	21.5	23.89
10	CSJ 724	25.5	41.38
11	H 09-96	22.0	56.25
12	Phule G 0302-10	23.5	38.41
13	GNG 2124	35.0	34.48
14	IPC 08-57	31.5	38.21
15	Phule G08108	27.5	58.13
16	GNG 2171	28.0	46.43
17	GJG 1001	34.5	36.24
18	IPC 06-126	34.0	47.06
19	GL 28295	33.0	30.30
20	AKG 1108	20.0	42.50
21	GL 28297	28.0	66.48
22	JG 29	16.5	85.11
23	GJG 1010	37.5	55.18
24	BG 3033	32.0	19.51
25	DKG 972	39.0	51.18
26	JG 30	24.0	87.14
27	BG 3034	39.5	32.85
28	DKG 1030	39.5	43.08
29	H 09-90	27.5	29.17
30	L 550 (susceptible check)	37.5	67.95

Table 4.13: Reaction of chickpea entries (Desi III) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score /reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	1	BG 3033
3.	20 – 30	3 (T)	3	NDG 11-12, RKG 020-22, H 09-90
4.	30 – 50	4 (S)	15	NDG 11-11, BGD 1073, NBEG 165, PG 0120, RKG 201-95, BGD 1074, CSJ 724, Phule G 0302-10, IPC 08-57, GNG 2171, GJG 1001, IPC 06-126, GL 28295, AKG 1108, BG 3034
5.	Above 50	5 (HS)	10	CSJ 563, H 09-96, GNG 2124, Phule G08108, GL 28297, JG 29, GJG 1010, DKG 972, JG 30, DKG 1030,
Total entries			29	
LSI			4.17	

Table 4.14: Screening of chickpea entries (Late sown I) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1	CSJ 697	22.0	45.96
2	NDG 11-5	14.0	35.42
3	BGD 1071	20.5	46.77
4	GJG 0921	17.5	63.07
5	IPC 06-77	34.0	44.12
6	H 08-75	22.5	38.45
7	BG 3032	23.5	41.02
8	Phule G 0204-16	32.0	35.59
9	GJG 0922	22.5	42.50
10	JG 14	34.5	45.00
11	L550 (susceptible check)	37.5	62.73

Table 4.15: Reaction of chickpea entries (Late sown I) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score / reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	5	CSJ 697, NDG 11-5, IPC 06-77, H 08-75, Phule G 0204-16,
5.	Above 50	5 (HS)	5	BGD 1071, GJG 0921, BG 3032, GJG 0922, JG 14
Total entries			10	
LSI			4.50	

Table 4.16: Screening of chickpea entries (Late sown II) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1.	NDG 11-21	29.5	57.74
2.	GL 29390	39.0	38.36
3.	IPC 0713	37.0	42.38
4.	NDG 11-24	22.5	33.40
5.	BG 3037	23.5	38.41
6.	PG 0100	34.0	38.27
7.	CSJ 513	32.0	41.86
8.	H 09-53	35.0	42.81
9.	IPC 08-69	39.5	41.83
10.	CSJ 692	35.0	37.01
11.	BG 3038	27.0	45.31
12.	GJG 1004	27.0	30.97
13.	RSG 963	27.0	59.74
14.	JG 31	26.0	30.36
15.	NBEG 3	32.0	36.70
16.	GNG 2127	18.0	52.78
17.	BGD 1075	33.5	44.80
18.	GJG 1013	31.0	44.33
19.	GNG 2144	25.0	50.00
20.	H 09-65	22.5	38.89
21.	PG 099	33.0	30.97
22.	GNG 1995 (R)	14.0	33.33
23.	JG 32	33.0	36.67
24.	GL 29389	37.0	38.68
25.	RKG 207-61	22.5	51.19
26.	RKG 206-31	22.0	43.68
27.	L 550 (susceptible check)	35.5	52.14

Table 4.17: Reaction of chickpea entries (Late sown II) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	20	GL 29390, NDG 11-24, BG 3037, PG 0100, CSJ 513, H 09-53, IPC 08-69, CSJ 692, BG 3038, GJG 1004, JG 31, NBEG 3, BGD 1075, GJG 1013, GNG 2144, H 09-65, PG 099, GNG 1995 (R), JG 32, GL 29389,
5.	Above 50	5 (HS)	6	NDG 11-21, IPC 0713, RSG 963, GNG 2127, RKG 207-61, RKG 206-31
Total entries			26	
LSI			4.23	

Table 4.18: Screening of chickpea entries (Rainfed I) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent Mortality
1.	GJG 0814	29.5	44.14
2.	CSJ 515(AVT-2)	30.0	48.33
3.	RSG 888	29.5	40.44
4.	H 08-71	16.5	36.03
5.	Vijay	39.0	44.74
6.	JG 25	40.0	36.25
7.	GJG 0904	29.5	39.17
8.	RSG 931	24.5	52.92
9.	CSJ 730	22.5	66.40
10.	IPC 07-09	29.5	64.14
11.	L 550 (susceptible check)	36.0	62.74

Table 4.19: Reaction of chickpea entries (Rainfed I) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score / reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	7	GJG 0814, CSJ 515(AVT-2), RSG 888, H 08-71, Vijay, JG 25, GJG 0904,
5.	Above 50	5 (HS)	3	RSG 931, CSJ 730, IPC 07- 09
Total entries			10	
LSI			4.30	

Table 4.20: Screening of chickpea entries (Rainfed II) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1.	NDG 11-41	18.0	36.69
2.	GNG 2146	29.5	74.54
3.	IPC 2009-21	33.0	37.93
4.	CSJ 079	30.5	53.83
5.	GJG 1012	35.5	43.67
6.	GL 28186	25.0	45.62
7.	JG 33	34.0	25.97
8.	CSJ 741	23.5	29.44
9.	Phule G 0752	4.0	18.75
10.	H 09-54	17.5	44.57
11.	RSG 888	25.5	54.77
12.	GJG 1003	32.5	53.98
13.	BG 3035	29.0	51.02
14.	H 09-32	34.0	54.51
15.	RSG 931	24.5	38.58
16.	IPC 2008-76	34.0	50.00
17.	BG 3036	26.5	62.57
18.	RKG 11-301	28.0	49.48
19.	GNG 2145	34.0	56.79
20.	Phule G 0305-3	5.0	37.50
21.	H 09-23	33.5	48.53
22.	NBEG 47-1	20.0	54.73
23.	Vijay	36.0	44.99
24.	RKG 160	28.0	53.51
25.	L 550(susceptible check)	37.0	58.33

Table 4.21: Reaction of chickpea entries (Rainfed II) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	1	Phule G 0752
3.	20 – 30	3 (T)	2	JG 33, CSJ 741
4.	30 – 50	4 (S)	11	NDG 11-41, IPC 2009-21, GJG 1012, GL 28186, H 09-54, RSG 888, RSG 931, IPC 2008-76, Phule G 0305-3, H 09-23, Vijay
5.	Above 50	5 (HS)	10	GNG 2146, CSJ 079, GJG 1003, BG 3035, H 09-32, BG 3036, RKG 11-301, GNG 2145, NBEG 47-1, RKG 160
Total entries			24	
LSI			4.25	

Table 4.22: Screening of chickpea entries (national nursery for elite lines) against dry root rot during 2012-2013

S.No.	Entries	Per cent emergence	Per cent mortality
1.	JG 38	75.5	37.94
2.	GNG 3099	36.0	39.82
3.	JG 130-51	67.0	41.00
4.	HIR 40	33.0	67.47
5.	JGK 90	75.5	40.53
6.	BBG 2	76.5	43.47
7.	GJK 80	25.5	32.87
8.	CSJ 80022	50.5	40.45
9.	RSG 9056	74.0	27.24
10.	JG 16	61.5	29.23
11.	JG 14	68.5	34.98
12.	CSK 8015	70.5	35.48
13.	JG 28	75.5	34.74
14.	CSJ 8023	75.5	23.23
15.	L 550 (susceptible check)	37.0	67.54

Table 4.23: Reaction of chickpea entries (national nursery for elite lines) against dry root rot during 2012-2013

S.No.	Per cent mortality	Score reaction	No. of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	3	RSG 9056, JG 16, CSJ 8023
4.	30 – 50	4 (S)	10	JG 38, GNG 3099, JG 130-51, JGK 90, BBG 2, GJK 80, CSJ 80022, JG 14, CSK 8015, JG 28
5.	Above 50	5 (HS)	1	HIR 40
Total entries			14	
LSI			3.85	

Table 4.24: Screening of chickpea entries (Desi I) against dry root rot during 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	Phule G 08108	13.5	65
2	GNG 2171	21.5	57.5
3	Vijay	22.5	67
4	NDG 11-12	20.5	46
5	H09-96	22.0	57
6	BDG 1074	20.0	44
7	H09-90	23.5	40.5
8	PG0120	21.5	67.5
9	GJG 1001	20.0	80
10	GL 28295	23.0	78.5
11	DCP 92-3	30.5	100
12	DJG 1010	23.5	77
13	GL 28297	23.0	93.5
14	GCP 101	23.5	79
15	IPC 2006-126	22.5	64
16	JG 315	21.5	82
17	GNG 469	6.5	100
18	DC 2012-3	17.0	35
19	GNG 1581	20.5	80
20	GCP 105	23.0	91
21	L 550 (susceptible check)	24.0	91.5

Table 4.25: Reaction of chickpea entries (Desi I) against dry root rot 2013-2014

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	4	NDG 11-12, BDG 1074, H09-90, DC 2012-3
5.	Above 50	5 (HS)	16	Phule G 08108,GNG 2171,vijay,H09-96 PG0120,GJG 1001,GL 28295,DCP 92-3 DJG 1010,GL 28297,GCP 101,IPC 2006-126,JG 315,GNG 469,GNG 1581,GCP 105,L 550
Total entries			20	
L.S.I.			4.80	

Table 4.26: Screening of chickpea entries (Desi II) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	Phule G 0405	19.0	97.0
2	GNG 2207	21.0	80.0
3	H 10-01	22.5	75.5
4	Phule G 0408	22.5	100.0
5	CSJ 513	24.0	83.5
6	GJG 1114	20.0	94.5
7	GNG 2216	17.5	72.0
8	H 10-21	16.5	87.0
9	CSJ 821	24.0	100.0
10	PG 0127	14.5	75.5
11	GJG 11007	25.0	100.0
12	GL 10006	21.0	75.5
13	IPC 2010-14	20.5	95.0
14	JG 35	23.0	100.0
15	NBEG 452	22.0	100.0
16	G 10023	24.0	68.0
17	IPC 2010-217	23.0	98.0
18	JG 36	21.0	100.0
19	RV SSG 20	11.0	64.5
20	JG 16	19.0	89.5
21	DCP 92-3	24.0	98.0
22	GNG 1958	16.5	100.0
23	L 550 (susceptible check)	25.0	75.0

Table 4.27: Reaction of chickpea entries (Desi II) against dry root rot 2013-2014

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	22	Phule G 0405,GNG 2207,H 10-01 Phule G 0408,CSJ 513,GJG 1114,GNG 2216,H 10-21,CSJ 821,PG 0127,GJG 11007,GL 10006,IPC 2010-14,JG 35,NBEG 452,G 10023,IPC 2010-217 JG 36,RV SSG 20,JG 16,DCP 92-3,GNG 1958
Total entries			22	
L.S.I.			5.00	

Table 4.28: Screening of chickpea entries (Late sown I) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	GNG 2144	24.5	94.0
2	NDG 11-21	22.5	82.5
3	Vijay	22.0	95.5
4	BG 3037	18.0	87.5
5	GJG 1013	23.5	73.5
6	GNG 2127	20.5	78.0
7	IPC 2008-69	21.0	88.0
8	Pant G 816	22.0	83.5
9	DC- 2012-4	19.0	82.0
10	NDG 11-24	17.0	91.5
11	RSG 963	24.0	77.5
12	BGD 1075	22.5	72.0
13	JG 315	22.5	84.5
14	IPC 2007-13	21.5	83.0
15	BG 3032	22.5	91.5
16	IPC 2006-77	23.5	51.5
17	JG 14	23.0	76.5
18	CSJ 687	23.5	92.0
19	L550 (susceptible check)	22.0	79.5

**Table 4.29: Reaction of chickpea entries (Late sown I) against dry root rot
2013-2014**

S. No	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	18	GNG 2144, NDG 11-21, Vijay, BG 3037 GJG 1013, GNG 2127, IPC 2008-69, Pant G 816, DC- 2012-4, NDG 11-24, RSG 963, BGD 1075, JG 315, IPC 2007-13, BG 3032, IPC 2006-77, JG 14, CSJ 687
Total entries			18	
L.S.I.			5.00	

**Table 4.30: Screening of chickpea entries (Late sown II) against dry root rot
2013-2014**

S.No	Entries	Per cent emergence	Per cent mortality
1	Phule G 0405	20.5	63.5
2	RSG 963	19.5	100.0
3	GJG 0820	20.0	97.5
4	Phule G 0511	9.5	79.0
5	IPC 2010-181	22.0	95.5
6	GJG 0831	20.0	100.0
7	GNG 2219	19.5	79.0
8	CSJ 564	13.5	96.0
9	JG 37	20.5	100.0
10	GNG 2215	18.0	91.5
11	JG 16	21.5	90.5
12	CSJ 769	21.5	95.5
13	NDG 12-23	21.0	81.0
14	JG 14	20.0	94.5
15	PG 0109	24.0	83.5
16	IPC 2010-62	16.5	100.0
17	NDG 12-24	13.0	79.0
18	JG 38	21.5	100.0
19	JL 29095	23.0	75.5
20	H 10 05	20.0	62.5
21	JG 315	22.5	71.0
22	Pant G 186	23.0	80.5
23	H 10-22	21.0	83.5
24	L 550 (susceptible check)	18.5	90.0

Table 4.31: Screening of chickpea entries (Late sown II) of their reaction against dry root rot 2013-2014

S. No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	23	Phule G 0405, RSG 963, GJG 0820 Phule G 0511, IPC 2010-181, GJG 0831, GNG 2219, CSJ 564, JG 37, GNG 2215, JG 16, CSJ 769, NDG 12-23, JG 14, PG 0109, IPC 2010-62, NDG 12-24, JG 38, JL 29095, H 10 05, JG 315, Pant G 186, H 10-22
Total entries			23	
L.S.I.			5.00	

Table 4.32: Screening of chickpea entries (Rainfed I) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	RKG 27-99	24.0	98.0
2	Phule G 101104	15.5	100.0
3	CSJ 515	22.0	84.0
4	JAG 1107	25.0	88.0
5	IPC 2010-72	22.5	77.0
6	Phule 11117	22.5	91.0
7	CSG 741	23.5	100.0
8	JG 39	21.5	89.0
9	Vijay	22.5	98.0
10	CSJ 778	22.5	100.0
11	Jaki 9218	14.0	100.0
12	GNG 2210	15.5	62.0
13	GNG 1115	21.0	67.5
14	NBEG 147	17.0	91.0
15	GNG 2226	21.0	83.5
16	JG 315	20.5	92.5
17	RKG 27-12	24.0	96.0
18	IPC 2009-191	23.0	93.5
19	H 10-41	20.5	75.5
20	JG 40	24.5	90.0
21	RSG 931	22.5	100.0
22	H 10-57	23.0	95.0
23	RVSSG 21	18.5	94.5
24	RSG 888	24.0	85.5
25	L 550 (susceptible check)	22.0	84.0

Table 4.33: Reaction of chickpea entries (Rainfed I) against dry root rot 2013-2014

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	24	RKG 27-99, Phule G 101104, CSJ 515, JAG 1107, IPC 2010-72, Phule 11117, CSG 741, JG 39, Vijay, CSJ 778, Jaki 9218, GNG 2210, GNG 1115, NBEG 147, GNG 2226 JG 315, RKG 27-12, IPC 2009-191, H 10-41, JG 40, RSG 931, H 10-57, RVSSG 21, RSG 888
Total entries			24	
L.S.I.			5.00	

Table 4.34: Screening of chickpea entries (Rainfed II) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	Vijay	11.0	95.5
2	GNG 2146	22.0	85.0
3	RSG 888	22.0	100
4	RSG 931	23.0	87.0
5	GJG 1012	23.5	100
6	JAKI 9128	20.0	100
7	JG 315	18.5	100
8	GNG 469	18.0	88.5
9	L 550 (susceptible check)	19.0	91.5

**Table 4.35: Reaction of chickpea entries (Rainfed II) against dry root rot
2013-2014**

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	8	Vijay ,GNG 2146, RSG 888,RSG 931GJG 1012, JAKI 9128, JG 315, GNG 469
Total entries			8	
L.S.I.			5.00	

**Table 4.36: Screening of chickpea entries (Kabuli I) against dry root rot
2013-2014**

S.No	Entries	Per cent emergence	Per cent mortality
1	VIHAR	2.0	100
2	KRIPA	5.0	100
3	CSJK 54	4.5	100
4	HK 2	3.5	100
5	BDNGK 798	3.5	100
6	HK4	4.0	100
7	CSJK 74	6.5	50.0
8	JGK1	2.0	100
9	BG 1035	3.5	83.5
10	BG 1003	8.0	82
11	L 550 (susceptible check)	19.0	85

Table 4.37: Reaction of chickpea entries (Kabuli I) against dry root rot 2013-2014

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	10	VIHAR, KRIPA, CSJK 54, HK 2, BDNGK 798, HK4, CSJK 74, JGK1BG 1035, BG 1003, L 550
Total entries			10	
L.S.I.			5.00	

Table 4.38: Screening of chickpea entries (Kabuli II) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	HK 09-201	2.0	83.5
2	BG 3042	6.0	91.5
3	IPCK 209-164	14.0	100
4	HK 09-206	6.0	69.0
5	BGD 1076	5.5	70.0
6	JGK 1	8.5	100
7	HK 09-219	3.5	80.0
8	BGD 1078	9.0	79.5
9	IPCK 2006-143	8.5	74.5
10	HK 2	3.5	83.5
11	BGD 1079	5.5	63.5
12	BG 1053	9.0	77.5
13	BG 3026	9.5	72.5
14	CSJK 27	7.0	65.0
15	HK 4	7.0	65.0
16	BG 3039	8.5	100
17	CSJK 72	9.0	77.5
18	BG 1003	10.0	94.0
19	BG 3040	8.5	100
20	JLK 27211	9.5	89.0
21	VIHAR	12.5	88.5
22	BG 3041	3.0	62.5
23	L 550 (susceptible check)	18.5	78.5

Table 4.39: Reaction of chickpea entries (Kabuli II) against dry root rot 2013-2014

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	1	Nil
5.	Above 50	5 (HS)	21	HK09-201,BG 3042,IPCK 209-164,HK 09-206,BGD 1076,JGK 1,HK 09-219,BGD 1078,IPCK 2006-143,HK 2 BGD 1079,BG 1053,BG 3026,CSJK 27HK 4 , BG 3039, CSJK 72 ,BG 1003,BG 3040, JLK 27211,VIHAR, BG 3041
Total entries			22	
L.S.I.			4.95	

Table 4.40: Screening of chickpea entries (Kabuli III) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	CSJK 4	5.0	85.5
2	Phule G 0625-6	1.0	100
3	PG 071	1.0	50.0
4	HK 2	1.0	100
5	Phule G 0625-9	4.0	100
6	CSJK 77	3.5	100
7	HK 4	4.0	58.5
8	GNG 2237	6.5	100
9	GLK 10082	2.5	58.0
10	NBEG 176	7.0	79.0
11	GNG 2228	2.5	100
12	JGK 1	10	100
13	RVSSG 23	5.0	62.5
14	HK 09-211	6.0	62.5
15	GLK 27234	5.0	69.0
16	GNG 1969	2.5	83.5
17	HK 10-103	4.0	56.5
18	RVSSG 24	10.0	81.5
19	NBEG 402	6.5	100
20	CSJK 6	8.5	100
21	L 550 (susceptible check)	14.5	82.5

**Table 4.41: Reaction of chickpea entries (Kabuli III) against dry root rot
2013-2014**

S.No.	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	20	CSJK 4 ,Phule G 0625-6, PG 071, HK 2, Phule G 0625-9, CSJK 77, HK 4, GNG 2237, GLK 10082,NBEG 176,GNG 2228JGK 1,RVSSG 23, HK 09-211,GLK 27234, GNG 1969, HK 10-103,RVSSG 24NBEG 402,CSJK 6
Total entries			20	
L.S.I.			5.00	

Table 4.42: Screening of chickpea entries (ELSK I) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	IPCK 2004-29	4	100.0
2	KRIPA	1.5	100.0
3	BG 3022	3.5	100.0
4	JGK 1	6.5	67.5
5	PKV 4	2.5	100.0
6	BG 3028	6	100.0
7	PKV 2	5	100.0
8	L 550 (susceptible check)	19.5	84.5

Table 4.43: Reaction of chickpea entries (ELSK I) against dry root rot 2013-2014

S. No	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	7	IPCK 2004-9,KRIPA,BG 3022,JGK 1,PKV 4,BG 3028,PKV 2
Total entries			7	
L.S.I.			5.00	

Table 4.44: Screening of chickpea entries (ELSK II) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	KRIPA	8	100
2	NBEG 119	9	100
3	JGK 1	3.5	100
4	JGK 18	11.5	93
5	IPCK 2002-29	5	100
6	L550 (susceptible check)	20.5	83

Table 4.45: Reaction of chickpea entries (ELSK II) against dry root rot 2013-2014

S.No.	Per cent mortality	Score reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	5	KRIPA, NBEG19, JGK 1, JGK 18IPCK 2002-29
Total entries			5	
L.S.I.			5.00	

Table 4.46: Screening of chickpea entries (ESLK III) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1	CSJK 46	3.0	100
2	CSJK 73	4.5	100
3	JGK 26	9.5	100
4	KRIPA	7.0	100
5	GLK 10100	5.5	100
6	JGK 1	6.5	100
7	HK 10- 124	4.5	100
8	GLK 10103	4.5	83.5
9	JGK 27	6.0	100
10	NBEG 27	5.5	100
11	Phule G 0627-3	6.0	100
12	L 550 (susceptible check)	21.5	86

Table 4.47: Screening of chickpea entries (ELSK III) of their reaction against dry root rot 2013-2014

S. No	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	0	Nil
3.	20 – 30	3 (T)	0	Nil
4.	30 – 50	4 (S)	0	Nil
5.	Above 50	5 (HS)	5	CSJK 46,CSJK73,JGK 26,KRIPA,GLK 0100,JGK 1, HK 10- 124,GLK 10103JGK 27,NBEG 27, Phule G 0627-3
Total entries			5	
L.S.I.			5.00	

Table 4.48: Screening of chickpea entries (national nursery for elite) against dry root rot 2013-2014

S.No	Entries	Per cent emergence	Per cent mortality
1.	JG 37	25.5	30.7
2.	IPCK 06-78	11.0	42.7
3.	Vijay	17.0	46.4
4.	JG 17	25.0	24.0
5.	GNG 1996	23.0	44.8
6.	CSJ 515	26.0	42.2
7.	JG 103 X ICC 11551	27.5	41.7
8.	GNG 2112	6.5	76.1
9.	JG 38	24.5	65.6
10.	HIR 60	6.5	45.0
11.	JG 14X ICC 4958	26.5	35.4
12.	HIR 70	2.0	63.4
13.	CSJ 808	26.5	21.3
14.	JGK 13	1.5	38.3
15.	JG 39	24.5	45.4
16.	BBG 1	4.0	31.2
17.	JGK 17	1.5	41.1
18.	JG 40	26.5	45.0
19.	CSJ 809	27.5	18.6
20.	JGK 18	3.0	41.5
21.	CSJ 811	27.0	41.1
22.	RSG 963	28.0	36.7
23.	JG 14	30.0	30.3
24.	CSJ 812	30.0	49.1
25.	CSJ 556	26.0	27.4
26.	JG 23	26.0	32.5
27.	CSJ 814	24.5	33.9
28.	GJG 0906	30.0	32.1
29.	JG 24	25.0	24.9
30.	CSJ 815	28.5	26.6
31.	GLK 26167	3.5	40.0
32.	JG 25	21.5	62.5
33.	CSJ 816	25.0	41.0
34.	JLK 28127	2.0	23.6
35.	JG 26	18.0	46.1
36.	CSJ 817	27.0	36.3
37.	JG 27	30.0	27.6
38.	JG 35	26.0	46.3
39.	CSJ 818	30.0	38.0
40.	JG 36	29.0	29.0
41.	CSJK 74	11.5	39.0
42.	L 550 (susceptible check)	14.5	70.1

Table 4.49: Reaction of chickpea entries (national nursery for elite) against dry root rot

S. No	Per cent mortality	Score / reaction	Number of entries	Name of entries
1.	Less than 10	1 (R)	0	Nil
2.	10 – 20	2 (MR)	1	CSJ 809
3.	20 – 30	3 (T)	8	JG 17, CSJ 808, CSJ 556, JG 24, CSJ 815, GLK 8127, JG 27, JG 36
4.	30 – 50	4 (S)	28	JG 37, IPCK 06-78, Vijay, GNG 1996, CSJ 515, JG 103 X ICC 11551,HIR 60, JG 14X ICC 4958, JGK 13, JG 39, BBG 1, JGK 17, JG 40, JGK 18, CSJ 811, RSG 963, JG 14, CSJ 556, JG 23, CSJ 814, CSJ 812GJG 0906, GLK 26167, JG 25, CSJ 816, JLK 28127, JG 26, CSJ 817, JG 27, JG 35 CSJ 818, JG 36, CSJK 74, JG 23, CSJ 814, CSJ 812GJG 0906, JG 24, CSJ 815, GLK 26167, CSJ 816, JG 26, CSJ 817, JG 35,CSJ 818, CSJK 74
5.	Above 50	5 (HS)	4	GNG 2112, JG 38, HIR 70, JG 25
Total entries			41	
L.S.I.			3.85	

4.7.2 *In vitro* screening

One thirty two entries were screened *in vitro* through blotter paper method against *R. bataticola* isolate RB 5 (Raigarh) which was most virulent under *in vitro* aggressiveness test. Variety L550 of chickpea was used as susceptible check on blotter paper technique (Nene *et al.*, 1981). The disease reaction of various entries varied from 1 – 9 on a scale of 1-9 (Table 2). Out of 132 entries, 21 entries categorized as resistant, were no infection on roots was observed in resistant lines as the root system was healthy, completely developed and exhibited white appearance, followed by 19 entries categorized as moderately resistance with exhibited very few small necrotic lesion on roots many lesions on roots, While, 21 entries grouped as tolerant, clear lesions was observed on roots but small, though new roots were free from infection. 53 entries were found to be susceptible with many lesions on roots but new roots generally free from lesions. However, 18 entries exhibited highly susceptible were the roots were completely infected, discoloured and rotten. Results are depicted in Table 4.50, 4.51 and Plate 12.

In relation to present finding Gupta *et al.* (2012) screened *in-vitro*, 170 genotypes, sixty eight were found resistant, seventy five genotypes were found moderately resistant. While remaining sixteen, four and seventeen germplasms were tolerant, susceptible and highly susceptible, respectively. In highly susceptible genotype BG 212 (check) which had a disease rating of 9 at eight days after incubation at 30°C. Similarly, Pande *et al.* (2004) evaluated twenty nine chickpea germplasm accessions for resistance against dry root rot caused by *R. bataticola* under *in vitro* condition and observed twenty two lines were either susceptible or highly susceptible to dry root rot whereas, seven germplasms were tolerant. Chickpea genotypes have been screened and identified as resistant to dry root rot by various workers on the basis of *in vitro* and field conditions (Nene *et al.*, 1981; Anand Rao and Haware, 1987; Reddy *et al.*, 1990; Gupta, 1997; Khalid and Ilyas, 2000 and Gupta *et al.*, 2003).

Table 4.50: *In vitro* screening of chickpea by blotter paper method gainst *R. bataticola* (RB5)

S.No.	Entry	Entry name	Disease rating
1.	P2	GNG 2171	3.0
2.	P3	Vijay	6.3
3.	P4	NDG 11-12	4.3
4.	P5	H09-96	5.0
5.	P6	BDG 1074	7.6
6.	P7	H09-90	8.3
7.	P8	PG0120	9.0
8.	P9	GJG 1001	9.0
9.	P10	GL 28295	9.0
10.	P11	DCP 92-3	8.3
11.	P12	DJG 1010	9.0
12.	P13	GL 28297	8.2
13.	P14	GCP 101	7.0
14.	P15	IPC 2006-126	6.3
15.	P16	JG 315	5.6
16.	P17	GNG 469	7.6
17.	P18	DC 2012-3	9.0
18.	P19	GNG 1581	9.0
19.	P20	GCP 105	9.0
20.	P25	Phule G 0405	8.3
21.	P26	GNG 2207	9.0
22.	P27	H 10-01	9.0
23.	P28	Phule G 0408	7.0
24.	P29	CSJ 513	4.3
25.	P30	GJG 1114	7.6
26.	P31	GNG 2216	8.3
27.	P32	H 10-21	8.3
28.	P33	CSJ 821	8.3
29.	P34	PG 0127	8.3
30.	P35	GJG 11007	6.3
31.	P36	GL 10006	5.0
32.	P37	IPC 2010-14	4.3

S.No.	Entry	Entry name	Disease rating
33.	P38	JG 35	5.0
34.	P39	NBEG 452	5.0
35.	P40	G 10023	5.0
36.	P41	IPC 2010-217	5.0
37.	P42	JG 36	4.3
38.	P43	RV SSG 20	5.0
39.	P44	JG 16	5.6
40.	P45	DCP 92-3	5.0
41.	P46	GNG 1958	5.6
42.	P51	GNG 2144	5.6
43.	P52	NDG 11-21	5.0
44.	P53	Vijay	5.6
45.	P54	BG 3037	2.3
46.	P55	GJG 1013	4.3
47.	P56	GNG 2127	3.0
48.	P57	IPC 2008-69	7.0
49.	P58	Pant G 816	1.0
50.	P59	DC- 2012-4	1.6
51.	P60	NDG 11-24	3.6
52.	P61	RSG 963	3.0
53.	P62	BGD 1075	2.3
54.	P63	JG 315	1.6
55.	P64	IPC 2007-13	1.0
56.	P65	BG 3032	3.6
57.	P66	IPC 2006-77	6.3
58.	P67	JG 14	1.6
59.	P68	CSJ 687	3.6
60.	P71	Phule G 0405	1.6
61.	P72	RSG 963	5.0
62.	P73	GJG 0820	5.6
63.	P74	Phule G 0511	7.0
64.	P75	IPC 2010-181	7.0
65.	P76	GJG 0831	7.0
66.	P77	GNG 2219	4.0

S.No.	Entry	Entry name	Disease rating
67.	P78	CSJ 564	8.3
68.	P79	JG 37	1.6
69.	P80	GNG 2215	1.6
70.	P81	JG 16	9.0
71.	P82	CSJ 769	9.0
72.	P83	NDG 12-23	8.3
73.	P84	JG 14	4.3
74.	P85	PG 0109	4.3
75.	P86	IPC 2010-62	1.6
76.	P88	JG 38	1.6
77.	P89	JL 29095	7.0
78.	P90	H 10 05	7.0
79.	P91	JG 315	5.0
80.	P92	Pant G 186	3.0
81.	P93	H 10-22	4.3
82.	P102	GNG 2146	1.0
83.	P104	RSG 931	1.0
84.	P105	GJG 1012	3.0
85.	P106	JAKI 9128	9.0
86.	P111	RKG 27-99	4.3
87.	P114	JAG 1107	2.3
88.	P115	IPC 2010-72	7.0
89.	P116	Phule 11117	8.3
90.	P117	CSG 741	8.3
91.	P118	JG 39	9.0
92.	P119	Vijay	9.0
93.	P120	CSJ 778	9.0
94.	P121	Jaki 9218	8.3
95.	P122	GNG 2210	7.6
96.	P123	GNG 1115	3.6
97.	P124	NBEG 147	7.0
98.	P125	GNG 2226	7.3
99.	P126	JG 3153	5.6
100.	P127	RKG 27-12	9.0

S.No.	Entry	Entry name	Disease rating
101.	P130	JG 40	8.3
102.	P131	RSG 931	8.3
103.	P132	H 10-57	3.6
104.	P134	RSG 888	5.0
105.	P135	VIHAR	5.6
106.	P136	KRIPA	7.0
107.	P138	HK 2	7.0
108.	P139	BDNGK 798	5.0
109.	P140	HK4	9.0
110.	P150	JGK 1	5.6
111.	P152	BGD 1078	5.6
112.	P156	BG 1053	9.0
113.	P157	BG 3026	8.3
114.	P160	BG 3039	8.3
115.	P162	BG 1003	5.6
116.	P163	BG 3040	3.6
117.	P164	JLK 27211	1.6
118.	P172	PG 071	1.6
119.	P176	HK 4	9.0
120.	P181	JGK 1	3.0
121.	P184	GLK 27234	6.3
122.	P185	GNG 1969	6.3
123.	P190	IPCK 2004-29	7.6
124.	P191	KRIPA	9.0
125.	P192	BG 3022	9.0
126.	P193	JGK 1	3.6
127.	P194	PKV 4	7.6
128.	P195	BG 3028	5.6
129.	P196	PKV 2	1.6
130.	P206	GLK 10100	1.6
131.	P207	JGK 1	3.6

Table 4.51: Reaction of the entries under *in vitro* condition by blotter paper method

S. No.	Reaction	Rating scale	Numbers of entries	Name of entires
1	Resistant	1.0 - 2.9	21	BG 3037, Pant G 816, DC- 2012-4, BGD 1075, JG 315, IPC 2007-13, BG 3032, JG 14, CSJ 687, Phule G 0405, JG 37, GNG 2215, GNG 2146, IPC 2010-62, JG 38, RSG 931, JAG 1107, JLK 27211, PG 071, PKV 2, GLK 10100
2	Moderately Resistant	3.0-4.9	19	GNG 2171, NDG 11-12, CSJ 513, IPC 2010-14, JG 36, GJG 1013, GNG 2127, NDG 11-24, RSG 963, JG 14, PG 0109, Pant G 186, H 10-22, GJG 1012, RKG 27-99, GNG 1115, H 10-57, BG 3040, JGK 1
3	Tolerant	5.0 – 6.9	21	Phule G 08108, Vijay, H09-96, IPC 2006-126, JG 315, JG 315, JG 3153, RSG 888, VIHAR, BDNGK 798, JGK 1, BGD 1078, BG 1003, BG 3026, BG 3039, KRIPA, BG 3022, BG 1053, BG 3028, GLK 27234, GNG 1969
4	Susceptible	7.0-8.9	53	BDG 1074, H09-90, DCP 92-3, GL 28297, GCP 101, GNG 469, Phule G 0405, Phule G 0408, GJG 1114, GNG 2216, H 10-21, CSJ 821, PG 0127, GJG 11007, GL 10006, JG 35, NBEG 452, G 10023, IPC 2010-217, RV SSG 20, JG 16, DCP 92-3, GNG 1958, GNG 2144, NDG 11-21, Vijay, IPC 2008-69, Phule G 0511, IPC 2010-181, GJG 0831, CSJ 564, NDG 12-23, IPC 2010-72, Phule JL 29095, H 10 0511117, CSG 741, Jaki 9218, GNG 2210, NBEG 147, GNG 2226, JG 40, RSG 931, KRIPA, HK 2, IPCK 2004-29 PKV 4,
5	Highly Susceptible	9.0	18	PG0120, GJG 1001, GL 28295, DJG 1010, DC 2012-3, GNG 1581, GCP 105, GNG 2207, H 10-01, JAKI 9128, JG 39, Vijay, CSJ 778, JG 16, CSJ 769, RKG 27-12, HK 4, HK4,
Total entries			132	

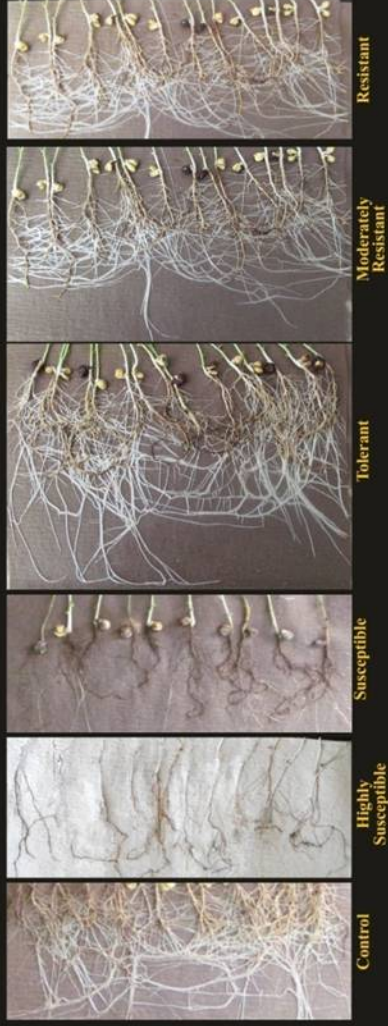


Plate 12: *In vitro* screening through blotter paper method

4.8 Soil factors affecting development of dry root rot caused by *R. bataticola* in chickpea

4.8.1 Effect of Soil pH on dry root rot of chickpea

An experiment was conducted to know which soil pH is favorable or unfavorable for dry root rot development caused by *R. bataticola* in chickpea as described in materials and methods. Result indicated significant difference, presented in Table 4.52, Fig. 4.10, Fig. 4.11 and Plate 13.

Significantly least mortality per cent (12.50%) was recorded at pH 8.0 followed by pH 4 with mortality per cent of 41.67%. pH 4.0 (41.67%), pH 7 (50.00%) and pH 6.0 (62.50%) which were next in order. However, significantly highest mortality was observed at pH 5.0 that is, 83.33%. All the treatment differ significantly in terms of mortality per cent with each other, except pH 4.0, pH 6.0 and pH 7.0. Extent of root damage varied from 3-9 grades and least disease grade was recorded at pH 8.0 with minimum root damage and infection that is, 3 rating, followed by pH 6.0 and pH 7.0 with disease rating 6.3 and 7.6 respectively. While, highest disease rate was recorded at pH 5.0 that is 9.

Table 4.52: Effect of different pH on disease development of dry root rot of chickpea

S.No.	pH	Disease rating	Per cent mortality
1	4.0	7.0	41.67 (40.15)
2	5.0	9.0	83.33 (66.17)
3	6.0	6.3	62.50 (52.39)
4	7.0	7.6	50.00(44.98)
5	8.0	3.0	12.50 (20.69)
SEm±			3.21
CD (5%)			10.12
Average of three replication			
Fig.ures in parentheses are angular transformation			

Soil pH also significantly influenced the germination per cent, root length and shoot length in chickpea when was inoculated with *R. bataticola* RB4, and data are presented in Table 4.53 and Fig. 4.12.

Highest germination was observed in pH 5.0, pH 6.0 and pH8.0 that is 90.00 per cent, followed by pH 7.0 and pH 4.0, 86.65 and 83.33 per cent respectively, which were at par with each other. Maximum root length was observed in pH 8.0 (18.6 cm) and pH 6.0 (15.6 cm) which were at par, but significantly better than other treatments. Least increased in root length was recorded in pH 7.0 that is 9.3 cm. Maximum shoot length was observed in pH 6.0 (16.4 cm) followed by pH 5.0 (15.2 cm) and pH 7.0 (15.0 cm) which were at par with each other. Whereas, least shoot length was measured in pH 4.0 (14.7 cm). There was significant difference among treatment viz., germination per cent, root length and shoot length.

Table 4.53: Effect of pH on germination, root and shoot length of chickpea in inoculated soil with *R.bataticola*

S.No.	pH	Germination (%)	Root length	Shoot length
		*	(cm) *	(cm) *
1	4.0	83.33 (9.20)	14.0	14.7
2	5.0	90.00 (9.53)	12.0	15.2
3	6.0	90.00 (9.53)	15.6	16.4
4	7.0	86.65 (9.36)	9.3	14.9
5	8.0	90.00 (9.53)	18.6	15.0
SEm±		0.058	1.05	0.97
CD (5%)		0.18	3.31	3.05

* Average of three replication

Figures in parentheses are square root transformation

Parmar, 2013 studied *in vitro* effect of different pH ranged from 6.0 to 8.0 on the growth of different isolates of *R. bataticola*. The maximum growth of the isolates of *R. bataticola* was recorded at 6.5 pH (84.92mm) and the minimum colony diameter was recorded at pH 8.0 (61.69mm).

Papavizas (1977) found reduced survivability in a very acid soil (pH 4.5) collected under a pine stand, 33% of the sclerotia survived for 1 yr. Soil pH had

very little or no effect on sclerotial survivability. In accordance to present finding Srivastava and Dhawan 1980 observed, maximum plant mortality was at pH 5.5 and disease intensity decreased with the increase in soil pH above 5.5. Plant mortality in soil of pH 4.5 and 8.5 was found to be poor. Epidemiological data of the disease revealed that disease incidence was greater in soils of low pH (Srivastava and Dhawan, 1979). Acidic soil reactions are known to favor the wilts (Garrett 1944, 1956; Walker 1975). According to Chapman (1965) soils which become acidic due to leaching of several soluble substances are characteristically deficient in many essential nutrients. This condition predispose the host to the attack of pathogen and thus, favours the development of disease. Satischandra *et al.* (1980) recorded variation in pH and addition of salts did not affect the saprophytic activity of fungus whereas, it is adversely affected by soil moisture.

Shanmugam and Govindswamy, 1973 studied the physiological parameter of *Macrophomina phaseolina* causing groundnut root rot and found that, it grew best at pH of 5.0. Sugha *et al.*, 1994 investigated that the soil characteristic soil pH in relation to fusarium wilt in chickpea, found that the maximum wilt occurred at pH 5.2, with slight decline toward neutrality and significant reduction of disease with increasing soil alkinity.

Shim and Star, 1999 examined the effect of soil pH on sclerotial germination and pathogenicity of two isolates of *Sclerotium rolfsii* on peanut and reported that sclerotial germination for both isolates was greater in acidic soil than at alkaline pH. Stems infection by *S. rolfsii* in greenhouse tests was greater at soil pH 5.6 than at alkaline soil pH 8.7 and 9.8. Singh and Gandhi (1991) reported maximum mortality to guar (*Cyamopsis tetragonoloba*) caused by *Sclerotium rolfsii* at soil pH 6.1 and and significantly decreases disease incidence at pH 8.4. Chattopadhyay and Mustafee, 1977 reported that *S. rolfsii* from jute, showed good growth at pH 5.0 and 6.0. Mathur and Sinha, 1968 studied the collar rot development in gram attacked with *Sclerotium rolfsii* observed that the infection was maximum at 5.7 (89.6%) and at alkaline condition disease reduced.

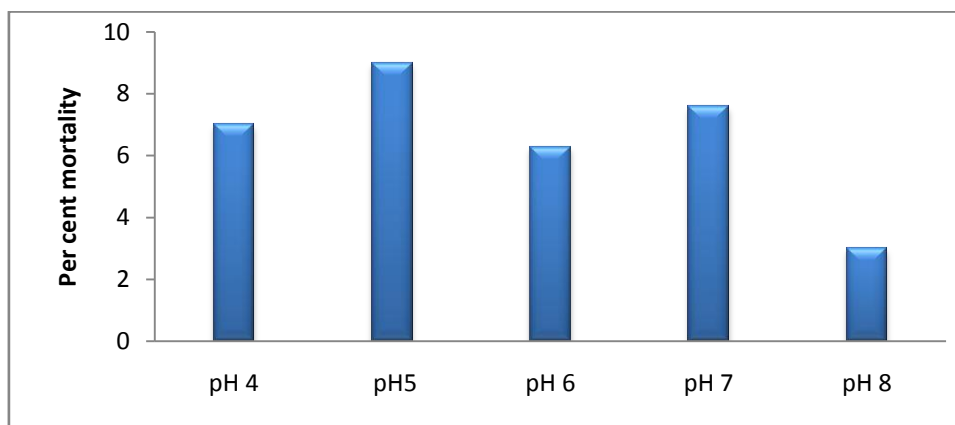


Fig 4.10: Per cent mortality of chickpea due to dry root rot under glass house condition at different soil pH

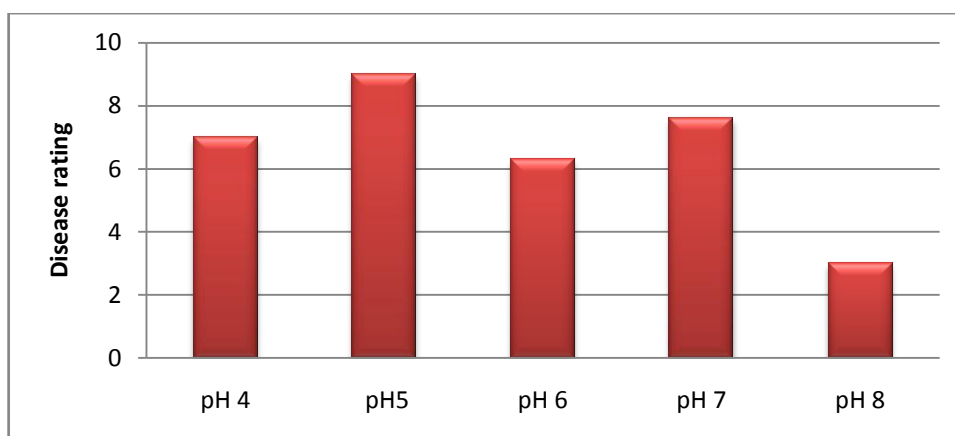


Fig 4.11: Disease rating of chickpea dry root rot in different pH

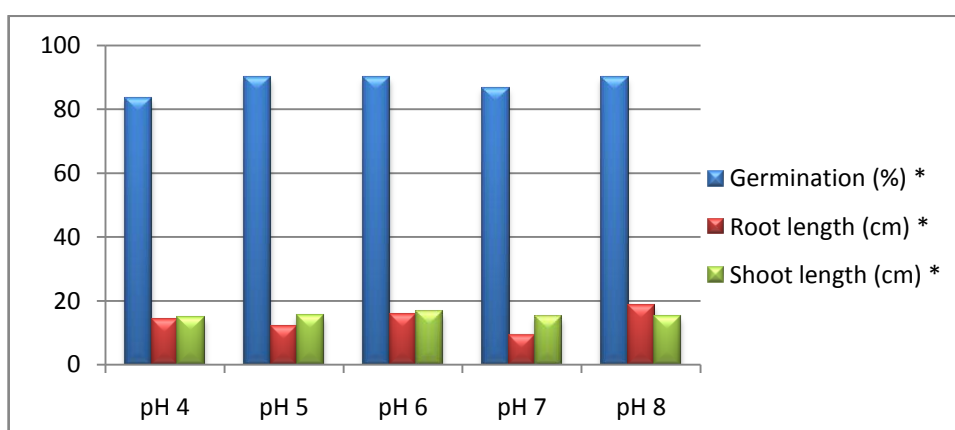


Fig 4.12: Effect of pH on germination, root and shoot length of chickpea in inoculated soil with *R. bataticola*



pH 4



pH 5



pH 6



pH 7



pH 8



Plate 13: Effect of different soil pH on disease development of dry root rot of chickpea

4.8.2 Effect of Soil type on dry root rot of chickpea

An experiment was conducted to know soil type favorable or unfavorable for dry root rot development caused by *R. bataticola* in chickpea as described in materials and methods. Result showed significant difference presented in Table 4.54, Fig. 4.13, Fig. 4.14 and Plate 14.

Four different soil types are found in Chhattisgarh region, that were taken for study. Significantly least mortality per cent (16.66%) was recorded in Matasi followed by Dorsa with mortality per cent of 20.83% which were at par with each other. However, highest mortality was observed in Bhata 62.5%, followed by Kanhar 54.16%. Extent of root damage varied greatly from 1-9 grades and lowest disease grade was recorded in Matasi with minimum root damage and infection that is, 1.6 rating, followed by Dorsa and Bhata with disease rating 5.6. While, the highest disease rate was recorded in Kanhar that is 8.3.

Table 4.54: Effect of different soil type on disease development of dry root rot of chickpea

S.No.	Soil type	Disease rating	Per cent mortality
1	Dorsa	5.6	20.83 (26.38)
2	Matasi	1.6	16.66 (23.79)
3	Kanhar	8.3	54.16 (47.39)
4	Bhata	5.6	62.5 (52.39)
SEm± 4.07			
CD(5%)		13.83	
Average of three replication			
Figures in parentheses are angular transformation			

Soil type also significantly influenced the germination per cent, root length and shoot length in chickpea when inoculated with *R. bataticola* RB4, was studied and data are presented in Table 4.55 and Fig. 4.15.

Highest germination was observed in Dorsa and Bhata (90 per cent) followed by Kanhar 86.65 per cent, which were found to be at par with each other. While, the lowest germination per cent was observed in Matasi that is 83.33 per cent. Maximum root length was observed in Matasi (16.8 cm) and Dorsa (15.9 cm)

which were at par, but significantly superior than other treatments. Minimum in root length was recorded in Bhata that is 13.7 cm and Kanhar (14.2 cm). Maximum shoot length was observed in Bhata (14.8 cm) followed by Matasi (14.6 cm) which were at par with each other. There was significant difference among treatment viz., germination per cent, root length and shoot length.

Table 4.55: Effect of soil type on germination, shoot and root length of chickpea in inoculated soil with *R. bataticola*

S.No.	Soil type	Germination (%) *	Root length (cm) *	Shoot length (cm) *
1	Dorsa	90.00 (9.53)	15.9	12.7
2	Matasi	83.33 (9.20)	16.8	14.6
3	Kanhar	86.65 (9.36)	14.2	13.3
4	Bhata	90.00 (9.53)	13.7	14.8
SEm		0.06	0.62	0.45
CD (5%)		0.21	2.02	1.48
* Average of three replication				
Figures in parentheses are square root transformation				

Nene 1980 reported vertisol soil seems to favour the disease more than alfisol soil. The mortality was high in vertisol which was significantly influenced by soil type when compared to alfisol. Chauhan, 1965 reported high wilt incidence in vertisol as it contain high clay content (64%). Krishna and Krishnappa, 1996 reported *Fusarium oxysporum* f.sp. *ciceri* spore was high in vertisol (1300 CFU/g soil) when compared to alfisol (1153 CFU/g soil). Total mortality was observed on 20-24 days after sowing in vertisol while the wilt incidence was only 73.4% in alfisol. Also infection by *F. oxysporum* f.sp. *ciceri* significantly affected all the plant growth characters resulted in a high percentage reduction of shoot fresh weight, shoot dry weight and root fresh weight more in vertisol when compared to alfisol. Hence, it could be possible that apart from *R. bataticola*, other soil borne pathogens are favoured in vertisol.

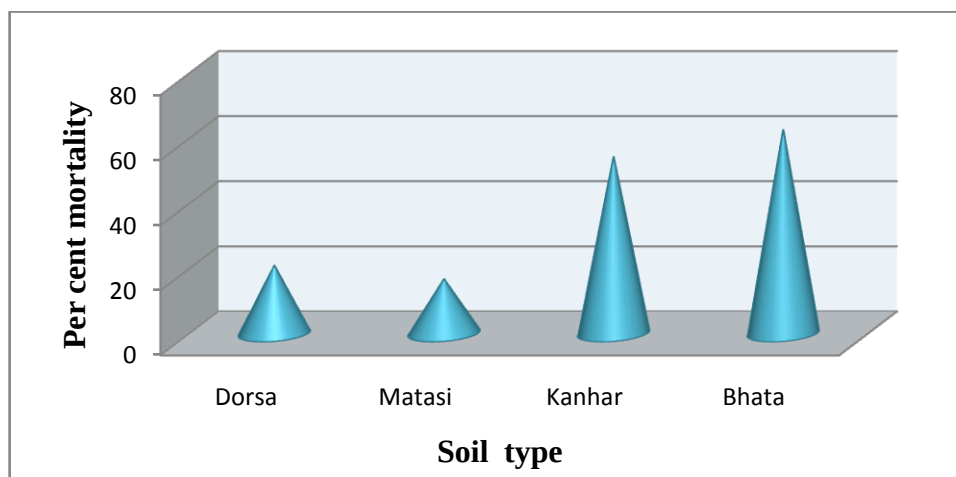


Fig 4.13: Per cent mortality of chickpea due to dry root rot under glass house condition at different soil type

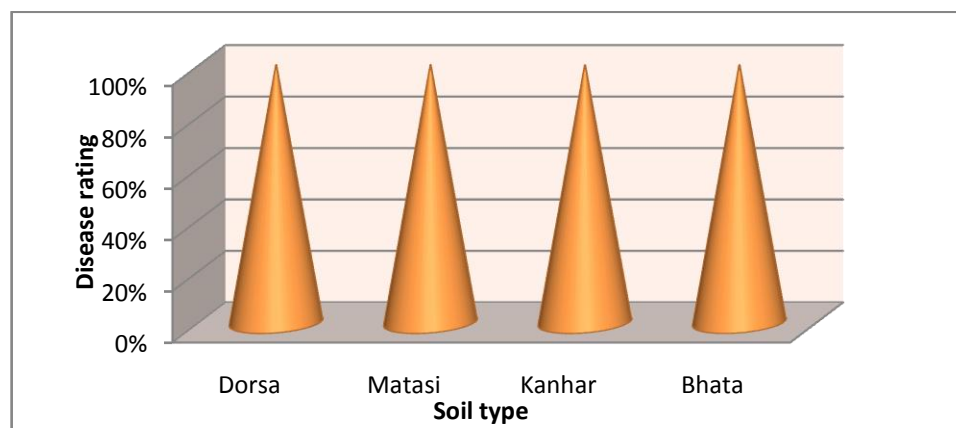


Fig 4.14: Disease rating of chickpea dry root rot in different soil type

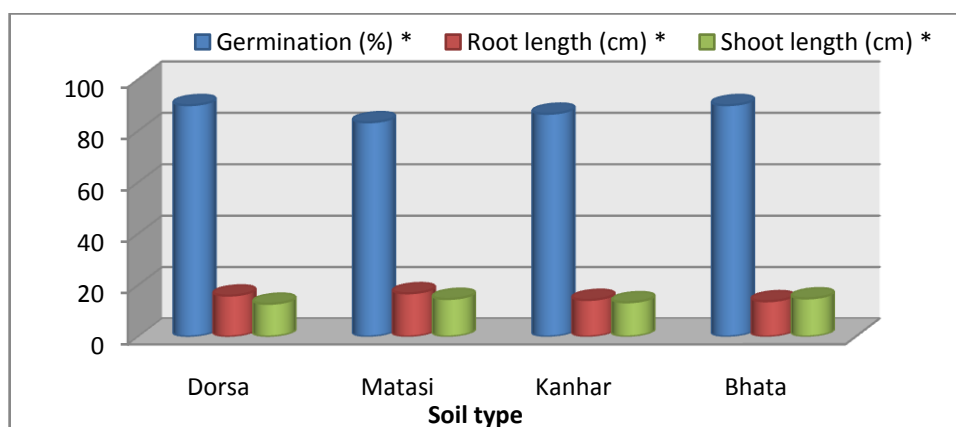


Fig 4.15: Effect of soil type on germination, root and shoot length of chickpea in inoculated soil with *R. bataticola*



Dorsa



Matasi



Kanhar



Bhata



Plate 14: Effect of different soil type on disease development of dry root rot of chickpea

4.8.3 Effect of Soil texture on dry root rot of chickpea

An experiment was conducted to know soil texture favorable or unfavorable for dry root rot development caused by *R. bataticola* in chickpea as described in materials and methods. Result showed significant difference presented in Table 4.56, Fig. 4.16, Fig. 4.17 and Plate 15.

Significantly least mortality per cent (20.8%) was recorded in sandy loam texture followed by sandy texture with mortality per cent of 25.0% which were at par with each other. However, highest mortality was observed in clay texture soil 62.5%, followed by silt clay 37.5%. Extent of root damage varied greatly from 1-7 grades and lowest disease grade was recorded in sandy loam with minimum root damage and infection that is, 1.6 rating, followed by sandy and loam with disease rating 3.6 and 5 respectively. While, the highest disease rate was recorded in clay texture soil that is 7.0.

Table 4.56: Effect of different soil texture on disease development of dry root rot of chickpea

S. No.	Soil texture	Disease rating	Per cent mortality
1	Sandy	3.6	25.0 (26.38)
2	Sandy loam	1.6	20.8 (23.79)
3	Loam	5.0	33.3 (47.39)
4	Clay	7.0	62.5 (52.39)
5	Silt clay	5.6	37.5(44.98)
SEm±6.18			
CD (5%)		19.47	
Average of three replication			
Figures in parentheses are angular transformation			

Soil texture also significantly influenced the germination per cent, root length and shoot length in chickpea when inoculated with *R. bataticola* RB4, was studied and data are presented in Table 4.57 and Fig. 4.18.

It is clear from the data which showed highest germination in sandy textured soil, 100.0 per cent which was significantly superior with other treatments. Sandy loam, loam and silt clay supported chickpea seed to germinate by 90.0 per cent,

while lowest germination per cent was observed in clay soil (80.0 per cent). Maximum root length was observed in sandy soil (17.9 cm), silt clay (16.5 cm) and loam (16.1 cm) which were at par, but significantly superior than other treatments. Least increased in root length was recorded in clay (14.7 cm) and sandy loam (15.6 cm). All the treatments were found at par in terms of shoot length, though maximum was observed loam soil (14.8 cm) followed by sandy loam (14.6 cm) and silt clay (14.5 cm). Significant difference was observed among all the treatments viz., germination per cent and shoot length.

Table 4.57: Effect of soil texture on germination, shoot and root length of chickpea in inoculated soil with *R. bataticola*

S.No.	Soil texture	Germination (%) *	Root length (cm) *	Shoot length (cm) *
1	Sandy	100.00 (10.05)	17.9	14.3
2	Sandy loam	90.00 (9.53)	15.6	14.6
3	Loam	90.00 (9.53)	16.1	14.8
4	Clay	80.00 (8.99)	14.7	13.1
5	Silt clay	90.00 (9.53)	16.5	14.5
SEm±		0.05	0.63	0.81
CD (5%)		0.18	1.87	2.39
* Average of three replication				
Figures in parentheses are square root transformation				

Jimenez (2006) which found that the soil texture had a significantly high impact on the *M. phaseolina* soil population dynamics. There was variation in the populations across soil types but this variation was significantly different in the sandy soil texture in which populations were reduced. It was found an increasing number of colonies of *M. phaseolina* as the amounts of silt loam in each soil were increased in order to form the artificial soil texture. In the soil *M. phaseolina* populations may be indirectly influenced by clay content. There is some evidence to suggest that increasing levels of organic matter contained in the silt loam soil used to form artificial soil textures can support antagonistic microorganisms that compete with microsclerotia in the soil having a negative impact on soil

populations. Decreasing soil populations in sandy soils also suggest that propagules require a certain level of soil nutrients to remain viable and active in soil.

In agreement to the above finding Sundar *et al.*, 2009 tested the effect of different soil type on fungicidal control of *Rhizoctonia solani* causing seedling mortality in mungbean. Three soil of different textural group viz. sandy, loam and silt clay were used. The result revealed that disease incidence was more in silt clay soil and loam soil than sandy soil. Taya *et al.* (1990) reported 41.66 % and 50% disease incidence in soil having clay as a textural component whereas, 28.33% and 65% disease incidence was exhibited in sandy soil when fungicide treated chickpea seed were sown (carbendazim alone and carbendazim + thirum, respectively). A similar observation was made when fungicide were applied on pre-sowing drench. Clay soil exhibited 46% and sandy 31.66% of disease incidence in case of carbendazim.

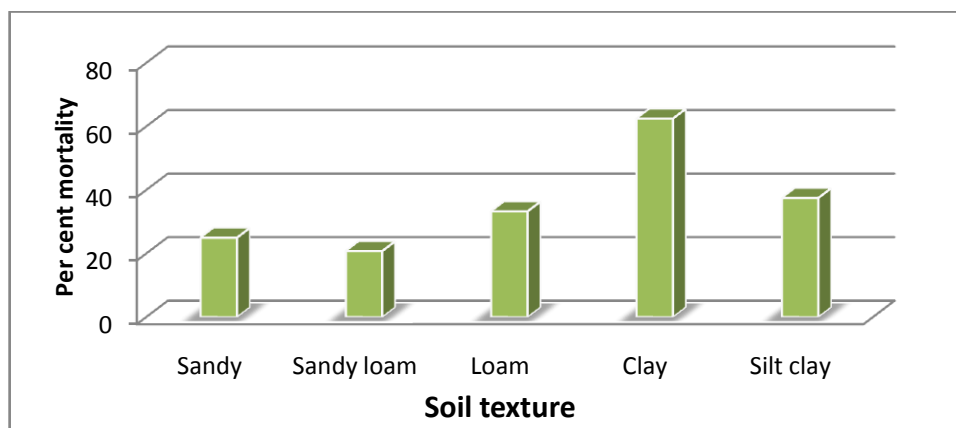


Fig 4.16: Per cent mortality of chickpea due to dry root rot under glass house condition at different soil texture

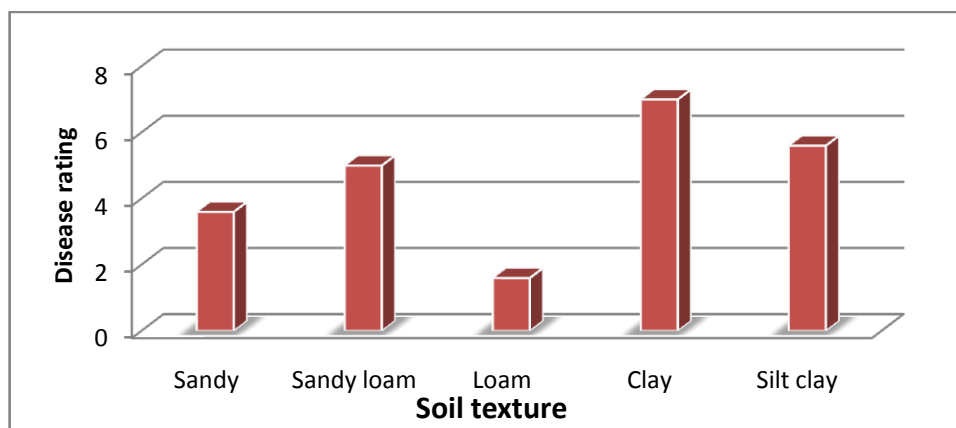


Fig 4.17: Disease rating of chickpea dry root rot in different soil texture

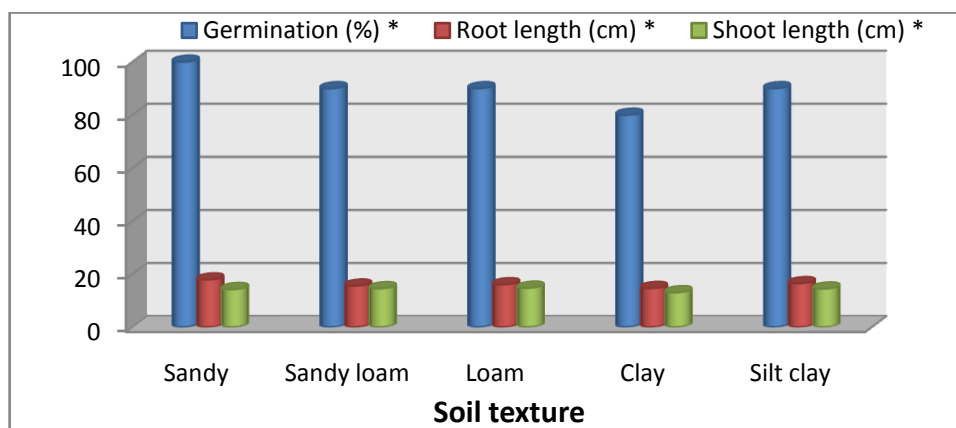


Fig 4.18: Effect of different soil texture on germination, root and shoot length of chickpea in inoculated soil with *R. bataticola*



Sandy



Sandy Loam



Loam



Clay



Silt Clay



Plate 15: Effect of different soil texture on disease development of dry root rot of chickpea

4.9 Potentiality of bio control agent *Trichoderma* spp. against *R. bataticola*

4.9.1 Isolation of *Trichoderma* isolates

*Trichoderma*spp. were isolated from the soil samples by serial dilution method using Potato Dextrose Agar (PDA) medium. The cultures were identified and confirmed as *Trichoderma*spp. through microscopic observation. In the present investigation, fifteen isolates of *Trichoderma* spp. were collected from different locations of Chhattisgarh given in Table 4.58. Out of fifteen *Trichoderma* isolates, seven isolates belongs to central region, three from northern and eastern region while, only one isolates each was isolated from western and southern region.

Table 5.58: Isolates of *Trichoderma* spp. collected from different locations of Chhattisgarh

S.No.	Area	District/Tehsil	Isolate
1.	Northern area	Ambikapur	T20
2.		Sitapur	T19
3.		Udaipur	T22
4.	Eastern area	Khadsiya	T14
5.		Lohara	T33
6.		Raigarh	T12
7.	Western area	Rajnangaon	T4
8.	Southern area	Kanker	T1
9.	Central area	Abhanpur	T7
10.		Bemetara	T40
11.		Dhamdha	T36
12.		Durg	T31a
13.		Raipur	T29
14.		Thankhamariya	T32
15.		Uattai	T31b

4.9.2 Cultural and Morphological characterization of *Trichoderma* isolates

The Petri dishes with *Trichoderma* culture were observed for colony colour, growth pattern (presence or absence of aerial mycelium or subdued growth), appearance, pustule, pigmentation and odour. Table 4.59 exhibit primary characterization on the basis of cultural and morphological characteristics (Plate 16).

4.9.2.1 Colony colour: Fifteen *Trichoderma* isolates can be categorized in 5 groups (Table 4.59). Yellow green colour of colony was observed in isolate T1 and T40, whitish green in isolates T4, T12, T20, T22 and T33, green in isolate T7, T14, T29 and T31a whereas, light green in isolates T19, T23, T31b and T36.

4.9.2.2 Growth pattern: *Trichoderma* isolates in terms of growth pattern are grouped in two, that is, effused and subdued. Isolates T4, T12, T22 and T23 exhibited effused growth pattern whereas, isolates T1, T7, T19, T20, T29, T31a, T31b, T33, T36 and T40 showed subdued pattern, as presented in Table 4.59.

4.9.2.3 Appearance: Colony appearance is grouped in 5 categories (Table 4.59). Uniform ringed velvety exhibited by isolate T1, velvety at periphery by isolate T4, T14 and T20, velvety ringed by several isolate namely, T7, T19, T29, T31a, T31b, T33 and T40 whereas, scattered appearance by isolates T12, T22, T23 and T36.

4.9.2.4 Pustules: One of the macroscopic features of *Trichoderma* is the ability to form pustules. Formation of pustules by *Trichoderma* is grouped into four categories, absence of pustules was found in isolate T1, T7, T14, T19, T20, T29, T31a, T31b, T33 and T40. Scattered pustules were observed in isolate T4, compact pattern pustules in T12 and T23 whereas, minute pustules were observed in T22 and T36 (Table 4.59).

4.9.2.5 Pigments: Pigmentation of *Trichoderma* isolates were studied on PDA plates. The production of diffusible yellowish orange pigments were found in T1, T7, T12, T19, T22, T29 and T40 isolates, whereas, in T4, T14, T20, T23, T31a, T31b, T33 and T36 isolates no pigmentation were found (Table 4.59).

4.9.2.6 Odour: No odour was found to produce from *Trichoderma* isolates (Table 4.59).

Table 4.59: Morphological characteristics of *Trichoderma* isolates based on key characters

Isolates	Colony colour	Growth pattern	Appearance	Pustule	Pigments	Odour
T1	Yellow green	Subdued	Uniform ringed velvety	Absent	Yellowish orange	Absent
T4	Whitish green	Effused	Velvety at periphery	Scattered	Absent	Absent
T7	Green	Subdued	Velvety ringed	Absent	Yellowish orange	Absent
T12	Whitish green	Effused	Scattered	Compact	Yellowish orange	Absent
T14	Green	Subdued	Velvety ringed at periphery	Absent	Absent	Absent
T19	Light green	Subdued	Velvety ringed	Absent	Yellowish orange	Absent
T20	Whitish green	Subdued	Velvety ringed at periphery	Absent	Absent	Absent
T22	Whitish green	Effused	Scattered	Minute	Yellowish orange	Absent
T23	Light green	Effused	Scattered	Compact	Absent	Absent
T29	Green	Subdued	Velvety ringed	Absent	Yellowish orange	Absent
T31a	Green	Subdued	Velvetyringed	Absent	Absent	Absent
T31b	Light green	Subdued	Velvety ringed	Absent	Absent	Absent
T33	Whitish green	Subdued	Velvety ringed	Absent	Absent	Absent
T36	Light green	Subdued	Scattered	Minute	Absent	Absent
T40	Yellow green	Subdued	Velvety ringed	Absent	Yellowish orange	Absent

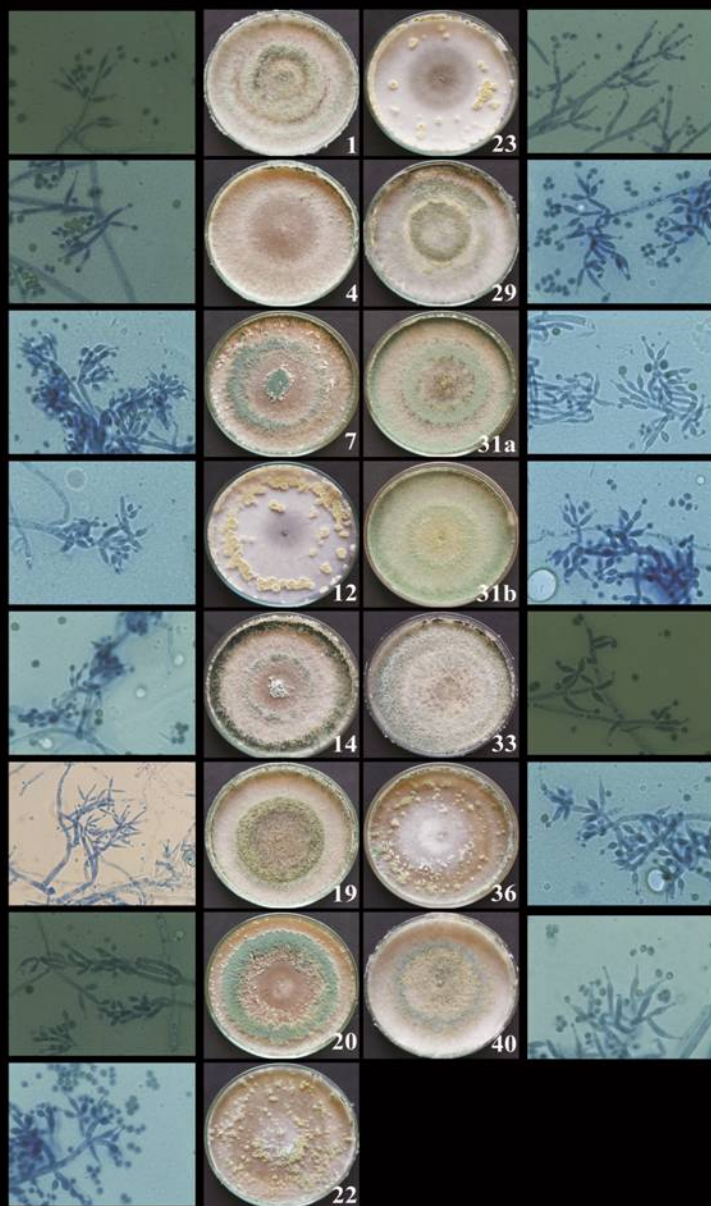


Plate 16: Cultural and phialide morphology of different *Trichoderma* isolates

4.9.3 Evaluation of antagonistic efficacy of *Trichoderma* spp. by dual culture method

The data are presented in Table 4.60 revealed that all the isolates of *Trichoderma* in dual culture significantly inhibited mycelial growth of *R. bataticola* and inhibition ranged from 35.00 to 90.00 per cent over control (Plate 17).

Fifteen isolates of *Trichoderma* were evaluated for their efficacy against nine isolates of *R. bataticola*. Isolate T40 was found significantly superior among all the isolates of *Trichoderma* in reducing the mycelial growth of *R. bataticola* by 67.65% over control, followed by T36 (64.81%), T14 (64.07%) and T29 (63.21%) which were at par with each other. Minimum inhibition was recorded in T31a (52.84%) followed by T20 (55.06%) which were statistically at par with each other.

In relation to another factor that is *R. bataticola* isolates significantly maximum mycelial inhibition was recorded in RB2 (63.61%) followed by RB1 (59.90%) and RB8 (56.81%) with significant difference. While, RB 8 (56.81%), RB6 (56.52%) and RB 9 (55.69%) were found statistically at par with each other. However, minimum inhibition was observed in RB7 (47.01%) followed by RB4 (48.33%).

In terms of antagonistic efficacy between *R. bataticola* and *Trichoderma* isolates. Isolate T40 proved significantly superior which inhibited the mycelial growth by 90.00% against RB2 followed by T40 (88.89%) against RB8. Next in order, isolate T40 exhibited to inhibit mycelial growth, 76.67% against RB1 followed by T33 against RB2 and T29 against RB4 76.67%. While, minimum mycelial inhibition was recorded in T33 (38.89%) against RB4 followed by T19 (42.22%) against RB4 and T1, T19 and T31a found to inhibit mycelial growth by 43.33% against RB4, RB7 and RB7, respectively (Plate 17).

In relation to present investigation, similar findings were reported by Kumar *et al.* (2013) they reported inhibition of the mycelium growth (per cent) of pathogen by *T. harzianum*, *T. viride*, *T. atroviride* varied from 61.1 to 70.1, 58.6 to 66.6, 52.0 to 63.1 respectively against *M. phaseolina* in moth bean. Similarly, Sharma *et al.*, (2012), Sharma *et al.* (2010) and Gurha (2007) reported antagonistic

effects of *T. harzianum* and *T. viride* against *M. phaseolina*. *T. viride* significantly inhibited the colony diameter as noted by Kolte and Raut (2007). *Trichoderma harzianum* was reported as effective bioagent, which inhibited mycelial growth by 71.85% of *M. phaseolina*. *T. harzianum* found most effective in inhibiting the mycelial growth of *R. bataticola* also by Haque *et al.*, (1990; Pant and Mukhopadhyay, (2001) and Ammajamma *et al.* (2009).

Singh *et al.* (2008) showed significant reduction in mycelial growth of *R. bataticola* by *Trichoderma* isolates when compared to control. Among 10 *Trichoderma* isolates, *Trichoderma* isolate-7 showed maximum inhibition of growth of *R. bataticola* (83.33%) followed by *Trichoderma* isolate-4 (81.11%) and *Trichoderma* isolate-10 (74.44%) which were at par with each other. Present results were in agreement with Bandyopadhyay *et al.* (2003) who reported that strain of *Trichoderma* inhibited the growth of *R. bataticola* by 51.1 per cent under *in vitro* conditions. Paul *et al.* (2008) reported that among 11 *Trichoderma* isolates, *Trichoderma harzianum* showed maximum inhibition of the growth of *R. solani* by 77 per cent under *in vitro* conditions. Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot.

T. viride showed significant antifungal activities by inhibiting the mycelial radial growth of *M. phaseolina* by 71.42% (Doley and Jite, 2012). Gejera (2012) also reported potential of *Trichoderma* spp. *T. koningi* (T4) inhibited mycelial growth (74.3%) of *M. phaseolina* followed by *T. harzianum* (T1) (61.4%) Similarly, Manjunatha *et al.* (2013) isolated six *T. viride* isolates and were evaluated their antagonistic efficacy against *M. phaseolina*, the cause of root rot of chickpea using the dual culture technique. All isolates were found to cause significant reduction in fungal growth when compared to the control. Such variation among isolates in their ability to control the growth of pathogens has been reported by several researchers (Laha *et al.*, 1992; Naik *et al.*, 2000; Upmanyu *et al.*, 2002; Singh *et al.*, 2008). *T. viride* exhibited reduction in mycelial growth of *M. phaseolina* by 44.94% (Ashwini *et al.*, 2014).

Table 4.60 Antagonistic efficacy of *Trichoderma* spp. against isolates of *R. bataticola* in dual culture technique

		<i>R. bataticola</i> isolates									
<i>Trichoderma</i> spp.		Per cent inhibition over control									
S.No.	Isolate	RB1	RB2	RB3	RB4	RB5	RB6	RB7	RB8	RB9	Average
1.	T1	56.67 (48.81)	71.11 (58.24)	55.56 (48.17)	43.33 (41.13)	62.22 (52.06)	60.00 (50.77)	58.89 (50.11)	61.11 (51.41)	57.78 (49.49)	58.64 (50.17)
2.	T4	64.44 (53.38)	65.56 (54.71)	55.56 (48.17)	52.22 (46.25)	64.44 (53.38)	54.44 (47.53)	46.67 (43.06)	54.44 (47.53)	54.44 (47.53)	56.91 (49.21)
3.	T7	72.22 (58.10)	63.33 (52.73)	50.00 (44.98)	50.00 (44.98)	56.67 (48.81)	65.56 (54.05)	55.56 (48.17)	54.44 (47.53)	60.00 (50.77)	58.64 (50.17)
4.	T12	48.89 (44.34)	75.56 (56.73)	54.44 (47.53)	52.22 (46.25)	68.89 (56.12)	62.22 (52.06)	50.00 (44.98)	56.67 (48.81)	63.33 (52.73)	59.14 (50.25)
5.	T14	71.11 (57.50)	74.44 (58.89)	71.11 (57.50)	53.33 (46.89)	57.78 (49.47)	63.33 (52.73)	64.44 (53.38)	55.56 (48.17)	65.56 (54.05)	64.07 (53.01)
6.	T19	68.89 (58.21)	67.78 (57.50)	66.67 (54.72)	42.22 (40.50)	35.56 (36.56)	55.56 (48.17)	43.33 (41.12)	57.78 (49.47)	58.89 (50.11)	55.19 (48.57)
7.	T20	62.22 (52.06)	47.78 (43.70)	53.33 (46.89)	53.33 (46.89)	47.78 (43.70)	60.00 (50.77)	46.67 (43.05)	58.89 (50.10)	65.56 (54.04)	55.06 (48.05)
8.	T22	65.56 (53.37)	57.78 (48.90)	57.78 (49.45)	50.00 (44.98)	62.22 (52.06)	61.11 (51.41)	46.67 (43.06)	56.67 (48.81)	57.78 (49.46)	57.29 (49.72)
9.	T23	66.67	75.56	62.22	44.44	65.56	62.22	48.89	57.78	52.22	59.91

		(56.79)	(56.86)	(52.06)	(41.79)	(54.04)	(52.06)	(44.34)	(49.46)	(46.25)	(50.40)
10.	T29	74.44	64.44	60.00	76.67	60.00	62.22	54.44	55.56	61.11	63.21
		(58.89)	(53.38)	(50.77)	(61.09)	(50.77)	(52.06)	(47.53)	(48.17)	(51.41)	(53.16)
11.	T31a	56.6	57.78	60.00	61.11	48.89	48.89	43.33	45.56	53.33	52.84
		(47.53)	(48.90)	(50.77)	(51.41)	(44.34)	(44.34)	(41.14)	(42.42)	(46.89)	(46.33)
12.	T31b	42.22	56.67	61.11	46.67	66.67	60.00	44.44	58.89	60.00	55.19
		(41.79)	(48.81)	(51.41)	(43.06)	(54.72)	(50.77)	(41.79)	(50.10)	(50.70)	(48.28)
13.	T33	56.67	76.67	66.67	38.89	54.44	61.11	42.22	72.22	56.67	58.40
		(48.81)	(61.25)	(54.73)	(38.53)	(47.53)	(51.42)	(40.50)	(58.24)	(48.81)	(49.36)
14.	T36	74.44	73.33	61.11	58.89	64.44	61.11	54.44	74.44	61.11	64.81
		(58.89)	(59.61)	(51.11)	(50.10)	(53.38)	(51.40)	(47.53)	(58.89)	(51.41)	(53.63)
15.	T40	76.67	90.00	62.22	50.00	58.89	66.67	52.22	88.89	63.33	67.65
		(61.85)	(71.52)	(52.06)	(44.98)	(50.11)	(54.72)	(46.25)	(70.52)	(52.71)	(55.98)
	Average	59.90	63.61	56.11	48.33	54.65	56.52	47.01	56.81	55.69	
		(50.08)	(52.47)	(47.54)	(43.05)	(46.69)	(47.76)	(42.25)	(48.15)	(47.28)	
Factors						SEm$\pm$		C.D. (5%)			
Factor (T)						0.67		1.89			
Factor (RB)						0.50		1.42			
Factor(T X RB)						2.03		5.69			

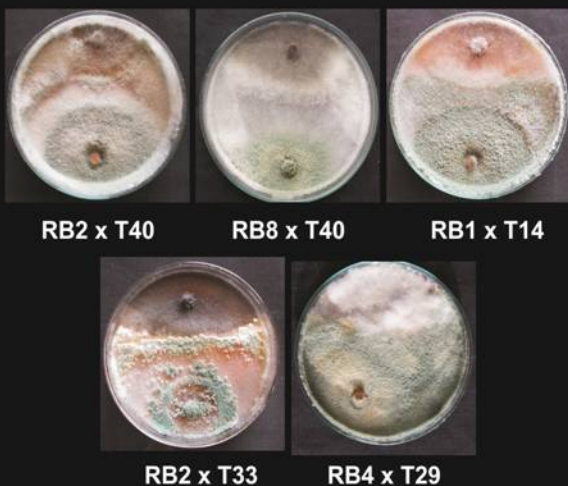


Plate 17: Best antagonistic efficacy of *Trichoderma* spp. against isolates of *R. bataticola* by dual culture technique

4.9.4 Screening for mycoparasitic activity *Trichoderma* spp. isolates against *R. bataticola*

Significant varying degree of antagonistic ability of *Trichoderma* isolates was found against different *R. bataticola* which were further grouped on 1-5 scale *via.*, dual culture interaction study is presented in Table 4.61.

The degree of antagonism (mycoparasitic activity) on a scale of class 1-5 by respective isolates of *Trichoderma* spp. was assessed as follows

- 1= *Trichoderma* completely overgrew the pathogen and covered the entire medium surface.
- 2= *Trichoderma* overgrew at least $2/3^{\text{rd}}$ of the medium surface.
- 3= *Trichoderma* and pathogen each colonized approximately $1/2$ of the medium surface (more than $1/3^{\text{rd}}$ and less than $2/3^{\text{rd}}$), and neither organism appeared to dominate the other.
- 4= The pathogen colonized at least $2/3^{\text{rd}}$ of the medium surface and appeared to withstand encroachment by *Trichoderma*.
- 5= The pathogen completely overgrew the *Trichoderma* and covered the entire medium surface.

The *Trichoderma* isolates, as a strong antagonistic potential were further grouped according to their mycoparasitic activity. Taking into account, 67 per cent inhibition scored as reaction type 2 and 50 per cent inhibition scored as reaction type 3 and 33 per cent inhibition scored as reaction type 4, which exhibit the extent of *in vitro* mycoparasitic activity assessed on 1-5 scale.

T14 and T40 found to inhibit 50 per cent of mycelial growth in five isolates of *R. bataticola* and 67 per cent in four isolates of *R. bataticola*. Next subsequent mycelial inhibition was observed in T36 isolate with 50 per cent mycelial inhibition in six isolates of *R. bataticola* and 67 per cent in three isolates of *R. bataticola*. Followed by T29 efficient in mycelial inhibition of 50 per cent in seven isolates of *R. bataticola* and 67 per cent in two isolates of *R. bataticola*. Mycelial inhibition in T4 and T12 was observed, 50 per cent in two isolates, 67 per cent in six isolates and 33 per cent in one isolate. Whereas, 50 per cent of mycelial inhibition in three isolates, 67 per cent also in three isolates and 33 per cent in two isolates was recorded in T23 and T33. Mycelial inhibition in T6 was observed 50 per cent in three isolates, 67 per cent in three isolates and 33 per cent in three isolates. Next *Trichoderma* isolate T31b was found to inhibit mycelium with 50

per cent in two isolates, 67 per cent in four isolates and 33 per cent in three isolates. Subsequent isolates *Trichoderma* T22 and T1 found to inhibit mycelium, by 50 per cent in one isolate, 67 per cent in seven isolates and 33 per cent in one isolate. However, comparatively low mycelial inhibition was recorded in T20 with 67 per cent in six isolates and 33 per cent three isolates and in T31a with mycelial inhibition of 67 per cent in five isolates and 33 per cent four isolates.

Among all the *Trichoderma* isolate screened T14 and T40 were found highly superior with maximum mycelial inhibition of *R. bataticola*. Based on percentage of inhibition radial growth values, 5 best antagonistic isolates in decreasing order were T14 and T40>T36>T29 and T7> T4 and T12 > T9 and T13.

Table 4.61: Screening for mycoparasitic activity of *Trichoderma* spp. isolates against *R. bataticola*

S.No.	Isolate	RB1	RB2	RB3	RB4	RB5	RB6	RB7	RB8	RB9
1	T1	3	2	3	4	3	3	3	3	3
2	T4	2	2	3	3	3	3	4	3	3
3	T7	2	3	3	3	3	2	3	3	3
4	T12	4	2	3	3	2	3	3	3	3
5	T14	2	2	2	3	3	3	3	3	2
6	T19	2	2	2	4	4	3	4	3	3
7	T20	3	4	3	3	4	3	4	3	3
8	T22	2	3	3	3	3	3	4	3	3
9	T23	2	2	3	4	2	3	4	3	3
10	T29	2	3	3	2	3	3	3	3	3
11	T31a	3	3	3	3	4	4	4	4	3
12	T31b	4	3	3	4	2	3	4	2	3
13	T33	3	2	2	4	3	3	4	2	3
14	T36	2	2	3	3	3	3	3	2	3
15	T40	2	2	3	3	3	2	3	2	3

4.9.5 Mechanism of antagonism of *Trichoderma* spp. on *R. bataticola* causing dry root rot of chickpea.

Interaction between the two fungi when documented through light microscopy indicated that the mode of action adopted to parasitize *R. bataticola* was through coiling (Plate 18). The coiling of the vegetative mycelium of *R. bataticola* by *Trichoderma* initiates and a sort of searching and recognition

mechanism between the two is speculated. The mycelial tip of *Trichoderma* runs parallel with that on the mycelium of *R. bataticola*, and with its mycelial tips it sticks on to the large hyphae of *R. bataticola*. This process is further followed by a very rapid and excessive coiling on the target fungus. The excessive coiling may also result in the production of several haustoria by the biocontrol agent at its surface which happen to interact with the large hyphae of *R. bataticola* followed by the production of enzymes, which functions by breaking down the complex polysaccharides, chitin, and beta glucans that are responsible for the rigidity of the fungal cell wall, thereby destroying the cell wall integrity. Minute bodies the sclerotia produced by *R. bataticola*, serves as its primary source of inoculum. *Trichoderma* isolates were observed to be capable of parasitizing these masses. Lysis of the pathogen too was observed.

The potential of *Trichoderma* species as bio-control agents of plant diseases was first recognized in the early 1930s, and in subsequent years, control of many diseases has been added to the list (Elad and Kapat, 1999; Harman, 2000; Lifshitz, 1986; Sharon, 2001 and Weindling, 1941). In the present investigation mechanism employed by *Trichoderma* for mycoparasitic activity was studied by pairing isolates of *Trichoderma* with that of the target organism *R. bataticola*. One of the mechanism which was observed to be adopted by *Trichoderma* to parasitize *R. bataticola* is by competition. The common mode of action adopted by the *Trichoderma* was through coiling around pathogen hyphae, penetration, and subsequent dissolution of the host cytoplasm.

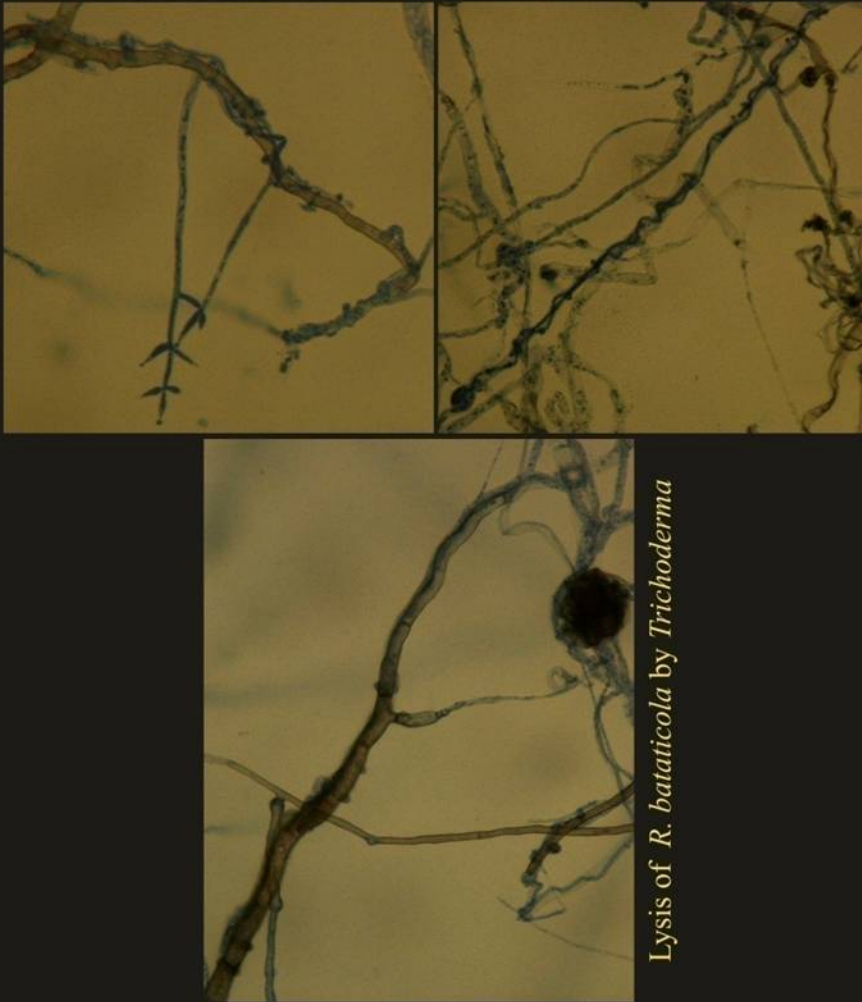
Weindling (1932) recognized *T. lignorum* as bio-control agent against citrus seedling disease, incited by *R. solani*. Also, described in detail the mycoparasitism of *R. solani*, hyphae by the hyphae of the bio-control agent, including coiling around pathogen hyphae, penetration and subsequent dissolution of the host cytoplasm a phenomenon which occurred regardless of the supply of external nutrients to the host or mycoparasite. Mycoparasitic behaviour as reported by Vipul *et al.* (2006) is of the similar kind and is in support to the present observations which were made and documented in the present investigation. Weindling (1932) reported that a strain of *T. lignorum* produced a “lethal principle” that was excreted into the surrounding medium, allowing parasitic activity by the bio-control agent. In 1941, he characterized the “lethal principle”, demonstrated that it was toxic to both *R. solani* and *Sclerotinia americana*, and

named it gliotoxin. *Trichoderma* species are often able to suppress the growth of endogenous fungi on an agar medium and therefore mask their presence (Weindling, 1934).

A case in point is that of *T. virens*-treated root segments taken from soil heavily infested with propagules of *Macrophomina phaseolina*, the pathogen that causes charcoal rot in a wide range of crops. One idea that has been advanced is that enzymes such as chitinases and/or glucanases produced by the bio-control agent are responsible for suppression of the plant pathogen. These enzymes function by breaking down the polysaccharides, chitin, and glucans that are responsible for the rigidity of fungal cell walls, thereby destroying cell wall integrity. A similar kind of observation was made in the present investigation that after coiling around pathogen hyphae, penetration, and subsequent dissolution of the host cytoplasm was observed. The process of dissolution involves the breaking down the polysaccharides, chitin, and glucans that are responsible for the rigidity of fungal cell and thus it was speculated that the *Trichoderma* involves the use of enzymes in the process of hyphal dissolution. The possible role of chitinolytic enzymes in bio-control is further supported by the work of Lorito *et al.* (1998), and several others (Poromarto, 1998; Elad and Kapat, 1999; Metcalf, 2001 and Sharon, 2001). It is speculated that a mixture of several enzymes might be necessary for efficient cell wall lysis.

The hyphae of *Trichoderma* penetrated the mycelium of the pathogen and grew luxuriantly within it and pathogenic hyphae were severely disrupted at the points of contact with hyphae of *Trichoderma* ultimately leading to hyphal lysis. These findings are in well agreement with that of findings made by Sen (2000) and Naik *et al.* (2000).

Mycoparasitism involves morphological changes, such as coiling and formation of appressorium-like structures, which serve to penetrate the host (McIntyre *et al.*, 2004). The first contact between *Trichoderma* spp. and pathogen *M. phaseolina* occurred after 2 to 3 days of inoculations, followed by growth inhibition. Differential antagonistic activity has even been observed for various *Trichoderma* spp. which demonstrates semi-specificity in the interaction of *Trichoderma* with its host (Schirmbock *et al.*, 1994). Microscopic study revealed that *T. koningi* MTCC 796 and *T. harzianum* NABII Th 1 showed effectively coiling on pathogen *M. phaseolina* at 14DAI, and may start the parasitism earlier



Coiling of the mycelium of
R. bataticola by *Trichoderma*

Lysis of *R. bataticola* by *Trichoderma*

**Plate 18: Intercoiling and lysis of mycelium of
R. bataticola with *Trichoderma***

during antagonism compared to other *Trichoderma* spp. Similar results, have been reported for *T. harzianum* against *Crinipellis pernicioso* (Marco *et al.*, 2000), *Sclerotium rolfsii* (El-Katany *et al.*, 2001) and *R.cerealis* (Innocenti *et al.*, 2003).

In agreement to present finding, Gejera *et al.* (2012) and Monteiro *et al.* (2010) found *Trichoderma* capable of overgrowing and degrading *M. phaseolina* mycelia, coiling around the hyphae. Manjunatha and Naik (2010) studied the hyphal interface between the antagonism (*Trichoderma* spp.) and the pathogen (*R. bataticola*), under microscopic and revealed that the interaction of the organism led to mycoparasitism in the form of coiling, penetration, parasitization of sclerotia by conidia of bioagent propagules on sclerotia and disruption of the sclerotia of *R. bataticola*. Similar observations were reported by Harman (2000) by *Trichoderma* spp. on various plant pathogens. The interactions lead to the possibility of mutual intermingling, inhibition, competition and antibiosis. There could be over growth of the bioagents leading to more competition for nutrients there by bioagent population i.e. hyphae and conidia may overlap the pathogen population. Naik *et al.* (2000) reported the same results on chilli Fusarium wilt pathogen by *Trichoderma* spp. When the bioagent reaches the host (pathogen), its hypha coils or attaches to it by forming a hook like structure and hyphae of the bioagent penetrates the host mycelium, apparently, by partially degrading its cell wall (Elad *etal.*, 1983).

4.9.6 Efficacy of *Trichoderma* spp. against *R. bataticola* under *in vivo* condition

The *Trichoderma* isolates under this study also releaved varying degree of difference in mortality per cent, germination per cent, root length and shoot length against RB 4 (Kawardha) isolate under *in vivo* conditions as presented in Table 4.62 Fig. 4.19, Fig. 4.20 and Plate 19.

All the treatments were significantly superior, in decreasing the incidence of dry root rot disease in chickpea, over control. Isolate T23 and T40 were found to produce significantly least mortality 20.83 per cent among all, which was 75 per cent reduced over control, subsequently followed by T4 and T36 with mortality 25.00 per cent, 69.99 per cent reduced over control, and T22 exhibited 29.16 per cent mortality, 65.00 per cent reduced over control, these isolates were statistically at par with each other. These isolates showed significantly

results, over other isolates and over control. Maximum mortality was observed in T7 that is 50.00 per cent whereas in control 83.33 per cent.

The root damage was access on 1-9 scale. The extent of root damage across all the *Trichoderma* isolates on sick pots when challenged with RB4 (Kawardha) *R. bataticola* isolate, it was found to vary from 3 -9 on 1-9 of disease rating scale. Lowest rating was scored by T4 and T31b as 3.0 followed by T1, T33 and T36 as 3.6 and T12, T14, T20, T29 and T31a scored as 4.3. Maximum damage was recorded in control (seeds treated with sterile water) as 9.0.

Table 4.62: Efficacy of *Trichoderma* spp. against dry root rot of chickpea

S.No.	Isolate	Disease rating	Per cent mortality	Per cent decrease over control
1	T1	3.6	37.50 (37.57)	54.99
2	T4	3.0	25.00 (24.19)	69.99
3	T7	7.8	50.00 (44.98)	39.99
4	T12	4.3	41.66 (40.15)	50.00
5	T14	4.3	45.83 (42.57)	45.00
6	T19	5.6	33.33 (34.98)	60.00
7	T20	4.3	37.50 (37.57)	54.99
8	T22	5.6	29.16 (26.96)	65.00
9	T23	8.3	20.83 (22.19)	75.00
10	T29	4.3	37.50 (37.57)	54.99
11	T31a	4.3	54.16 (44.98)	35.00
12	T31b	3.0	41.66 (39.98)	50.00
13	T33	3.6	33.33 (34.98)	60.00
14	T36	3.6	25.00 (24.19)	69.99
15	T40	4.3	20.83 (22.19)	75.00
16	Control	9.0	83.33 (66.17)	00.00
(untreated)				
SEm±	4.45			
CD (5%)	12.89			
* Average of three replication				
Figures in parentheses are angular transformation				

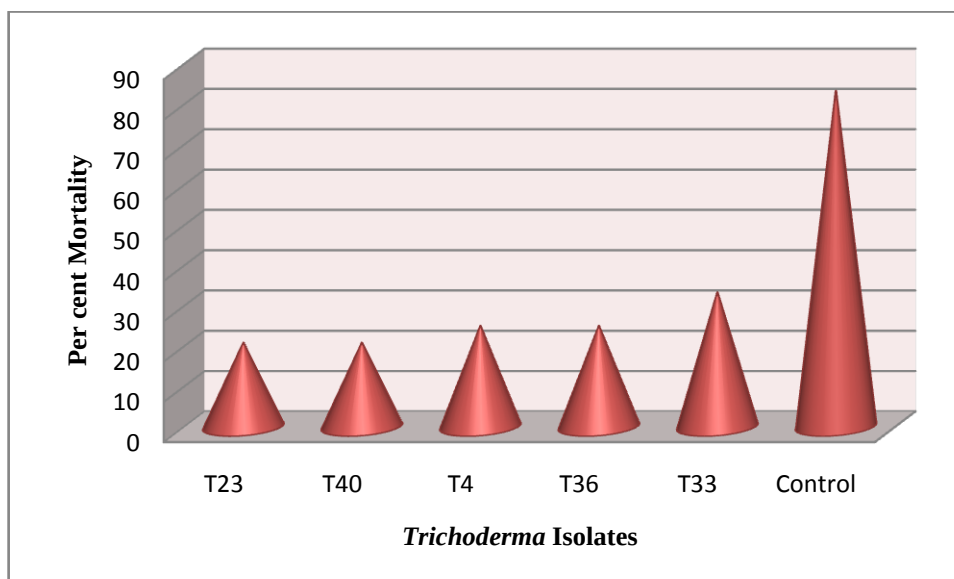


Fig 4.19: Efficacy of *Trichoderma* spp. against dry root rot of chickpea

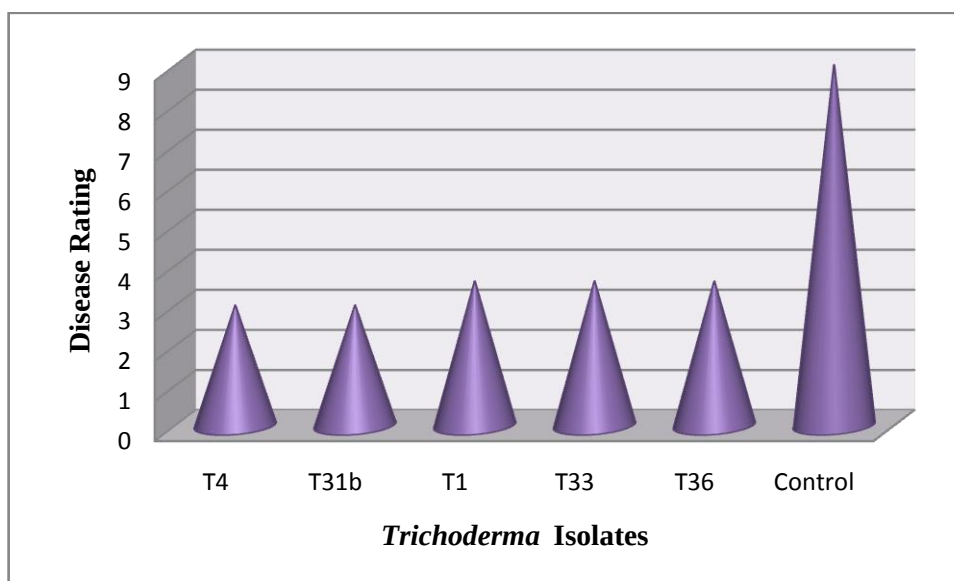


Fig 4.20: Disease rating scored from different *Trichoderma* isolates against *R. bataticola* (RB4) causing dry root rot of chickpea



T23



T40



T4



T36



Plate 19: Best results of chickpea seeds bioprimered with *Trichoderma* in sick pots

However, all the *Trichoderma* isolates in this study releaved variation in germination per cent, root length and shoot length and also among themselves over control, when challenge with RB4 (Kawardha).

The influence of different *Trichoderma* spp. on germination per cent, root length and shoot length in chickpea when challenge inoculated with *R. bataticola* RB4, was studied and data are presented in Table 4.63 and Fig. 4.21.

Highest germination per cent was observed in eight isolates like T12, T14, T19, T22, T29, T31a, T31b and T40 that is 90 per cent which was found significantly superior with other isolates and over control (86.66 per cent). Followed by T1, T20, T33 and T36 with 86.66 germination per cent which was at par with control.

All the *Trichoderma* isolates were significantly superior in increasing the root length over the control (9.33 cm), when challenged inoculated with *R. bataticola* RB4 (Kawardha) isolate. Maximum root length was recorded in T29 isolate (32.6 cm) (250.53 per cent increase over control) and T33 isolate 31.8 cm (241.93 per cent increase over control), which was at par and significantly superior over all the other treatments and control (9.33 cm). Followed by T40, 28.5 cm (206.45 per cent increase over control) and T23, 26.2 cm (181.72 per cent increase over control). Least increased in root length was recorded T4, 11.4 cm (22.58 per cent increase over control) and T1, 15.0 cm (61.29 per cent increase over control).

All the treatments in terms of shoot length, observed were at par with each other and with control. However, highest shoot length was exhibited by T7, T20 and T40 that is 17.2 cm, 15.3 cm and 15.0 cm (12.41, 1.29 and 1.96 per cent increase over control), respectively.

Table 4.63: Effect of *Trichoderma* spp. on germination, shoot and root length of chickpea

S.No.	Isolate	Germination (%) *	Root length (cm) *	Per cent increase over control	Shoot length (cm) *	Per cent increase over control
1	T1	86.66 (9.36)	15.0	61.29	13.6	11.11
2	T4	86.66 (9.36)	11.4	22.58	13.0	15.03
3	T7	83.33 (9.18)	15.9	70.96	17.2	12.41
4	T12	90.00 (9.53)	17.4	86.09	13.7	10.45
5	T14	90.00 (9.53)	20.2	117.20	14.5	5.22
6	T19	90.00 (9.53)	23.6	153.76	13.8	9.80
7	T20	86.66 (9.36)	24.0	158.06	15.3	1.29
8	T22	90.00 (9.53)	25.6	175.26	11.6	24.18
9	T23	86.66 (9.36)	26.2	181.72	14.2	7.18
10	T29	90.00 (9.53)	32.6	250.53	13.6	11.11
11	T31a	90.00 (9.53)	22.4	140.86	13.2	13.72
12	T31b	90.00 (9.53)	19.2	106.45	14.2	7.18
13	T33	86.66 (9.36)	31.8	241.93	11.8	22.87
14	T36	86.66 (9.36)	21.3	129.03	13.4	12.41
15	T40	90.00 (9.53)	28.5	206.45	15.0	1.96
16	Control (untreated)	83.33 (9.18)	9.33	00.00	15.5	00.00
	SEm±	0.05	0.67		0.47	
	CD(5%)	0.15	1.69		2.13	

* Average of three replication

Figures in parentheses are square root transformation

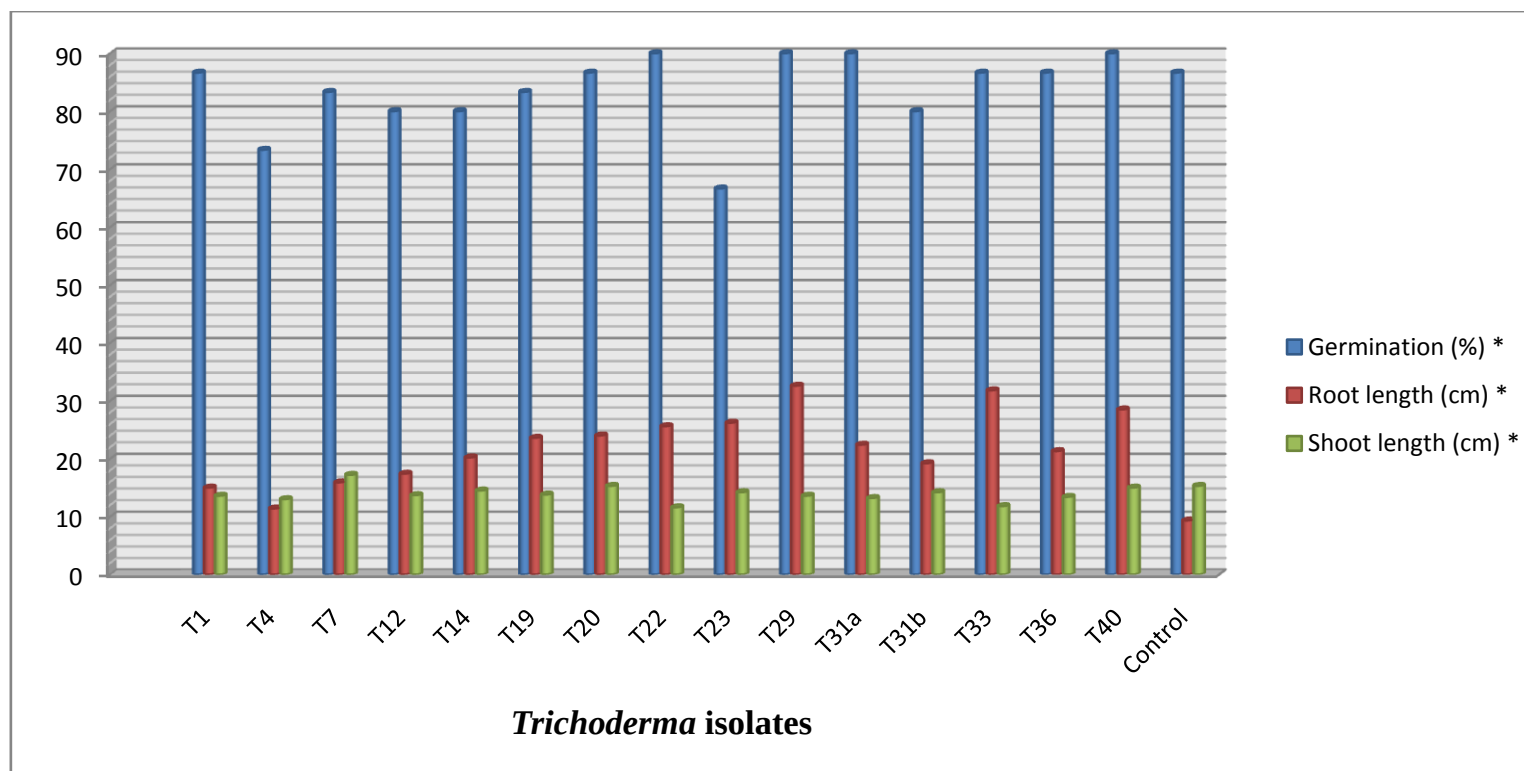


Fig 4.21: Effect of *Trichoderma* spp. on germination, root and shoot length of chickpea

Lokesh *et al.* (2004) found 78.22% mycelial inhibition of *M. phaseolina* 73.88 per cent through *T. harzianum*. Deshmukh and Raut (1992) showed *T. harzianum* and *T. viridae*, efficiently inhibiting the mycelial growth of *M. phaseolina* and reducing the disease incidence in pot experiment. Hussain *et al.* (1990) observed that *T. harzianum* was effective in controlling the infection of *M. phaseolina* in mungbean. Manczinger *et al.* (2002) reported that *T. harzianum*, *T. viride* and *T. polysporum* have a strong antagonistic against soil borne pathogens. *T. viride* was most effective among the antagonists in minimizing the disease, followed by *T. harzianum*, *T. atroviride* and *B. subtilis*. Gupta *et al.* (2006) also reported similar results on chickpea. A direct positive correlation was found between the population of pathogen and seedling mortality. Kumaret *al.* (2013) revealed that the *T. harzianum*, *T. viride* and *T. atroviride* antagonists reduced root rot incidence in a susceptible moth bean cv. RMO- 225 up to 69.0, 65.0 and 58.3%, respectively, as compared to control where 100 % mortality was observed under green house condition.

4.10 Potentiality of bio control agent fluorescent *Pseudomonas* against *R. bataticola*

4.10.1 Isolation of fluorescent *Pseudomonas* isolates

Fluorescent *Pseudomonas* were isolated from the soil samples by serial dilution method using Kings medium 'B', thereafter, examined under UV light. Colonies with yellow - green and blue-white color pigmentation were marked and recorded. Twenty two fluorescent *Pseudomonas* isolates were isolated from collected soil sample from different location of (Table 4.64). Out of twenty two isolates, fluorescent *Pseudomonas* twelve were isolated from central region, four, three, two and one isolates were isolated from northern, western, eastern and southern region of Chhattisgarh respectively.

4.10.2 Screening of fluorescent *Pseudomonas*

Twenty two isolates which showed fluorescent against ultra violet light were identified as the putative fluorescent *Pseudomonas*. Also, Gram staining for bacterial cell (fluorescent *Pseudomonas*) found to retain the pinkish red colour of counter stain called safranine, resulted the conformation of fluorescent *Pseudomonas*.

Table 4.64: Isolates of fluorescent *Pseudomonas* collected from different locations of Chhattisgarh

S.No.	Area	District/Tehsil	Isolate
1	Northern area	Ambikapur	P20
2		Bilaspur	P23
3		Sitapur	P18
4		Udaipur	P22
5	Eastern area	Raigarh	P8
6		Khadsiya	P14
7	Western area	Karawdha	P25
8		Lohara	P33
9		Rajnandgaon	P3
10	Southern area	Kanker	P2
11	Central area	Abhanpur	P4
12		Abhanpur	P7
13		Balod	P38
14		Bemetara	P40
15		Bhatapara	P19
16		Dhamdha	P36
17		Dhamtari	P27
18		Patan	P39
19		Raipur	P29
20		Saragaon	P34
21		Thankhamariya	P35
22		Uattai	P41

They were further, screened for following biochemical test gelatin hydrolysis test, nitrate test, starch hydrolysis test, casein hydrolysis test, lypolytic test, growth test at 4°C and 42° C and antibiotic sensitivity test, triple sugar iron agar test and hydrogen sulphide test.

4.10.3 Biochemical tests for characterization of fluorescent *Pseudomonas*

4.10.3.1 Gelatin hydrolysis test

In the present investigation twenty two isolates were categorized in two group based on liquification/hydrolysis of gelatine media nineteen isolates releaved positive whereas, three isolates were negative to gelatin hydrolysis test (Table 4.65). Positive result means liquification of media, P2, P3, P8, P14, P19, P20, P22, P23, P25, P27, P29, P33, P34, P35, P36, P38, P39 and P40 isolates may be considered as *Pseudomonas fluorescens* or *Pseudomonas aeruginosa*. Negative result means solidification of media and may considered as *Pseudomonas putida* (P4, P7 and P41). Gelatin is a protein resulted from

collagen hydrolysis. Gelatinase breakdown protein to polypeptide, peptide, and fatty acid so that they could be absorbed by organism cell membrane. This character to hydrolyze gelatin was also indicated by several *P. fluorescens* showing positive test to hydrolyze gelatin (Arwiyanto *et al.*, 2007).

4.10.3.2 Nitrate test

On the basis of nitrate utilization nineteen isolates were assign positive (+) (P2, P3, P4, P8, P14, P18, P19, P20, P22, P23, P25, P27, P29, P33, P35, P36, P38, P39 and P40) and may considered as *P. fluorescens* whereas three showed negative (-) response (P7, P34 and P41) unable to utilized nitrate and regarded as *P. putida* (Table 4.65).

4.10.3.3 Starch Hydrolysis Test

In starch hydrolysis test isolates were grouped in two categories, negative (-) giving blue colour and positive (+) with no colour change. This was grouped on the basis of blue colour formation when iodine indicator was added to the Petri plates. Nineteen isolates (P2, P4, P7, P8, P14, P18, P19, P20, P22, P23, P29, P33, P34, P35, P36, P38, P39, P40 and P41) exhibited negative (-) response which confirms the presence of *P. fluorescens* whereas three isolates (P3, P25 and P27) showed positive result (Table 4.65).

The positive result of starch hydrolysis test was mentioned by clear zone surrounding the bacterial colony. The zone showed that starch in the media could be hydrolyzed because of an enzymatic reaction, *i.e.*, amylase, secreted by the antagonist (Karkalas, 1985). The hydrolyzed starch could absorb purple colour from Iodine because of an interaction between polysaccharide and Iodine. Karkalas (1985) also mentioned that amylase is an enzyme that could disperse glycoside in starch and change it. The ability of fluorescent *Pseudomonas* in hydrolyzing starch was a specific character of most gram negative bacteria, supported by Arwiyanto *et al.* (2007) that fluorescent *Pseudomonas* is a gram negative bacteria and can hydrolyze starch.

4.10.3.4 Casein hydrolysis

In casein hydrolysis test seventeen isolates (P2, P3, P8, P14, P18, P19, P20, P22, P25, P29, P33, P34, P35, P36, P38, P39 and P40) exhibited positive (+) response and they may be considered as *P. fluorescens* or *P. aeruginosa* whereas, five isolates (P4, P7, P23, P27 and P41) showed negative result may assign as *P. putida* (Table 4.65). Some microorganisms have the ability to degrade (isolate shown positive results) the protein casein by producing proteolytic exoenzyme called caseinase which break the peptide bond CO-NH by introducing water into the molecule, liberating smaller chains of amino acid called peptides which are later broken down into free amino acid by extracellular or intracellular peptidases.

4.10.3.5 Lypolytic activity

In lypolytic test four isolates produce positive (+) to test (P2, P3, P7 and P40). These isolates may be considered as *P. fluorescens* or *P. aeruginosa* whereas, rest of the eighteen isolates (P4, P8, P14, P18, P19, P20, P22, P23, P25, P27, P29, P33, P34, P35, P36, P38, P39 and P41) exhibited negative (-) response (Table 4.65). In negative result there was no clear zone around the bacterial colony growth. The activity of lypolysis is related to enzyme production that could disintegrate fat. In this test, the media was supplemented with Tween 80 (polyhydroxybutirate) as indicator of lipase enzyme formation. The enzyme formed by lypolysis microorganisms could change fat to fatty acid and glycerol (Abouseoud *et al.*, 2007).

4.10.3.6 Growth at 4°C and 42°C

Isolates when inoculated in King's B medium and incubated at 4°C for 48 hours. Nine isolates (P2, P3, P4, P7, P8, P14, P29, P34 and P35) showed growth therefore they may be considered as *P. fluorescens* whereas, rest of the isolates (P18, P27, P33, P36, P38, P39, P40 and P41) may be *P. aeruginosa* (Table 4.65).

Also when these same set of the isolated were subject to 42°C, no growth was observed at in five isolates (P8, P19, P22, P27 and P41) they may considered as *P. fluorescens* or *P. putida* whereas, isolates resulted in growth may be assign as *P. aeruginosa* (Table 4.64).

4.10.3.7 Antibiotic sensitivity test

Antibiotic sensitivity test was used to distinguish different species of fluorescent *Pseudomonas*. Isolates (P2, P4, P3, P7, P8, P14, P18, P19, P20, P22, P23, P25, P27, P29, P33, P34, P35, P36, P38, P39 and P40) resistance to kanamycin may be considered as *P. aeruginosa*, except P41 which exhibited sensitivity towards kanamycin can assign as fluorescent *Pseudomonas* (Plate 29). On other hand in carbenicillin test isolates showed resistance response may be considered as fluorescent *Pseudomonas* or, *P. putida* (P2, P3, P4, P18, P19, P20, P22, P25, P29, P33, P34, P35, P36, P38, P39 and P41) while isolates sensitive to carbenicillin may be *P. aeruginosa* (P7, P8, P14, P23, P27 and P40)(Table 4.66).

4.10.3.8 Triple sugar iron agar test

Triple Sugar Iron agar media is used for the differentiation of microbes by ability to determine carbohydrate fermentation and hydrogen sulphide production. Organisms that ferment glucose produce a variety of acids, turning the colour of the medium from red to yellow. More amounts of acids are liberated in butt (fermentation) than in the slant (respiration). Growing bacteria also form alkaline product from the oxidative decarboxylation of peptone and these alkaline products neutralize the large amount of acid present in the butt. Thus the appearance of the alkaline (red) slant and an acid (yellow) butt after incubation indicates that the organism is a glucose fermenter but is unable to ferment lactose and or sucrose. Nineteen isolates (P4, P3, P7, P8, P14, P18, P19, P20, P22, P25, P27, P29, P33, P34, P35, P36, P38, P39, P40 and P41) produced red colour after on slant and butt after incubation whereas isolate (P2 and P23) produced yellow colour on slant (Table 4.67).

4.10.3.9 Hydrogen sulphide test

The slants were observed for the presence or absence of black coloration along the line of stab inoculation. The black colour indicated positive result. The black colour is due to the production of H₂S from sodium thiosulphate present in the medium that then combines with ferrous ammonium sulphate resulting in the formation of black insoluble ferrous sulphide. This was found in ten isolates (P3, P4, P8, P14, P18, P19, P35, P36, P39 and P40) (Table 4.67).

Table 4.65: Biochemical tests for characterization of fluorescent *Pseudomonas*

S.No.	Isolate	Gelatine hydrolysis test	Nitrate test	Starch hydrolysis test	Casein hydrolysis test	Lypolytic test	4 ° C	42° C
1.	P2	+	+	-	+	+	+	+
2.	P3	+	+	+	+	+	+	+
3.	P4	-	+	-	-	-	+	+
4.	P7	-	-	-	-	+	+	-
5.	P8	+	+	-	+	-	+	+
6.	P14	+	+	-	+	-	+	+
7.	P18	+	+	-	+	-	-	+
8.	P19	+	+	-	+	-	-	-
9.	P20	+	+	-	+	-	-	+
10.	P22	+	+	-	+	-	-	-
11.	P23	+	+	-	-	-	-	+
12.	P25	+	+	+	+	-	-	+
13.	P27	+	+	+	-	-	-	-
14.	P29	+	+	-	+	-	+	+
15.	P33	+	+	-	+	-	-	+
16.	P34	+	-	-	+	-	+	+
17.	P35	+	+	-	+	-	+	+
18.	P36	+	+	-	+	-	-	+
19.	P38	+	+	-	+	-	-	+
20.	P39	+	+	-	+	-	-	+
21.	P40	+	+	-	+	+	-	+
22.	P41	-	-	-	-	-	-	-

- (Negative) and + (positive)

Table 4.66: Antibiotic sensitivity test to confirm the identity of fluorescent *Pseudomonas*

S.No.	Isolate	Mycelial growth (mm)	
		Kanamycin	Carbenicillin
1.	P2	6	4
2.	P3	23	5
3.	P4	3	5
4.	P7	5	2
5.	P8	24	1
6.	P14	4	0
7.	P18	8	4
8.	P19	25	7.5
9.	P20	31.5	4
10.	P22	28	10
11.	P23	5	0
12.	P25	22.5	3
13.	P27	5	0
14.	P29	21	5
15.	P33	21	3
16.	P34	9	5
17.	P35	22	11
18.	P36	17	21
19.	P38	22.5	6
20.	P39	32.5	4
21.	P40	32	1
22.	P41	0	6

Table 4.67: Biochemical test to confirm the identity of fluorescent *Pseudomonas* through Triple Sugar Iron Agar test and Hydrogen Sulphide Test

S.No.	Isolate	Triple Sugar Iron Agar Test		Hydrogen Sulphide Test
		Slant/Butt colour red	Slant/Butt colour yellow	Slant/Butt colour black
1.	P2	-	+	-
2.	P3	+	-	+
3.	P4	+	-	+
4.	P7	+	-	-
5.	P8	+	-	+
6.	P14	-	-	+
7.	P18	+	-	+
8.	P19	+	-	+
9.	P20	+	-	-
10.	P22	+	-	-
11.	P23	-	+	-
12.	P25	+	-	-
13.	P27	+	-	-
14.	P29	+	-	-
15.	P33	+	-	-
16.	P34	+	-	-
17.	P35	+	-	+
18.	P36	+	-	+
19.	P38	+	-	-
20.	P39	+	-	+
21.	P40	+	-	+
22.	P41	+	-	-

- (Negative) and + (positive)

4.10.4 Evaluation of antagonistic efficacy of fluorescent *Pseudomonas* by dual culture method

Twenty two fluorescent *Pseudomonas* isolates were collected from different geographical locations of Chhattisgarh presented in Table 4.64. They were screened *in vitro* for their antagonistic activity against different *R. bataticola* isolates causing dry root rot of chickpea. Significant varying degree of antagonistic ability of fluorescent *Pseudomonas* isolates was found under *in vitro* condition by dual culture technique presented in Table 4.68 Plate 20. Fluorescent *Pseudomonas* inhibited the mycelial growth by the antagonistic activity across all isolates of *R. bataticola* ranged from 4.00 to 85.00 per cent over control.

Twenty two fluorescent *Pseudomonas* isolates were evaluated for their efficacy against nine isolates of *R. bataticola*. Isolate P33 was found significantly efficient in reducing the mycelial growth of *R. bataticola* by 63.21% over control, followed by P40 (62.35%), P36 (61.91%), P35 (60.25%), P18 (60.00%), P34 (59.94%), P39 (59.63%), P19 (58.89%), P22 (58.95%) and P38 (57.10%) which were statistically at par with each other. Minimum inhibition was recorded in P2 (4.63%) followed by P4 (7.78%).

In relation to other factor fluorescent *Pseudomonas*. Among all the isolate of *R. bataticola*, significant maximum mycelial inhibition was noted in RB3 (53.21%) followed by RB7 (53.19%) and RB8 (50.89%) which were statistically at par with each other. Significant minimum mycelial inhibition was recorded in RB6 (35.17%) followed by RB2 (35.48%) and RB5 (35.82%).

In case of antagonistic efficacy between fluorescent *Pseudomonas* and *R. bataticola* isolates, maximum inhibition was recorded in P19 (83.33%) against RB7 followed by P38 (75.56%) against RB1, 75.00% inhibition was observed in P35, P36 against RB7. Minimum inhibitory effect was recorded as 16.67% in P4 and P41 against RB2 and RB4 respectively, followed by P3 (18.33%) against RB2.

The result showed that all the antagonistic strains of *Pseudomonas* used in the present study, restricted the mycelial growth of *R. bataticola* significantly. The best of all, significant inhibitory effect was recorded by P33 (63.21%) followed by P40 (62.35%), P36 (61.91%) and P35 (60.25%) with no significant difference.

Significant variation was observed between P41 (25.19%) and P7 (11.48%), rest *Pseudomonas* isolate did not vary significantly. Thus, the result revealed that fluorescent *Pseudomonas* can efficiently inhibit the mycelial growth of all the isolates of *R. bataticola* taken in this study.

Present study is on conformity to the finding made by (Krishnamurthy and Gnanamanickam, 1998) they reported that antagonistic activity of *Pseudomonas* spp. against several fungus both *in vivo* and *in vitro* condition. Similar results were also obtained by Samanta *et al.* (2004) by *Pseudomonas fluorescens* against *Macrophomina phaseolina*, *R.solani*, *Sclerotinia sclerotiorum* and *Sclerotium rolfsii*. Lokeshet *al.* (2004) found 76.66% mycelial inhibition of *M. phaseolina* through *Pseudomonas fluorescens*. The mechanism involved in the inhibition of growth may be due to the agglutination potential of *P. fluorescens* and colonization of sclerotia of *M. phaseolina* (Jana *et al.*, 2000). Similar finding was reported by Manjunatha *et al.* (2013) isolated eight isolates of *P. fluorescens*, which were evaluated for their antagonists efficacy against *M. phaseolina*, causing root rot of chickpea through the dual culture technique. All isolates were found to cause significant reduction in fungal growth when compared to the control. Fluorescent *Pseudomonas* exhibited inhibitory effect under *in vitro* condition against *R.bataticola* was also reported by Nautiyal (1997); Kumar *et al.* (2002). Dewangan *et al.* (2014) reported 55.56% of mycelia inhibition of *R.bataticola* against fluorescent *Pseudomonas*. In agreement to above finding Patel *et al.* (2011) reported the same results, conducted an experiment in which four strains of fluorescent *Pseudomonas*.

Tab 4.68: Antagonistic efficacy of fluorescent *Pseudomonas* against isolates of *R. bataticola* by dual culture technique

		<i>R. bataticola</i> isolates									
Fluorescent <i>Pseudomonas</i>		Per cent inhibition over control									
S.No.	Isolate	RB1	RB2	RB3	RB4	RB5	RB6	RB7	RB8	RB9	Average
1.	P2	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	41.67 (40.13)	00.00 (00.00)	00.00 (00.00)	4.63 (4.45)
2.	P3	55.00 (47.86)	18.33 (24.34)	62.22 (52.06)	67.78 (55.40)	43.33 (40.91)	43.89 (41.46)	52.78 (46.59)	57.22 (49.16)	63.89 (53.24)	51.60 (45.67)
3.	P4	00.00 (00.00)	16.67 (24.03)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	53.33 (46.89)	00.00 (00.00)	00.00 (00.00)	7.78 (7.88)
4.	P7	00.00 (00.00)	42.78 (40.82)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)	60.56 (51.08)	00.00 (00.00)	00.00 (00.00)	11.48 (10.21)
5.	P8	56.67 (48.81)	00.00 (00.00)	60.56 (51.14)	61.67 (51.76)	46.11 (42.75)	46.11 (42.74)	51.11 (45.61)	65.00 (53.78)	49.44 (44.66)	48.52 (42.36)
6.	P14	62.22 (52.06)	69.44 (56.75)	46.11 (42.56)	71.11 (57.50)	00.00 (00.00)	62.78 (52.39)	44.44 (41.74)	68.89 (56.15)	65.00 (53.75)	54.44 (47.53)
7.	P18	65.56 (54.37)	61.67 (51.85)	68.33 (55.75)	66.67 (54.95)	38.33 (38.23)	38.33 (38.19)	73.33 (58.93)	61.11 (51.41)	66.67 (54.73)	60.00 (50.94)
8.	P19	63.33 (53.03)	49.44 (44.64)	65.56 (54.05)	66.67 (54.73)	40.56 (39.47)	40.56 (39.53)	83.33 (66.06)	65.56 (54.08)	55.00 (47.85)	58.89 (50.38)
9.	P20	66.67 (55.81)	42.22 (40.50)	51.67 (45.94)	62.22 (52.06)	44.44 (41.79)	42.22 (40.50)	73.89 (59.27)	16.11 (22.90)	19.44 (25.88)	46.54 (42.84)
10.	P22	70.00 (56.78)	60.00 (50.77)	69.44 (56.48)	62.22 (52.06)	60.00 (50.77)	22.22 (28.10)	55.56 (48.17)	68.89 (56.51)	62.22 (52.06)	58.95 (51.18)
11.	P23	00.00 (00.00)	64.44 (53.38)	69.44 (46.57)	00.00 (00.00)	57.78 (49.46)	63.33 (52.72)	41.11 (39.85)	52.78 (46.58)	66.67 (54.73)	46.17 (39.25)
12.	P25	65.56	51.67	68.33	66.67	23.89	23.89	58.33	72.22	58.89	54.38

13.	P27	(54.51)	(45.93)	(55.83)	(54.71)	(28.17)	(28.89)	(49.81)	(58.33)	(50.14)	(47.37)
		60.56	00.00	46.11	66.67	00.00	00.00	61.11	65.00	14.44	34.88
		(51.14)	(00.00)	(42.74)	(54.90)	(00.00)	(00.00)	(51.42)	(53.73)	(21.76)	(30.63)
14.	P29	66.11	41.67	62.22	66.67	43.89	47.22	25.00	55.56	55.00	51.48
		(54.53)	(39.56)	(52.06)	(54.95)	(41.12)	(43.39)	(29.57)	(48.17)	(47.58)	(45.2)
15.	P33	60.56	62.22	71.11	70.00	71.11	68.89	44.44	59.44	61.11	63.21
		(51.07)	(52.06)	(57.50)	(56.77)	(57.50)	(56.43)	(41.72)	(50.44)	(51.41)	(52.78)
16.	P34	62.22	44.44	70.56	65.56	66.67	61.11	42.22	63.33	63.33	59.94
		(52.06)	(41.78)	(57.28)	(54.05)	(54.71)	(51.41)	(40.50)	(52.72)	(52.73)	(50.82)
17.	P35	60.00	35.56	70.56	65.00	55.56	46.67	75.00	68.89	65.00	60.25
		(50.77)	(35.27)	(57.11)	(53.78)	(48.17)	(52.87)	(60.04)	(56.15)	(53.42)	(50.90)
18.	P36	64.44	41.67	71.67	64.44	52.78	51.67	75.00	66.67	68.89	61.91
		(53.38)	(40.13)	(57.84)	(53.38)	(46.57)	(45.93)	(60.01)	(54.90)	(56.15)	(52.04)
19.	P38	75.56	22.22	72.78	60.56	43.89	43.89	61.11	69.44	64.44	57.10
		(64.14)	(26.72)	(58.66)	(51.16)	(41.44)	(41.46)	(51.46)	(56.46)	(53.38)	(49.43)
20.	P39	73.89	38.33	61.11	63.89	71.11	42.22	58.33	67.22	60.56	59.63
		(59.34)	(37.88)	(51.41)	(53.30)	(57.50)	(50.46)	(49.83)	(55.06)	(51.07)	(50.67)
21.	P40	59.44	53.33	68.89	66.67	64.44	63.89	63.89	66.67	53.89	62.35
		(50.80)	(46.89)	(56.15)	(55.07)	(53.38)	(53.07)	(53.08)	(54.72)	(47.21)	(52.36)
22.	P41	00.00	00.00	67.22	16.67	00.00	00.00	27.78	60.56	54.44	25.19
		(00.00)	(00.00)	(55.06)	(24.07)	(00.00)	(00.00)	(31.57)	(51.08)	(47.53)	(23.26)
		47.30	35.48	53.21	49.18	35.82	35.17	53.19	50.89	46.45	
		(39.61)	(32.76)	(46.23)	(41.08)	(31.82)	(32.15)	(46.20)	(42.69)	(40.10)	
Factors							SEm±				
Factor(P)							2.07				
Factor(RB)							1.29				
Factor(P X RB)							6.21				
								C.D. (5%)			
								4.08			
								2.55			
								12.25			

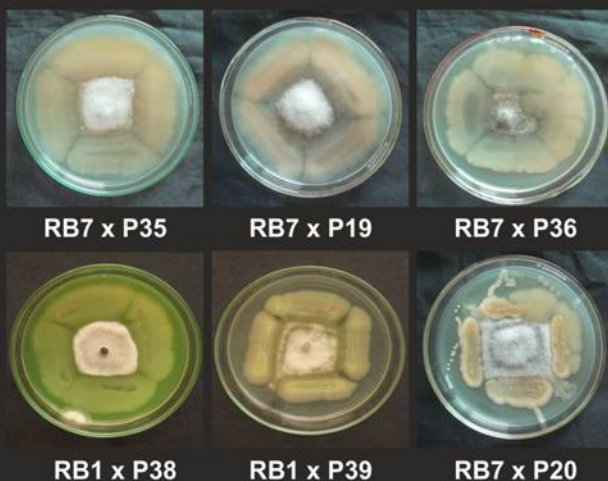


Plate 20: Best antagonistic efficacy of *Trichoderma* spp. against isolates of *R. bataticola* by dual culture technique

4.10.5 Efficacy of fluorescent *Pseudomonas* against *R. bataticola* under *in vivo* condition

The data on effect of antagonistic efficacy of fluorescent *Pseudomonas* on *R. bataticola* are presented in Table 4.69, Fig. 4.22, Fig. 4.23, Plate 21.

In vivo experiment was conducted to evaluate the efficacy of fluorescent *Pseudomonas* on the incidence of dry root rot of chickpea. The average mortality of chickpea in sick pots after 30 days of challenge inoculation with RB4 was recorded. All the treatments were significantly superior, in decreasing the incidence of dry root rot disease in chickpea, over control. Isolate P3, P8, P29 and P40 exhibited significant minimum mortality 12.5 per cent (84.20 percent decreased over control) followed by P14, P33 and P38 (mortality 16.66 per cent) with 78.95 per cent decreased over control, P18 (mortality 20.83 per cent) with 73.68 per cent decreased over control subsequently followed by P35 (mortality 25.00 per cent) and P36 (mortality 29.16 per cent) (68.41 and 63.16 per cent decreased over control, respectively) which were statistically at par with each other. These isolates showed significantly better results, over other isolates. Maximum mortality was observed in P4 by 62.50 per cent whereas in control 79.16 per cent.

Taking into account, the diseases rating was assigned, lowest rating was scored by P23 as 2.33 followed by P3, P18, P19 and P20 as 3.66, and P14, P25, P34 and P35 scored 4.33. Maximum damage was recorded in control (seed treated with sterile water) 9.00.

The influence of different fluorescent *Pseudomonas* isolate on germination per cent, root length and shoot length in chickpea when challenge inoculated with *R. bataticola* (RB4), was studied and data are presented in Table 4.70 and Fig. 4.24.

Highest germination per cent was observed in most of the isolates like P8, P19, P20, P23, P34 and P35 that is 90 per cent which were at par with each other. The above isolates were found to be significantly superior with all the other treatment and over control (66.65 per cent).

Table 4.69: Efficacy of fluorescent *Pseudomonason* the incidence of dry root rot of chickpea

S. No.	Isolate	Disease rating	Per cent mortality	Per cent decrease over control
1	P2	8.33	33.33 (34.98)	57.87
2	P3	3.66	12.50 (20.69)	84.20
3	P4	7.00	62.50 (52.39)	21.04
4	P7	8.33	54.16 (47.39)	31.58
5	P8	5.66	12.50 (20.69)	84.20
6	P14	4.33	16.66 (23.79)	78.95
7	P18	3.66	20.83 (22.19)	73.68
8	P19	3.66	29.16 (26.57)	63.16
9	P20	3.66	33.33 (34.47)	57.87
10	P22	5.66	41.66 (40.15)	47.37
11	P23	2.33	41.66 (40.15)	47.37
12	P25	4.33	50.00 (44.98)	36.83
13	P27	7.00	54.16 (47.39)	31.58
14	P29	5.00	12.50 (20.69)	84.20
15	P33	5.00	16.66 (23.79)	78.95
16	P34	4.33	33.33 (34.47)	57.87
17	P35	4.33	25.00(24.19)	68.41
18	P36	6.33	29.16 (26.96)	63.16
19	P38	7.66	16.66 (23.79)	78.95
20	P39	5.00	33.33 (34.47)	57.87
21	P40	5.66	12.50 (20.69)	84.20
22	P41	8.33	41.66 (40.15)	47.37
23	Control (untreated)	9.00	79.16(63.58)	00.00
SEm±			4.22	
CD(5%)			12.05	
Average of three replication				
Figures in parentheses are angular transformation				

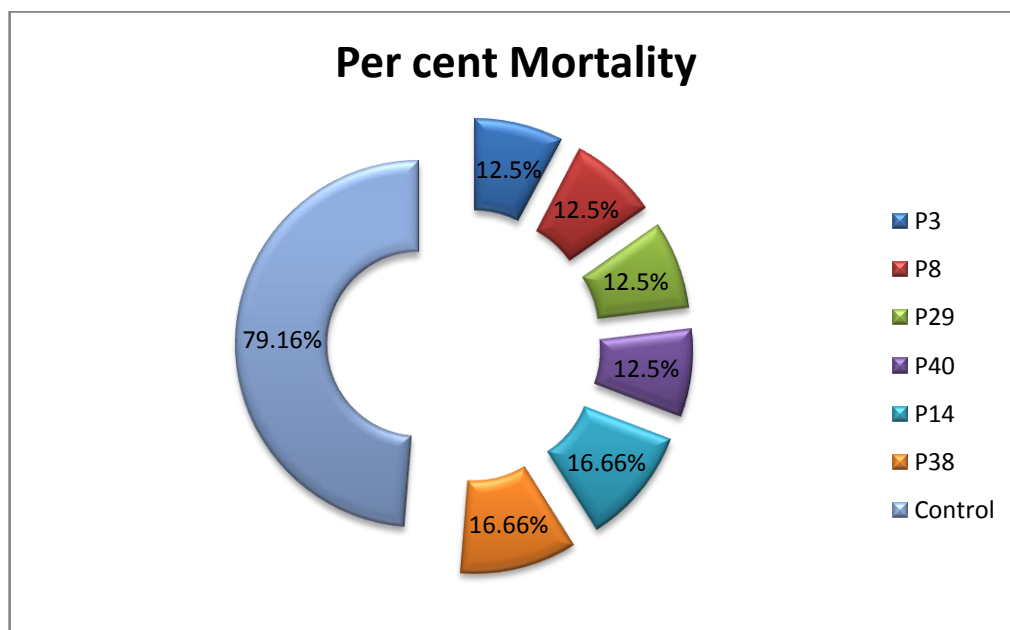


Fig 4.22: Efficacy of fluorescent *Pseudomonas* on the incidence of dry root rot of chickpea

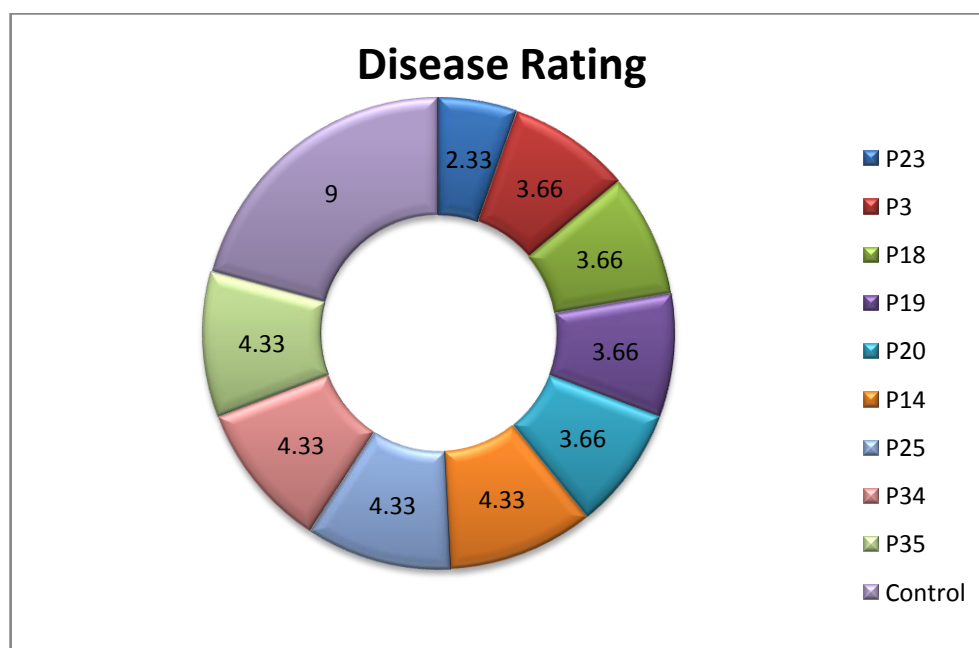


Fig 4.23: Disease rating scored from different fluorescent *Pseudomonas* isolates against *R. bataticola* (RB4) causing dry root rot of chickpea

All the *Pseudomonas* isolates were significantly superior in increasing the root length over the control (10.2 cm), when challenged inoculated with *R. bataticola* RB4 (Kawardha) isolate. Maximum root length was recorded in P23, 30.2 cm (196.07 per cent increase over control), which was significantly superior over all the other treatments. Followed by P34 and P40, 21.4 cm (109.80 per cent increase over control) and P20, 19 cm (86.27 per cent increase over control), statistically at par with each other. Lowest root length was recorded in P36 (11.0 cm) and P39 (11.9 cm) which were at par with each other.

Maximum shoot length was observed in P14 (16.6 cm) which was 38.33 per cent increase over control followed by P40 and P41 (15.5 cm) (29.16 per cent increase over control) and P34 (15 cm) (25.00 per cent increase over control) which were at par with each other. Minimum shoot length was observed in P36 and P7 that is 11.5 cm. However, significant difference was observed among the treatments in terms of shoot length.

[illegible]

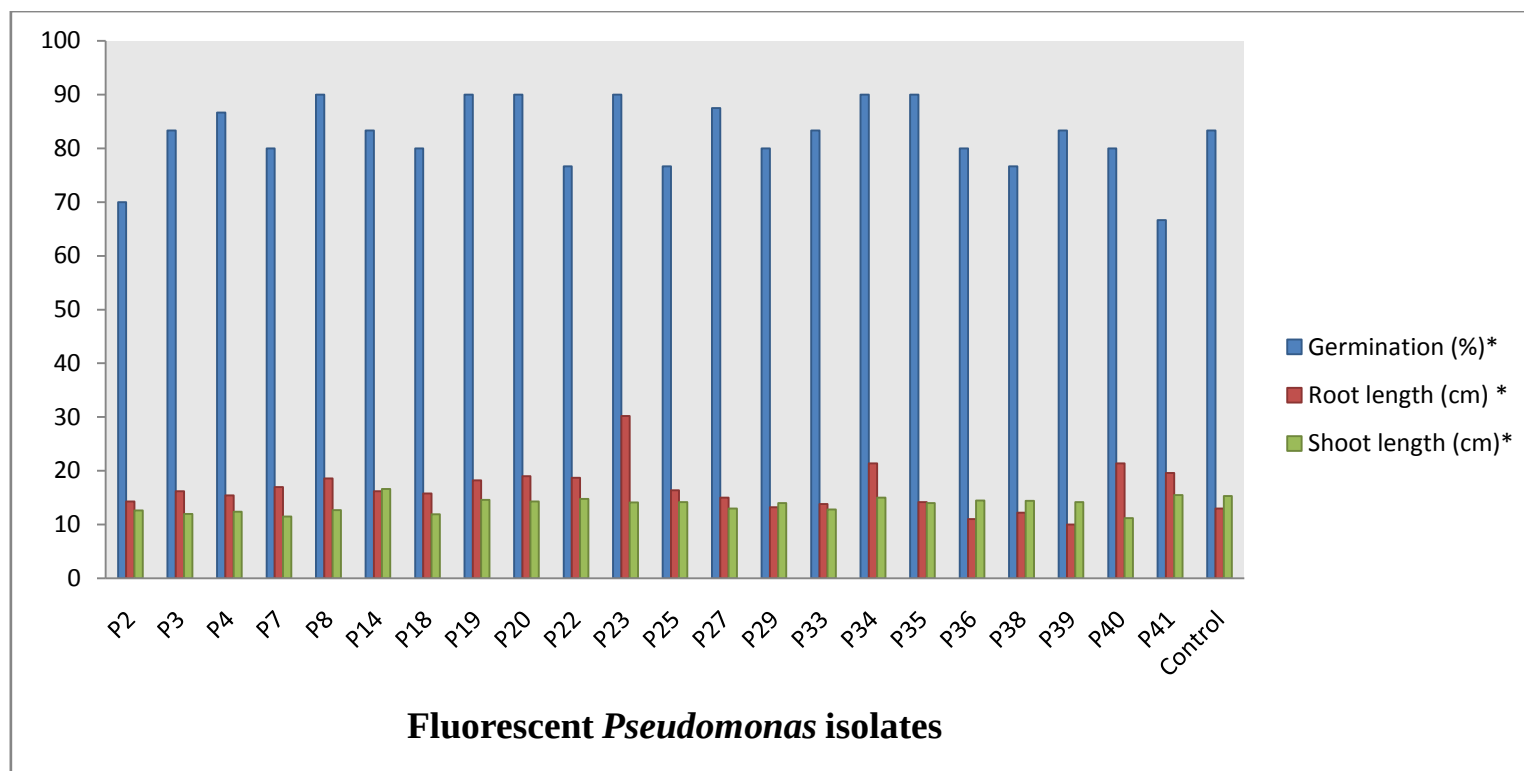


Fig 4.24: Effect of fluorescent *Pseudomonas* on germination, root and shoot length of chickpea

However, all the fluorescent *Pseudomonas* isolates in this study revealed variation in germination per cent, root length and shoot length, also among themselves and over control, when challenge inoculated with *R. bataticola* RB4 (Kawardha) isolate. Successful control of stem rot and root rot by *Pseudomonas* was demonstrated by several workers Sharma (1994) and Sharma *et al.* (1999). Inhibition of chickpea root rot by *Pseudomonas* was also reported by Selvarajan and Jeyarajan (1996). In agreement to present finding Manjunatha *et al.* (2013) found significantly higher seed germination per centage and lower root rot incidence. There was 4-13% increase in germination, 78-86% reduction in root rot incidence and 19-32% increase in yield in the treatments when compared to the control.

In seed pelleting treatment, the antagonistic organism readily multiplies on seed surface, which in turn prevents the entry of the pathogen (Raguchander *et al.*, 1998). *P. fluorescens* against *M. phaseolina* further seed treatment along with soil application of these bioagents enhanced the germination and better crop stand. Seed treatment was found to be better than soil application of antagonist is in soil applied with *T. harzianum*, the bioagents at first multiply in the introduced environment and then it has to move near the root region to prevent the entry of pathogen, hereby reducing the severity of dry root rot.

The biocontrol agents not only controlled dry root rot but also promoted plant growth, and this gives them an advantage over the use of chemical fungicides against root rot in disease management prolonged use of the bio-control agents will increase their rhizosphere population (Suriachandraselvan *et al.*, 2004 and Naik and Sen, 1995) and thereby enhancing the disease suppressive characteristics of the soil (Singh and Nema, 1987). The bacteria belonging to the fluorescent *Pseudomonas* which colonize roots of a wide range of crop plants are reported to be antagonistic to soil-borne plant pathogens (Siddiqui and Ehteshamul- Haque, 2000). The production of certain antibiotics (Raajimaker *et al.*, 2002) and siderophores (De Meyer & Hofte, 1997) by *P. aeruginosa* has been regarded as one of the mechanism involved in antagonism. Raajimakers & Weller, (1998) reported role of 2, 4- diacetylphloroglucinol an antifungal metabolite from species of fluorescent *Pseudomonas* in plant root disease suppression. Raajimakers and

Weller, (1998) reported role of 2, 4- diacetylphloroglucinol an antifungal metabolite from species of fluorescent *Pseudomonas* in plant root disease suppression. Biocontrol agents have a direct antagonistic activity not only by producing various metabolites, but also by inducing defense enzymes, which have recently been found to be a new way whereby plants defend themselves from pathogen attack (Bharathi *et al.*, 2004).



P3



P8



P29



P40



Plate 21: Best results of chickpea seeds bioprimered with fluorescent *Pseudomonas* in sick pots

CHAPTER-V

SUMMARY AND CONCLUSION

Among the several soil borne fungal diseases of chickpea, dry root rot caused by *Rhizoctonia bataticola* Taub (Butler) is the most severe disease. Distributed worldwide, *Rhizoctonia* fungi occur in agricultural and non-agricultural fields, living as plant pathogens, symbionts or saprophytes. Because some groups in *Rhizoctonia* spp. can severely damage major crops control of diseases resulting from infection by *Rhizoctonia* spp. is a major concern for farmers and plant pathologists. The low yield of chickpea can be attributed of biotic and abiotic constraints. Among biotic factors, diseases are the most dominant.

Compare with other genera of plant pathogenic fungi that have physiological forms or pathogenic races, mechanism of host specificity in *R. bataticola* seems to be vaguer. Understanding the pathogen variability helps in understanding the role pathogens playing in shaping the genetic structure of plant population and communities are requires an understanding of the pathogen diversity and origin. A variety of useful approach has been developed for identifying many groups of *Rhizoctonia* spp. based on ultrastructural features, information from telomorph, anastomosis behaviour and molecular biology. The anastomosis group concept has represented the single most important advance towards understanding the genetic diversity in *Rhizoctonia* spp. but yet not in case of *Rhizoctonia bataticola*.

Various edaphic conditions are considered to be the foremost factors governing the development of soil-borne plant diseases as they greatly affect the activity of pathogens in soil. Severity of the disease is significantly affected by various soil factors. The use of chemicals for controlling disease is a growing concern to environmentalists. Many effective fungicides have been tested against soil borne pathogens, but are not considered as long term solutions because of concerns about exposure risks, health and environmental hazards,

high cost, residue persistence, the development of resistance to pesticides and the elimination of natural enemies. Biological control is a potential non-chemical means for plant disease management by reducing the harmful effects of a parasite or pathogen through the use of other living entities.

The findings of the present investigation on “Studies on dry root rot (*Rhizoctonia bataticola* Taub (Butler)) of chickpea (*Cicer arietinum*)” were carried out in the Department of Plant Pathology, College of Agriculture, IGKV, Raipur (C.G.). The investigation mainly consists of four objectives- A: Isolation and characterization (pathogenicity, AG grouping) of *R. bataticola* isolates from different geographical locations of Chhattisgarh. B: Identification of potential candidate isolate / bio-control agent against *R. bataticola*. C: Screening of chickpea varieties / wild relatives for resistance to dry root rot. D: Study of soil factors affecting disease development of dry root rot in chickpea. The summary and conclusion of findings of the present investigations are given below:

The survey for the occurrence and severity of chickpea dry root rot were undertaken during crop season 2012-13. The maximum mean per cent disease incidence was recorded in Kawardha 19.1% followed by Gandai 16.9% and Patan 14.8% however, minimum disease incidence was observed in Rajnandgoan 2.8%.

Isolation of the test fungus was made from infected plants of chickpea, collected from the major chickpea growing region of Chhattisgarh (Balod, Dhamtari, Durg, Kanker, Kawardha, Raigarh, Raipur, Rajnandgoan, and Saragoan). Nine different isolates of the fungus *R. bataticola* were obtained, and were maintained on PDA at $27 \pm 2^{\circ}\text{C}$.

The symptoms of dry root rot were observed under natural field conditions. The characteristic symptoms found were blackening of the tap root with signs of extensive rotting and loss of lateral and finer roots. The dead roots were quite brittle and accompanied with shredding of bark and lateral roots.

Nine isolates were characterized for cultural and morphological characteristics on PDA medium. All the isolates shared characteristics of *R. bataticola*. The isolates showed varied pattern of mycelial growth fluffy mycelial growth (RB1, RB6 and RB8), partially submerged (RB3, RB4 and RB7) and

Submerged (RB2, RB5 and RB9). Colony colour of the isolates varied like, black (RB2, RB5, RB7 and RB9) and greyish (RB1, RB3, RB4, RB6 and RB8).

Branches of mature hyphae that served to indentify isolated pathogen *R. bataticola* arose at right and acute angle that is near to 45 degree to main hyphae. Variation in hyphal width among the isolates was observed, the pathogen was categorized in three groups. 5-6 μm of hyphal width (RB1, RB3 and RB6, RB8 and RB9), 6-7 μm (RB2 and RB7) and hyphal width of more than 7 μm (RB4 and RB5).

On the basis of microscopic observations the size of sclerotia varied from 68-103 x 59-97 μm . Where, RB6 found to have largest sclerotial size 103 x 97 μm and RB2 smallest sclerotial size 68 x 59 μm . Whereas, sclerotial shape of the isolates was also found to vary round (RB3, RB4, RB5, RB6, RB8 and RB9) and oval (RB1, RB2 and RB7). Mature sclerotia were much less closely interwoven and there were no well dentified zones.

For mycelial growth of different *R. bataticola* isolates, pathogen was incubated for four days. Isolates RB1, RB4, RB5, RB6, RB7 and RB8 exhibited full growth (fast, 90mm) and least (slow) growth in RB3 (56.6 mm).

Pathogenicity of the fungus was tested, all the isolates were found to be pathogenic under artificially inoculated condition and expressed typical symptoms of dry root rot. *In vitro* blotter paper technique revealed that the isolates were variable in their aggressiveness, it varied between 5-9 disease grade. The most aggressiveness was found to be RB5 isolate which scored 8.33 disease grade and least aggressive was found to be RB2 isolate scored 5.66 disease rate grade. *In vivo* sick pot method also exhibited the variability in virulence of the isolates. Morality per cent varied between 55 to 100%. Highest mortality was observed in RB4 with 100 % which scored 9 disease rate followed by RB8 87.5 % scored 6.3. However, lowest mortality was observed in RB3 with 58.33% scored 5.6 disease rate.

Anastomosis behavior of different isolates of *R. bataticola* with each other collected from different location of Chhattisgarh was investigated in all possible combinations. All isolates exhibited self-compatibility that is fusion of wall and membrane and showed homologous anastomosis (reaction type 3).

In case of screening of entries against dry root rot of chickpea none of the entries were found highly resistance and resistance. During first *rabi* season 2012-2013 in Desi III only one entry was moderately resistant (BG 3033), three entries NDG 11-12, RKG 2020 22 and H 09-90 were tolerant. Whereas, in rainfed (R.F. II) sets of entries, Phule G 0752 was moderately resistant and JG 33 and CSJ 741 was tolerant. Three entries namely, RSG 9056, JG 16 and CSJ 8023 were found tolerant in elites lines, however, remaining all the entries were susceptible to dry root rot. While during second *rabi* season (2013-14) CSJ 809 was found moderately resistant whereas, JG 17, CSJ 808, CSJ 556, JG 24, CSJ 815, GLK 28127, JG 27, JG 36 were tolerant.

One thirty two entries were screened *in vitro* through blotter paper method against *R. bataticola* isolate RB 5 which was most virulent under *in vitro* aggressiveness test. Out of 132 entries, 21 entries categorized as resistant, were no infection on roots was observed in resistant lines as the root system was healthy, completely developed and exhibited white appearance, followed by 19 entries categorized as moderately resistance with exhibited very few small necrotic lesion on roots many lesions on roots, While, 21 entries grouped as tolerant, clear lesions was observed on roots but small, though new roots were free from infection. 53 entries were found to be susceptible with many lesions on roots but new roots generally free from lesions. However, 18 entries exhibited highly susceptible were the roots were completely infected, discoloured and rotten.

An experiment was conducted to know which soil pH is favorable or unfavorable for dry root rot development in chickpea. Significantly least mortality per cent (12.5%) was recorded at pH 8.0 followed by pH 4 with mortality per cent of 41.67%. No significant difference was observed among mortality at pH 4.0 (41.67%), pH 7.0 (50.0%) and pH 6.0 (62.5%) which were next in order. However, significantly highest mortality was observed at pH 5.0 that is, 83.3%. Extent of root damage varied from 3-9 grades and least disease grade was recorded at pH 8.0 with minimum root damage and infection. Highest germination was observed in pH 5.0, pH 6.0 and pH 8.0 that is 90.00 per cent. Maximum root length was observed in pH 8.0 (18.6 cm) and pH 6.0 (15.6 cm) which were at par, but

significantly better than other treatments. Maximum shoot length was observed in pH 6.0 (16.4 cm) followed by pH 5.0 (15.2 cm) and pH 7.0 (15.0 cm).

Four different soil types are found in Chhattisgarh region, that were taken for study in which, significantly least mortality per cent (16.66%) was recorded in matasi followed by dorsa with mortality per cent of 20.83% which were at par with each other. However, highest mortality was observed in Bhata 62.5%, followed by Kanhar 54.16%. Extent of root damage varied greatly from 1-9 grades and lowest disease grade was recorded in Matasi with minimum root damage and infection that is, 1.6 rating, followed by Dorsa and Bhata with disease rating 5.6. Highest germination was observed in Dorsa and Bhata (90 per cent) followed by Kanhar 86.65 per cent, which were found to be at par with each other. Maximum root length was observed in Matasi (16.8 cm) and Dorsa (15.9 cm), while maximum shoot length was observed in Bhata (14.8 cm) followed by Matasi (14.6 cm) which were at par with each other.

In case of soil texture, significantly least mortality per cent (20.8%) was recorded in sandy loam texture followed by sandy texture with mortality per cent of 25.0% which were at par with each other. However, highest mortality was observed in clay texture soil 62.5%, followed by silt clay 37.5%. Extent of root damage varied greatly from 1-7 grades and lowest disease grade was recorded in sandy loam with minimum root damage and infection that is, 1.6 rating, followed by sandy and loam with disease rating 3.6 and 5 respectively. While, the highest disease rate was recorded clay texture soil that is 7.0. Highest germination percentage was exhibited in sandy textured soil, 100.0 per cent which was significantly superior with other treatments. Maximum root length was observed in sandy soil (17.9 cm), silt clay (16.5 cm) and loam (16.1 cm) which were at par, but significantly superior than other treatments. Least increased in root length was recorded in clay (14.7 cm) and sandy loam (15.6 cm). All the treatments were found at par in terms of shoot length, though maximum was observed loam soil (14.8 cm) followed by sandy loam (14.6 cm) and silt clay (14.5 cm).

Fifteen *Trichoderma* spp. were isolated from the soil samples collected from different location of Chhattisgarh and were characterized for cultural and morphological characteristics on PDA medium. All the isolates shared

characteristics of *Trichoderma*. The isolates showed variation in colony colour, yellow green (isolate T1 and T40), whitish green (isolate T4, T12, T20, T22 and T33), green (isolate T7, T14, T29 and T31a), light green (isolates T19, T23, T31b and T36) and green colony (T29 and T31a).

Trichoderma isolates varied in growth pattern, that is, effused (T4, T12, T22 and T23) and subdued (T1, T7, T14, T19, T20, T29, T31a, T31b, T33, T36 and T40). Difference was observed in colony appearance like, uniform ringed velvety (T1), velvety at periphery (T4, T14 and T20), velvety ringed (T7, T19, T29, T31a, T31b, T33 and T40) and scattered (T12, T22, T23 and T36).

Trichoderma varied in the ability to form pustules. No pustules were found in isolate T1, T7, T19, T20, T29, T31a, T31b, T33 and T40. Scattered pustules were observed in isolate T4, compact pattern pustules in T12 and T23 whereas, minute pustules were observed in T22 and T36.

Pigmentation of *Trichoderma* isolates were studied on PDA plates. The production of diffusible yellowish orange pigments were found in T1, T7, T12, T19, T22, T29 and T40 isolates, whereas, in T4, T14, T20, T23, T31a, T31b, T33 and T36 isolates no pigmentation were found. However, no odour was found to produce from *Trichoderma* isolates.

Fifteen isolates of *Trichoderma* were evaluated for their efficacy against nine isolates of *R. bataticola*. Isolate T40 was found significantly superior among all the isolates of *Trichoderma* in reducing the mycelial growth of *R. bataticola* by 67.65% over control, followed by T36 (64.81%), T14 (64.07%) and T29 (63.21%) which were at par with each other. Minimum inhibition was recorded in T31a (52.84%) followed by T20 (55.06%) which were statistically at par with each other.

In *in-vivo* mycoparasitic activity against different *R. bataticola* isolates were evaluated. T40 isolate was highly effective with significant inhibition of 67.65 % across all *R. bataticola* isolate taken in this study, this was followed by T36 (64.8%) and T14 (64.07%).

Among the fifteen *Trichoderma* isolates tested for their efficacy in overcoming dry root rot disease of chickpea via, seed treatment, T23 and T40 were found to produce least mortality 20.83 % among all, subsequently followed by T4 and

T36 with mortality 25.00%. Maximum mortality was observed in T7 that is 50.00% whereas in control 83.33%. The extent of root damage was found to vary from 3-9 on 1-9 of disease rating scale. Lowest rating was scored by T4 and T31b as 3.0 followed by T1, T33 and T36 as 3.6. Highest germination per cent was observed in eight isolates like T12, T14, T19, T22, T29, T31a, T31b and T40 that is 90 per cent which was found significantly superior with other isolates and over control (86.66 per cent). All the *Trichoderma* isolates were significantly superior in increasing the root length over the control (9.33 cm), when challenged inoculated with *R. bataticola* RB4 (Kawardha) isolate. Maximum root length was recorded in T29 isolate (32.6 cm) and T33 isolate 31.8 cm, which was at par and significantly superior over all the other treatments and control. However, all the treatments in terms of shoot length, observed were at par with each other and with control. However, highest shoot length was exhibited by T7, T20 and T40 that is 17.2 cm, 15.3 cm and 15.0 cm respectively.

Mechanism of antagonism of *Trichoderma* spp. on *R. bataticola* was studied. Interaction between the two fungi when documented through light microscopy indicated that the mode of action adopted to parasitize *R. bataticola* was through coiling. The coiling of the vegetative mycelium of *R. bataticola* by *Trichoderma* initiates a sort of searching and recognition mechanism between the two is speculated. The mycelial tip of *Trichoderma* runs parallel with that on the mycelium of *R. bataticola*, and with its mycelial tips it sticks on to the large hyphae of *R. bataticola*. This process is further followed by a very rapid and excessive coiling on the target fungus. Lysis of the pathogen was also observed.

Twenty two fluorescent *Pseudomonas* isolates were evaluated for their efficacy against nine isolates of *R. bataticola*. Isolate P33 was found significantly efficient in reducing the mycelial growth of *R. bataticola* by 63.21% over control, followed by P40 (62.35%), P36 (61.91%), P35 (60.25%), P18 (60.00%), P34 (59.94%), P39 (59.63%), P19 (58.89%), P22 (58.95%) and P38 (57.10%) which were statistically at par with each other. Minimum inhibition was recorded in P2 (4.63%) followed by P4 (7.78%). Thus, the result revealed that fluorescent *Pseudomonas* can efficiently inhibit the mycelial growth of all the isolates of *R. bataticola* taken in this study.

Among the twenty two fluorescent *Pseudomonas* isolates tested for their efficacy in overcoming dry root rot disease of chickpea via., seed treatment, it was found that P3, P8 and P40 exhibited minimum mortality 12.5% followed by P14, P33 and P38 (mortality 16.66%). Maximum mortality was observed in P4 by 62.50% whereas in control 79.16%. Diseases rating was assigned, lowest rating was scored by P23 as 2.33. Highest germination per cent was observed in most of the isolates like P8, P19, P20, P23, P34 and P35 that is 90 per cent. Maximum root length was recorded in P23, 30.2 cm (17.2 cm more over control), which was significantly superior over all the other treatments, while largest shoot length was observed in P14 (16.6 cm) followed by P40 (15.5 cm) and P34 (15 cm) which were at par with each other.

CONCLUSION

In the light of the findings of present investigation, some conclusion could be drawn like-

- Isolates of *R. bataticola* exhibited a high degree of variability for its cultural, morphological characteristic and aggressiveness.
- Aggressiveness of the isolates (*in vitro* (blotter paper method) the Raigarh isolate (RB 5) was found more virulent and *in vivo* (sick soil method)) the Kawardha isolate (RB 4) was highly virulent.
- All isolates of *R. bataticola* exhibited self-compatibility that is fusion of wall and membrane.
- Source of disease resistance has been identified from the available material of chickpea.
- Soil factors (soil pH, soil type and soil texture) strongly influenced the development of dry root rot of chickpea. Soil pH 5, Kanhar and clay soil exhibited highest mortality in chickpea.
- *Trichoderma* spp. and fluorescent *Pseudomonas* is good biocontrol agent against *R. bataticola* as it showed strong antagonistic activity when tested in dual culture techniques (T40 and P33) and *in vivo* potcondition (T23, T40 and P3, P8, P40) respectively.

SUGGESTIONS FOR FUTURE RESEARCH WORK

- Meteorological data should also be recorded to determine its relationship to the occurrence of wilt and disease forecasting.
- A regular survey for the assessment of dry root rot in chickpea growing areas is suggested to be carried out each year, particularly in the months of December to March.
- More advanced techniques such as, SSR and RFLP can be applied for better understanding of genetic diversity and pathogenic variability among the isolates of *R. bataticola*.
- The genotypes, resistant at seedling and reproductive stage, should be utilized in breeding programmes to build disease resistant pyramids due to complex nature of *R. bataticola*
- Continuous systematic breeding programmes blending the advantages of the molecular tools may help in evolving the new promising varieties.
- More intensive studies need to be conducted at the molecular level in order to understand the nature of mechanism by which the resistant host defend against infection.
- More attention needs to be focussed on more microbes for identifying their resistance imparting and growth promoting abilities.
- Fully understanding the functional role of organism in these natural environment
- Condition influencing the performance and adoptability of the *Trichoderma* and *Pseudomonas* isolates to be worked out.
- *Trichoderma* genomes are a source of candidate genes for developing transgenic plants that are tolerant to biotic and abiotic stresses. Development of engineered plants will be benefited by *Trichoderma* genome mining.

REFERENCES

- Abouseoud, M., Maachi, R. and Amrane, A. 2007. Communicating Current Research and Educational Topics and Trends. *Applied Microbiology*, 340-347.
- Aghakhani, M. and Dubey, S.C. 2009. Determination of genetic diversity among Indian isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea. *J. Mycol. Pl. Pathol.*, 96: 607-619.
- Agharkar, S.P. 1991. Medicinal plants of Bombay presidency. Scientific Publishers, Jodhpur. India.
- Ahmed, Q. and Mohammad, A. 1986. Losses in yield due to *Rhizoctonia* root rot of chickpea in Bihar. *Indian Phytopathology*, 39: 590-592.
- Ali, S.M. and Dennis, J. 1992. Host range and physiologic specialization of *Macrophomina phaseolina* isolated from field peas in South Australia. *Australian J. of Experi. Agric.*, 32(8): 1121-1125.
- Allen, D.J. and Lenne, J.M. eds. CAB International, Wallingford, UK, p. 473-516.
- Aly, A.A., Mohamed, A.A., Moawad, R.O. and Kamel A.A. 2007. Differential antagonism of *Trichoderma* spp. against *Macrophomina Phaseolina*. *Journal of Plant Protection Research*, 47(2): 381-384.
- Ammajamma, R., Hegde, Y.R., Lingaraju, S., Sreedevi, S.C. and Hedge, L. 2009. Studies on biological management of *Rhizoctonia bataticola* infecting *coleus forskolii*. *Biomed.*, 4(1): 28-32.
- Anand, R.P.K. and Haware, M.P. 1987. Inheritance of dry root-rot *Rhizoctonia bataticola* resistance in chickpea. *Plant Breeding*, 98: 349-352.
- Aneja, K.K. 1993. Experiments in Microbiology plant pathology and tissue culture, Wishwa Prakasam (A division of Wiley Eastern Ltd.), 117-121.
- Anju, G. and Varma, R.K. 2007. Non-target effect of agro-chemicals and bio-controls agents on important soil-borne pathogen of soybean. *JNKVV. Res. J.*, 4(1): 65-70.
- Anonymous, 1965. Plant disease survey for the year ending 30th June 1964. 34th Annual Report N S W Department of Agricultural Biology Branch. p. 38.

- Arwiyanto, T., Maryudani R., and Azizah, A.A. 2007. Sifat-sifat Fenotipik *Pseudomonas fluorescens*, Agensia Pengendalian Hayati Penyakit Lincat pada Tembakau Temanggung. J. Biodiversitas, 8(2):147-151.
- Ashby, S.F. 1927. *Macrophomina phaseolina* (Mabul.) Com. Nov. the pycnidial stage of *Rhizoctonia bataticola* (Taub.) Butl. Trans Brit. Mycol. Soc., 12:141-147.
- Ashwini, C., Giri, G.K. and Halgekar, N.Y. 2014. Efficacy of bioagents against seed borne fungi of black gram. International Journal of Applied Biology and Pharmaceutical Technology, 5: 56-57.
- Baker, M.A. and Ahmed, F. 1991. Additional sources of resistance of will and root rot of chickpea in Bangladesh. Intern. Chickpea Newsl., 25: 28-29.
- Baldev, B., Ramanujam, S. and Jain, H.K. 1988. Pulse crops. Fungal diseases of pulse crops. p. 266-267.
- Bandyopadhyay, S., Sharma, N.D. and Dutta, S. 2003. Screening of potential *Trichoderma* strains against major root pathogens. Annals of plant protection sciences, 11:163.
- Bateman, D.F. 1963. The macerating enzyme of *Rizoctonia solani*, Phytopathology. 53: 1178-1186.
- Beas-Fernandez, R., de Santiago-de A.S., Hernandez-Delgado, S., and Mayek-Perez, N. 2006. Characterization of Mexican and non-Mexican isolates of *Macrophomina phaseolina* based on morphological characteristics, pathogenicity on bean seeds and endoglucanase genes. Journal of Plant Pathology, 88: 53–60.
- Belkar, Y. K. And Gade, R. M. 2013. Management of Root Rot and Collar Rot of Soybean by Antagonistic Microorganism. J. Pl. Dis. Sci., 8(1): 39-42.
- Bell, D., Wells, H. and Markham, C. 1982. *In vitro* antagonism of *Trichoderma* species against six fungal plant pathogens. Phytopathology, 72: 379-382.
- Bharathi, R., Vivekananthan, R.H., Ramanathan, A. and Samiyappan, R. 2004. Rhizobacteria-based bioformulations for the management of fruit rot infection in chillies. Crop Protection, 23: 835–843.
- Bhuiyan, M.K.A. and Arai, K. 1993. Physiological factor affecting hyphal growth and fusion of *Rhizoctonia oryzae*. Trans. Mycol. Soc. Jpn., 32: 389–97.

- Bondaroy, 1785.cited in J. C. Walkers: *Plant Pathology*, IIIrd edition. Teta Mcgrath-hill publication compony Ltd., New Delhi.
- Bracker, C.K. and Butler, E.E. 1963. The ultra structure and development of septa in hyphae of *Rhizoctonia solani* associated with the plant debris. *Phytopathology*. 49: 192-198.
- Briton- Jones, H.R.1925. Mycological work in Egypt during the period 1920-22. *Tech. Sci. Bull.*, 49-129.
- Buller, A.H.R. 1933. *Researches on Fungi*. London: Longman, 5: 416.
- Bulliard, P. 1791. *Histoire des champignons dela France* L. 81. Paris,
- Butler, E.J. 1918. *Fungi and Diseases in plants*.Thacker Spink and Co. Calcutta.p 547.
- Butler, E.J. 1925. *Fungi of India*. ICAR, New Delhi.
- Byadgi, A.S. and Hegde, R.K. 1985. Variation among the isolates or *Rhizoctonia bataticola* from different host plants.Indian Pytopath., 38: 297-301.
- Byadgi, A.S. and Hegde, R.K.1988. Factors affecting survival of *Rhizoctonia bataticola* in soil.India Phytopath.,41: 122-127.
- Carling, D.E., Baird, R.E., Gtaitis, R.D., Brainard, K.A. and Kuninaga, S. 2002.Characterization of AG-13, a newly reported anastomosis group of *Rhizoctonia solani*.*Phytopathology*., 92: 893-899.
- Caten, C.E. and Jinks, J.L. 1966. Heterokaryosis: its significance in wild homothallic ascomycetes and fungi imperfecti. *Trans. Br. Mycol. Soc.*, 49: 81–93.
- Chan, Y.H. and Sackston, W.E. 1971. Polygalacturonate transeliminase and cellulose production by virulent and avirulent isolates of *S. bataticola* in culture and in sunflower. *Cand. J. Bot.*, 49: 483-486.
- Chan, Y.H. and Sackston, W.E. 1972. Production of pectolytic and cellulolytic enzymes by virulent and avirulant isolate of *S. bataticola* during diseases development in sunflower. *Cand. J. Bot.*, 50: 2249-2453.

- Chapman, H.D. 1965. Chemical factors of the soil as they affect microorganism, *Ecology of soil borne plant pathogens*, p. 120-141.
- Charles, M.T., Dominique, R., Kumar, J. and Dangi, O.P. 2002. A preliminary study of the functional properties of chickpea leaves. In: Annual Meeting of the Canadian Society of Food and Nutrition, May 2002, Edmonton, Alberta, Canada.
- Chattopadhyay, S.B. and Mustafee, T.P., 1977, Behaviour of *Macrophomina phaseoli* and *Sclerotium rolfsii* in relation to soil texture and soil pH. *Current Science*, 46: 226-228.
- Chaturvedi, S.K. and Dua, R.P. 2009. Improved varieties of chickpea in India. *Bull, Kanpur.*, p. 1-10.
- Chauhan, V.B., Singh, V.B. and Singh, A.K. 1962. Status of phytophthora blight of pigeon pea in eastern Uttar Pradesh. *Ann. Pl. Protec. Sci.*, 10: 402-404.
- Coppin, E., Debuchy, R., Arnaise, S. and Picard, M. 1997. Mating types and sexual development in filamentous ascomycetes. *Microbiol. Mol. Biol. Rev.*, 61:411–28.
- Cubero, J.I. 1987. Morphology of chickpea. In: MC Saxena and KB Singh (eds.), *The Chickpea*. CAB. International, Wallingford, Oxon, OX10 8DE, UK. p. 35-66.
- Dandnaik, B.P., Garud, T.B. and Zote, K.K. 1986. Suitability of different grains as medium for mass culturing of sorghum charcoal rot organism. *India. J. Mycol. Pl. Pathol.*, 16: 293-294.
- Davey, C.B., 1996. Nursery soil management—organic amendments. In: Landis, T.D., Douth, D.B. (Tech. Coordinators), *National Proceedings, Forest and Conservation Nursery Associations. General Technical Report PNW-GTR-389*. USDA Forest Service PNWRS, p. 6–18.
- Davidson, F.A., Sleeman, B.D., Rayner, A.D.M., Crawford, J.W. and Ritz, K. 1996. Context dependent macroscopic patterns in growing and interacting mycelial networks. *Proc. R. Soc. London Ser. B.*, 263:873–880.
- De Candolle, A.P. 1815. *Memorie ur les Rhizoctonia madme museum d' historie. Natureue*. 2: 209-219.

- De Meyer, G. and Hofte, M. 1997. Salicylic acid produced by rhizobacterium *Pseudomonas aeruginosa* 7NSK2 induced resistance to leaf infection by *Botrytis cinerea* on bean. *Phytopath.*, 87: 588-593.
- Debets, A.J.M. and Griffiths, A.J.F. 1998. Polymorphism of *het* genes prevents resource plundering in *Neurospora crassa*. *Mycol. Res.*, 102:1343–49.
- Debets, F.Y.X. and Griffiths, A.J.F. 1994. Vegetative incompatibility in *Neurospora*, its effect on horizontal transfer of mitochondrial plasmids and senescence in natural populations. *Curr. Genet.*, 26:113–119.
- Deshmukh, D.P. and Raut, J.G. 1992. Antagonism by *Trichoderma* species on five plant pathogenic fungi. *New agriculturist*, 3(2):127-130.
- Devadason, A. and Subramanian, S. 2012. Effects of biocontrol agents and plant products on *Macrophomina phaseolina* and colchicine content in *Gloriosa superba*. *Plant Protect. Sci.*, 48(3): 110–115.
- Dewangan, P.K., Koma, B., Baghel, S., Khare, N. and Singh, H.K. 2014. Characterization of *Pseudomonas fluorescens* in different media and its antagonistic effect on phytopathogenic fungi. *The Bioscan*, 9(1): 317-321.
- Dhingra, O.D. and Sinclair, J.B. 1973. A location of *Macrophomina phaseoli* on soybean plants related to cultural characteristics and virulence. *Phytopath.*, 63 : 934-936.
- Dhingra, O.D., Schneider, R. W. and Sinclair, J.B. 1974. Cellulolytic and pectolytic enzyme associated with virulent and avirulent isolates of *Macrophomina phaseoli* *in vitro* and in soyabean seedlings. *Phytopathology.*, 80: 324-329.
- Dhingra, O.D. and Sinclair, J.B. 1977. An annotated bibliography of *Macrophomina phaseolina* 1905-1975. *Brazil Univer- Stdade Feederal de Vucisa University of Illionois Urbana*, U.S.A. p. 224.
- Dhingra, O.D. and Sinclair, J.B. 1978. Biology and pathology of *Macrophomina phaseolina*. *Phytopathology.*, 44: 196-202.
- Dinakaran, D., Ramakrishnan, D., Sridhar, R. and Jeyarajan, R., 1995. Management of sesamum root rot with biocontrol agents. *J. Oilseeds Res.* 12(2): 262-263.

- Doley, K. and Jite, P.K. 2012. *In-Vitro* Efficacy of *Trichoderma viride* against *Sclerotium rolfsii* and *Macrophomina phaseolina*. Not. Sci. Biol., 4(4): 39-44.
- Du Hamel, H.L. 1728. Explication physique d'une maladie que fait petire plantes dans Le getinois et particul erment Le safram. Hist. Acda. Roy. Sci. Paris, p. 100-112.
- Duby, J.E. 1830. Botanicon gallicum, 2: 867.
- Duke, J.A. 1981. Handbook of legumes of world economic importance. Plenum Press, New York. p.52-57.
- Durai, M. 2004. Management of charcoal rot of sesame (*Sesamum indicum* L.) incited by *Macrophomina phaseolina* (Tassi) Goid, M.Sc. (Ag.) Thesis submitted to Acharya N.G. Ranga Agricultural University Hyderabad, Andhra Pradesh.
- Elad, Y., Chet, I. and Hennis, Y. 1983. Parasitism of *Trichoderma* spp. on *Rhizoctonia solani* and *Sclerotium rolfsii*: Scanning electron microscopy and fluorescent microscopy. Phytopathology., 73: 85-88.
- Elad, Y. and Kapat, A.1999. The role of *Trichoderma harzianum* protease in the bio-control of *Botrytis cinerea*. Eur. J. Plant Pathol., 105:177-189.
- El-Katany, M.H., Gudelj, M., Robra, K.H., Enaghy, M.A. and Gubitz, G.M. 2001. Characterization of a chitinase and an endo-1,3-glucanase from *Trichoderma harzianum* Rifae T24 involved in control of the phytopathogen *Sclerotium rolfsii*. Appl. Microbiol. Biotechnol., 56: 137–143.
- Esser, K and Blaich, R. 1994. Heterogenic incompatibility in fungi. In *The Mycota I: Growth, Differentiation and Sexuality*, ed. JGH Wessels, F Meidhardt, p. 211–32.
- Fries, E. 1823. "Systema mycologicum". 2: 265-266.
- Galun, M., Malki, D. and Galun, E. 1981. Visualization of chitin-wall formation in hyphal tips and anastomoses of *Diplodia natalensis* by fluorescein-conjugated wheat germ agglutinin and N-acetyl-Dglucosamine. Arch. Microbiol., 130:105–10.
- Gams, W. and Bissett, J., 2002. Morphology and identification of *Trichoderma*. In: Kubicek, C. P. and Harman, G. E. (eds.). *Trichoderma and Gliocladium: Basic biology, taxonomy and genetics*, 9. 3-31.

- Gangopadhyay, S., Agarwal, D.K., Sarbhoy, S.K. and Wadhi, S.R. 1973. Charcoal rot disease of soyabean in India. *Indian Phytopath.*, 26: 730-732.
- Garnjobst, L. and Wilson, J.F. 1956. Heterocaryosis and protoplasmic incompatibility in *Neurospora crassa*. *Proc. Natl. Acad. Sci., USA* 42: 613–618.
- Garrett, S.D. 1944. Root disease fungi (Walthen, Mass, Chronica Botanica Co.), p. 159.
- Garrett, S.D. 1956. *Biology of Root-infecting Fungi* Cambridge University Press, p. 293.
- Gejera, H.P., Bambharolia, R.P., Patel, S.V., Khatrani, T.J. and Goalkiya S. 2012. Antagonism of *Trichoderma* spp. against *Macrophomina phaseolina*: Evaluation of Coiling and Cell Wall Degrading Enzymatic Activities. *BA J. Plant Pathol Microb.*, 3:7.
- Girbardt, M. 1957. Der Spitzenkorper von *Polystictus versicolor* (L.). *Planta*, 50: 47–59.
- Glass, N.L. and Kulda, G.A. 1992. Mating type and vegetative incompatibility in filamentous ascomycetes. *Annu. Rev. Phytopathol.*, 30:201–224.
- Goidanich, G. 1947. Review of genus *Macrophomina* Petrak, *Ann. Sper. Agr. Rome. N.S.*, 1: 449-461.
- Gonzalez, D., Carling, D.E., Kuninaga, S., Vilgalys, R. and Cubeta, M.A. 2001. Ribosomal DNA systematics of *Ceratobasidium* and *Thanatephorus* with *Rhizoctonia* anamorphs. *Mycologia*, 93(6): 1138-1150.
- Gooday, G.W. 1975. Chemotaxis and chemotropism in fungi and algae. In *Primitive Sensory and Communication Systems: The Taxes and Tropisms of Microorganisms and Cells*, ed. MJ Carlile, p. 155–204.
- Goodman, R.N., Kiraly, Z. and Zatlin, M. 1967. The biochemistry and physiology of infectious plant disease. Princeton, New Jersey, Toronto, London, Melbourne, D. Van Norst and Co. p. 89-91.
- Govindappa, M., Lokesh, S., Raib, V. R., Naik, V. R. and Raju, S.G. 2010. Induction of systemic resistance and management of safflower *Macrophomina phaseolina* root-rot disease by biocontrol agents. *Archives of Phytopathology and Plant Protection*, 43(1): 26–40

- Govindappa, M., Ravishankar, R.V. and Lokesh, S. 2011. Screening of *Pseudomonas fluorescens* isolates for biological control of *Macrophomina phaseolina* root-rot of safflower. African Journal of Agricultural Research, 6(29): 6256-6266.
- Gregory, P.H. 1984. The fungal mycelium: an historical perspective. Trans. Br. Mycol. Soc., 82:1-11.
- Grover, R.K. and Sukhuja, P.K. 1981. Some pathological studies on *Rhizoctonia bataticole* leaf bight of mungbean. Indian Phytopath., 34: 24-29.
- Gupta, O. 1997. Occurrence of combined resistance to wilt and dry root rot in chickpea. Indian Journal of Pulses Research, 10:125-126.
- Gupta, O., Jharia, H.K. and Sharma, N.D. 2003. *Bacillus subtilis*: An effective antagonist of *Rhizoctonia bataticola* (Taub) Butler causing dry root rot of chickpea. Indian Journal of Pulses Research, 16: 42-46.
- Gupta, S.B., Thakur, K.S., Tedia, K. and Singh, A.K. 2006. Influence of *Trichoderma viride* on performance of chick pea in wilt complex area. Ann. Pl. Protec. Sci. 14: 120-124.
- Gupta, O., Patel, S. and Mishra, M. 2012. Diversity in isolates of *Rhizoctonia bataticola* causing dry root rot in chickpea from central India. JNKVV. Res. J., 46(3): 376-381.
- Gupta, O., Rathi, M. and Mishra, M. 2012. Screening for resistance against *Rhizoctonia bataticola* causing dry root-rot in Chickpea. Journal of Food Legumes, 25(2): 139-141.
- Gurha, S.N. and Trivedi, S. 2008. Status of soil borne pathogens infecting chickpea in Karnataka state. Annual Journal of Plant Protection Sciences, 16(1): 257-258.
- Gurha, S.N., Srivastava, M., Trivedi S. and Narain, U. 2007. Prospects of ecofriendly management of wilt and dry root rot in chickpea. Ecofriendly management of plant disease. p. 215-221.
- Haffmaster, D.E., Mc Laughlin, J.H., Roy, W.W. and Chester, K.S. 1943. The problem of dry root rot caused by *Macrophomina phaseolina*. Phytopathology., 33:1113-1114.
- Haigh, V.K. 1930. *Macrophomina phaseoli* (Maubl.) Ashby, the pycnidial stage of *Rhizoctonia bataticole* (Taub.) Butter. Trop. Agriculturist, 70: 77-79.
- Halsted, B.D. 1890. Fungal diseases of sweet potato. N.J. Agri. Exp. Sta. Bull., 76: 7-14.

- Hancock, J.G. and Miller, R.L. 1965. Association of cellulolytic proteolytic and xylolytic enzyme with southern anthracnose spring black stem and *stemphylium* leaf spot of alfalfa. *Phytopathology*, 55: 356-364.
- Hansford, C.G. 1943. Tomato diseases and their control. Jamaica Department of Agricultural Microbiology, p. 12.
- Haque, S.E., Ghaffar, A. and Zaki, M.J. 1990. Biological control of root rot disease of okra, sunflower, soybean and mungbean. *Pakistan J. Bot.*, 22(2): 121-124.
- Harman, G.C., 2000. Myths and dogmas of biocontrol-Changes in perceptions derived from research on *Trichoderma harzianum* T22. *Plant Disease*, 84: 377-393.
- Haware, M.P. 1990. *Fusarium* wilt and other important diseases of chickpea in the Mediterranean area. *Options mediterraneennes-serie seminaires*, 9: 61-64.
- Haware, M.P. 1998. Diseases of chickpea. In: *The Pathology of Food and Pasture Legumes*.
- Hay, F.S. 1995. Unusual germination of spores of *Arthrobotrys conoides* and *A.cladodes*. *Mycol. Res.*, 99: 981-82.
- Hegde, Y.R., Rajalaxmi, S.K. and Tippeshi, L.C. 2013. Biological management of *jatropha curcas* root rot of caused by *Rhizoctonia bataticola*. *Bioinfolet.*, 10(1A): 89 – 91.
- Hildebrand, A.A., Miller, J.J. and Koch, L.W. 1945. Some studies on *Macrophomina phaseolina* (Maubl.) Ashby in Ontario. *Sci. Agric.*, 25: 690-706.
- Hooda, I. and Grover, R.K. 1982. Studies on different isolates, age and quantity of inoculum of *Rhizoctonia bataticola* in relation to disease development in mung bean. *Indian-Phytopathology*, 35(4): 619-623.
- Hussain, S., Ghaffar, A and Aslam, M. 1990. Biological control of *Macrophomina phaseolina* charcoal rot of sunflower and mung bean. *Phytopathology*, 130: 157-160.
- Hwang, S.F., Gossen, B.D., Chang, K.F., Turnbull, G.D., Howard, R.J. and Blade, S.F. 2003. Etiology, impact and control of *Rhizoctonia* seedling blight and root rot of chickpea on the Canadian prairies. *Canadian Journal of Plant Sciences*, 83(4): 959-967.

- Ibrikci, H., Knewton, S. and Grusak, M.A. 2003. Chickpea leaves as a vegetable green for humans: Evaluation of mineral composition. *J. Sci. Food Agric.*, 83: 945-950.
- Ingle, V.N., Deshmukh, V.V., Jiotode, D.J., Chore, N.S. and Dhawad, C.S. 2004. Screening of safflower varieties against root rot (*Rhizoctonia bataticola*). *Res. on crops.*, 5(1): 113-114.
- Innocenti, G., Roberti, R., Montanari, M. and Zakrisson, E. 2003. Efficacy of microorganisms antagonistic to *Rhizoctonia cerealis* and their cell wall degrading enzymatic activities. *Mycol. Res.*, 107: 421-427.
- IIPR, 2013. Rabi 2013-2014 technical programme, annual meet, AICRP on chickpea pathology, IIPR at Durgapur, Jaipur (Rajasthan), p. 48.
- Jain, N.K., Khare, M.N. and Sharma, H.C. 1973. Variation among the isolates of *Rhizoctonia bataticola* from Urdbean plant parts and soil. *Mysore Journal of Agricultural Science*, 6: 461-465.
- Jana, T. K., Srivastava, A. K., Csery, K. and Arora, D. K. 2000. Agglutination potential of *Pseudomonas fluorescens* in relation to energy stress and colonization of *M. phaseolina*. *Soil Biology and Biochemistry*, 32: 511-519.
- Jayalakshmi, S.K., Usharani, S., Benagi, V.I. and Mannur, D.M. 2008. Sources of resistance to dry root rot of chickpea caused by *Rhizoctonia bataticola*. *Agric. Sci. Digest*, 28(2): 147 – 148.
- Jayanti, S. and Bhatt, J. 1993. Reaction of chickpea cultivars to *R. bataticola* (Taub.) Butler. *Indian J. Pulss Res.*, 6: 118-119.
- Jayashree, K., Shanmugam, V., Raguchander, T., Ramanathan, A. and Samiyappan, R. 2000. Evaluation of *pseudomonas fluorescens* (pf-1) against blackgram and sesame root-rot disease. *J. Biol. Control.*, 14(2): 55-61.
- Jharia, H.K. and Khare, M.N. 1985. Biological control of *Rhizoctonia bataticola* causing disease in soybean. *Indian Phytopathology*, 39: 148-153.
- Jimenez, D.R.C. 2006. Influence of soils, nutrition, and water relations upon charcoal rot disease processes in Kansas. B.S., Universidad Nacional de Colombia, Thesis, p. 56-95.

- Jyothirmai, T., Rao, V.K., Reddy, P.N. and Sankar, A.S. 2011. Effect of Plant Growth Promoting Microorganisms on Fungi Associated With Seedling Diseases in Groundnut. *Indian Journal of Plant Protection*, 39(3): 200-203.
- Kaisar, S.A.K.M. and Das, S.N. 1988. Physical factors that influence the growth and spread of charcoal rot pathogen (*Macrophomina phaseolina*) infecting maize. *J. Phytopathol.*, 123: 47-51.
- Kanchan, C. and Biswas, S.K. 2009 Morphological and Pathogenic Variability of *Rhizoctonia bataticola* (Taub) Butler, causal agent of Leaf Spot and Blight Disease of Pigeon pea. *Ann. Pl. Protec. Sci.*, 17(1): 124-126.
- Karthikeyan, V., Sankaralingam, A. and Nakkeeran, A. 2006. Management of groundnut root rot with biocontrol agents and organic amendment. *Archives of phytopath. and Pl. Protec.*, 39(3): 215 – 223.
- Karunanithi, K. 1996. Studies on root rot of sesamum (*Sesamum indicum* L.) caused by *Macrophomina phaseolina* (Tassi) Goid. *Ph. D. Thesis*, Tamil Nadu Agricultural University, Coimbatore, India.
- Kaswate, N.S., Shinde, S.S. and Rathod, A. 2003. Effect of biological agents against different isolates of *Rhizoctonia bataticola* (Taub.) Butler *in vitro*. *Annals-of-Plant-Physiology*, 17(2): 167-168.
- Kataria, H.R. and Grover, R.K. 1977. Comparison of fungicides for the control of *Rhizoctonia solani* causing damping off of mungbean (*Phaseolus aureus*). *Annals of Applied Biology*, 83: 79-85.
- Katariya, L., Gaur, V.K. and Sharma, R. 2007. Assessment of genetic variability in *Rhizoctonia bataticola* infecting chickpea isolates using pathogenicity and RAPD markers. *Indian Journal of Mycology and Plant Pathology*, 37(3): 491-494.
- Kaushal, R.P. 2008. Management of root rot in chickpea. *Journal of Food Legumes*, 21(3): 178-181.
- Kaushik, C.D., Chand, J.N. and Satyavir, R. 1987. Seed borne nature of *Rhizoctonia bataticola* causing leaf blight of mungbean. *Indian J. Mycol. Pl. Pathol.*, 17: 154-157.

- Kernkamp, M.F., De Zeeum, D.J., Chen, S.M., Ortega, B.C., Tisang, C.T. and Khan, A.M. 1952. Investigation on physiologic specialization and parasitism of *Rhizoctonia solani*. *Minnesota Agr. Expt. Sta. Tech. Bull.*, 200: 36.
- Khalid, I. and Ilyas, M.B. 2000. Screening of chickpea germplasm against dry root rot disease (*Macrophomina phaseolina*) in pots/glass house. *Pakistan Journal of Phytopathology*, 12:66-70.
- Khan, R.A., Bhat, T. A. and Krishan, K. 2012. Management of Chickpea (*Cicer arietinum* L.) Dry root Rot Caused by *Rhizoctonia bataticola* (Taub.) Butler. *International Journal of Research in Pharmaceutical and Biomedical Sciences.*, 3 (4): 1539-1548.
- King, E., Ward, M.K. and. Raney, D.E. 1954. Two simple media for the demonstration of pyocyanin and fluorescin. *J.lab. din. Med.*, 44: 301-307.
- Kolte, V.S. and Raut, S.P. 2007. Efficacy of *Trichoderma* spp. against *Sclerotium rolfsii* inciting collar dry rot of orchids. *Ann. Pl. Protec. Sci.*, 15: 516-517.
- Krishna, R. and Krishnappa, K. 1996. Interaction of *Fusarium oxysporum* f.sp.*ciceri* with *Meloidogyne incognita* on chickpea in two soil types. *Indian Phytopathology*, 49(2):142-147.
- Krishnamohan, G., Arjunan, G., Gangadharan, K. Kausalya-Ganga-dharan, Shammugam, N., Jeyarajan, R. and Vidhyasekaram, P. 1981. Reaction of different Bengal gram types to root rot caused by (Tassi). *Goid. Proc. of the National seminar on disease resistance in crop plants.* p. 76-77.
- Krishnamurthy, K. and Gnanamanickam, S.S. 1998. Bicontrol of rice sheath blight with formulated *Pseudomonas putida*. *Indian Phytopath.*, 51(3): 233-236.
- Kuhn, J. 1858. *Krankheiten der kulturgewachse.* p. 224-228.
- Kulkarni, V.R. 2007. Epidemiology and integrated management of Potato wilt caused by *Sclerotium rolfsii* Sacc. Ph.D. Thesis, University of Agricultural Sciences, Dharwad.
- Kumar, M., Gaur, V.K., Kant, K. and Sharma, A. 2013. Host range and pathogenic variation in isolates of *Macrophomina phaseolina* incitant of dry root rot in Moth bean. *Ann. Pl. Protec. Sci.*, 21 (2): 333-336.

- Kumar, N.R., Thirumalai, A.V. and Gunasekaran, P. 2002. Genotyping of antifungal compounds producing plant growth-promoting rhizobacteria, *Pseudomonas fluorescens*. Curr. Sci., 82(12): 1463-1466.
- Kumar, R. and Singh, S.B. 1996. Changes in biochemical constituents of sunflower leaves in relation to Alternaria blight development. Indian J. Mycol. and Pl. Pathol., 26: 234-236.
- Kumar, S., Sharma, S., Pathak, D.V. and Beniwal, J. 2011. Integrated management of jatropha root rot caused by Rhizoctonia bataticola. Journal of Tropical Forest Science, 23(1): 35-41.
- Kuninaga, S. 1980. Hyphal anastomosis behaviour of a single basidiospore isolates in Thanatephora cucumeris (Frank) Donk. Higashi Nippon Gakuen J. Liberal Art and Science., 6: 95-106.
- Kuninaga, S. 2002. Current situation of the taxonomy of the genus Rhizoctonia and the R. solani species complex. Jpn. J. Phytopathol., 68:3-20.
- Kuninaga, S. 2003. Current situation of the taxonomy of Rhizoctonia solani. Plant Protect., 57:219-222.
- Kuninaga, S. and Yokosawa, R. 1980. A comparison of DNA base compositions among anastomosis groups in Rhizoctonia solani Kuhn. Annals of the phytopathological society of Japan, 46(2): 150-158.
- Ladizinsky, G. 1975. A new Cicer from Turkey. Notes of the Royal Botanic Garden Edinburgh, 34: 201-202.
- Laha, G.S., Singh, R.P. and Verma, J.P. 1992. Biocontrol of R. solani in cotton by fluorescent Pseudomonads. Indian Phytopathol., 45: 412-415.
- Lee, I.H., Walline, R.G. and Plamann, M. 1998. Apolar growth of Neurospora crassa leads to increased secretion of extracellular proteins. Mol. Microbiol., 29: 209-18.
- Leslie, J.F. 1993. Fungal vegetative compatibility. Annu. Rev. Phytopathol., 31: 127-150.
- Leveille, J.H. 1843. Memoire sur le genre sclerotium. Ann. Sci. Nat. Bot., 20: 233.
- Lifshitz, R., Windham, M.T. and Baker, R. 1986. Mechanism of biological control of pre-emergence damping off of pea by seed treatment with Trichoderma spp. Phytopathol., 6:720-725.

- Lokesha, N.M. and Benagi, V.I. 2007. Studies on cultural variability of isolates of *Macrophomina phaseolina* (Tassi) Goid. Karnataka J.Agric.Sci., 17(4):721-724.
- Lokesha, N.M., Benagi, V.I. and Kulkarni, S. 2004. Reaction of pigeonpea genotype to *Macrophomina phaseolina* (Tassi) Goid. The incitant of dry root rot of pigeonpea. Indian Phytopath., 5(3): 387.
- Lorito, M., Woo, S.L., Fernandez, I.G., Collucci, G., Harman, G.E., Pintor-Toros, J.A., Filippone, E., Muccifora, S., Lawrence, C.B., Zoina, A., Tuzun, S. and Scala, F. 1998. Genes from mycoparasitic fungi as a source for improving plant resistance to fungal pathogens. *Proc Natl Acad Sci.*, 95: 7860- 7865.
- Manczinger, L., Antal, Z. and Kredics, L. 2002. Ecophysiology and breeding of mycoparasitic *Trichoderma* strains. *Acta Microbiol Immunol Hung.*, 49: 1-14.
- Mandal, R., Sinha, M.K., Ray, M.K.G., Mishra, C.B.P. and Chakrabarty, N.K. 1998. Variation in *M. phaseolina* causing stem rot of jute. *Environment and Ecology*, 16: 424-426.
- Mandhare, V.K. and Suryawanshi, A.V. 2008. Efficacy of some botanicals and *Trichoderma* species against soil borne pathogens infecting chickpea. *J. Food legumes*, 21(2): 122-124.
- Manjunatha, S.V. and Naik, M.K. 2010. Mechanism of antagonism of *Trichoderma* spp. on *Rhizoctonia bataticola* causing dry root rot of chickpea. *J. Pl. Dis. Sd.*, 5(1):163-165.
- Manjunatha, S.V. and Naik, M.K. 2011. Cultural and Morphological Diversity in Isolates of *Rhizoctonia bataticola* Causing Dry Root Rot of Chickpea. *J. Mycol. Plant Pathol.*, 41(2): 279-281.
- Manjunatha, S.V., Naik, M.K., Khan, M.F.R and Goswami, R.S. 2013. Evaluation of bio-control agents for management of dry root rot of chickpea caused by *Macrophomina phaseolina*. *Crop Protection*, 45:147-150.
- Marco, J.L.D., Lima, L.H.C., Souza, M.V.D. and Felix, C.R. 2000. A *Trichoderma harzianum* chitinase destroys the cell wall of the phytopathogen *Crinipellis perniciosa*, the causal agent of witches broom disease of cocoa. *World J Microbiol Biotechnol.*, 16: 383–386.

- Marisa, R., Pastor, N., Niederhauser, M. and Rosas, S.B. 2014. Evaluation of *Pseudomonas chlororaphis* subsp. *aurantiaca* SR1 for growth promotion of soybean and for control of *Macrophomina phaseolina*. *Biocontrol Science and Technology*, 24(9): 1012-1025.
- Martin, W.H. 1917. *Sclerotium bataticola*- the caus *Sclerotium bataticola* of fruit rot of peppers. *Phytopath.*, 7: 308-312.
- Mathur, S.B. and Sinha, S. 1968. Disease development in guar (*Cyamopsis psoraloides* D.C.) and gram (*Cicer arietinum* L.) attacked with *Sclerotium rolfsii* under different soil pH conditions. *Phytopathology Zeitschrift*, 62:319-322.
- Matsumoto, T., Yamamoto, W. and Hirane, S. 1932. Physiology and parasitism of the fungi generally referred to as *Hypochnus sasakii* Shirai. Differentiation of strains by means of hyphal fusion and culture in differential media. *J. Soc. Trop. Agr.*, 4: 370-388
- Maublanc, A. 1905. *Especies noveues de.Champignous inferieurs. Soc. Muco. De. France. Bull.*, 21: 81-94.
- Maurice S.M., Kamel K.S., Moawad, R.O., Amal A.A. and Khaled K.K. 2010. Current *Rhizoctonia solani* anastomosis groups in Egypt and their pathogenic relation to cotton seedling. *African Journal of Microbiology Research*, 4(5), p. 386-395.
- Mayee, C.D. and Garud, T. B. 1978. Mass culturing of *M. phaseolina*. *Indian Phytopath.*, 31:129.
- Mayer, M.S., Tullu. A., Simon, C.J., Kumar, J., Kaiser, W.J., Kraft, J.M. and Muehlbauer, F.J. 1997. Development of a DNA marker for *Fusarium* wilt resistance in chickpea. *Crop Sci.*, 37: 1625-1629.
- Mazzola, M. and Gu, Y. 2002. Wheat genotype-specific induction of soil microbial communities suppressive to disease incited by *Rhizoctonia solani* anastomosis group (AG)-5 and AG 8. *Phytopathology*, 92: 1300-1307.
- Mc Rae, 1929. New diseases reported during the year 1928 from India. *International Bulletin of Plant protection*, 3: 21-22.

- Mc Cabe, P.M., Gallagher M.P., and Deacon, J.W. 1999. Microscopic observation of perfect hyphal fusion in *Rhizoctonia solani*. Mycol. Res., 103: 487–90.
- McIntosh, G.H. and Topping, D.L. 2000. Food legumes in human nutrition. In: Linking Research and Marketing Opportunities for Pulses in the 21st Century, *Proceedings of the Third International Food Legumes Research Conference*, 22–26 September 1997, Adelaide, Australia. Current Plant Science and Biotechnology in Agriculture, Vol. 34 (ed. by R. Knight), Kluwer Academic Publishers, London, UK. p. 655-660.
- McIntyre, M., Nielsen, J., Arnau, J., Brink, V.H. and Hansen, K. 2004. Proceedings of the 7th European Conference on Fungal Genetics. Copenhagen, Denmark.
- Mehrotra, R.S. 1988. Plant pathology. Tata Mc Graw-Hill Publishing Company Ltd., New Delhi, p. 771.
- Mehta, S.M. 2004. Epidemiology and management of leaf blight of mungbean (*Vigna radiata* Wilczek.) caused by *Macrophomina phaseolina* (Tassi) Goid. p. 56-90.
- Metcalf, D.D. and Wilson, C.R. 2001. The process of antagonism of *Sclerotium cepivorum* in white rot affected onion roots by *Trichoderma koningii*. Plant Pathol., 50: 249-257.
- Mikhail, M.S., Kamel, K.S., Moawad, R.O., Amal, A.A. and Khaled, K.K. 2010. Research Paper Current *Rhizoctonia solani* anastomosis groups in Egypt and their pathogenic relation to cotton seedlings. African Journal of Microbiology Research, 4(5): 386-395.
- Mirza, M.S. 1984. Occurrence of sunflower diseases in Pakistan, *In Proceedings of the National Sunflower Workshop, PARC*, p. 31-32.
- Mishra, A.N., Prasad, S.N. and Ram, R.M. (2005). Search for source of resistance of dry root rot in chickpea. Ann. Pl. Protec. Sci., 13(2): 465-529.
- Mohammadi, M., Banihashemi, M., Hedjaroude, G. and Rahimian, H. 2003. Genetic diversity among Iranian isolates of *Rhizoctonia solani* Kühn anastomosis Group1 subgroups Based on Isozyme Analysis and Total Soluble Protein Pattern. Journal of Phytopathology, 2(6):89-93.
- Mohan, R. and Balabaskar, P. 2012. Survey on the incidence of groundnut root rot disease in cuddalore district of Tamil nadu and assessing the cultural characters and

- pathogenicity of *Macrophomina phaseolina* (Tassi.) Gold. Asian Journal of Science and Technology, 3(4): 90-94, 2012.
- Monga, D. and Sheo, R. 1995. Cultural and pathogenic variation in the isolates of *Rhizoctonia* species causing root rot of cotton. Indian Phytopath., 47: 403–408.
- Monteiro, V.N., Silva, R.N., Steindorff, A.S., Costa, F.T. and Noronha, E.F. 2010. New insights in *Trichoderma harzianum* antagonism of fungal plant pathogens by secreted protein analysis. Curr Microbiol., 61: 298–305.
- Moreno, M.T. and Cubero, J.I. 1978. Variation in *Cicer arietinum* L. Euphytica, 27: 465-485.
- Mungan, G. and Yildiz, M. 1990. Investigations on the incidence of tobacco charcoal rot disease (*Macrophomina phaseolina*) in the Aegean region; its pathogenecity and susceptibility of Turkish tobacco cultivars. J. Turk. Phytopath., 19: 13-19.
- Muthukrishnan, K., Arjunan, G. and Raguchander, T. 1995. Factors affecting survival of *Macrophammina phaseolina* in soil. Indian J. Pulses Res., 8(2): 156-161.
- Muthusamy, S. and Mariappan, V. 1991. Disintegration of sclerotia of *Macrophomina phaseolina* (Soybean isolate) by oil cake extract. Indian Phytopahology, 44(2): 271-273.
- Mylyk, O.M. 1976. Heteromorphism for heterokaryon incompatibility genes in natural populations of *Neurospora crassa*. Genetics, 83: 275–84.
- Nagamani, P., Viswanath, K. and Kiran Babu T. 2014. Management of dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler in chickpea. Current Biotica, 5(3): 364-369.
- Naik, M.K. and Sen, B. 1995. Bio-control of plant disease caused by *Fusarium* spp. In: Mukherjee, K.G. (Ed.), Recent Developments in Bio-control of Plant Diseases. Aditya Publishing House, India. p. 32.
- Naik, M.K., Singh, S.J. and Sinha, P. 2000. Mechanism of biocontrol of wilt of chilli caused by *Fusarium oxysporum* f. sp. *capsid*. Proceedings of Golden Jubilee International Conference on Integrated Plant Disease Management for Sustainable Agriculture, Indian Phytopathological Society, IARI, New Delhi, p. 665.

- Nair, R., Thakur, K.D., Shubhangi, B.J.V. and Tirthakar S. 2006. Control of root rot (*Rhizoctonia bataticola*) of sesamum by different bioagents. J. Pl. Dis. Sci., 1(2): 236-237.
- Naito, S. and Sugimoto, T. 1978. Basidio-spore infection and lesion development on sugar beet leaves by *Thanatephorus cucumeris* (Frank) Donk. Ann Phytopath Soc Japan, 44: 426–431.
- Narain, U. 1983. Some New diseases of sunflower from India. India Phytopath., 36-196.
- Nautiyal, C.S. 1997. Selection of Chickpea-Rhizosphere-Competent *Pseudomonas fluorescens* NBRI1303 Antagonistic to *Fusarium oxysporum* f. sp. *ciceri*, *Rhizoctonia bataticola* and *Pythium* spp. Curr. Microbiol., 35(1): 52-58.
- Nees, V.E.T. 1817. System der pilzeund schuiamme wurzhurg negoc da agr, Indus. E. Com. Bol., p. 99.
- Nene, Y.L., Haware, M.P. and Reddy, M.V. 1978. Diagnosis of some wilt like disorders of chickpea (*Cicer arietinum* L.) *Information Bulletin* No. 3, ICRISAT. p. 44.
- Nene, Y.L., Kannaiyan, J., Haware, M.P. and Reddy, M.V. 1979. Review of the work done at ICRISAT on soil-borne diseases of peageon pea and chickpea. p. 14.
- Nene, Y.L. 1979. *Proceeding international workshop on chickpea improvement*. p.171-178.
- Nene, Y.L. 1980. ICRISAT (International Crops Research institute for the Semi -Ar id Tropics). 1980. *Proceedings of the international Workshop on Chickpea Improvement*, 28Feb-2 Mar 1979, Hyderabad, A.P., India.
- Nene, Y.L., Haware, M.P. and Reddy, M.V. 1981. Chickpea disease resistance screening techniques. *Information Bulletin* ICRISAT, 10: 10-11.
- Nene, Y.L., Reddy, M.V., Haware, M.P., Ghanekar, A.M. and Amin K.S. 1991. Field Diagnosis of Chickpea Diseases and their control. *Information Bulletin*, ICRISAT 10:29- 30.
- Nene, Y.L., Sheila, V.K. and Sharma, S.B. 1996. A world list of chickpea and pigeon pea pathogens. Fifth edition, ICRISAT, Patencheru 502324, Andhra Pradesh, India (Semi-formal publication). p. 27.
- Nene, Y.L., Sheila, V.K., and Sharma, S.B. 1996. A world list of chickpea (*Cicer arietinum* L.) and pigeonpea (*Cajanus cajan* (L.) Mill) pathogens. International

- Crops Research Institute for the Semi-Arid Tropics, Patancheru 502324, Andhra Pradesh, India 5: p. 3.
- Newhouse, J.R. and MacDonald, W.L. 1991. The ultrastructure of hyphal anastomoses between vegetatively compatible and incompatible virulent and hypovirulent strains of *Cryphonectria parasitica*. Can. J. Bot., 69:602–14.
- Ogoshi, A. 1975. Grouping of *Rhizoctonia solani* Kühn and their perfect stages. Rev. Plant. Protect. Res., 8: 93-103.
- Ogoshi, A. 1978. Interspecific groups of *Rhizoctonia solani* and their significance. 9th Symp. Soil borne plant disease. Morioka: Phytopathol. Soc. Jpn. pp., 43-50.
- Ogoshi, A. 1985. Anastomosis and interspecific groups of *Rhizoctonia solani* and binucleate *Rhizoctonia*. Fitopathologia Brasileira., 18: 372-390.
- Ogoshi, A. 1987. Ecology and pathogenicity of anastomosis and intraspecific groups of *Rhizoctonia solani* Kuhn. Annu. Rev. Phytopathol., 25:125–143.
- Oniki, M., Ogoshi, A., Araki, T., Sakai, R. and Tanaka, S. 1985. The perfect state of *Rhizoctonia oryzae* and *R. zae* and the anastomosis groups of *Waitea circinata*. Transactions of the Mycological Society of Japan, 26: 189-198.
- Padwick, G.W. 1948. Plant protection and food crops of India, plant pest and disease at rice, wheat, sorghum and gram. Imperial J. Exp. Agri., 16: 55-64.
- Pande, S., Kishore, G.K. and Rao, J.N. 2004. Evaluation of chickpea lines for resistance to dry root rot caused by *Rhizoctonia bataticola*. SO: International-Chickpea-and-Pigeonpea Newsletter, (11): 37-38.
- Pandit, A and Maheshwari, R. 1996. A demonstration of the role of *het* genes in heterokaryon formation in *Neurospora* under simulated field conditions. Fungal Genet. Biol., 20: 99–102.
- Pant, R. and Mukhopadhyay, A.N. 2001. Integrated management of seed and seedling rot complex of soybean. Indian phytopath., 54(3): 346-350.
- Papavizas, G.C. 1977. Some factors affecting survival of sclerotia of *Macrophomina phaseolina* in soil. Soil Biology and Biochemistry, 9(5): 337-341.
- Parmar, R. 2013. Studies on cultural and morphological variability in *Rhizoctonia bataticola* (Taub.) causing dry root rot in soybean-chickpea system.

- M.Sc. thesis, Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior (M.P.), p. 55-59.
- Parmeter, J.R.J. 1970. *Rhizoctonia solani*: Biology and Pathology. Based on American Phytopathology Society, Symposium on *Rhizoctonia solani* held at the Miami meeting of the society. October 1965. Edited by J.R. Parmeter. University of California Press, Berkley, USA. p. 32-44.
- Parmeter, J.R.J. and Whitney, H.S. 1970. *Rhizoctonia solani*: Biology and Pathology. In Taxonomy and nomenclature of the perfect state. Edited by J.R. Parmeter. University of California Press, Berkley, USA. p. 6-19.
- Parmeter, J.R.J. and Whitney, H.S. 1970. Taxonomy and nomenclature of the imperfect state. Pages 7-19 in: J. R. Parmeter Jr., (ed.) Biology and Pathology of *Rhizoctonia solani*. University of California Press, Berkeley. p. 255.
- Parmeter, J.R.J., Sherwood, R.T. and Platt, W.D. 1969. Anastomosis grouping among isolates of *Thanatephorus cucumeris*. Phytopathology, 59: 1270-1278.
- Patel, D.P., Singh, H.B., Shroff, S. and Sahu, J. 2011. Antagonistic efficiency of pseudomonas strains against soil borne disease of chickpea crop under *in vitro* and *in vivo*. Elixir Agriculture, 30: 1774-1777.
- Paul, Y.S., Devi, M. and Kapoor, A.S. 2008. Integrated organic management of wilt-root rot complex of pea. Journal of Mycology and Plant Pathology, 38(3): 571-576.
- Peltier, G.L. 1916. Parasitic Rhizoctonias in America. *Uni. Ill. Agri. Ex. St. Bull.* No.189.
- Perkins, D.D., Radford, A. and Sachs, M.S. 2000. The *Neurospora compendium: Chromosomal Loci*. San Diego: Academic. In press
- Persoon, C.H. 1801. "Synopsis methodica fungurum". p. 190-120.
- Philip, C.T., Kartha, K.K., Joshi, R.K. and Nema, K.G. 1969. A new *Rhizoctonia* disease of mungbean in Madhya Pradesh. JNKVV. Reseach Journal, 3: 40-43.
- Pineda, J., Nass, H. and Rodriguez, H. 1985. Effect of the inoculum density of *Macrophomina phaseolina* on infection of sesame seedlings. Agronomia Tropical, 35(4):133-138.

- Pineda, J.B. and Avila, M.J. 1993. Losses caused by *Macrophomina phaseolina* on sunflower crop *Helianthus annuus*. *Agronomia Tropical Maracay*, 43(5-6): 241-251.
- Pontecorvo, G. 1956. The parasexual cycle in fungi. *Annu. Rev. Microbiol.*, 10: 393–400.
- Poromarto, S.H., Nelson, B.D., Freeman, T.P. 1998. Association of binucleate *Rhizoctonia* with soybean and mechanism of biocontrol of *Rhizoctonia solani*. *Phytopathol.*, 88: 1056-1067.
- Prajapati, R.K., Gangwar, R.K. and Srivastava, A. 2003. Resistant source of chickpea against dry root rot. *Farm-Science-Journal*, 12(1): 86
- Prameela, T. and Singh, R.H. 1998. Cultural variation of *Macrophomina phaseolina* isolates collected from *Vigna mungo*. *Indian Phytopathol.*, 51(1): 292–293.
- Prosser, J.I. 1994. Kinetics of filamentous growth and branching. In *The Growing Fungus*, ed. NAR Gow, GM Gadd, p. 301–318.
- Puhalla, J.E. and Speith, P.T. 1985. A comparison of heterokaryosis and vegetative incompatibility among varieties of *Gibberella fujikuroi* (*Fusarium moniliforme*). *Exp. Mycol.*, 9: 39–47.
- Purkayastha, S., Kaur, B. and Chaudhury, A. 2004. Cultural and pathogenic variation in the Charcoal rot pathogen from clusterbean. *Ann. Agri. Biol. Res.*, 9: 217–221.
- Raaijmakers, J.M. and Weller, D.M. 1998. Natural plant protection by 2,4-diacetylphloroglucinol producing *Pseudomonas* spp. In take all decline soils. *Mol. Plant-Microbe Interact.*, 11: 144-152.
- Raaijmakers, J.M., Vlami, M., Mand J.T. de Souza. 2002. Antibiotic production by bacterial biocontrol agents. *Antonie van Leeuwenhoek*, 81: 537-547.
- Raguchander, T., Rajappan, K. and Samiyappan, R. 1998. Influence of biocontrol agents and organic amendments on soybean root rot. *International Journal of Tropical Agriculture*, 16(1): 247-252.
- Rajeswari, B., Rao, K.C. and Kumar, C.P.C. 1999. Efficacy of antagonists and carbendazim against dry root rot of mungbean (*Vigna radiata* (L.) Wilczek) incited by *Macrophomina phaseolina* (Tassi) Goid under glasshouses conditions. *J. of Biological Control*, 13: 93-99.

- Rajpurohit, T.S. 2004. *Trichoderma viride* biocontrol agent effective against *Macrophomina* and root rot of sesame. J. Eco-physiol., 7(3/4): 187-188.
- Ramezani, H. 2008. Biological control of root rot of eggplant caused by *Macrophomina phaseolina*. American Eurasian J. Agric. Env. Sci., 4(2): 218-220.
- Rangaswamy, G. 1996. Diseases of crop plants. III rd ed. p. 257.
- Rashad, Y.M., Abdel-Fattah, G. M., Hafez, E. E. and El-Haddad, S. A. 2012. Diversity among some Egyptian isolates of *Rhizoctonia solani* based on anastomosis grouping, molecular identification and virulence on common bean. African Journal of Microbiology Research, 6(37): 6661-6667.
- Rasu, T., Duraisamy S., Nachimuthu R. and Ramasamy S. 2007. A combination of biocontrol agents improves the management of dry root rot (*Macrophomina phaseolina*) in greengram. Phytopathol. Mediterr., 46: 157–167.
- Ratan, V. and Biswas, S.K. 2010. Influence of date of sowing on incidence of dry root rot of chickpea. Annuals of Plant Protection Science, 18 (1): 258-259.
- Ratnoo, R.S., Jain, K.L. and Bhatnagar, M.K. 1997. Variations in *Macrophomina* isolates of ash-grey stemblight of cowpea. J. Mycol. Pl. Pathol., 27: 91-92.
- Rauf, C.A. I., Ahmad and Ashraf, M. 2007. Anastomosis groups of *Rhizoctonia Solani* Kuhn isolates from potato in Pakistan. Pak. J. Bot., 39(4): 1335-1340.
- Raut, J.G. and Ingle, R.W. 1989. Variations in isolates of *Rhizoctonia bataticola*. Indian Phytopathology, 42(4): 506-508.
- Rayner, A.D.M. 1996. Interconnectedness and individualism in fungal mycelia. In *A Century of Mycology*, ed. BC Sutton, p.193–232.
- Reddy, M.V., Raju, R.N. and Pundir, R.P.S. 1990. Additional sources of resistance to wilt and root rot in chickpea. International Chickpea Newsletter, 22: 36-38.
- Riker, A.J. and Riker, R.S. 1936. Introduction of Research on Plant Disease. John Swift Co., St. Louis, pp.117.
- Riquelme, M., Reynaga-Pena, C.G., Gierz, G. and Bartnicki-Garcia, S. 1998. What determines growth direction in fungal hyphae? *Fungal Genet. Biol.*, 24: 101–9.

- Riyaz, A.K. and Towseef, A.B. 2012. Management of chickpea dry root rot caused by *Rhizoctonia bataticola*. I. J. Res. P. B., Sci., 3: 1539-1548.
- Sajeena, A., Salalrajan, F., Seetharaman, K. and Mohan Babu, R. 2004. Evaluation of biocontrol agents against dry root rot of blackgram (*Vigna mungo*). Journal of Mycology and Plant Pathology., 34(2): 341-343.
- Samanta, S.K. and Dutta, S. 2004. Potential of native plant growth promoting rhizobacteria in the management of *Sclerotinia* stem rot of mustard. J. Mycol. Pl. Pathol., 34(3): 761- 768.
- Samuels, G.J. 1996. *Trichoderma*: A review of biology and systematics of the genus. Mycol Res., 100: 923–935.
- Sanecha, P. and Srivastava, H.P. 1982. Leaf blight of *Vigna aconitifolia* caused by *Macrophomina phaseolina*. Indian phytopath., 35: 160-161.
- Sankar, S. and Jeyarajan, R. 1995. Biological control of sesame root rot by seed treatment with *Trichoderma* sp. and *Bacillus subtilis*. Indian Journal of Mycology and Plant Pathology, 26: 217-220.
- Satischandra, K.M. and Hegde, R.K. 1980. Factors affecting the survival of *R. bataticola* in black soil. Plant and soil, 54: 307-312.
- Sawada, K. 1916. A new stem rot of jute plant caused by *M. carchori* sp. nov. Butl. Agric. Expt. Stn. Formosa: 107.
- Saxena, M.C. 1990. Problems and potential of chickpea production in the nineties. In: Chickpea in the Nineties. *Proceedings of the Second International Workshop on Chickpea Improvement*, 4-8 December 1989. Patancheru, India: ICRISAT. p.13-27.
- Schirmbock, M., Lorito, M., Wang, Y.L., Hayes, C.K. and Arisan-atac, I. 1994. Parallel formation and synergism of hydrolytic enzymes and peptaibol antibiotics, molecular mechanisms involved in the antagonistic action of *Trichoderma harzianum* against phytopathogenic fungi. Appl. Environ. Microbiol., 60: 4364-4370.
- Schneider, R. 1953. Untersuchungen über feuchtigkeitsansprüche parasitischer Pilze. Phytopathol. Z., 21: 63-78.

- Schultz, H. 1936. Vergleichende untersuchungen zur okologie, morphologie and systematik des vermehrungspilzes. Arb. Boil. Reichsanst. *Land –u-b Forst- Wirts.*, 22: 1-141.
- Selvarajan, R. and Jeyarajan, R. 1996. Inhibition of chickpea root rot pathogens (*Fusarium solani* and *Macrophomina phaseolina*) by antagonists. Indian J. Mycol. Pl. Pathol., 26: 248-251.
- Sen, B. 2000. Biological control: a successes story. Indian Phytopath., 53(3): 243-249.
- Sen, R., Ari, M. H. and Carla, D.Z. 1999. Common anastomosis and internal transcribed spacer RFLP groupings in binucleate *Rhizoctonia* isolates representing root endophytes of *Pinus sylvestris*, *Ceratorhiza* spp. from orchid mycorrhizas and a phytopathogenic anastomosis group, 144(2): 331–341.
- Senecha, P. and Srivastava, H.P. 1982. Leaf blight of *Vigna aconitifolia* caused by *Macrophomina phaseolina*. Indian phytopath., 35: 160-161.
- Shahzad, S. and Ghaffer, A. 1986. *Macrophomina phaseolina* on some new hosts FAO *Plant Protection Bulletin*. 34: 163.
- Shalini, S. and Kotasthane, A.S. 2007. Parasitism of *Rhizoctonia Solani* by strains of *Trichoderma* spp. *EJEAF Chemistry*, 6: 2272-2281.
- Shamarao, J., Patil, M. S. and Indira, S. 2001. Biological control of charcoal rot of sorghum caused by *Macrophomina phaseolina*. Agri. Sci. Digest, 21 (3): 153-156.
- Shanmugam, N. and Govindaswamy, C.V. 1971. Irregular leaf spot of groundnut caused by *Macrophomina phaseoli* (Maubl) Ashby. Madras Agric. J., 58: 30-32.
- Shanmugam, N. and Govindswamy, C.V. 1973. Control of *Macrophomina phaseolina* root rot of groundnut. Madras Agric. J., 60: 500-503.
- Sharma, B.K. 1994. Efficacy of biocontrol agents for the control of chickpea stem rot. J. Biol. Control., 8:1.
- Sharma, S.K., Verma, B.R. and Sharma, B.K. 1999. Biocontrol of *Sclerotinia sclerotiorum* causing stem rot of chickpea. Indian Phytopath., 50: 315-317.
- Sharma, K.K., Zaidi, N.W., Pundhir, V.S. and Singh, U.S. 2010. Study of genetic diversity in rhizospheric *Trichoderma* isolates from Uttarakhand. Ann. Pl. Protec. Sci., 18: 403-410.

- Sharma, K.K., Zaidi, N.W., Pundhir, V.S. and Singh, U.S. 2012. Biological control of phytopathogens using *Trichoderma* isolated from rhizospheric soil of Uttarakhand. *Ann. Pl. Protec. Sci.*, 20: 392-396.
- Sharma, M., Ghosh, R. Sharma, T.R. and Pande, S. 2012. Intra population diversity in *Rhizoctonia bataticola* causing dry root rot of chickpea (*Cicer arietinum* L.) in India. *African Journal of Microbiology Research*, 6(37): 6653-6660.
- Sharon, E., Bar-Eyal, M., Chet, I., Herra-Estrella, A., Kleifeld, O. and Spiegel, Y. 2001. Biological control of the rootknot nematode *Meloidogyne javanica* by *Trichoderma harzianum*. *Phytopathol.*, 91: 687-693.
- Sharon, M., Kuninaga, S., Hyakumachi, M. and Sneh, B. 2006. The advancing identification and classification of *Rhizoctonia* spp. using molecular and biotechnological methods compared with the classical anastomosis grouping. *Mycoscience*, 47: 299–316.
- Sharon, M, Kuninaga, S, Hyakumachi, M, Naito, S., and Sneh, B. 2008. Classification of *Rhizoctonia* spp. using rDNA-ITS sequence analysis supports the genetic basis of the classical anastomosis grouping. *Mycoscience*, 49: 93–114.
- Shaw, F.J.F. 1912. The morphology and parasitism of *Rhizoctonia*. *Indian Bot. Sci.*, 4:6.
- Shekhar, M., Sharma, R.C., Singh, L. and Dutta, R. 2006. Morphological and pathogenic variability of *Macrophomina phaseolina* (Tassi.) Goid, incitant of charcoal rot of maize in India. *Indian Phytopath.*, 59(4) : 453-459.
- Shim M.Y and Star J. L. 1999. Effect of Soil pH on Sclerotial Germination and Pathogenicity of *Sclerotium rolfsii*. *Peanut Science*, 24(1):17-19.
- Shukla, D.N. and Bhargava, S.N. 1976. Some pathogenic fungi from pulse and oil seed crops. *Proc.Nat.Acad.Sci.India*, 46(B): 530-531.
- Siddiqui, I.A. and Ehteshamul-Haque, S. 2001. Suppression of the root rot-root knot disease complex by *Pseudomonas aeruginosa* in tomato: The influence of inoculum density, nematode population, moisture and other plant associated bacteria. *Plant and Soil.*, 237: 81-89.
- Sinclair, J.B. 1982. Compendium of soybean diseases. *American Phytopathological Society*. p. 104.

- Sindhan, G.S., Hooda, I. and Karwasra, S.S. 2002. Biological control of dry root rot of chickpea caused by *Rhizoctonia bataticola*. Pl. Dis. Res., 17(1): 68-71.
- Singh, R.S. 1953. A leaf blight of guar. Sci. Cult., 19: 155-156.
- Singh, P.J. and Mehrotra, R.S. 1982. Biological control of *Rhizoctonia bataticola* on gram by coating seeds with *Bacillus* and *Streptomyces* sp. and their influence on plant growth. Plant and Soil, 56: 575-583.
- Singh, D. and Nema, K.G. 1987. Effect of soil amendment on *Rhizoctonia bataticola* causing dry root rot of chickpea. *Int. Chickpea Newsl.*, 17:23-25.
- Singh, K.B. 1987. Chickpea breeding. In: MC Saxena and KB Singh (Eds.), The Chickpea, C.A.B. International Wallingford, UK. p. 127-162.
- Singh, S.K., Nene, Y.L. and Reddy, M.V. 1990, Effect of age on susceptibility of chickpea to *R. bataticola*. *International Chickpea Newsletter*, 23: 25-26.
- Singh, R.P. and Gandhi, S.K. 1991. Effect of soil pH and temperature on seedling mortality of guar caused by *Sclerotium rolfsii* and its fungicidal control. *Indian Phytopathology*, 44: 360-365.
- Singh, S.K., Rahman, S.J., Gupta, B.R. and Kelha, C.S. 1992. Integration of pesticide application schedules for disease and insect pest management in chickpea under dryland conditions. *Indian J. Pl. Protec.*, 20: 158-161.
- Singh, S.D. 1999. Soil borne diseases of chickpea. File under plant disease B 10 (field crops). Issued March 1999. 4782000399 M.S.S.D. Singh @Cgiar.org/http://grep.icrisat.cgiar.org.
- Singh, A. and Sirohi, A. 2003. Status of chickpea diseases in Himachal Pradesh, India. *International Chickpea and Pigeonpea News letter*, 10: 29-31.
- Singh, A.S. and Agarwal, S.N. 2002. Prevalence of chickpea wilt and dry root rot diseases in Vindhyan plateau zone of Madhya Pradesh. *Indian Journal of Pulses Research*, 15(2): 14-19.

- Singh, B. and Dubey, S.C. 2005. Cultural and morphological variability of *Phytophthora drechsleri* sub.sp.*cajanica* causing pigeon pea blight. Ann. Pl. Protec. Sci., 13: 508-510.
- Singh, S., Chand, H. and Varma, P.K. 2008. Screening of bioagents against root rot of mung bean caused by *Rhizoctonia solani*. Legume Res., 31: 75-76.
- Small, W. 1924. A *Rhizoctonia* causing root disease in Uganda. Trans. Brit. My. Sec., 9: 152-166.
- Small, W. 1926. *S. bataticola* (Taub.). Trop. Agri., 67: 237-241.
- Smiley, R.W., Ogg, A.G. and Cook, R.J. 1992. Influence of glyphosate on *Rhizoctonia* root rot, growth, and yield of barley. Plant Disease, 76: 937-942.
- Smith, W.H. 1969. Germination of *Macrophomina phaseolina* sclerotia as affected by *Pinus lamberitina* root exudes. Canadian J. Microbiol., 15: 387-391.
- Sobti, A.K. and Sharma, L.C. 1992. Cultural and pathogenic variation in isolates of *Rhizoctonia bataticola* from groundnut in Rajasthan. Indian Phytopath., 45: 117-119.
- Srivastava, S.K. and Dhawan, A. 1979. Epidemiology of *Macrophomina* stem and root rot of *Brassica juncea* in northern India, Proc. Indian Nat. Sci. Acad. B., 45: 617-622.
- Srivastava, S.K. and Dhawan, S. 1980. Pathogenicity of *Macrophomina phaseolina* isolates causing stem and root rot of *Brassica juncea* effect of varying soil texture, soil reaction and soil moisture. Proceedings of the Indian National Science Academy Part B Biological Sciences, 46: 723-727.
- Subramanian, S., Subramaniam, T. and Devadason, A. 2011. Exploration of molecular variability in *Rhizoctonia bataticola*, the incitant of root rot disease of pulse crops. Journal of plant protection research, 51(2): 289-293.
- Sugha, S.K., Kapoor, S.K and Singh, B.M. 1994. Factors influencing Fusarium wilt of chickpea (*Cicer arietinum* L.). Indian Journal of Mycology and Plant Pathology. 24: 97-102.
- Sulaiman, M. and Patil, B.C. 1966. Existence of physiological race of *Macrophomina phaseolina* causing root rot of cotton. Beitrage, tropicale subtropical land wirtsch Tropen lenterinarmed, 4: 291-298.

- Sundar, S., Singh, R., Dodan, D.S. and Kataria, H.R. 2009. Effect of soil type and amount of inoculums on fungicidal control of *Rhizoctonia solani* incited seedling mortality of mungbean. *J. Mycol Pl. Pathol.*, 3(1): 48-51.
- Sundravadana, S., Alice, D. and Thirumurugan, S. 2012. Exploration of variability in colony morphology and virulence of *Rhizoctonia bataticola* isolates causing dry root rot of pulses. *G.J.B.B.*, 1(1): 91-97.
- Suriachandraselvan, M., Salarajan, F., Aiyathan, K.E.A. and Seetharaman, K. 2004. Inhibition of sunflower charcoal rot pathogen, *Macrophomina phaseolina* by fungal antagonists. *J. Mycol. Plant Pathol.*, 34: 364-366.
- Sutton, J.C. and Peng, G. 1993. Biocontrol of *Botrytis cinerea* in strawberry leaves. *Phytopathology*, 83: 615-621.
- Talbot, P.H.B. 1970. Taxonomy and nomenclature of the perfect state. In: *Biology and Pathology of Rhizoctonia solani*. J. R. Parmeter Jr., (ed.) University of California Press, Berkeley.
- Taubhanus, J.J. 1913. The black rots of the sweet potato. *Phytopathology*. 3: 159-165.
- Taya, R.S., Tripathi, M.H. and Pawan, M.S. 1990. Influence of soil type, soil moisture and fertilizers on the severity of chickpea dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler. *Indian J. Pl. Pathol.*, 18: 133-136.
- Than, H., Thein, M. and Mvint, S. 1991. Relationship among *Rhizoctonia bataticola* isolates in rice based cropping system based on colony fusion types. *Inter. Chickpea Newsl.* 25: 29-31.
- Thilagavanthi, R., Saravankumar, D., Raghupati, N. and Samiyappan, R. 2007. A combination of bio-control agents improve the management of dry root rot (*Microphomina phaseolina*) in greengram. *Phytopathol.*, 46: 157-167.
- Tu, C.C. and Kimbrough, J.W. 1978. Systematic and phylogeny of fungi in the *Rhizoctonia* compl. *Botanicals gazette.*, 139: 454-466.
- Tulasne, L.R. and Tulasne, C. 1862. "Fungi Hypogaei". p. 188-195.
- Tuncer, T.S. and Eken, C. 2013. Anastomosis Grouping of *Rhizoctonia solani* and Binucleate *Rhizoctonia* spp. Isolated from Pepper in Erzincan. *Plant Protect. Sci.*, 4(3): 127-131.
- Udit, N. 1983. Some New diseases of sunflower from India. *IndiaPhytopath.*, 36-196.

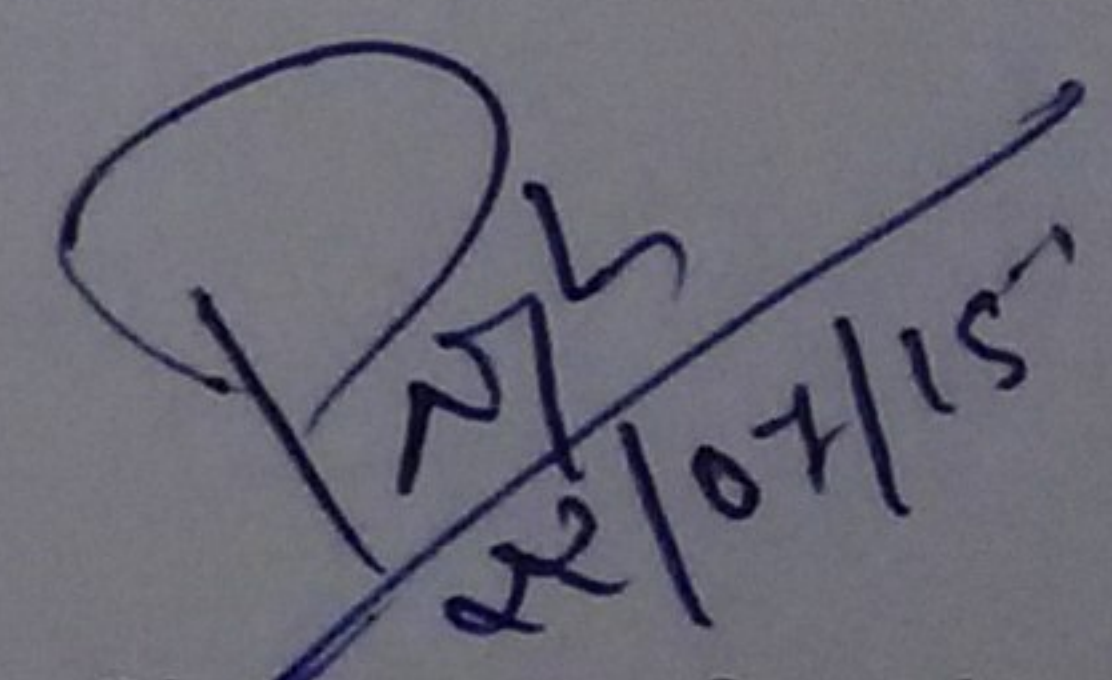
- Upmanyu, S., Gupta, S.K., Shyam, K.R. 2002. Innovative approaches for the management of root rot and web blight (*R. solani*) of French bean. *J. Mycol. Plant Pathol.*, 32: 317-331.
- Uppal, B. N., Kolhatkar, K. G. and Patel, M. K. 1936. Blight and hollow stem of sorghum. *Indian Journal of Agricultural Sciences*, 6: 1323-1334.
- USDA, 2005. National nutrient database for standard reference, release 18. U.S. Department of Agriculture, *Agric. Res. Ser.* Nutrient Data Laboratory.
- Usha Rani, S., Udayakumar, R. and John Christopher, D. 2009. Bio-efficacy of plant extracts and bio-control agents against *macrophomina phaseolina*. *Ann. Pl. Protec. Sci.*, 17(1): 389-393.
- Van der Maesen, L.J.G. 1987. *Cicer* L. Origin, history and taxonomy of chickpea. In: MC Saxena and KB Singh (ed.), *The Chickpea*. C.A.B. International Cambrian News Ltd, Aberystwyth, UK. p. 11-34.
- Varma, R.K. and Pathe, A. 2013. Morphological and cultural variability of *Rhizoctonia bataticola* responsible for charcoal rot of soybean. *JNKVV. Res. J.*, 47(1): 88-94.
- Veena, G.A., Reddy, N.P.E., Reddy, B.V.B. and Prasanthi, L. 2014. Potential of *Trichoderma Spp.* as biocontrol agents against *Rhizoctonia bataticola* causing dry root rot of chickpea. *International Journal of Plant, Animal and Environmental Science*, 4: 78-81.
- Vipul, G., Anil, S., Maisuria, V., Phadnis, A. and Chhatpar, H.S. 2006. Bio prospecting and antifungal potential of chitinolytic microorganisms, *African Journal of Biotech.*, 5: 54-72.
- Walker, J. C. and Snyder, W.C. 1933. Pea wilts and root rots: *Wis.Agr. Exptl.Sta.Bull.* 424.
- Walker, J.C. 1975. *Plant Pathology* 2nd edition (New York: McGraw book Co.). pp707.
- Weindling, R. 1934. Studies on a lethal principle effective in the parasitic action of *Trichoderma lignorum* on *Rhizoctonia solani* and other soil fungi. *Phytopathol.*, 24: 1153-1179.
- Weindling, R. 1941. Experimental consideration of the mold toxin of *Gliocladium* and *Trichoderma*. *Phytopathol.*, 31: 991-1003.

- Weindling, R. 1932. *Trichoderma lignorum* a parasite of other soil fungi. *Phytopathol.*, 22: 837-845.
- Weller, D.M., Raaijmakers, J.M., Gardener, B.B.M. and Thomashow, L.S. 2002. Microbial population responsible for specific soil suppressiveness to plant pathogens. *Annu. Rev. Phytopathol.*, 40: 309-348.
- Whitney, H.S. and Parmeter, J.R.J. 1963. Synthesis of heterokaryons in *Rhizoctonia solani*. Kuhn. *Can. J. Botany*, 41: 879-886.
- Williams, P.C. and Singh, U. 1987. The chickpea nutritional quality and evaluation of quality in breeding programs. In: Saxena MC, and Singh KB. (eds), *The Chickpea*, CAB Int., Walling-ford, P. 329-356.
- Windels, C.E. and Nabben, D.J. 1989. Characterization and pathogenicity of anastomosis Groups of *Rhizoctonia Solani* isolated from *Betavulgaris*. *Phytopathology*, 79: 83-88.
- Wood R. K. S. 1965. *Physiological plant pathology*. Black well scientific publication, Oxford and Edinburg, p. 570.
- Wyllie, T., and Calvert, O. 1969. Effect of flower removal and pod set on formation of scerotia and infection of *Glycine max* by *Macrophomina phaseolina*. *Phytopathology*, 59: 1243.
- Xiang, X. and Morris, N.R. 1999. Hyphal tip growth and nuclear migration. *Curr. Opin. Microbiol.*, 2: 636-40.
- Yang, C.Y. 1977. Mungbean disease and control. First international Mung Symposium AVRDC, Taiwan. p. 141-146.

The author of this thesis *Ms. Pushpanjali Wagh*, D/o Mr. *Sudhir Wagh* was born on 16th May, 1987 at Durg, district- Durg (C.G.). She passed his high school (10th) in the year 2003 from St. Xavier High School, Durg (C.G.) with 52.7% marks and higher secondary in the year 2005 from St. Xavier High School, Durg (C.G.) with 61.9% marks.

She successfully passed B.Sc. (Agriculture) in the year 2009 from Bhartiya College of Agriculture, Durg (C.G.), affiliated to Indira Gandhi Krishi Vishwa Vidyalaya, Raipur (C.G.) with 74.7 per cent marks and post graduation studies from the Department of Plant Pathology, College of Agriculture, Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur (M.P.).

The author thereafter, carried her Ph.D. (Doctor of Philosophy) from the Department of Plant Pathology, College of Agriculture, Indira Gandhi Krishi Vishwavidyalaya, Raipur (C. G.) This thesis is a bonafide research being submitted by her as a partial fulfillment for the award of Ph.D. degree.


Signature of student